

Machine Learning

Dmytro Fishman (dmytro.fishman@ut.ee)



UNIVERSITY OF TARTU
Institute of Computer Science

BIIT

September 6, 2021

INDUSTRIAL AI APPLICATIONS

HEALTHCARE

Caption Health theater insitro
Recursion OVERJET OWKIN
UNLEARN Olive

SMART HOME

ORIGIN

CONSUMER DEVICES

Audio Analytic
FRITZ.AI

RETAIL & CPG

MSIGHT AIFI
syte Vue.ai

MEDIA

descript

SUPPLY CHAIN & LOGISTICS

covariant OSARO
INCEPTIO Outrider

WASTE MANAGEMENT

AMP Robotics
greyparrot

FINANCE & INSURANCE

zesty.ai AKUR8 TRACTABLE

FOOD & AGRICULTURE

prospera
Aquabyte
BEEWISE

DEFENSE

DEEPSIG

EDUCATION

ELSA
Riiid

LEGAL

LegalForce
lexion

TRANSPORTATION

parallel domain Aurora algolux
KONUX DEEPMAP GHOST
momenta

MANUFACTURING

drishti
LANDING AI

CONSTRUCTION

OPENSOURCE

Mining

KoBold Metals

GAMING

mod.ai

ENERGY

BRAINBOX AI
myst.ai

AI 100 2021

CBINSIGHTS

COMPUTING, DATA PROCESSING, AND AI DEPLOYMENT

AI PROCESSORS

Horizon Robotics Tenstorrent bloize
SYNTIAN GRAPHCORE

DATA SCIENCE PLATFORMS

dataiku

DEEP LEARNING ACCELERATORS

deci run:ai LatentAI

AIOPS (IT & DEVOPS AUTOMATION)

harness snyk
StormForge

DATA ANNOTATION

ALBANY scale
SuperAnnotate

FEATURE STORES & MLOPS PLATFORMS

ALGORITHMIA OF ONEFLOW
SELDON TACTON
SPELL DARWIN AI

ML EXPERIMENT TRACKING

Weights & Biases neptune.ai

AI MODEL MONITORING

fiddler Arthur

CROSS-INDUSTRY AI APPLICATIONS

MACHINE TRANSLATION

LILT

NLP & CONVERSATIONAL AI

Hugging Face RASA

DOCUMENT ANALYSIS

(h[s]) HYPERSCIENCE ROSSUM Eigen Technologies
cinnamon AI

CLIMATE RISK SCORING

JUPITER

SPEECH RECOGNITION

DEEPRGRAM

ENTERPRISE SEARCH

coveo jine

SALES & CRM

Clari tact.ai
循环智能 Recurrent
PolyAI
CRESTA

COMPUTER VISION

Matroid

REMOTE INSPECTION

PERCEPTO

CYBERSECURITY

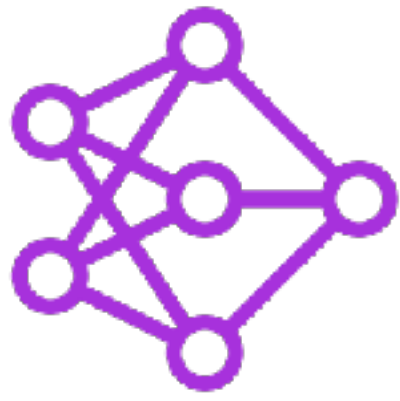
INKY ROBUST INTELLIGENCE
SentinelOne BLUE HEXAGON
securiti

WATER LEAK DETECTION

wint

OTHER R&D

PRYON
COGENT LABS
EXAWIZARDS
InstaDeep

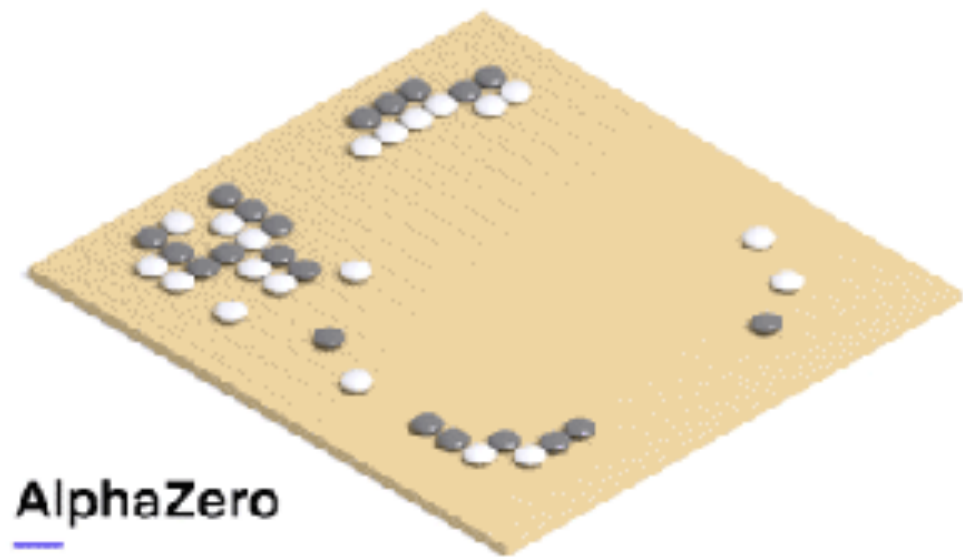


DeepIndex

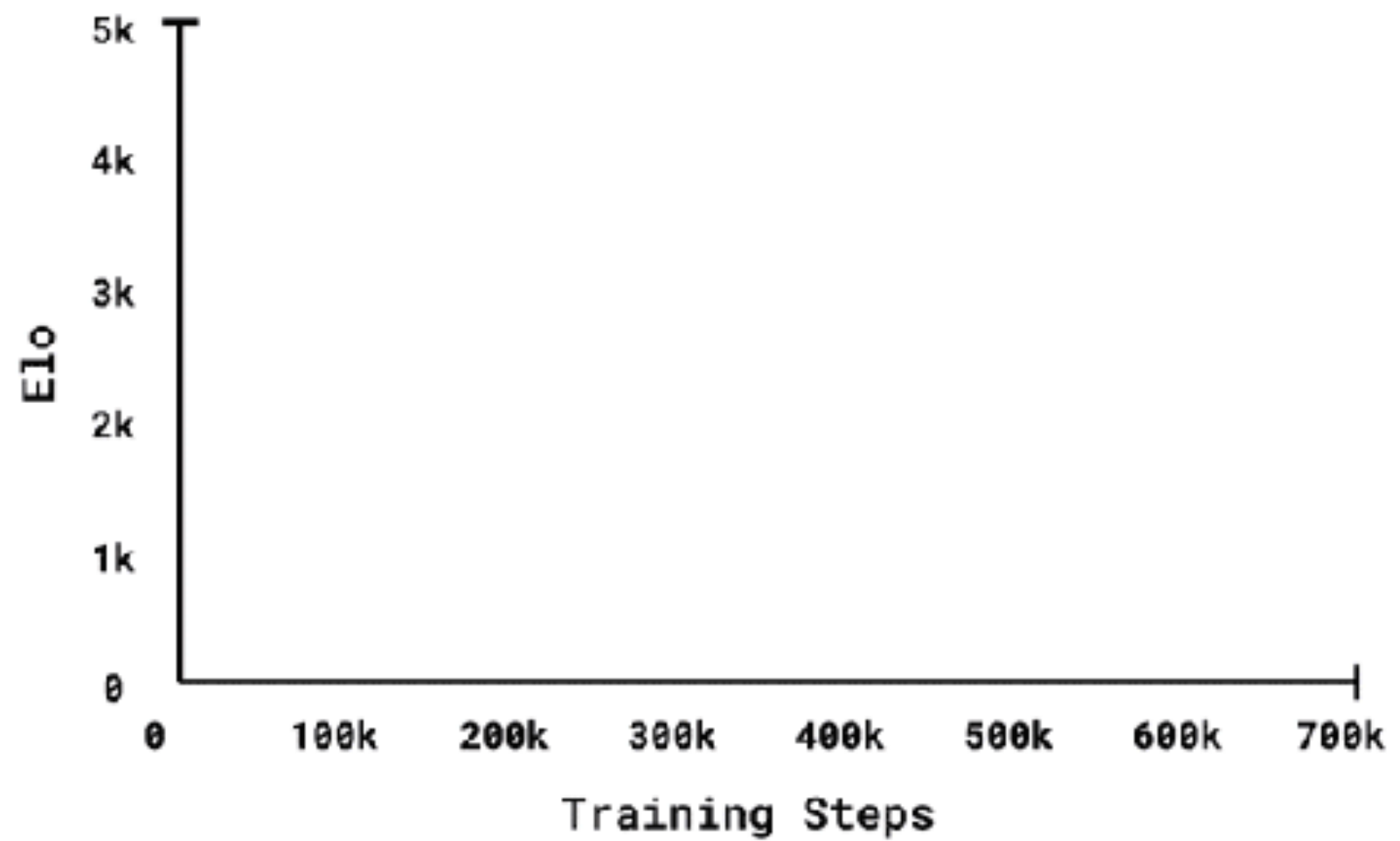
Games	Professional	Industrial	Government & nonprofit	Science	Speech & language
Creative	Finance	Agriculture	Education	Security	Vision
Home & lifestyle	Admin	Transport	Medicine	Computing	Robotics

Find examples of AI in action at <https://deepindex.org/>

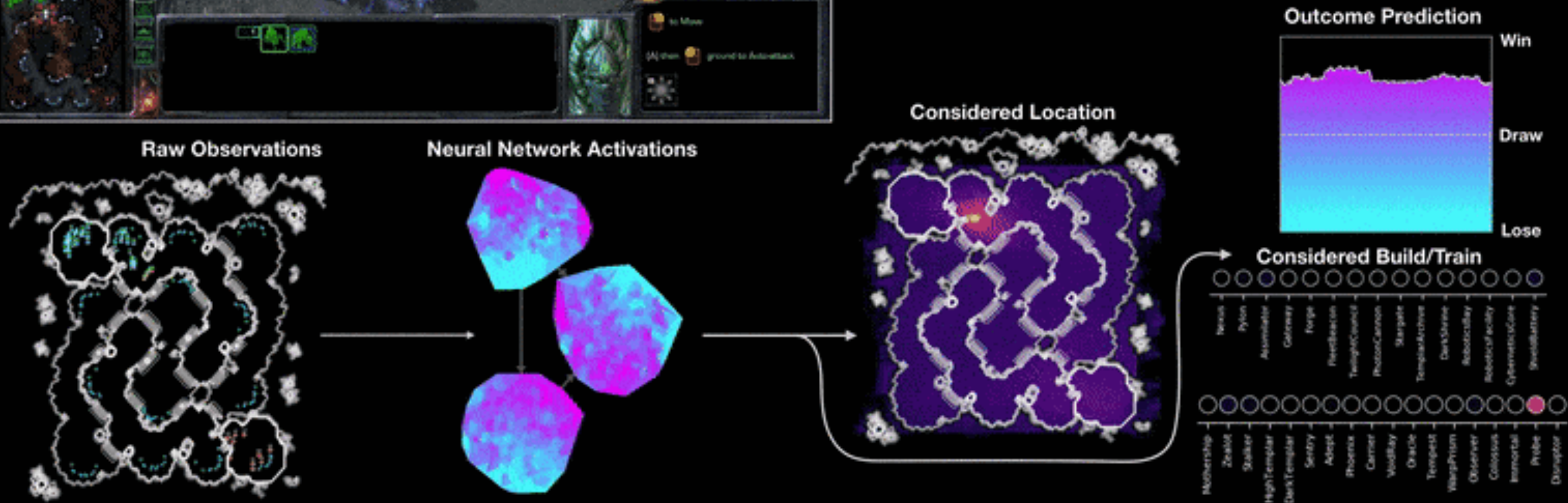
AlphaZero



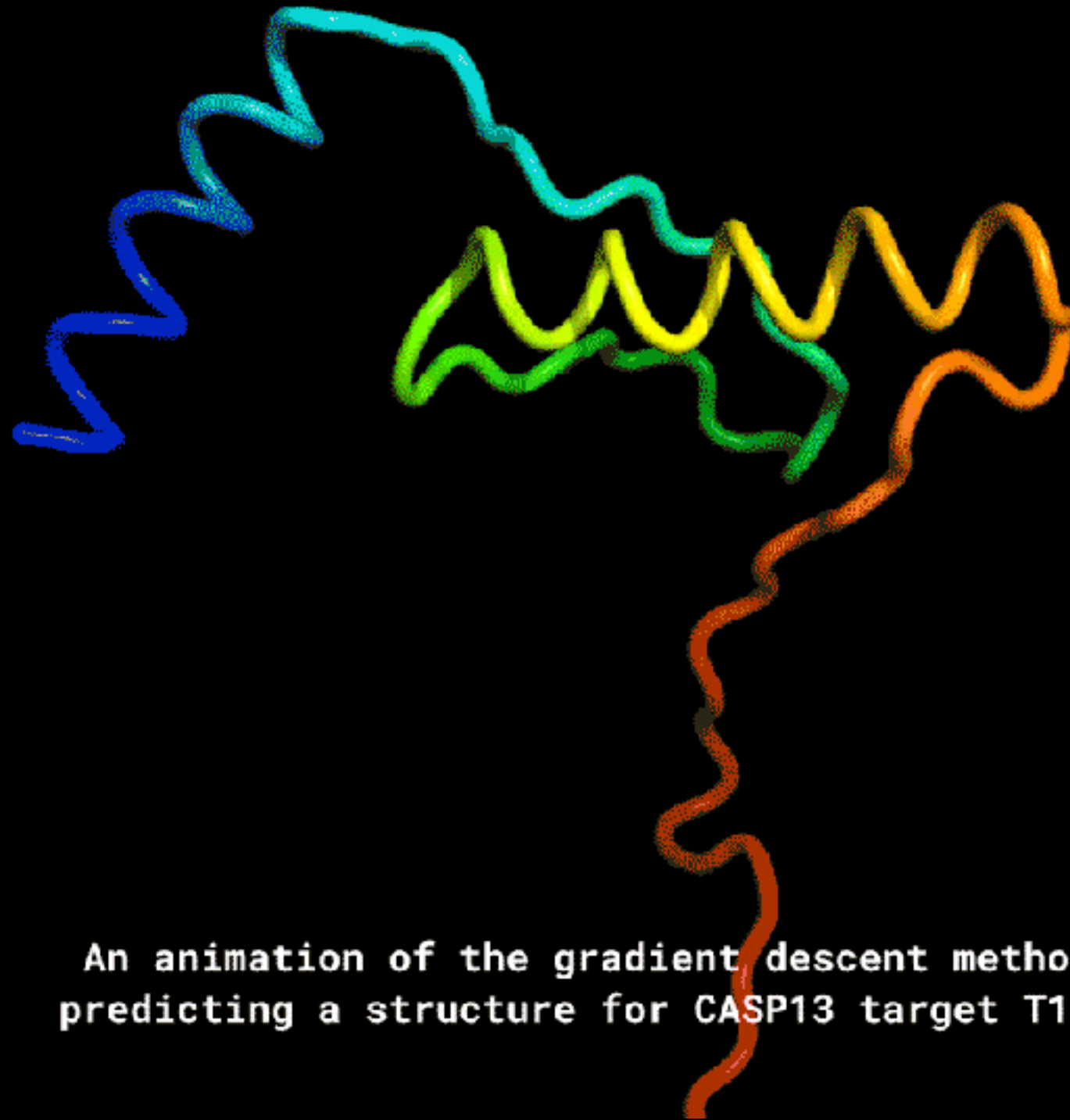
AlphaZero



AlphaStar



AlphaFolding



An animation of the gradient descent method
predicting a structure for CASP13 target T1008

Diabetic Retinopathy



NORMAL VISION
Vision remains intact

Diabetic Retinopathy



NORMAL VISION
Vision remains intact



DIABETIC RETINOPATHY
Vision is obstructed by macular edema

Diabetic Retinopathy



NORMAL
Vision

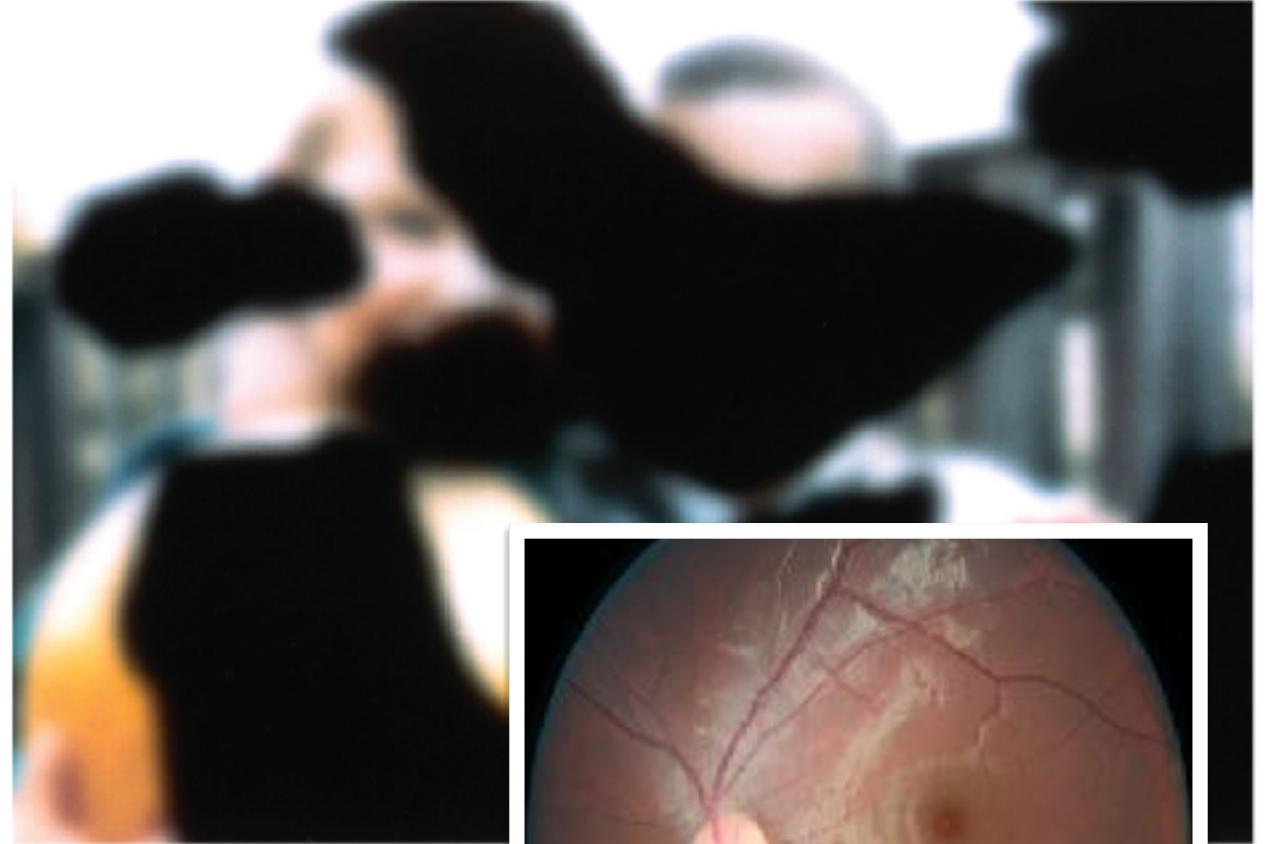


DIABETIC RETINOPATHY
Vision is obstructed by macular edema

Diabetic Retinopathy



NOR
Vision



DIABE
Vision is obstr



Living Portraits





Which face is real?

<http://www.whichfaceisreal.com>



Which face is real?

<http://www.whichfaceisreal.com>



Towards a Human-like Open-Domain Chatbot

Common sense:

Q: What is your favorite animal?

A: My favorite animal is a dog.

Q: Why?

A: Because dogs are loyal and friendly.

Q: What are two reasons that a dog might be in a bad mood?

A: Two reasons that a dog might be in a bad mood are if it is hungry or if it is hot.

Q: How many eyes does a giraffe have?

A: A giraffe has two eyes.

Q: How many legs does a frog have?

A: A frog has four legs.

Q: Are there any animals with three legs?

A: No, there are no animals with three legs.

Q: Why don't animals have three legs?

A: Animals don't have three legs because they would fall over.

Non sense:

Q: How do you sporgle a morgle?

A: You sporgle a morgle by using a sporgle.

Q: How many bonks are in a quoit?

A: There are three bonks in a quoit.

Q: How many rainbows does it take to jump from Hawaii to seventeen?

A: It takes two rainbows to jump from Hawaii to seventeen.

Q: Which colorless green ideas sleep furiously?

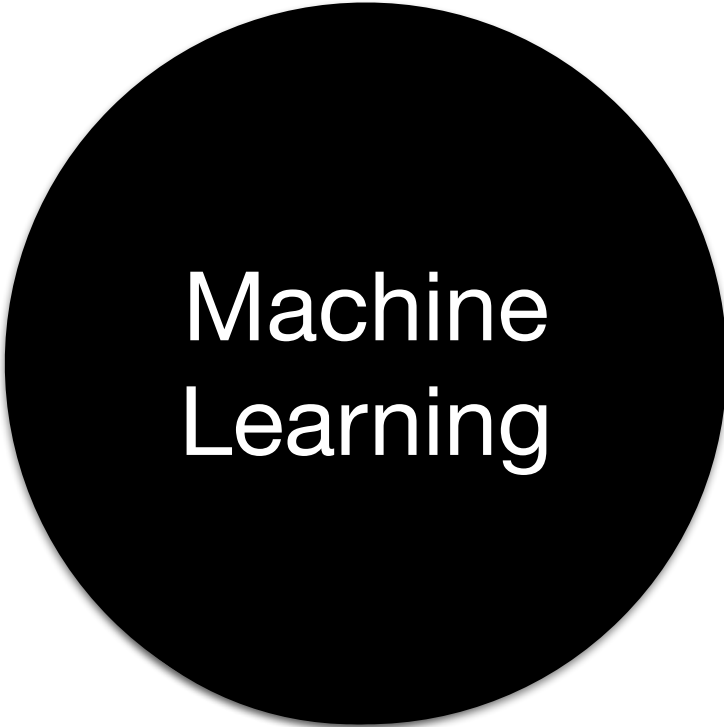
A: Ideas that are colorless, green, and sleep furiously are the ideas of a sleep furiously.

Q: Do you understand these questions?

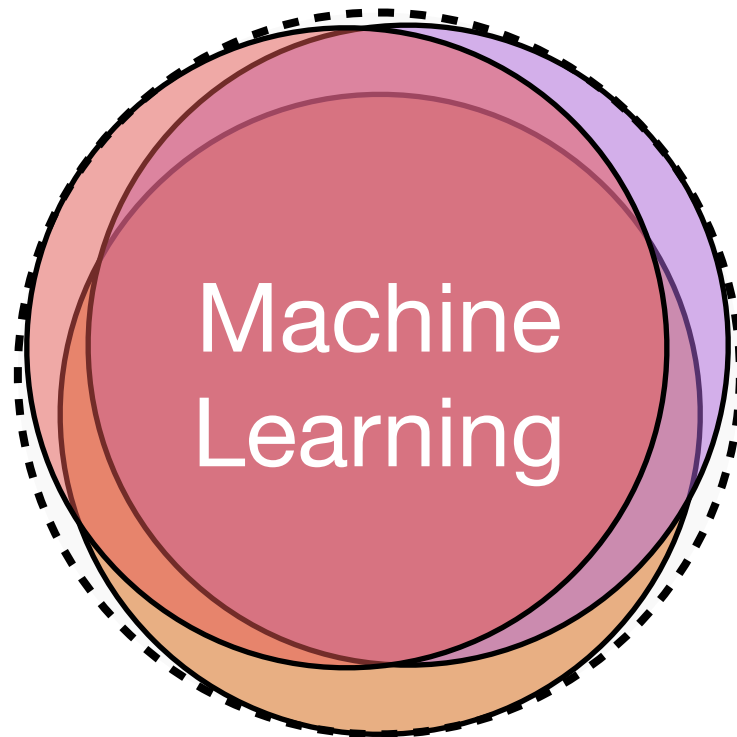
A: I understand these questions.

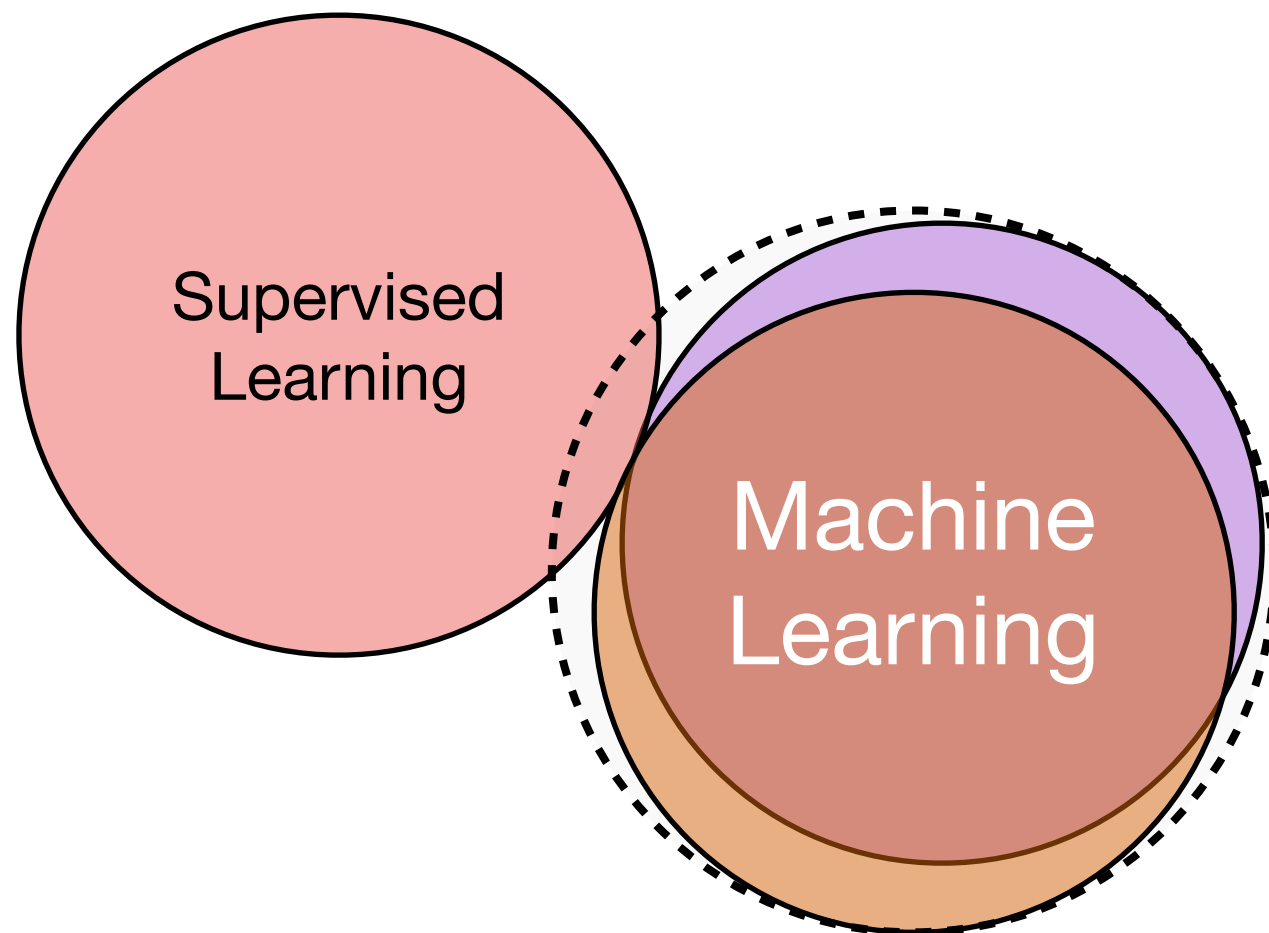


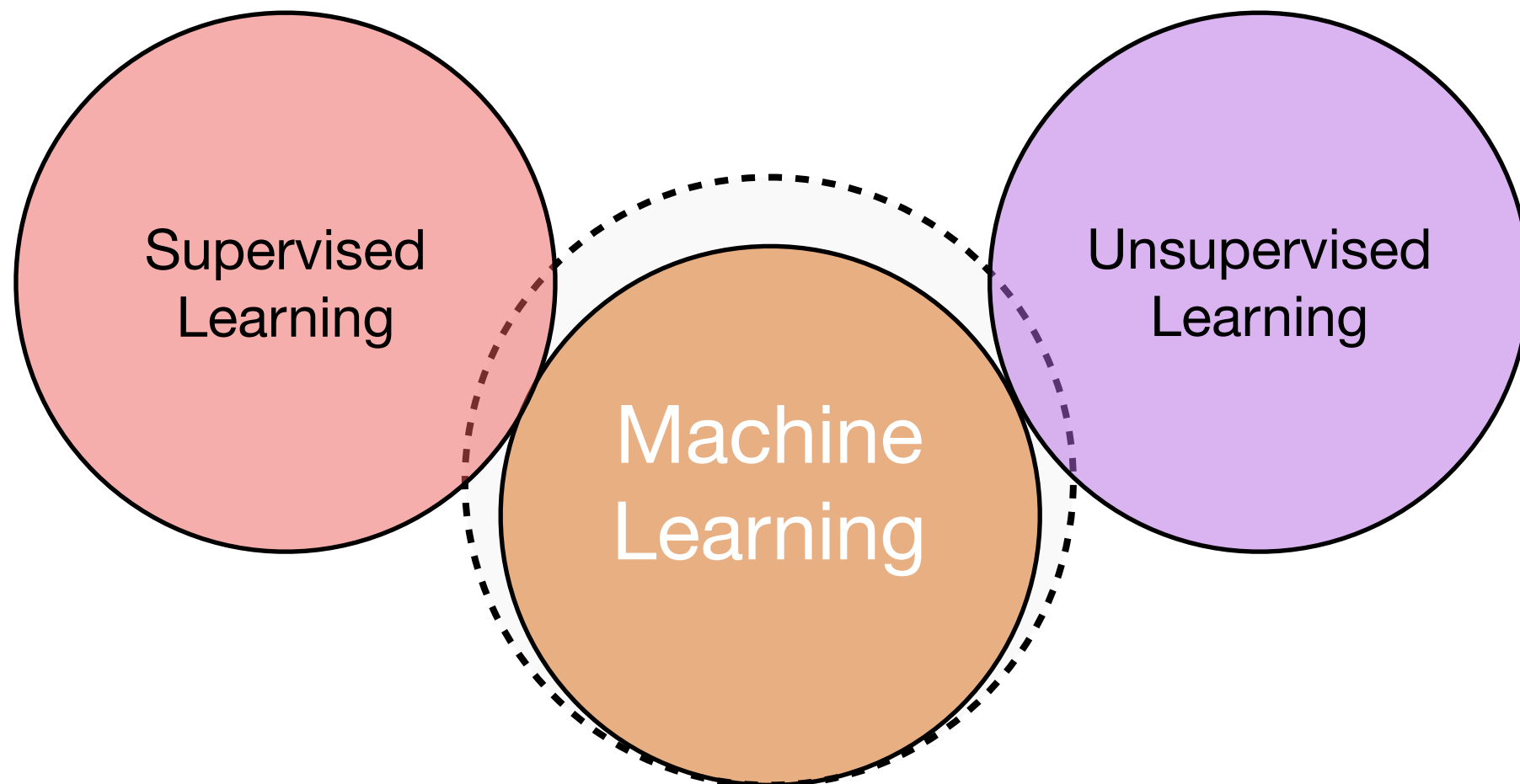
<https://www.synthesia.io/>

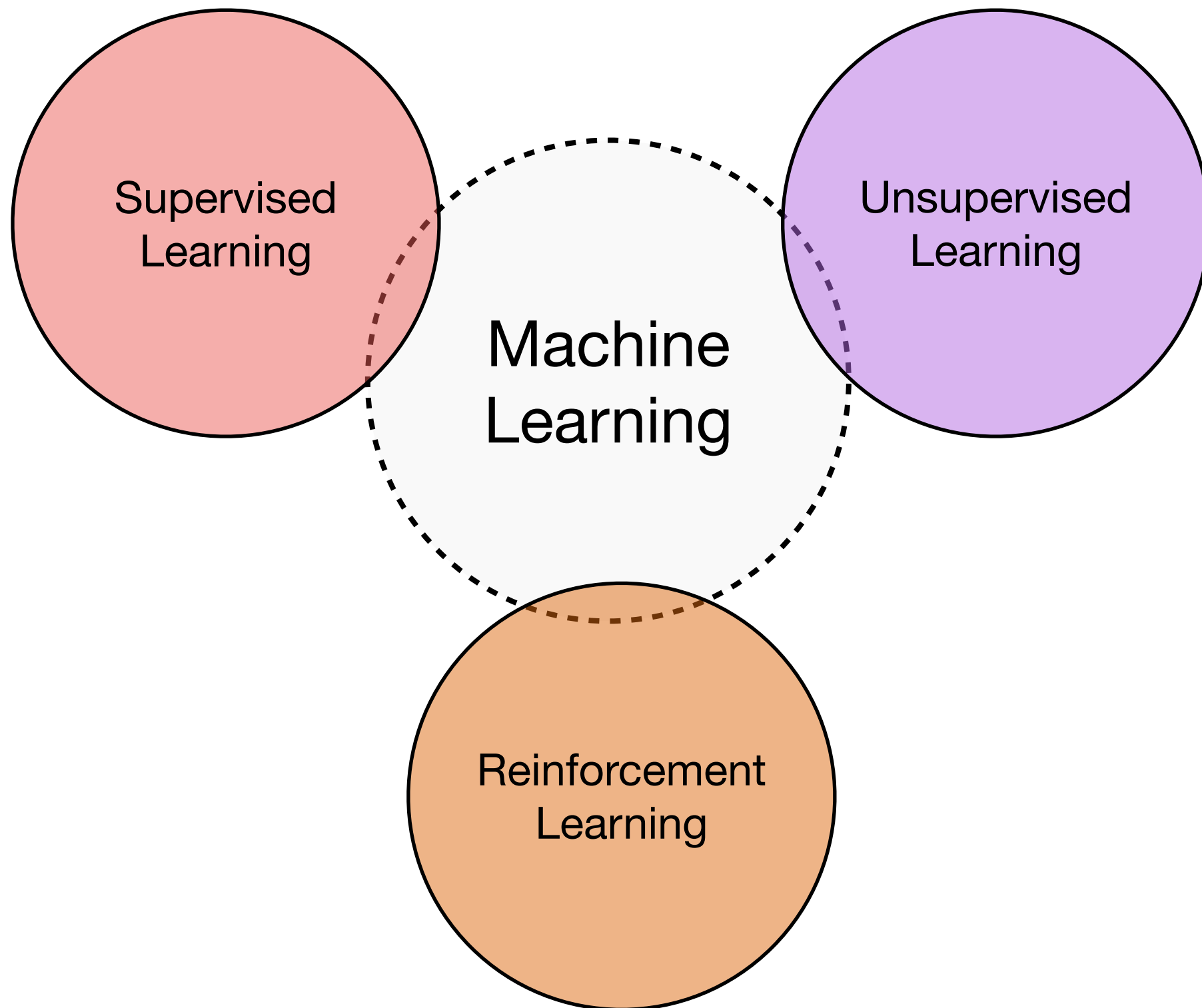


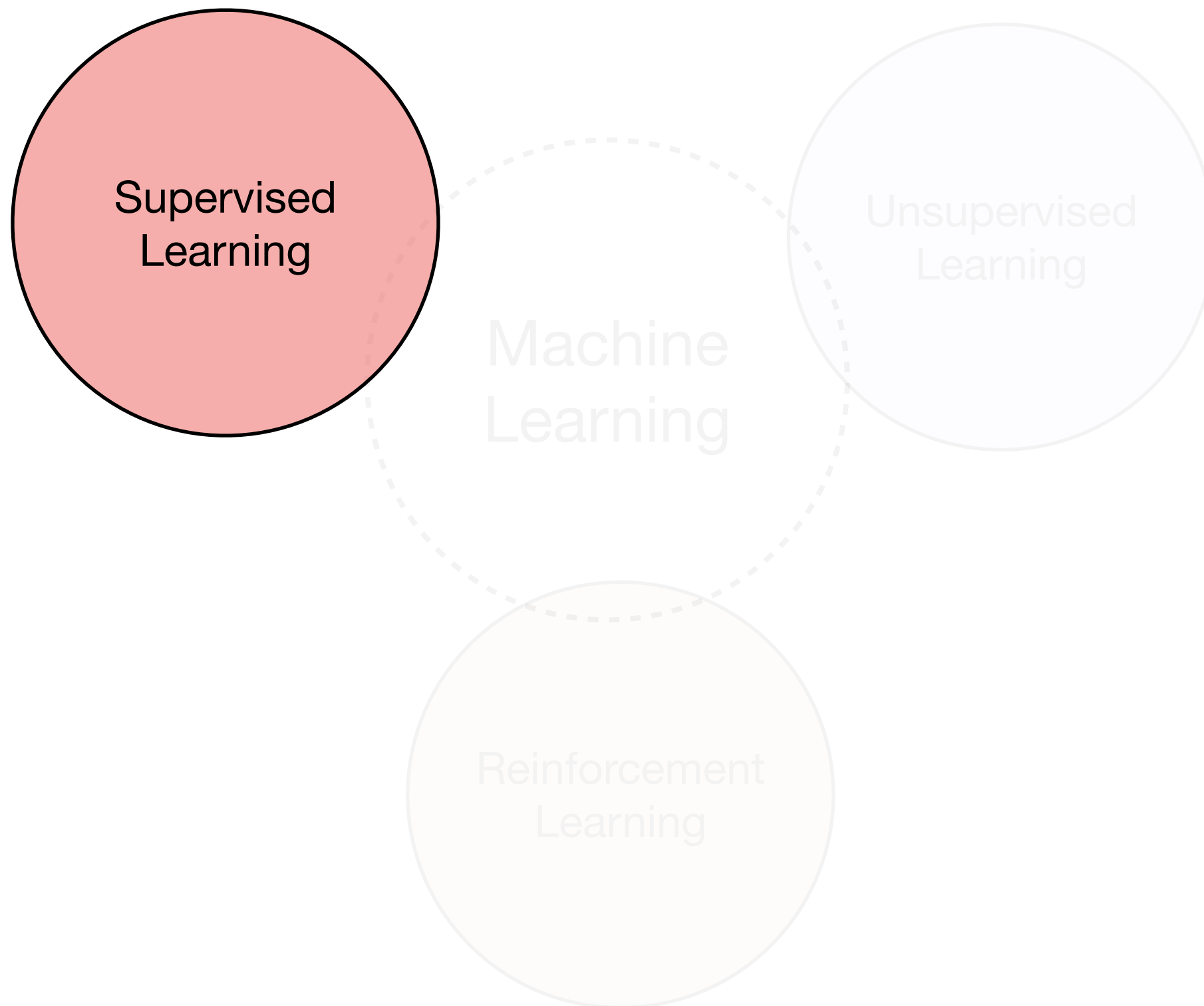
Machine
Learning

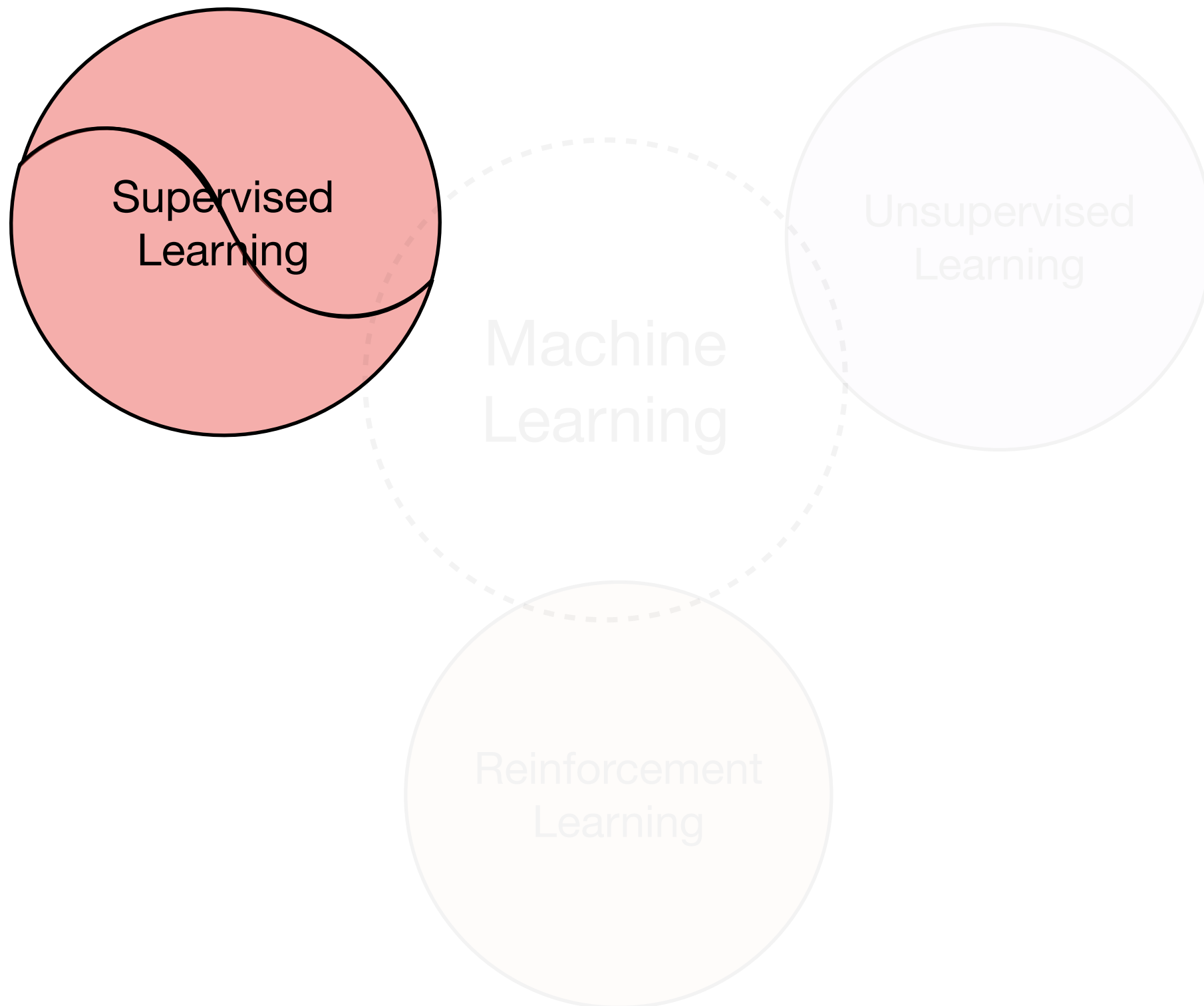




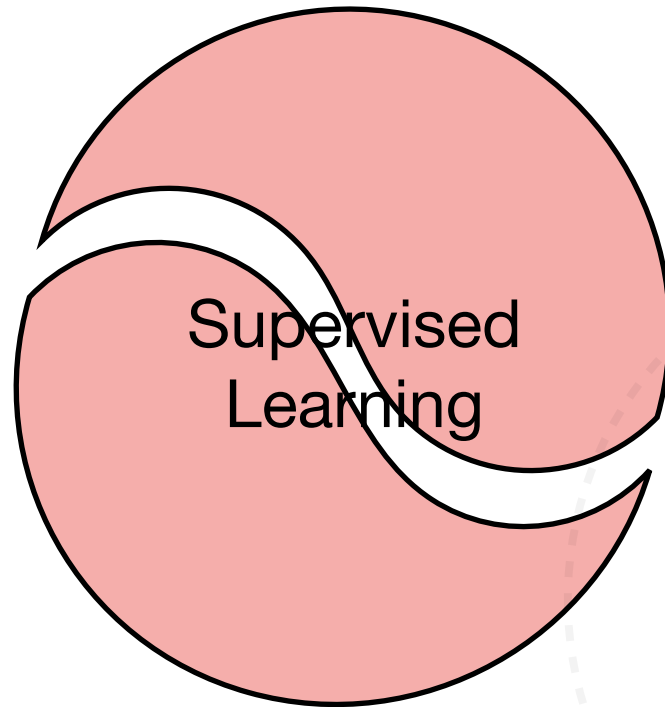








Classification



Regression

Machine
Learning

Unsupervised
Learning

Reinforcement
Learning

Classification

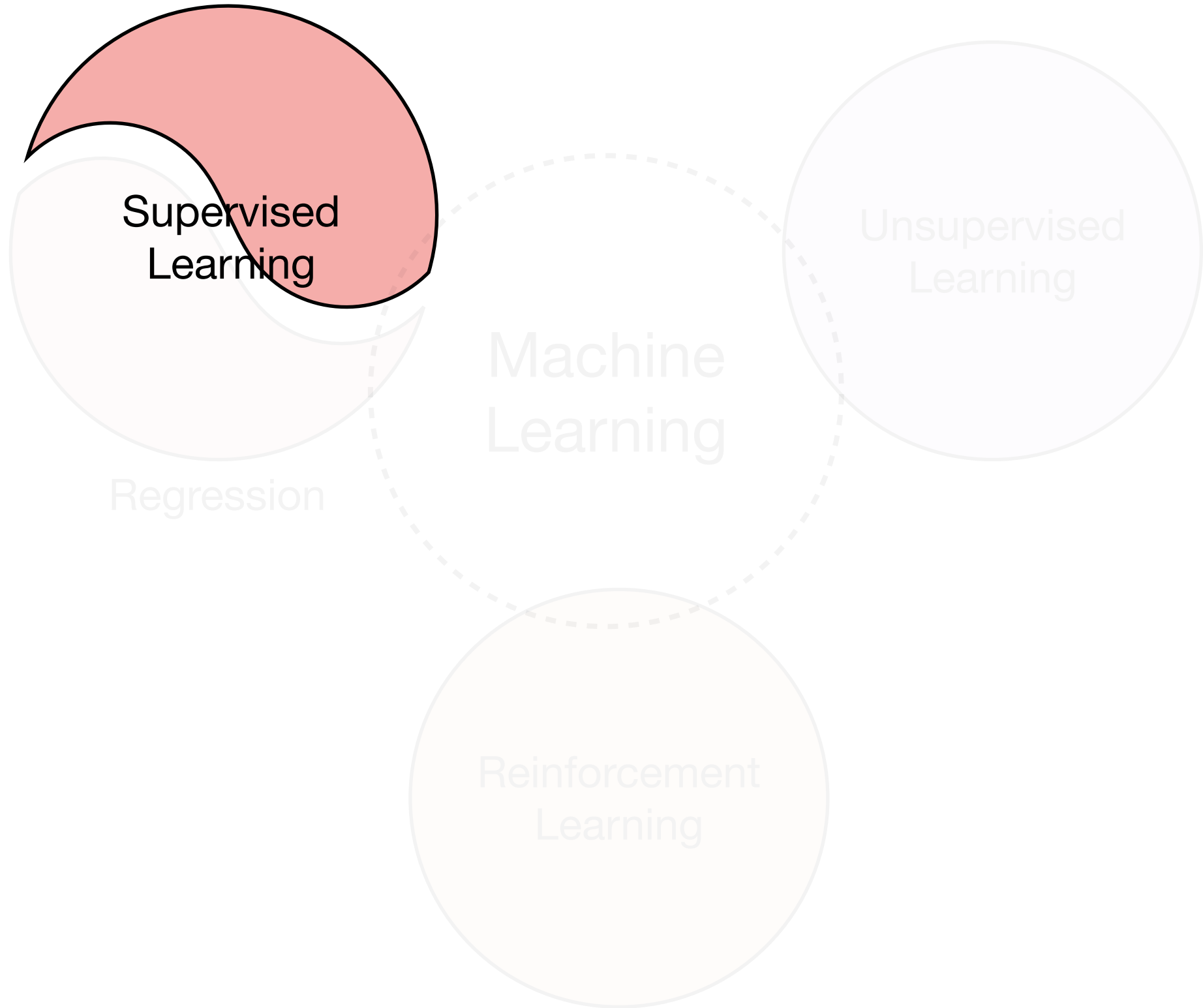
Supervised
Learning

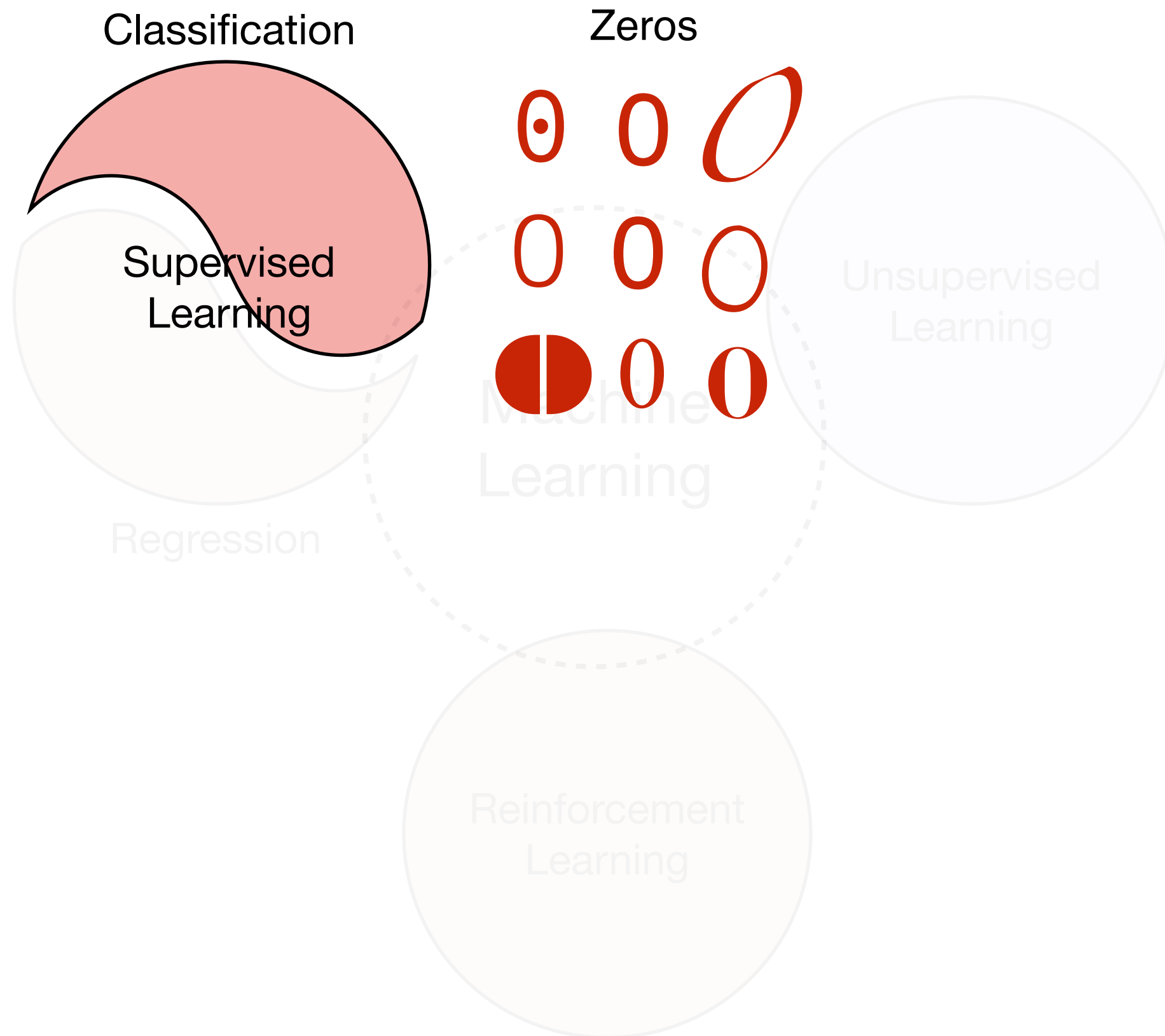
Regression

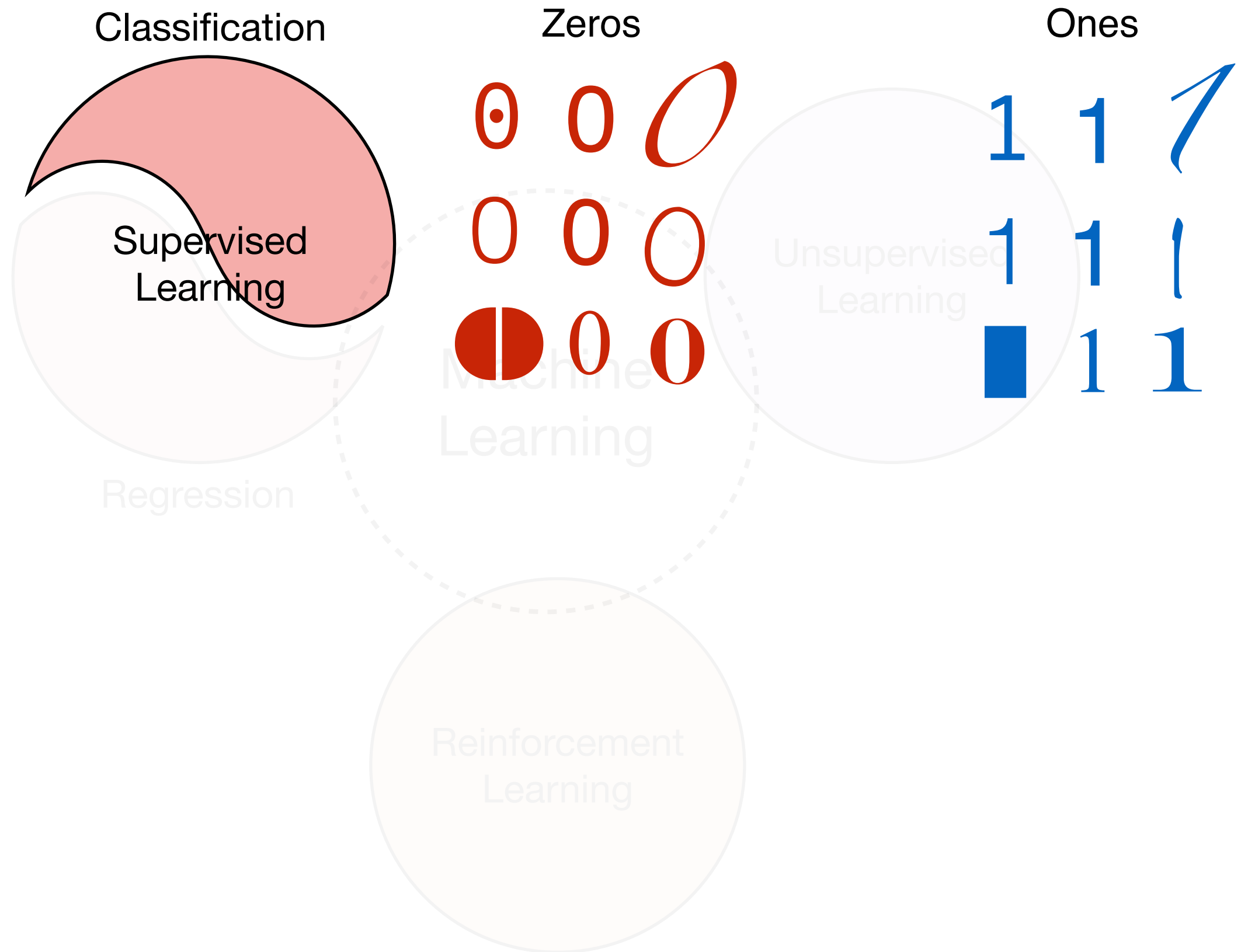
Machine
Learning

Unsupervised
Learning

Reinforcement
Learning



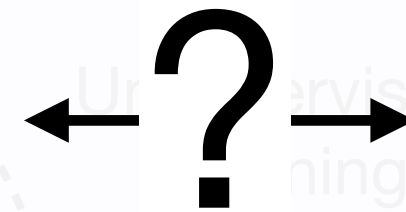
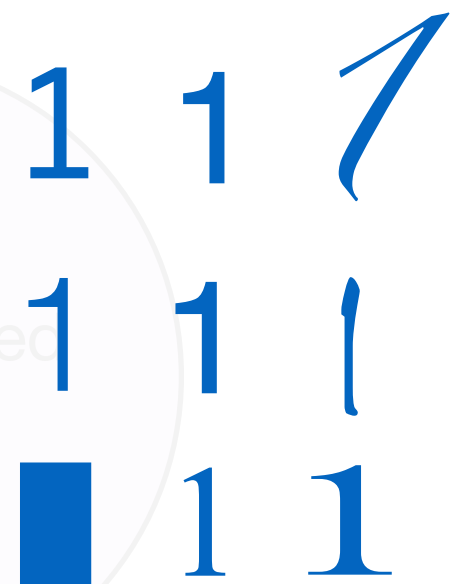
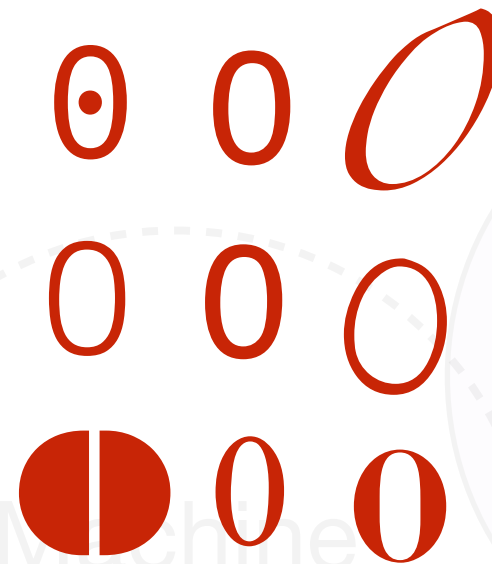
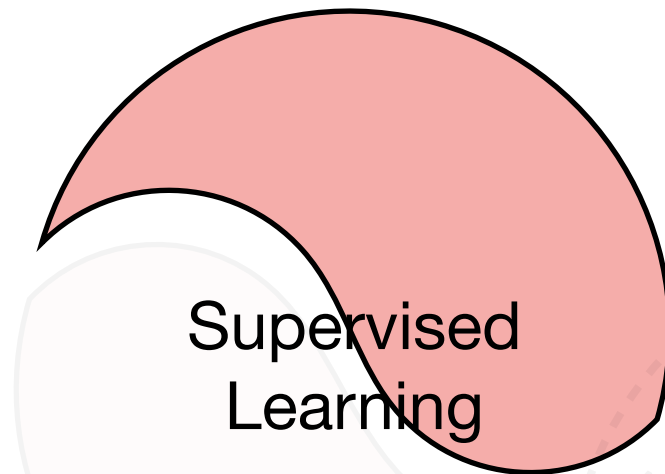




Classification

Zeros

Ones



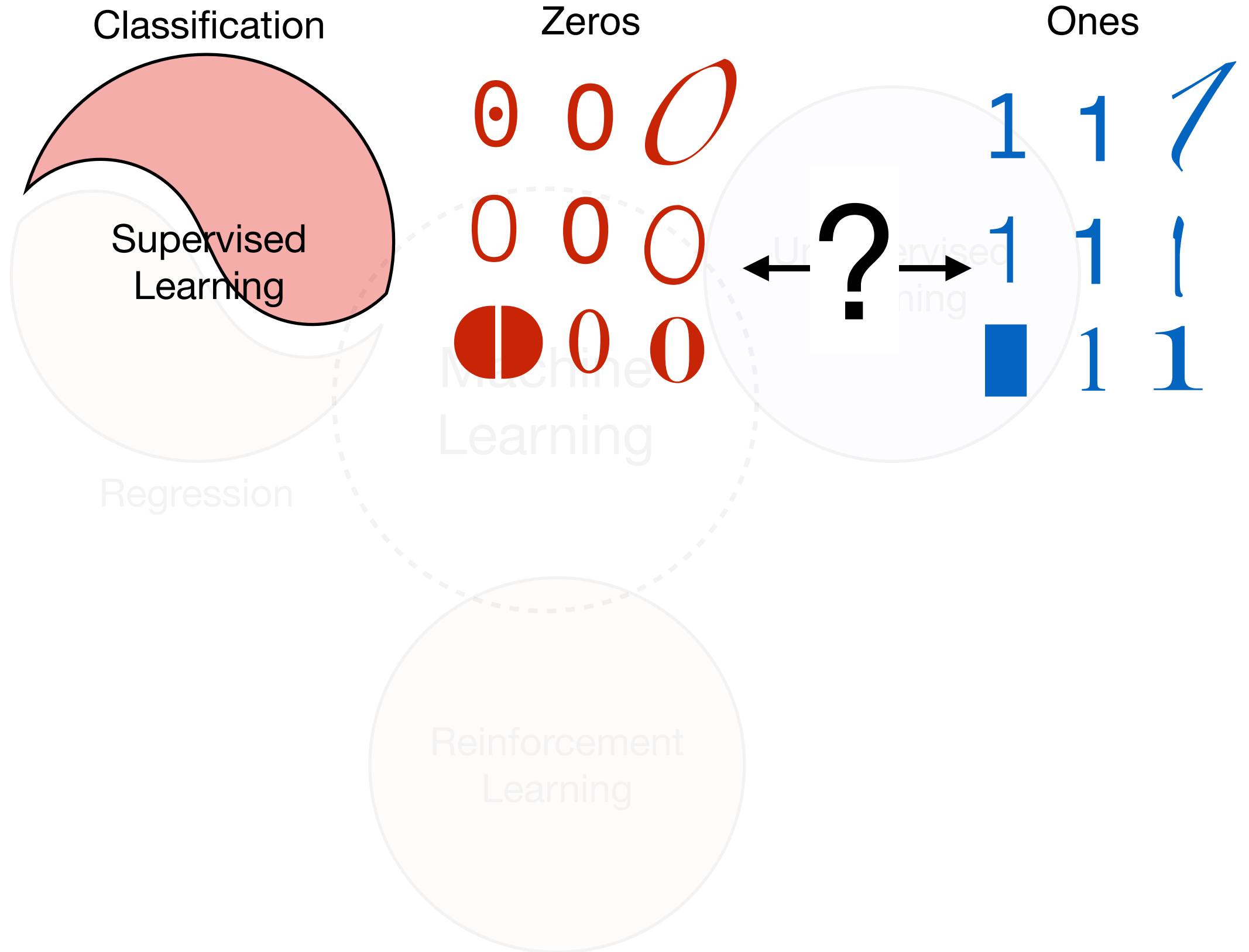
Supervised
Learning

Regression

Machine
Learning

Reinforcement
Learning

Binary Classification

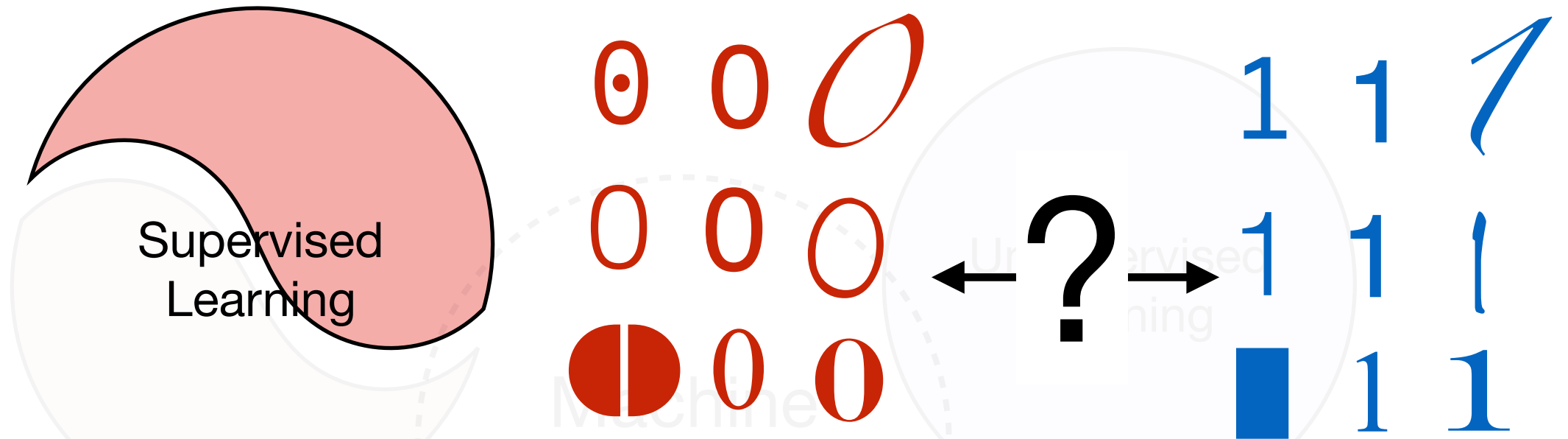


Binary Classification

Classification

Zeros

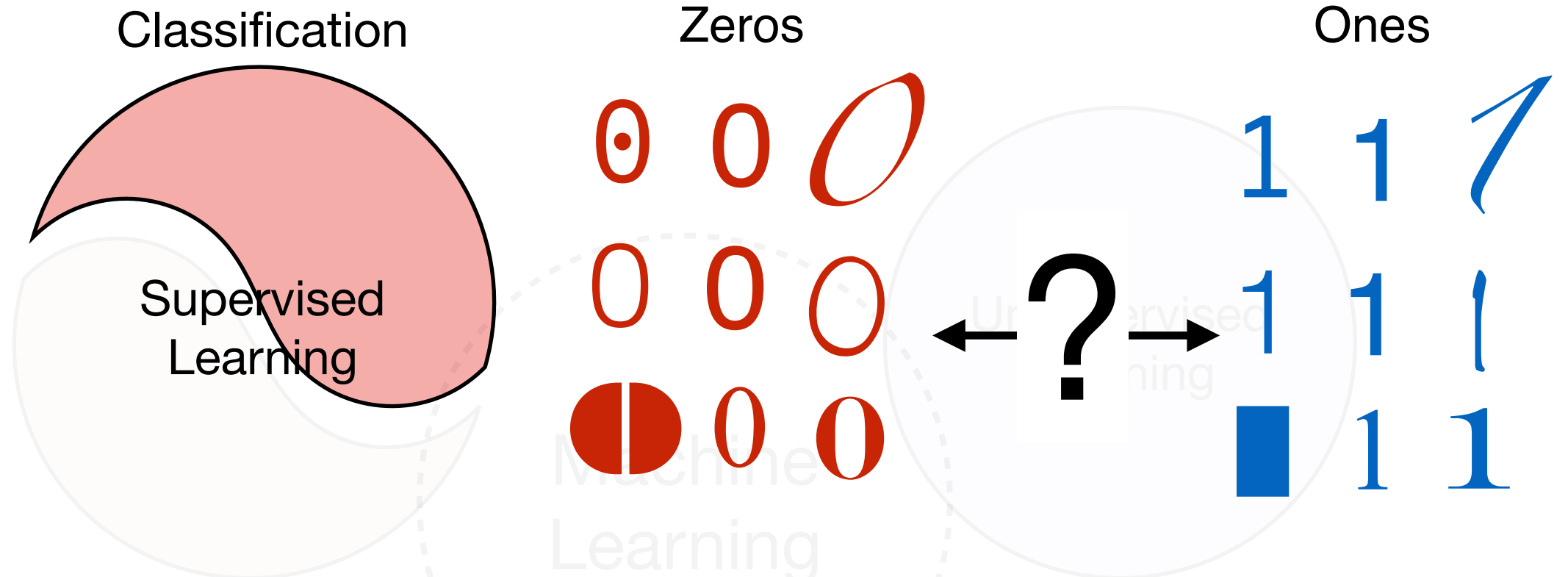
Ones



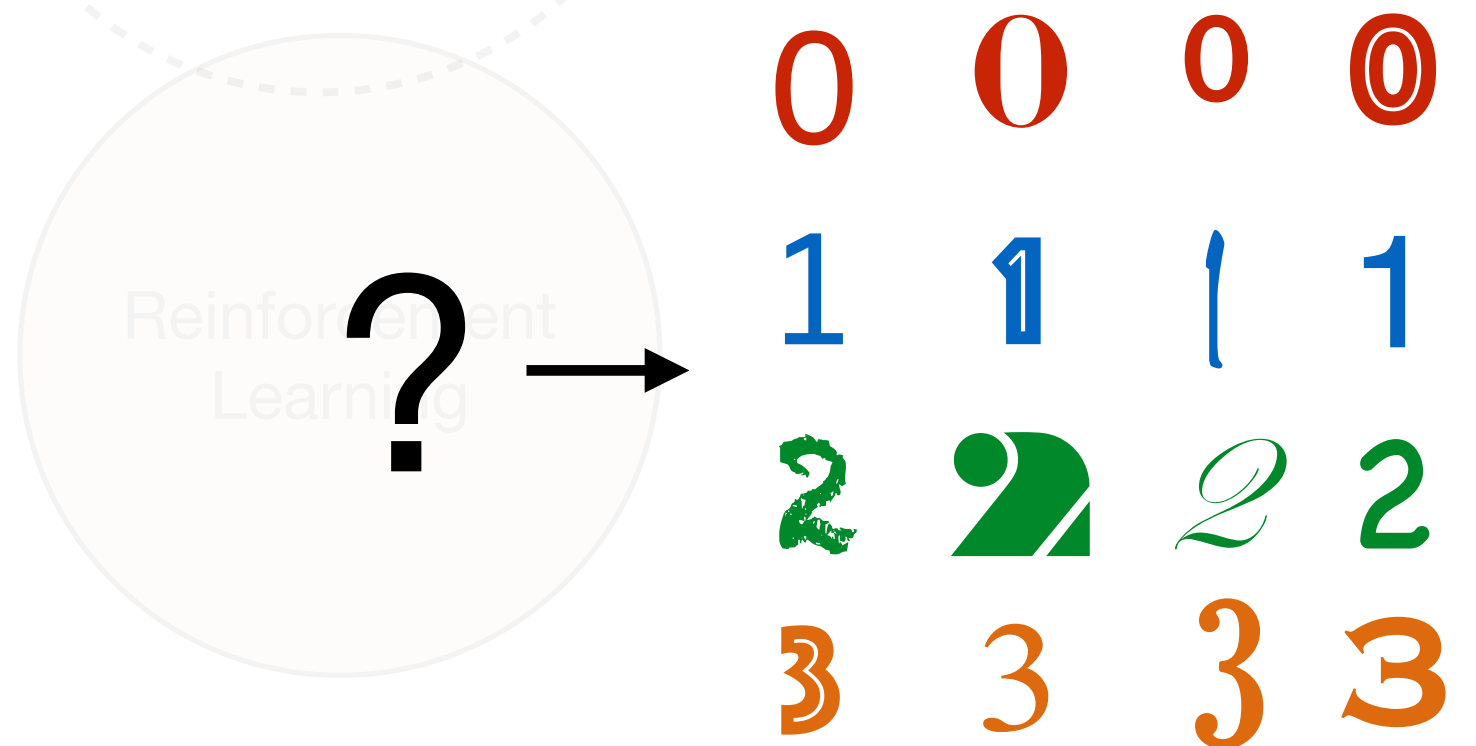
Multiclass Classification

Reinforcement Learning

Binary Classification

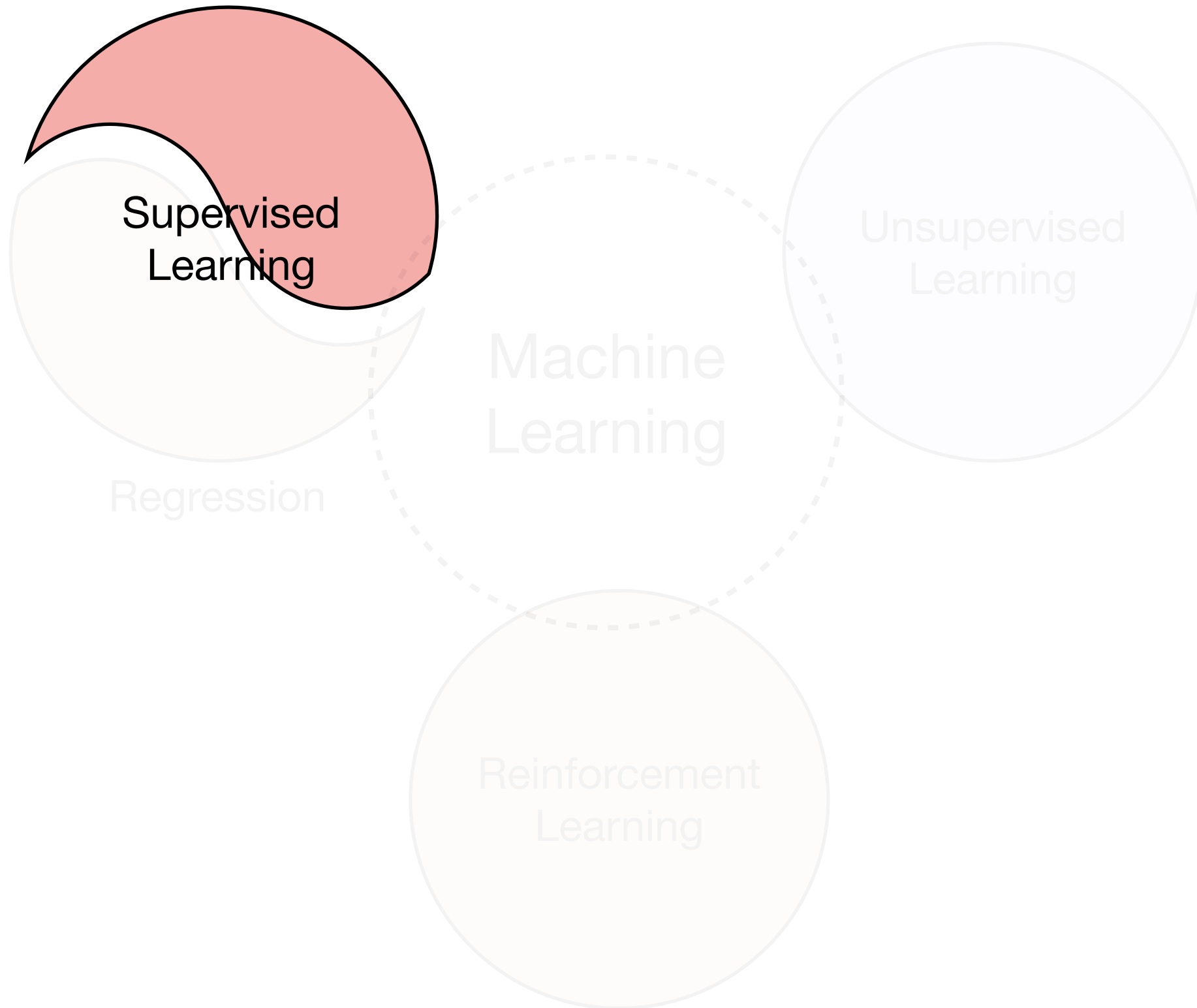


Multiclass Classification



0 ← ? → 1

Classification



0 $\longleftrightarrow$? $\longleftrightarrow$ 1

Classification

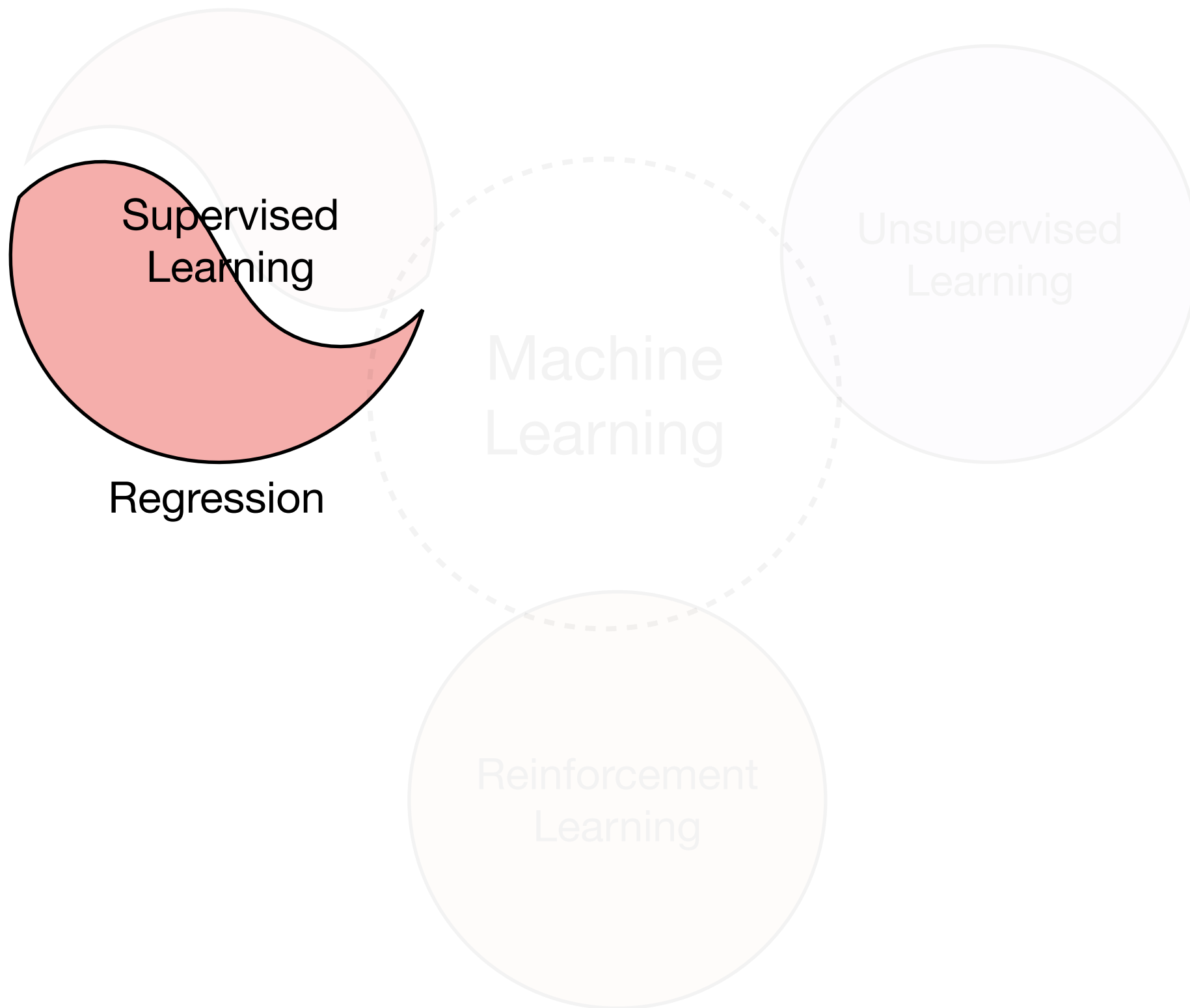
Supervised
Learning

Regression

Machine
Learning

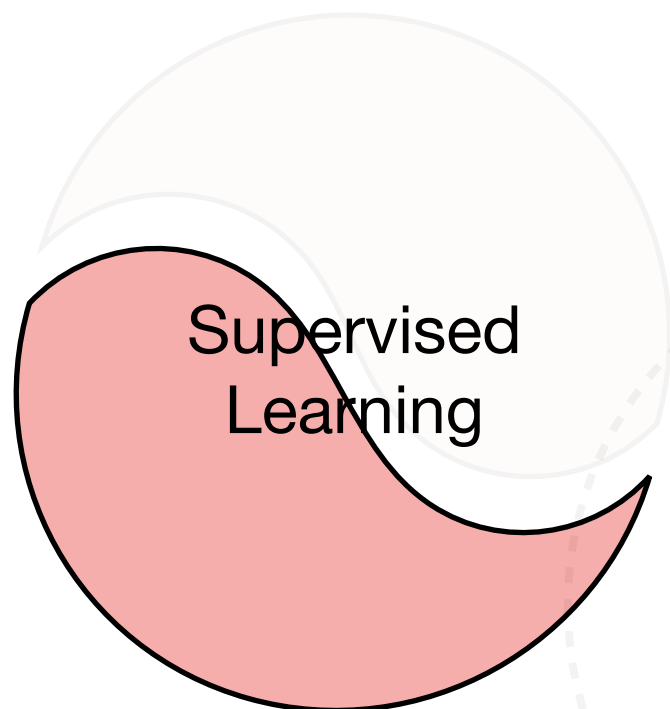
Unsupervised
Learning

Reinforcement
Learning



0 $\longleftrightarrow$? $\longleftrightarrow$ 1

Classification



Supervised
Learning

Regression

Machine
Learning

Unsupervised
Learning

Reinforcement
Learning



0 $\longleftrightarrow$? $\rightarrow$ 1

Classification

Supervised
Learning

Regression

Unsupervised
Learning

Machine
Learning

Reinforcement
Learning



5\$



0 $\longleftrightarrow$? $\rightarrow$ 1

Classification

Supervised
Learning

Regression

Unsupervised
Learning

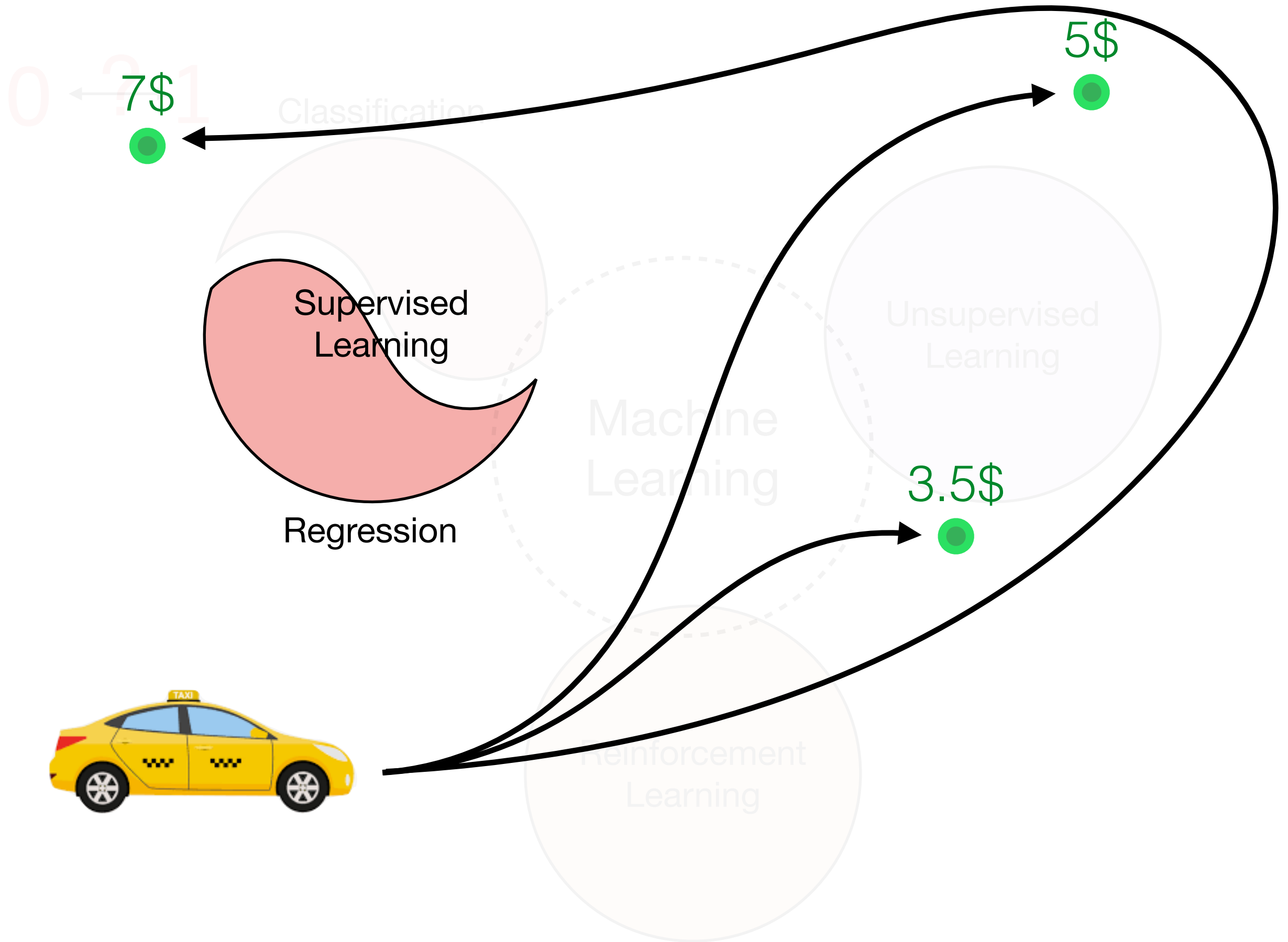
Machine
Learning

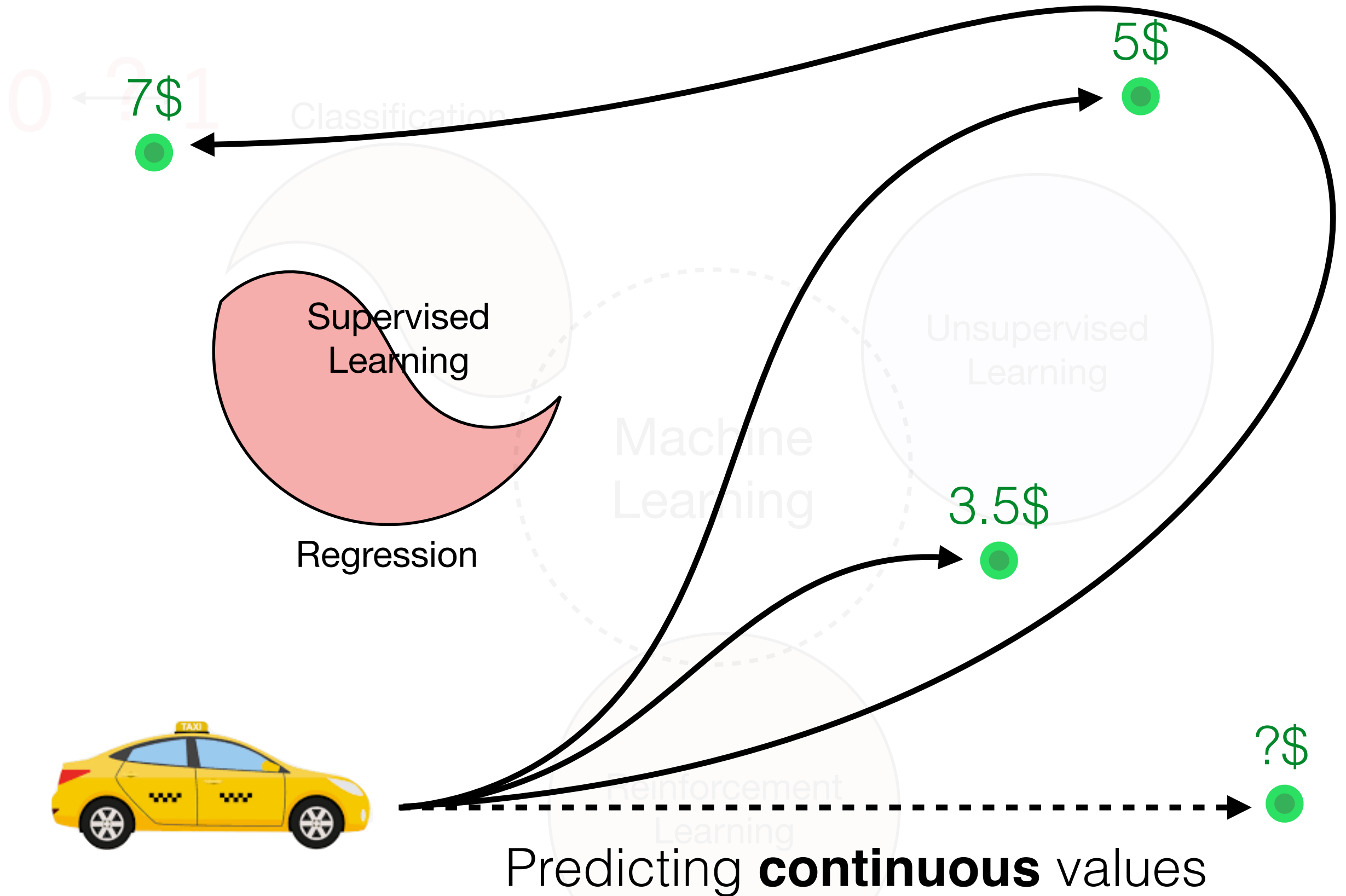
Reinforcement
Learning

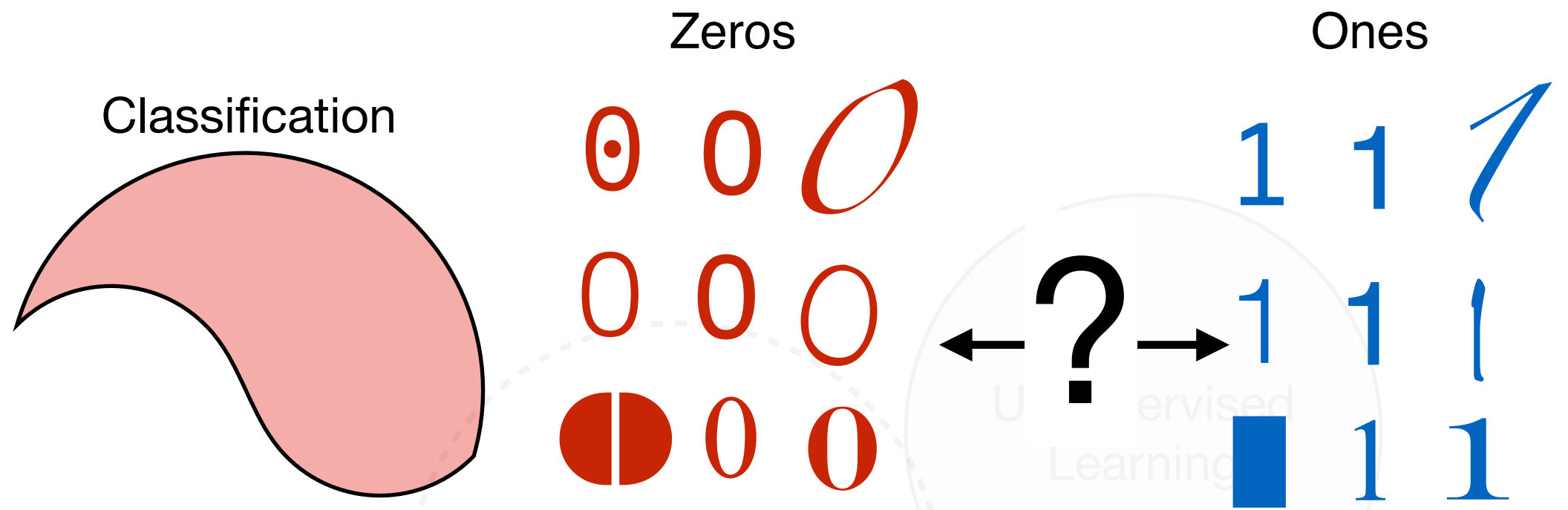


5\$

3.5\$







What is the main similarity/difference between these two classes of **supervised learning**?



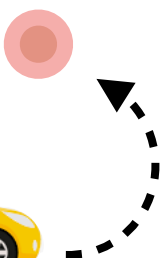
0 ← ? → 1

Classification

Supervised
Learning

Regression

?\$



Machine
Learning

Unsupervised
Learning

Reinforcement
Learning

0 ← ? → 1

Classification

Supervised
Learning

Regression

Unsupervised
Learning

Machine
Learning

Reinforcement
Learning

?\$



0 $\longleftrightarrow$? 1

Classification

Supervised
Learning

Regression

Machine
Learning

Unsupervised
Learning

Reinforcement
Learning



0 $\longleftrightarrow$? 1

Classification

Supervised
Learning

Regression

Machine
Learning

Unsupervised
Learning

Reinforcement
Learning



0 $\longleftrightarrow$? 1

Classification

Supervised
Learning

Regression

Machine
Learning

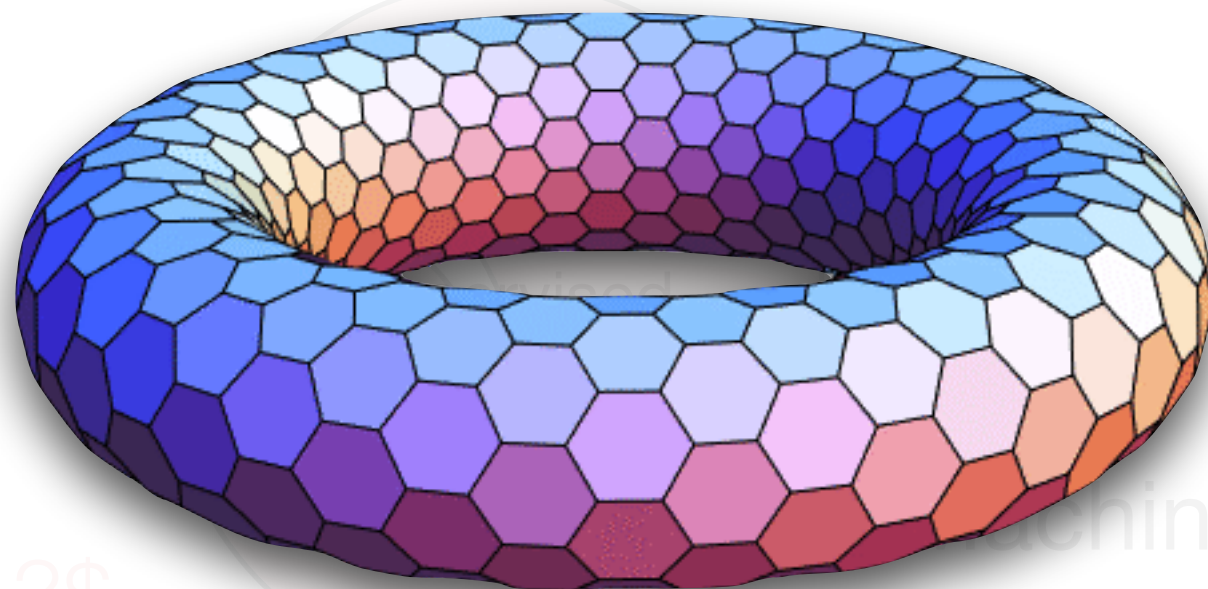
Reinforcement
Learning

Dimensionality
reduction

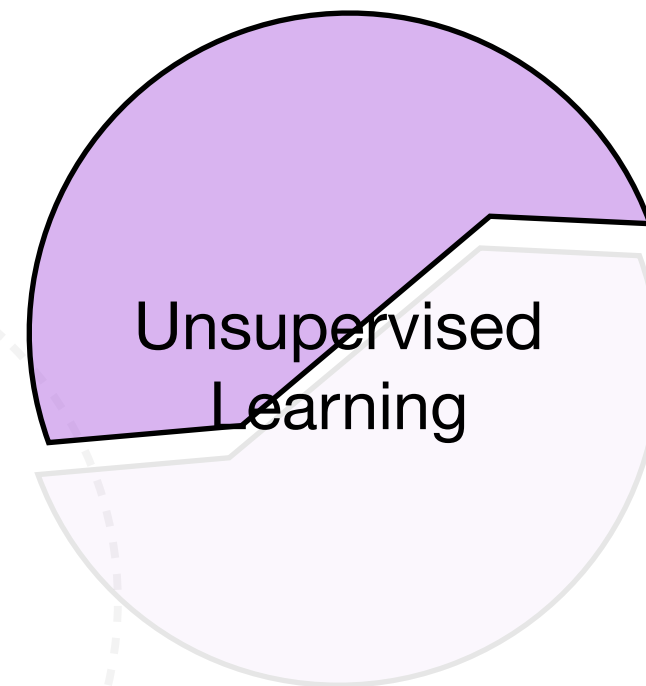
Unsupervised
Learning



0 ← ? 1 **High Dimensional Space**



Dimensionality
reduction

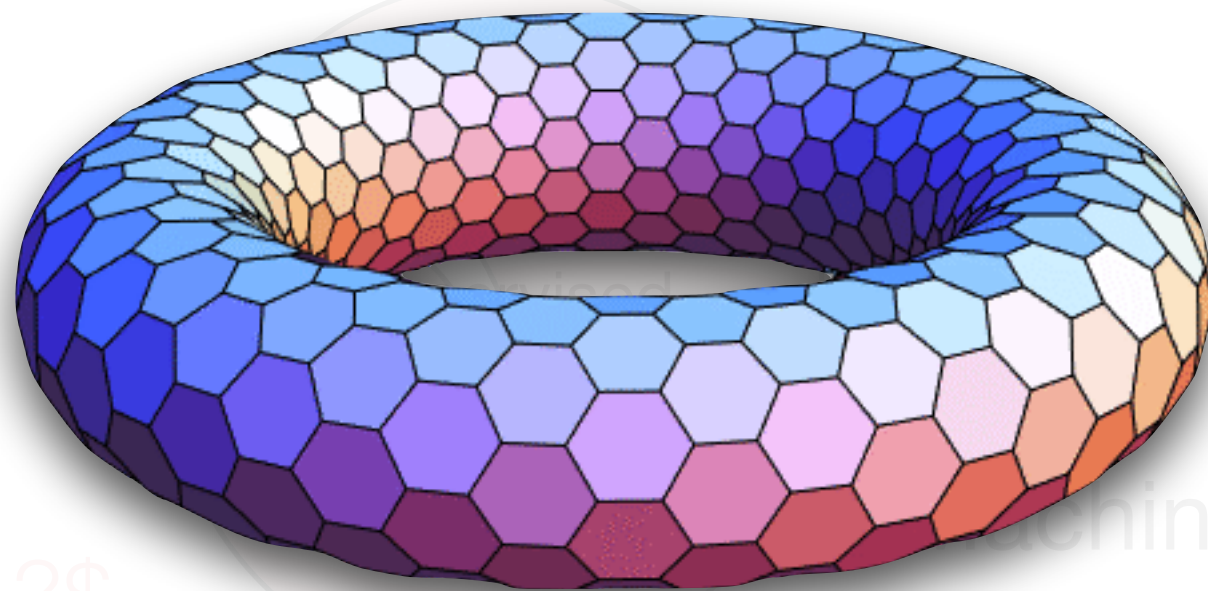


Regression

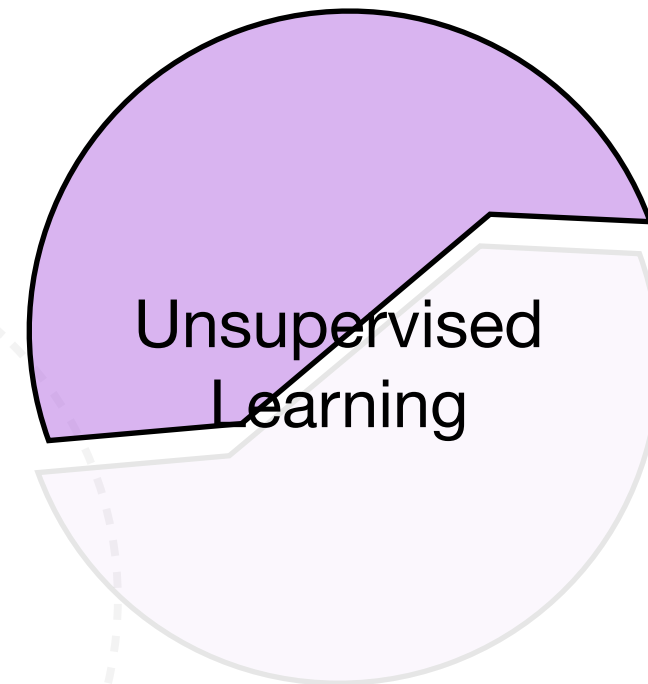


Reinforcement
Learning

0 ← ? 1 **High Dimensional Space**



Dimensionality
reduction

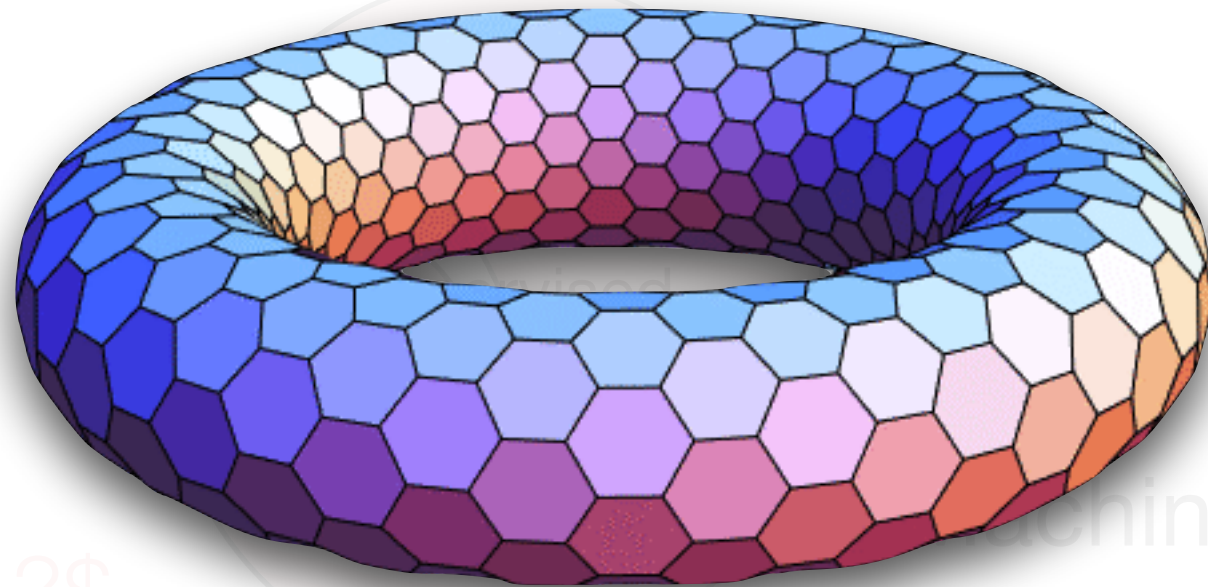


Dimensionality
reduction



Reinforcement
Learning

High Dimensional Space



Dimensionality
reduction

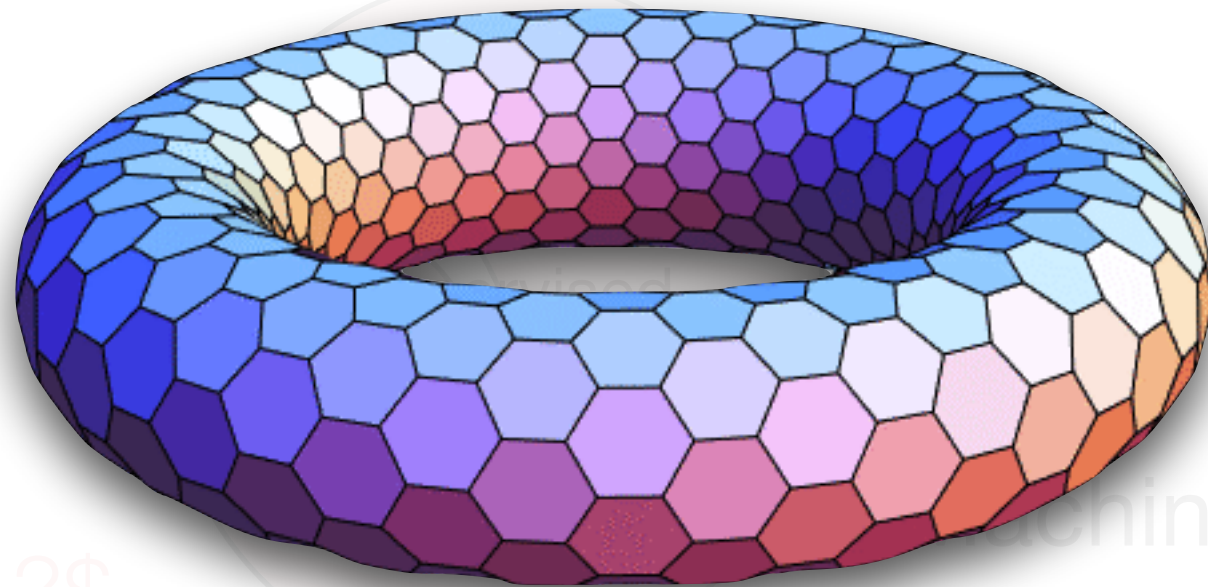
Unsupervised
Learning

Dimensionality
reduction

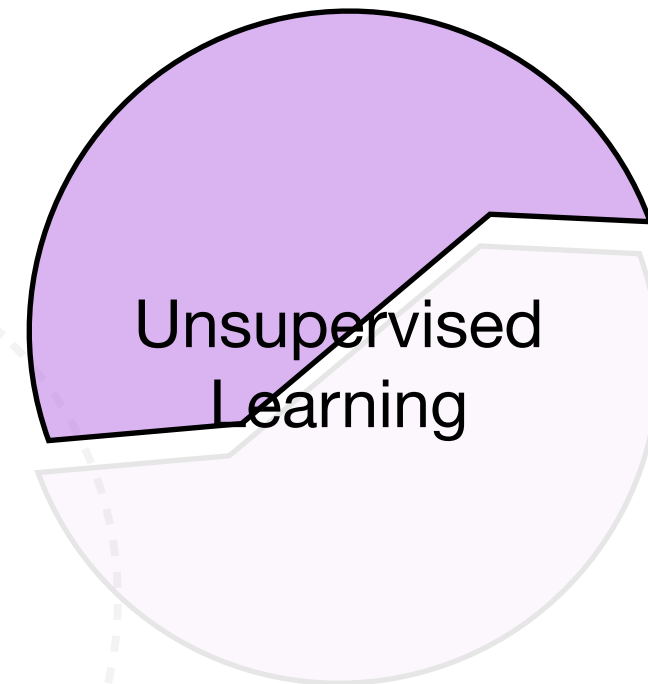


Low Dimensional Space

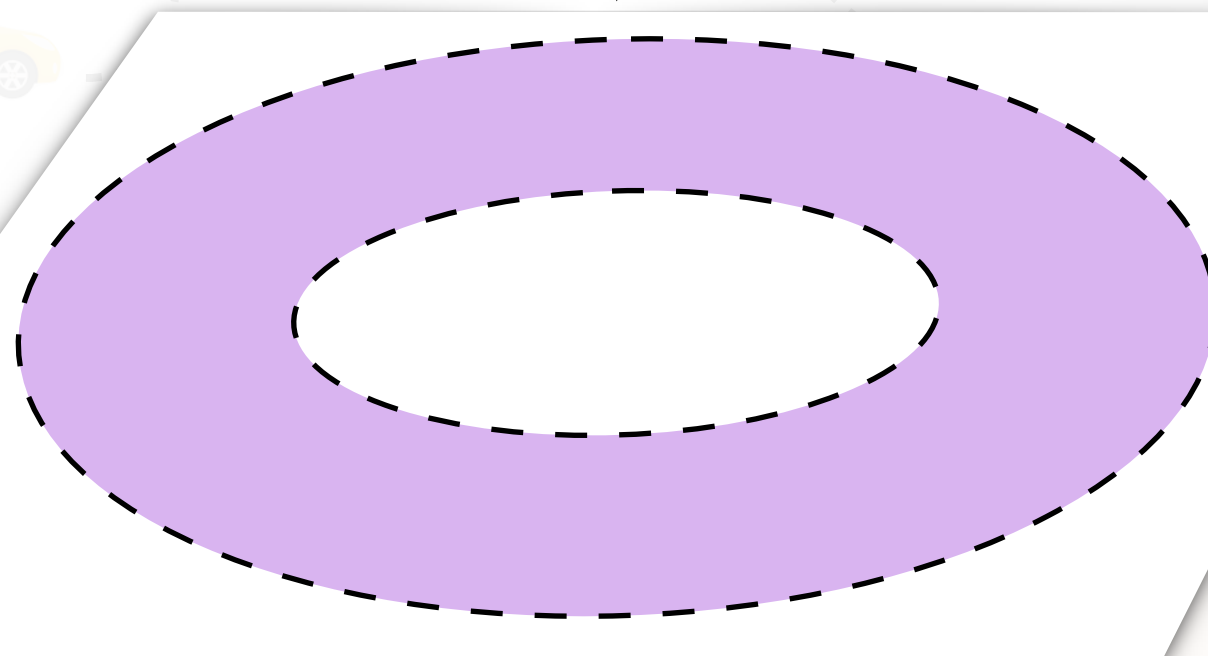
High Dimensional Space



Dimensionality
reduction



Dimensionality
reduction



Low Dimensional Space

0 $\longleftrightarrow$? $\longleftrightarrow$ 1

Classification

Supervised
Learning

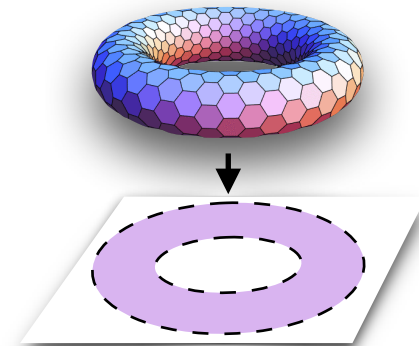
Regression

Machine
Learning

Reinforcement
Learning

Dimensionality
reduction

Unsupervised
Learning



0 $\longleftrightarrow$? $\rightarrow$ 1

Classification

Supervised
Learning

Regression

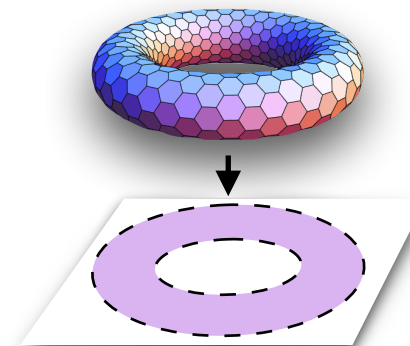
Machine
Learning

Reinforcement
Learning

Dimensionality
reduction

Unsupervised
Learning

Clustering



0 $\longleftrightarrow$? 1

Unlabelled
data

Classification

Supervised
Learning

Regression

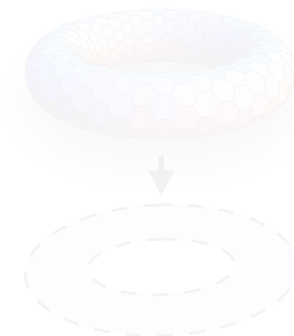
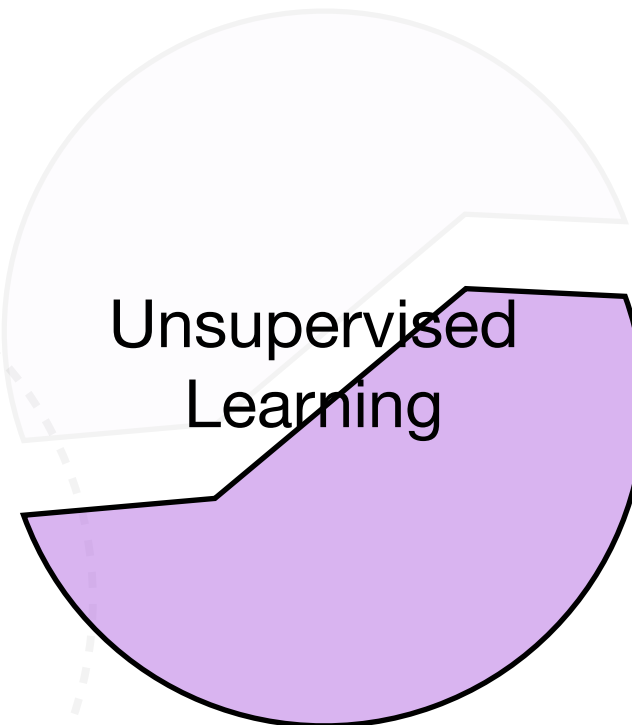
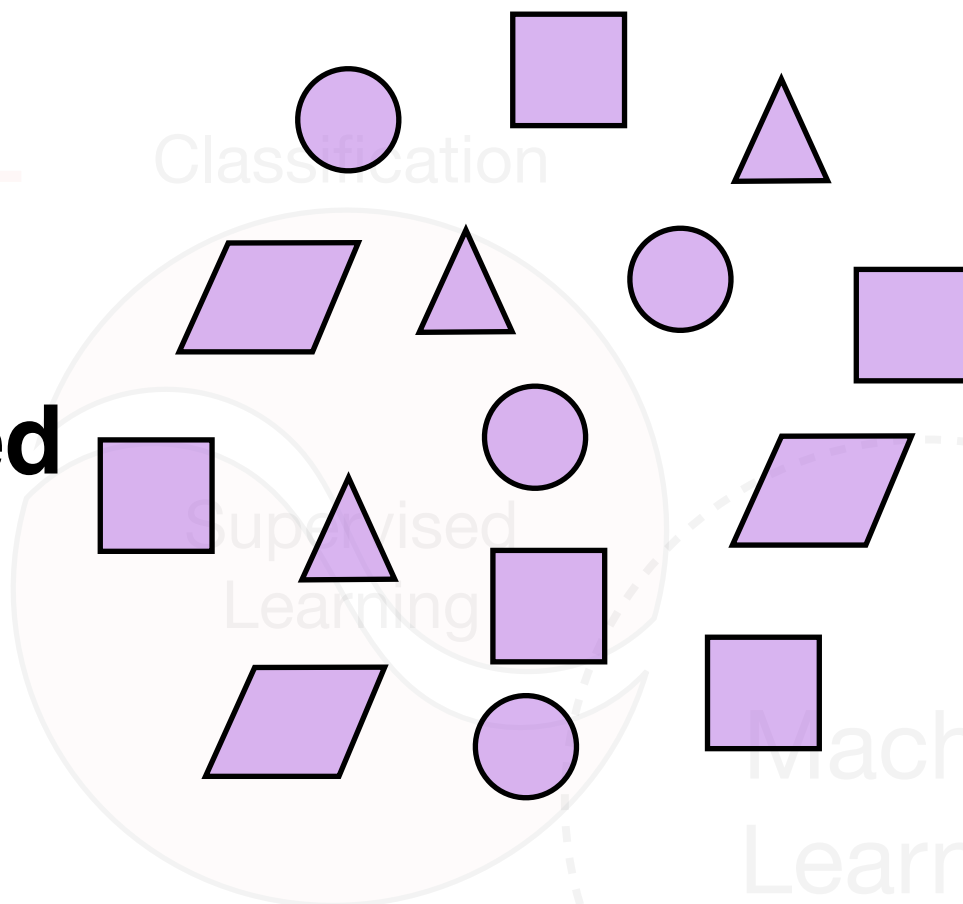
Machine
Learning

Dimensionality
reduction

Unsupervised
Learning

Clustering

Reinforcement
Learning



0 $\longleftrightarrow$? $\longleftrightarrow$ 1

Unlabelled
data

Classification

Supervised
Learning

Machine
Learning

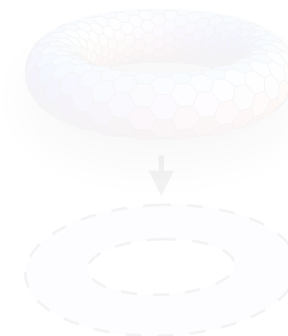
Dimensionality
reduction

Unsupervised
Learning

Clustering

Clustering
algorithm

Reinforcement
Learning



0 $\longleftrightarrow$? $\longleftrightarrow$ 1

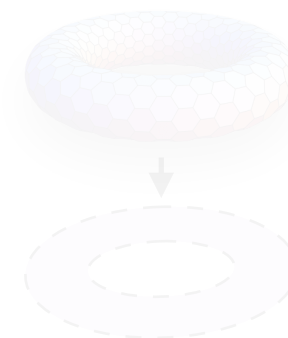
Unlabelled
data

Classification

Supervised
Learning

Machine
Learning

Dimensionality
reduction



Unsupervised
Learning

Clustering

Clustering
algorithm

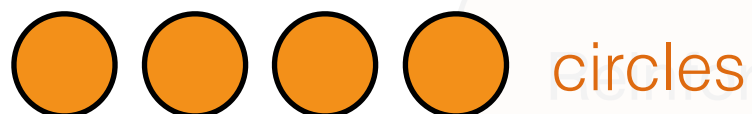


?\$

Clusters



squares



circles



diamonds



triangles

Reinforcement
Learning

0 $\longleftrightarrow$? $\rightarrow$ 1

Classification

Supervised
Learning

Regression

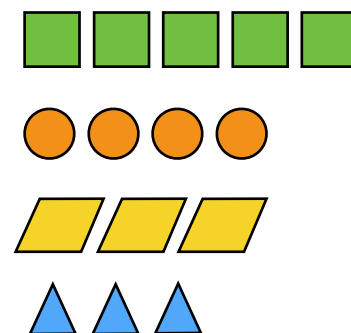
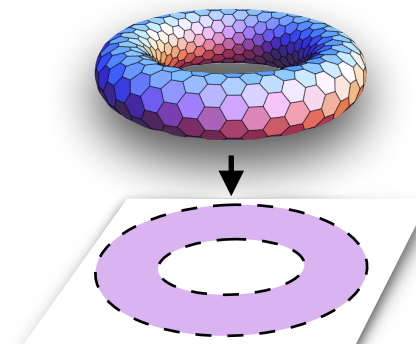
Machine
Learning

Reinforcement
Learning

Dimensionality
reduction

Unsupervised
Learning

Clustering



0 ← ? → 1

Classification

Supervised
Learning

Regression

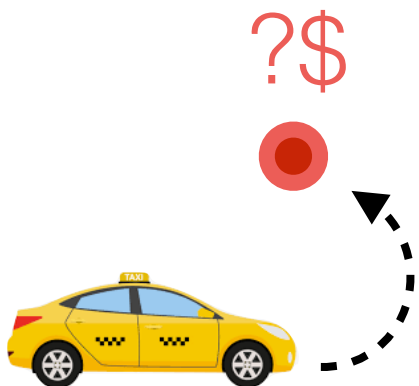
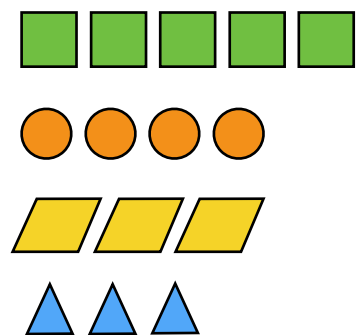
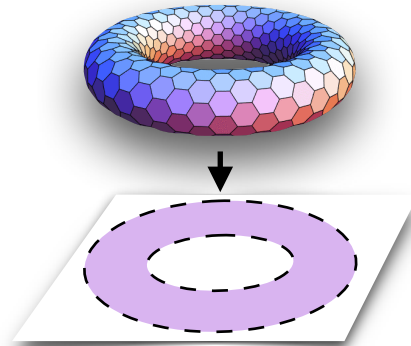
Dimensionality
reduction

Unsupervised
Learning

Clustering

Machine
Learning

Reinforcement
Learning



0 ← ? → 1

Classification

Supervised
Learning

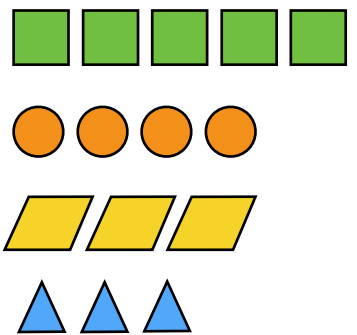
Regression

Machine
Learning

Dimensionality
reduction

Unsupervised
Learning

Clustering



What is the main similarity/difference between
classification and **clustering**?

0 ← ? → 1

Classification

Supervised
Learning

Regression

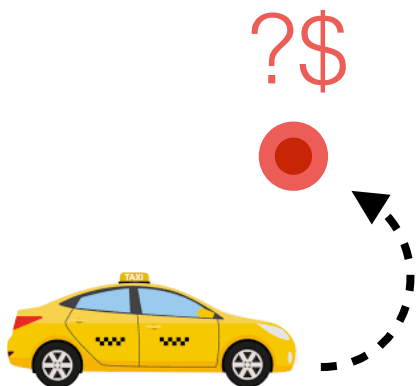
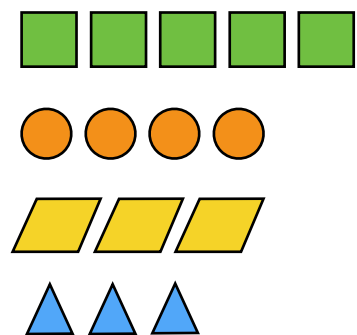
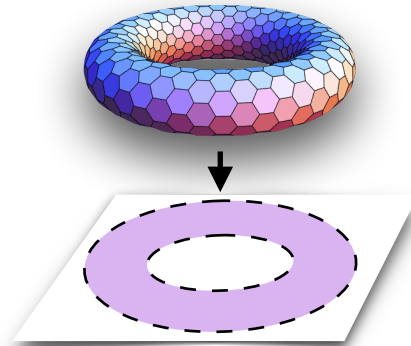
Dimensionality
reduction

Unsupervised
Learning

Clustering

Machine
Learning

Reinforcement
Learning



0 $\longleftrightarrow$? $\longleftrightarrow$ 1

Classification

Supervised
Learning

Regression

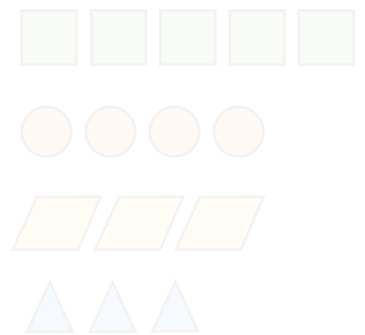
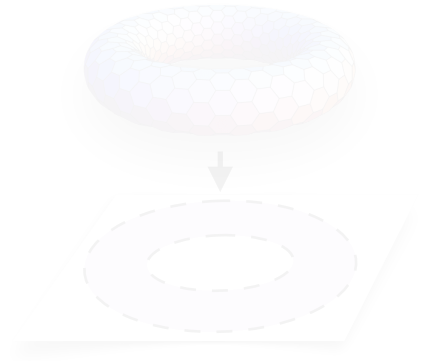
Dimensionality
reduction

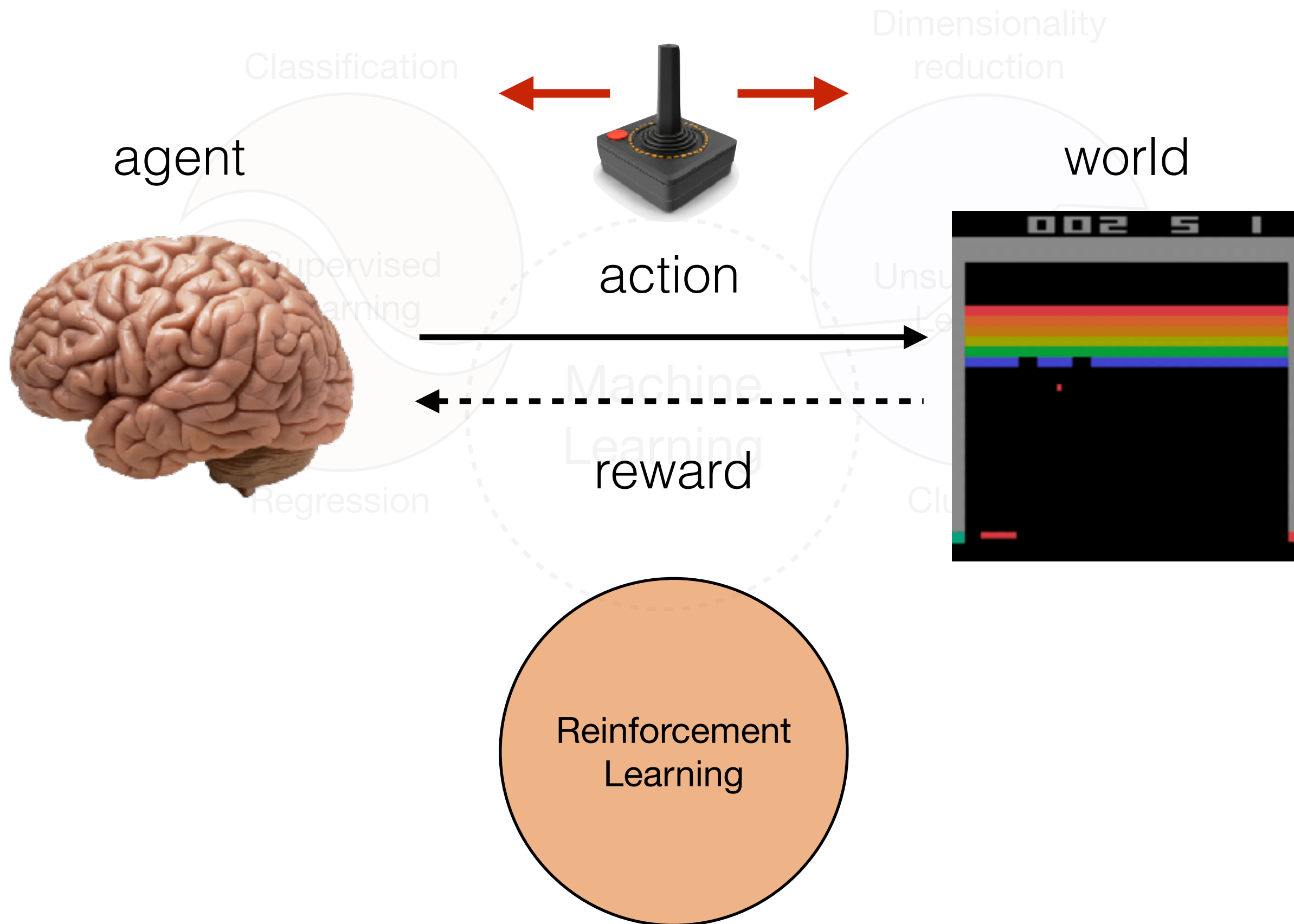
Unsupervised
Learning

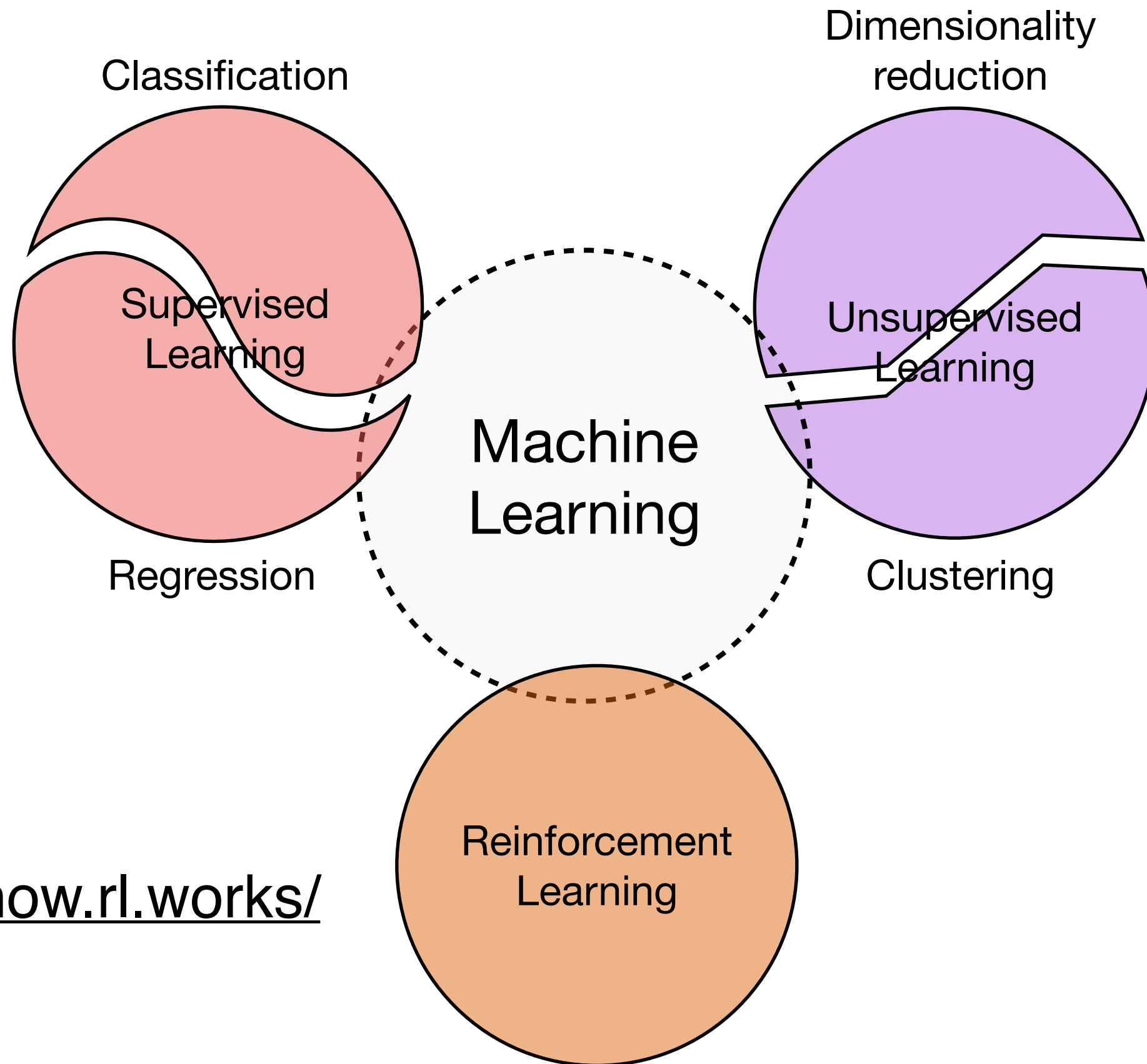
Clustering

Machine
Learning

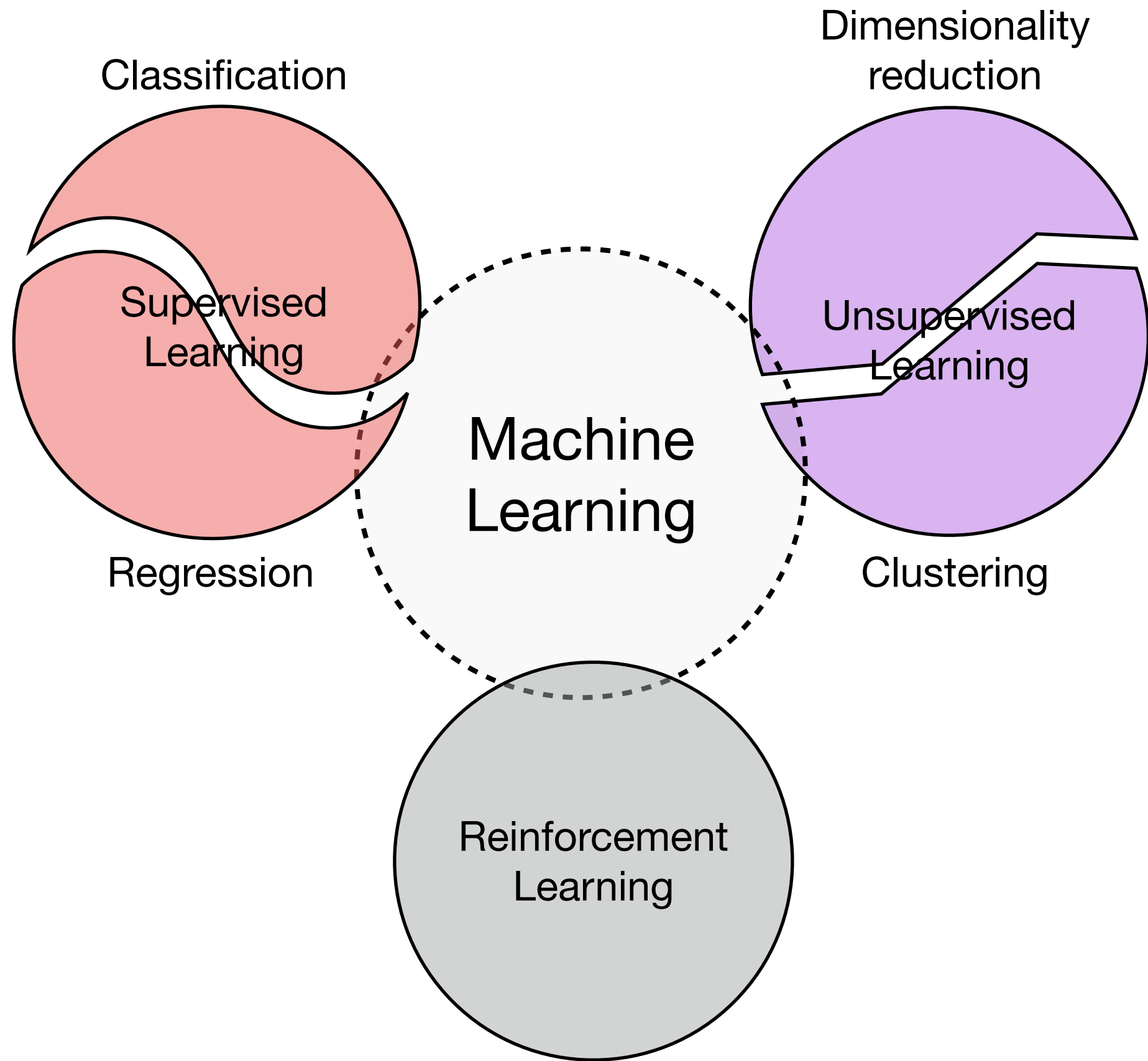
Reinforcement
Learning





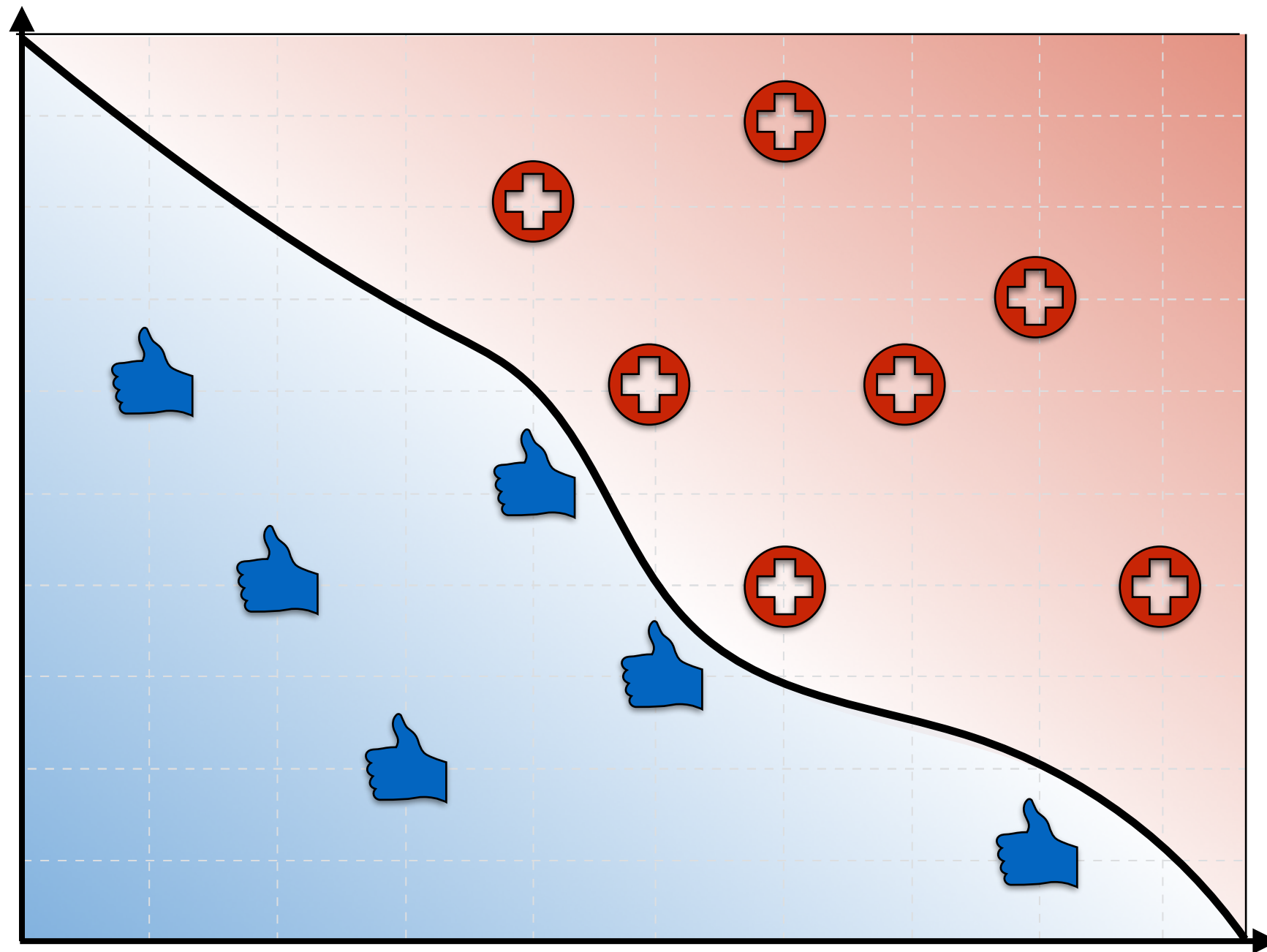


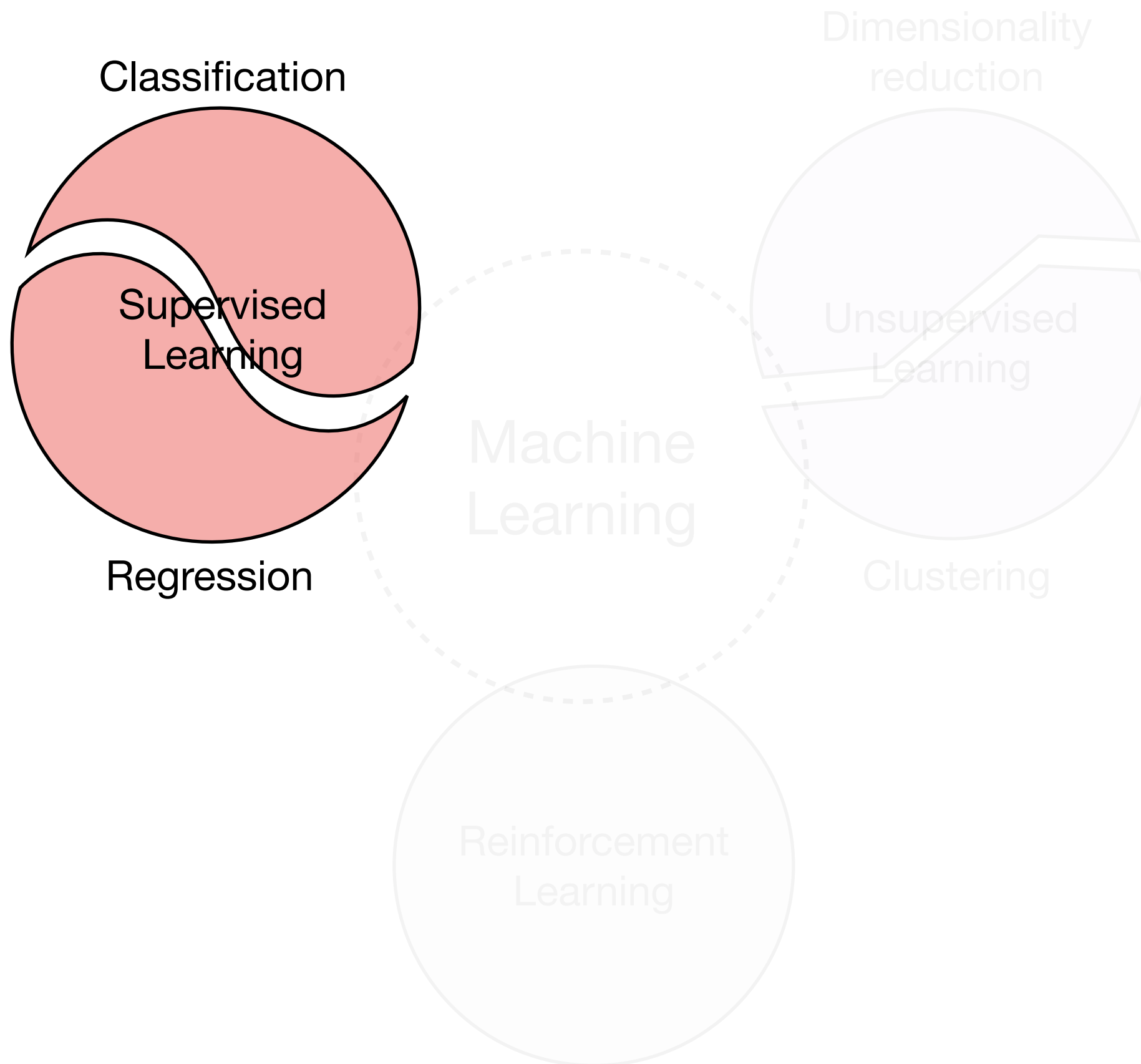
<https://how.rl.works/>



Supervised Learning

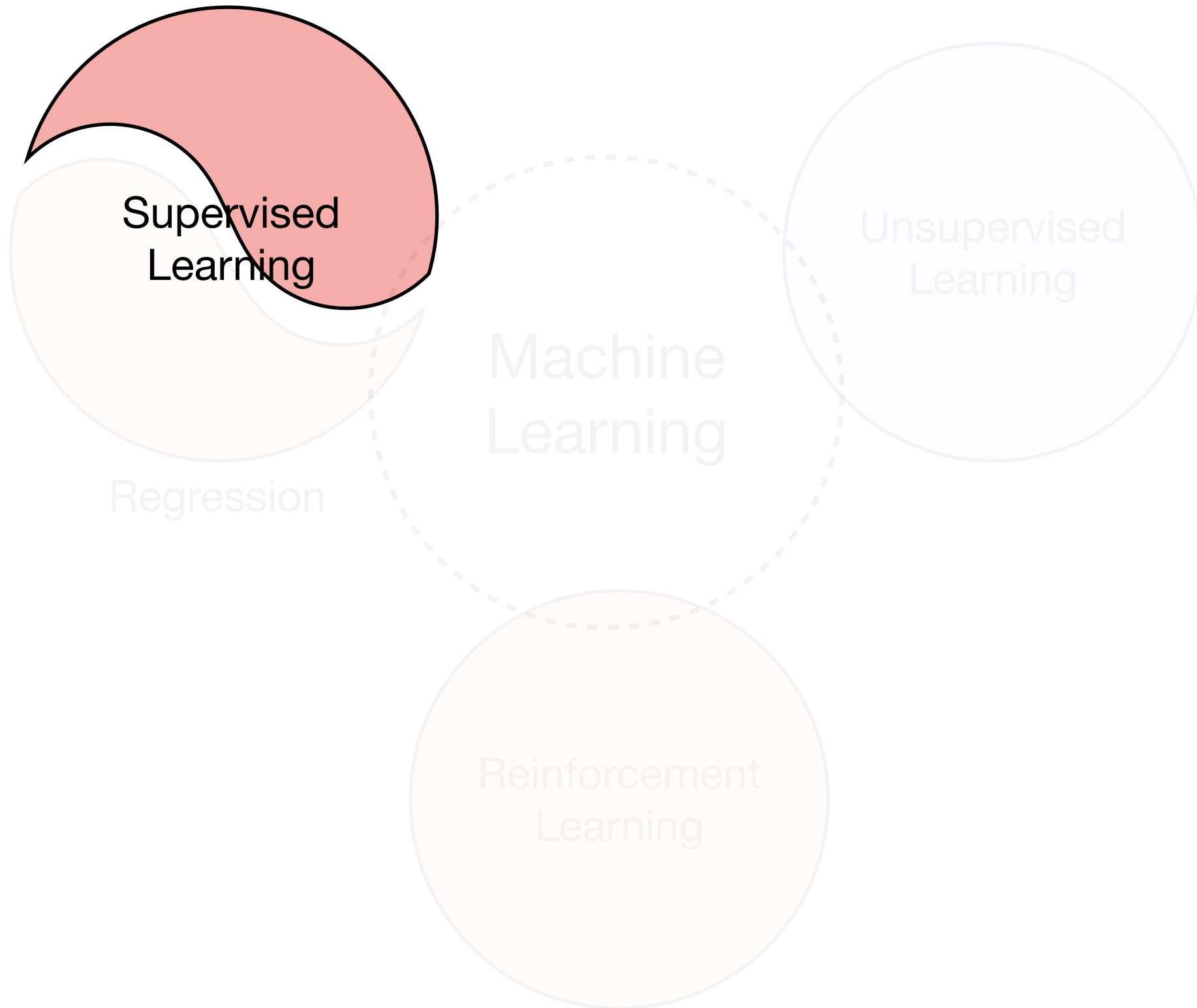
Lecture I



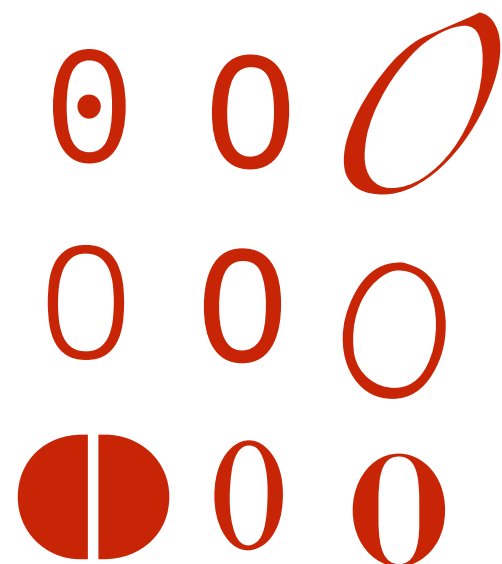


0 ← ? → 1

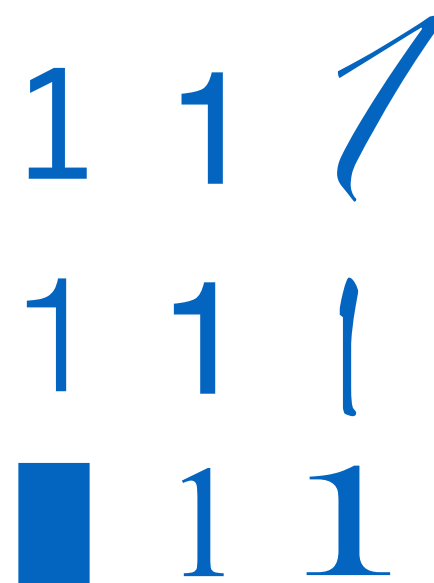
Classification



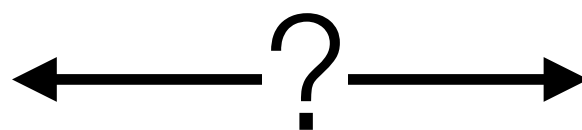
Zero



One

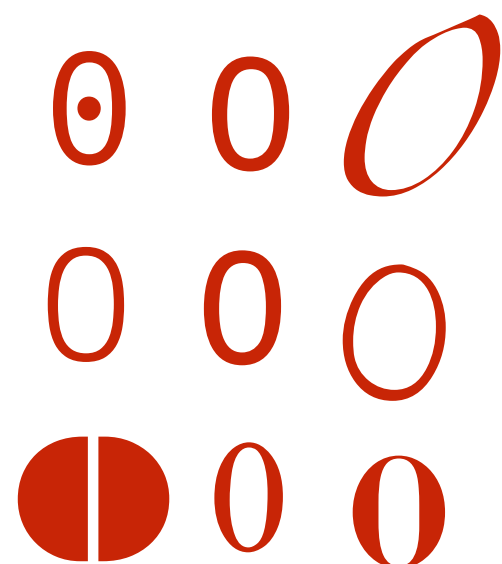


new digit

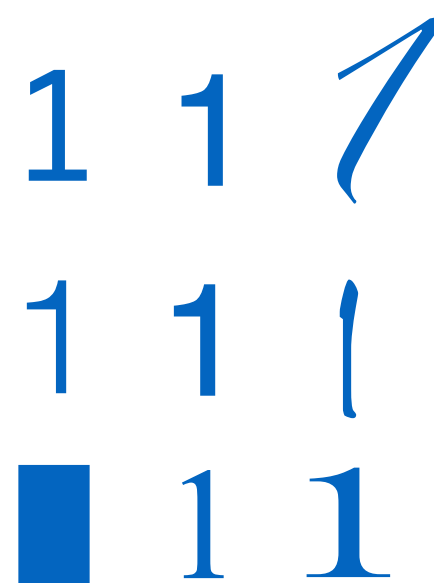
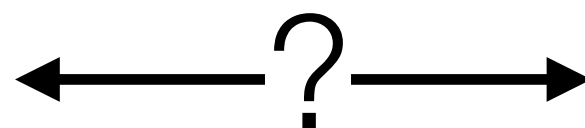


Zero

One



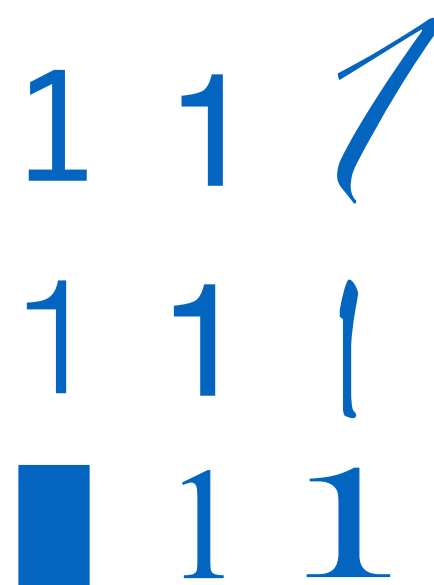
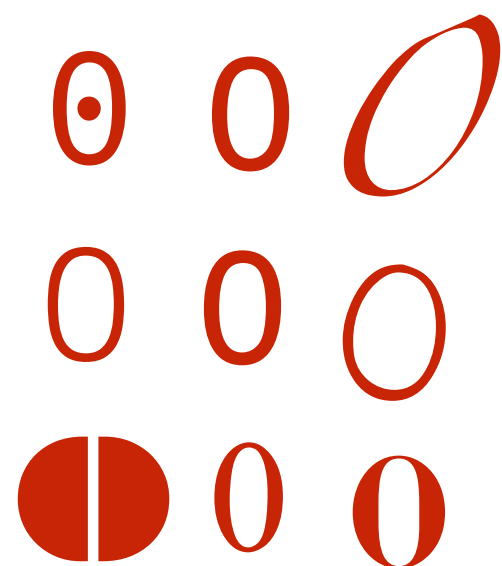
new digit



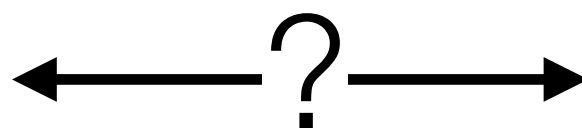
height

Zero

One

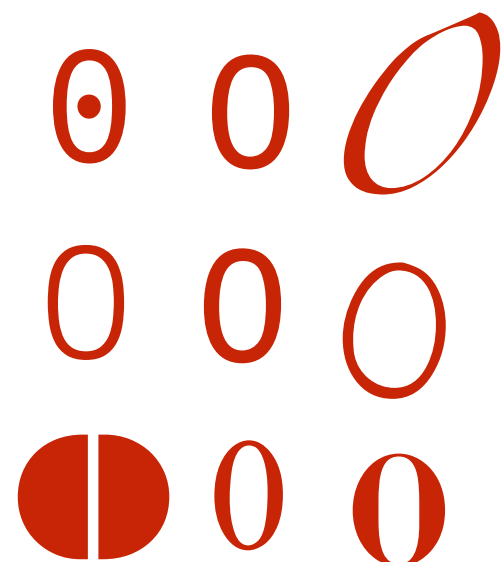


new digit

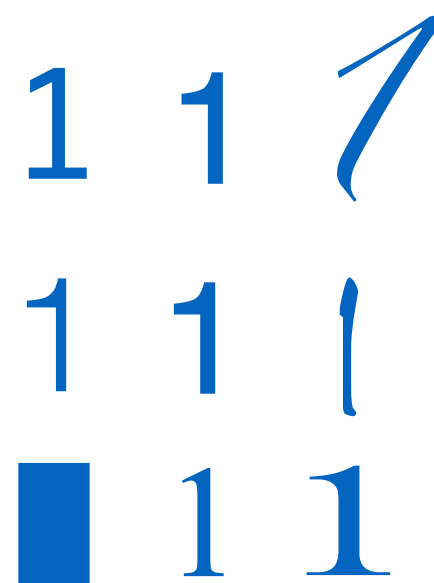


height

Zero

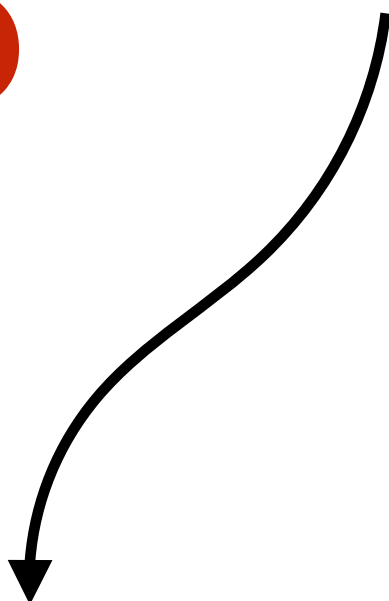


One

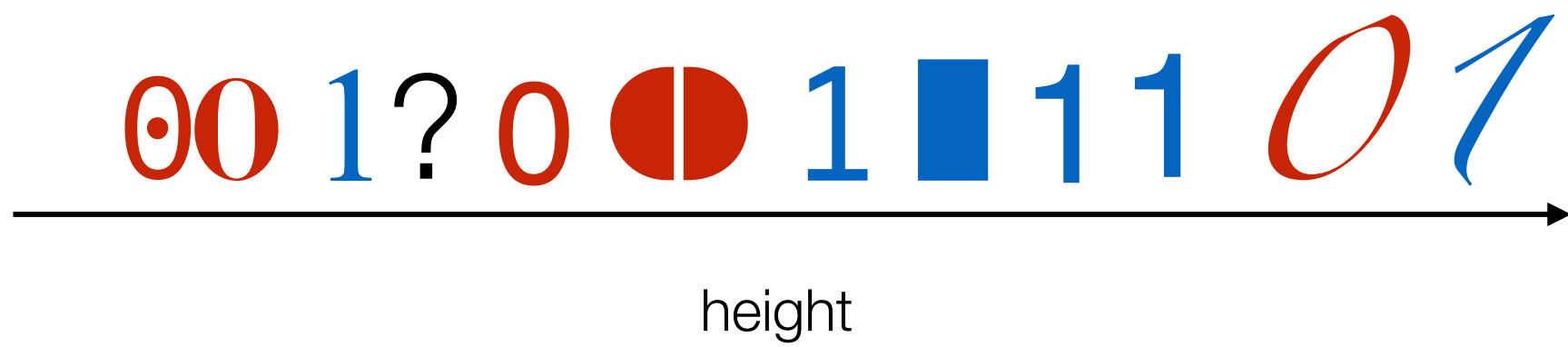


new digit

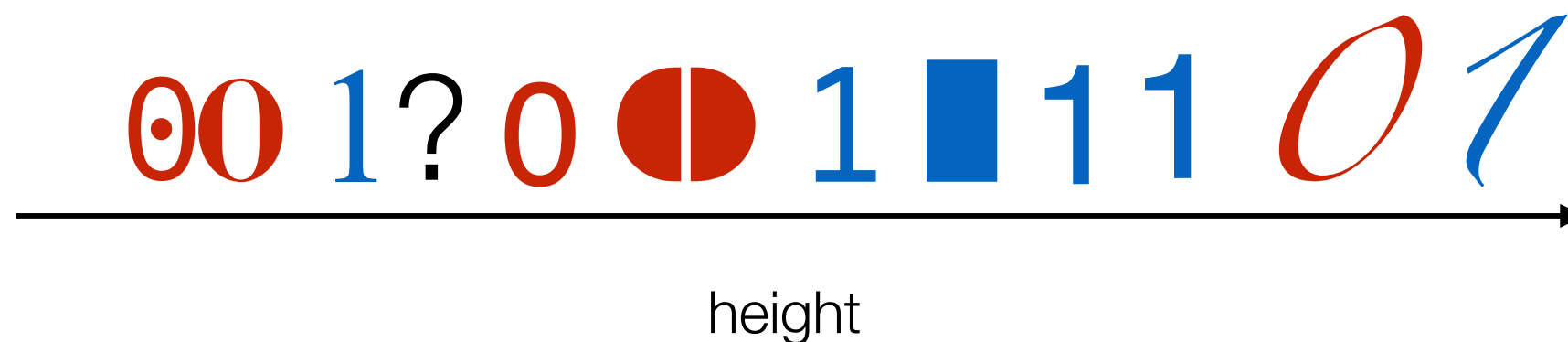
?



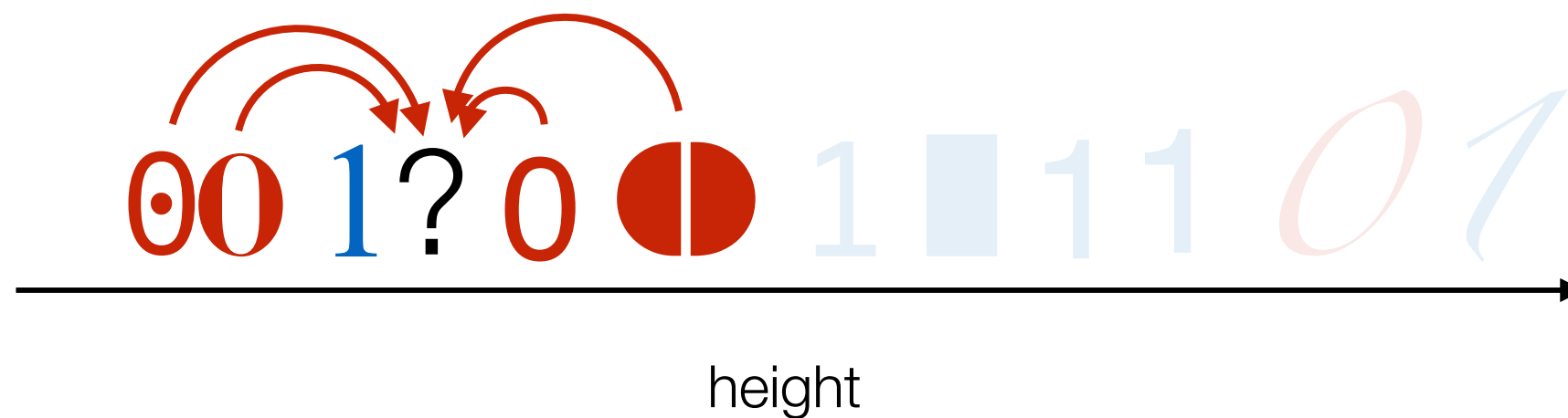
height



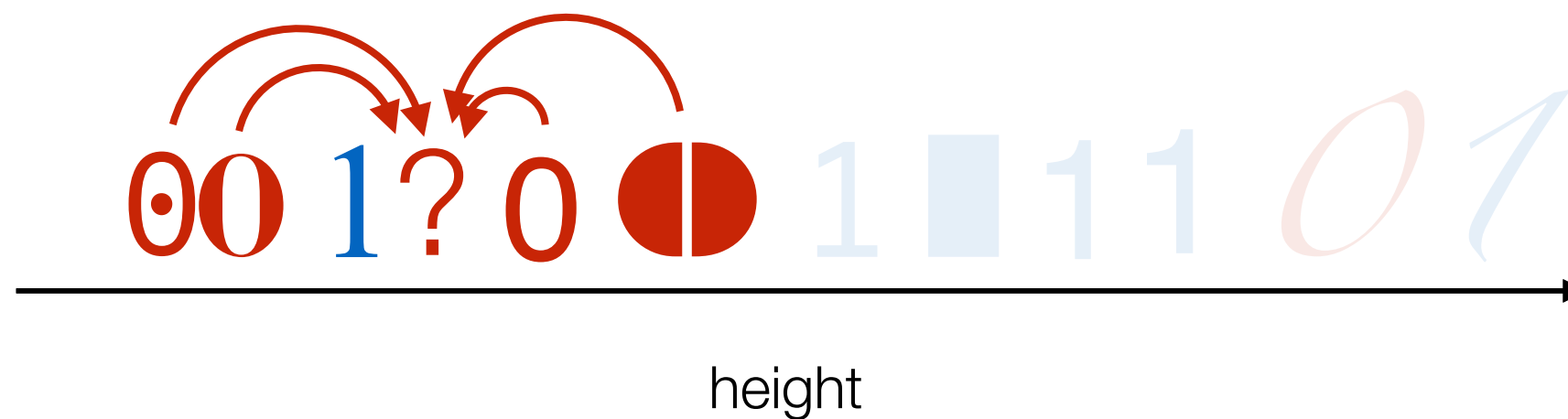
If we would be forced to decide now, what would be **the class** of the new digit?

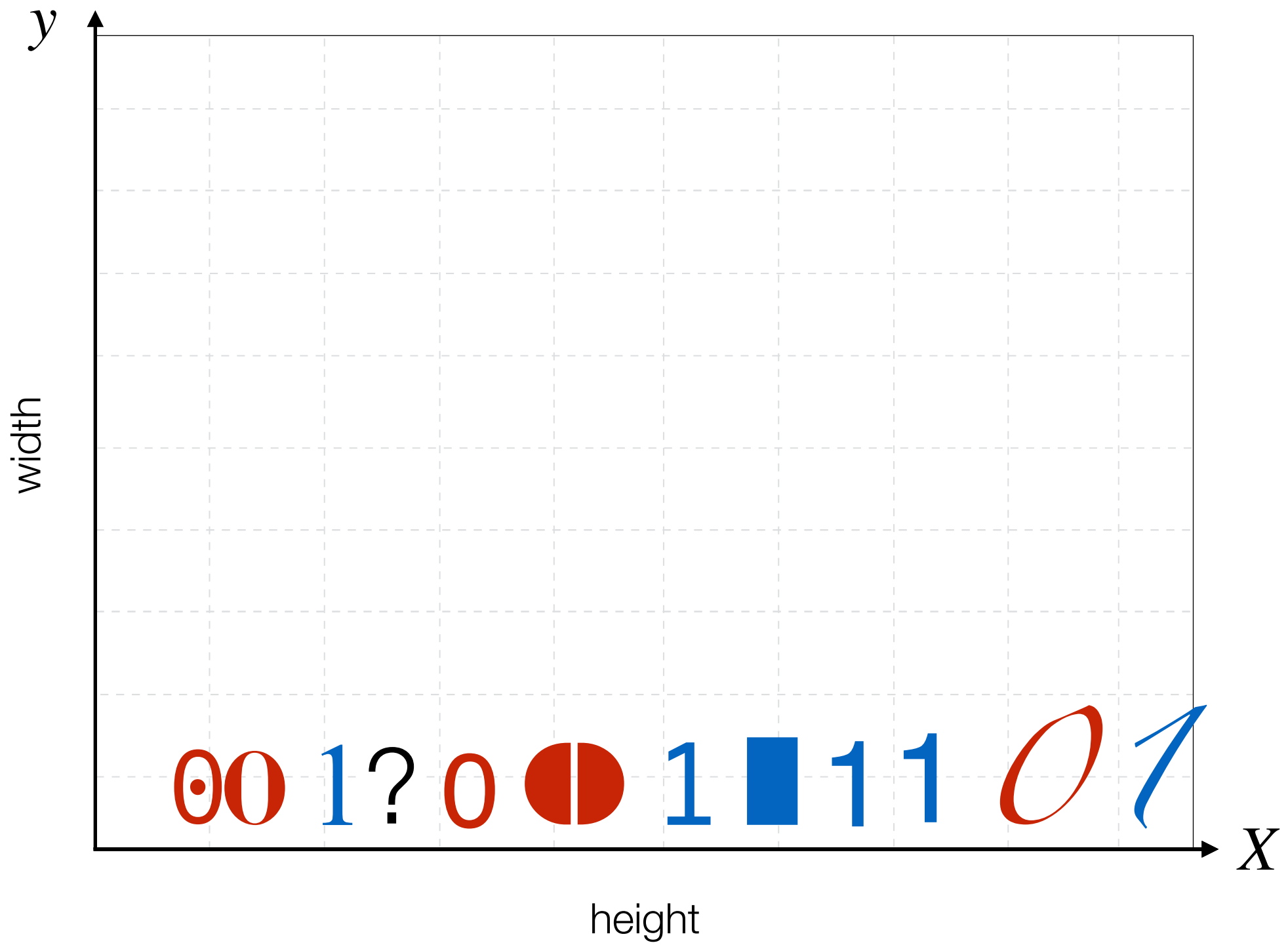


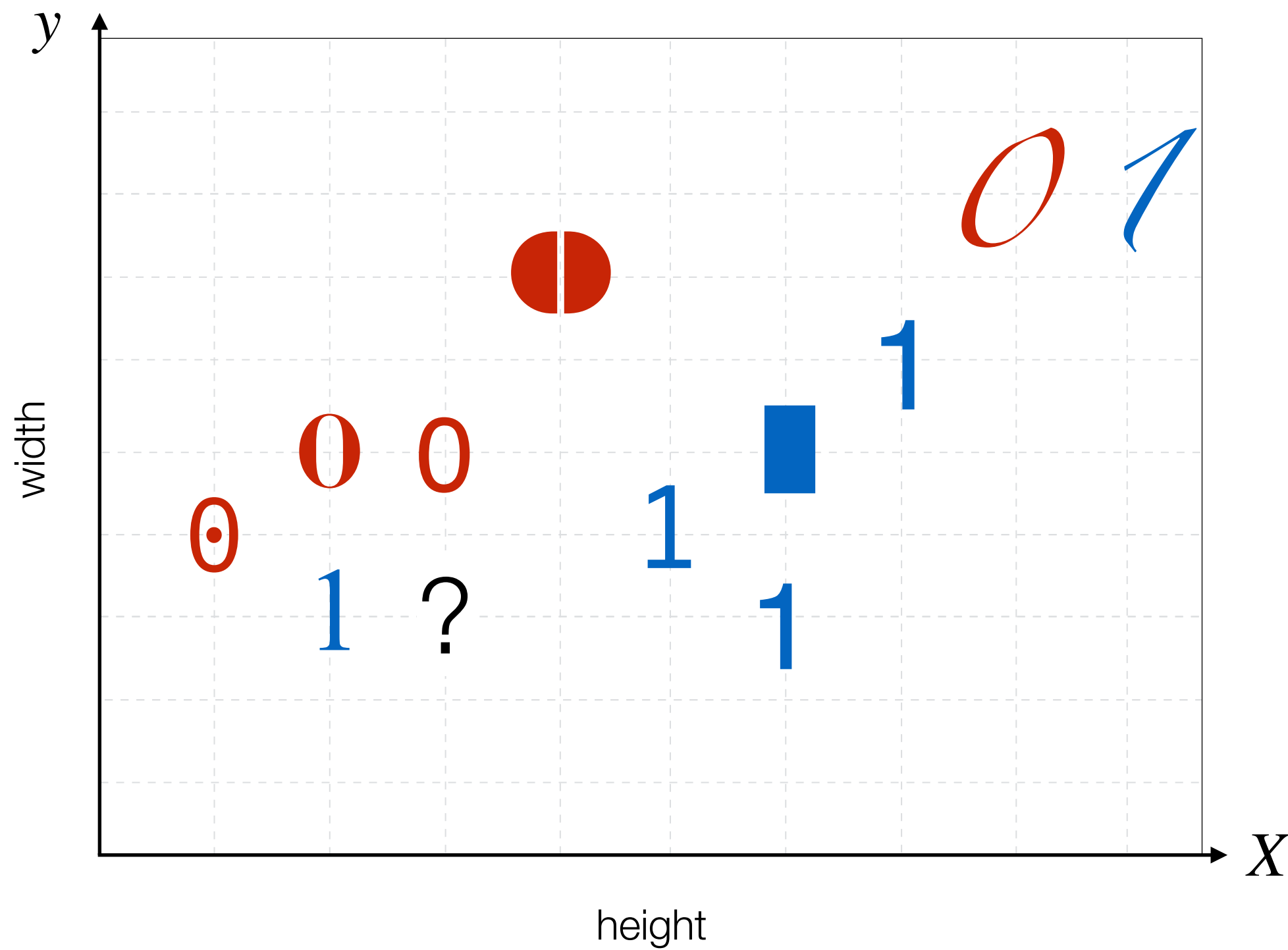
If we would be forced to decide now, what would be **the class** of the new digit?



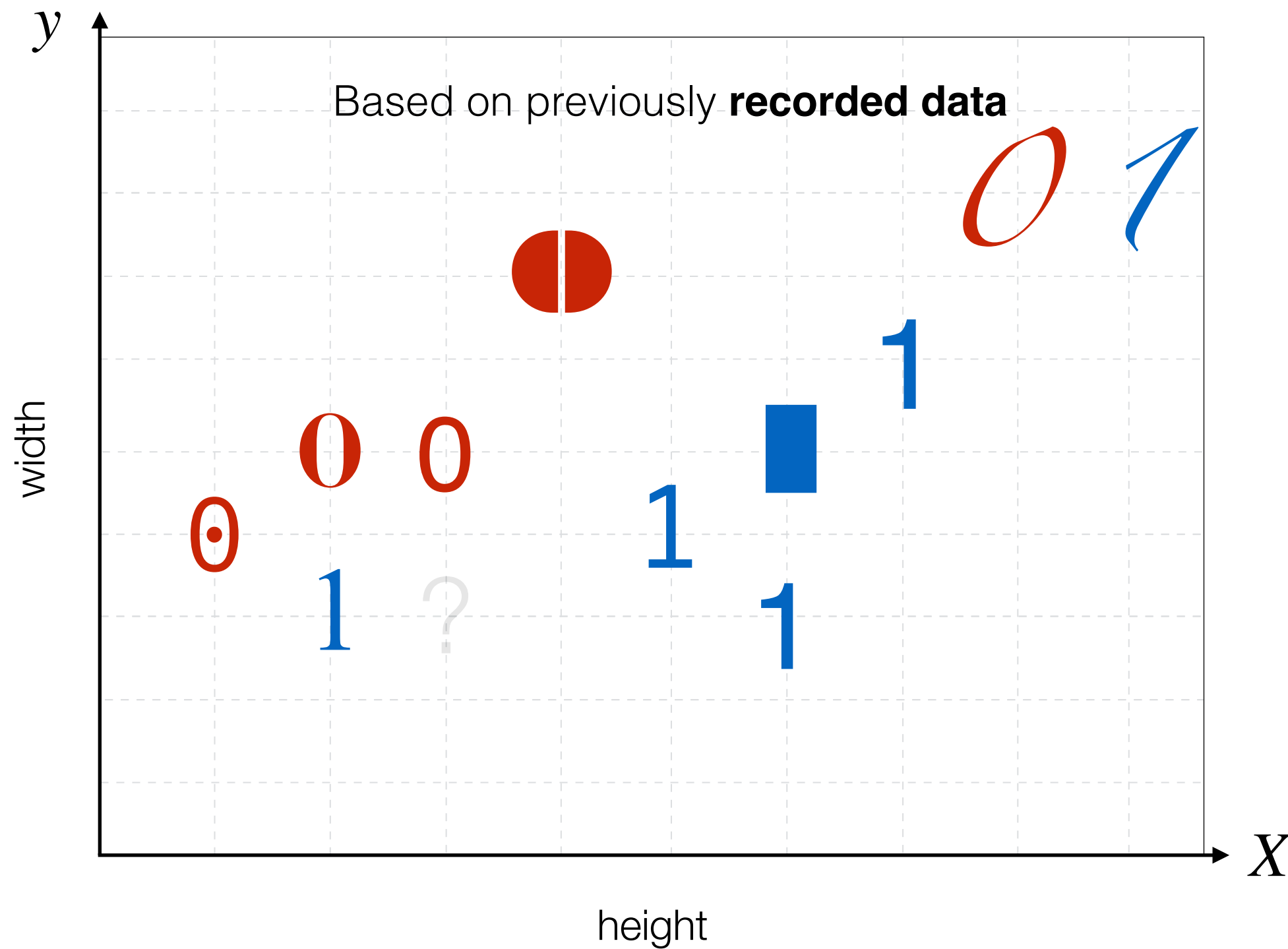
If we would be forced to decide now, what would be **the class** of the new digit?



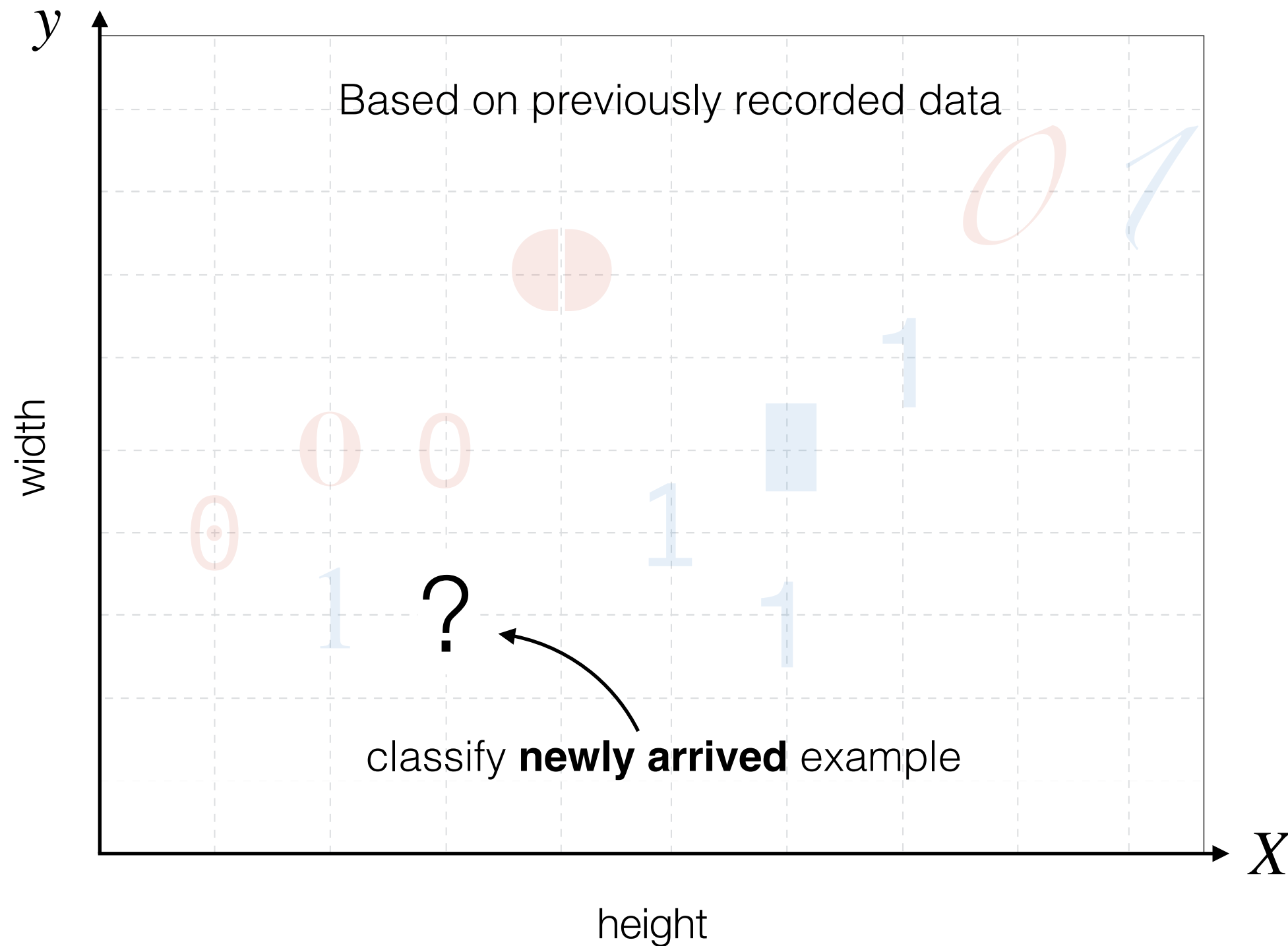




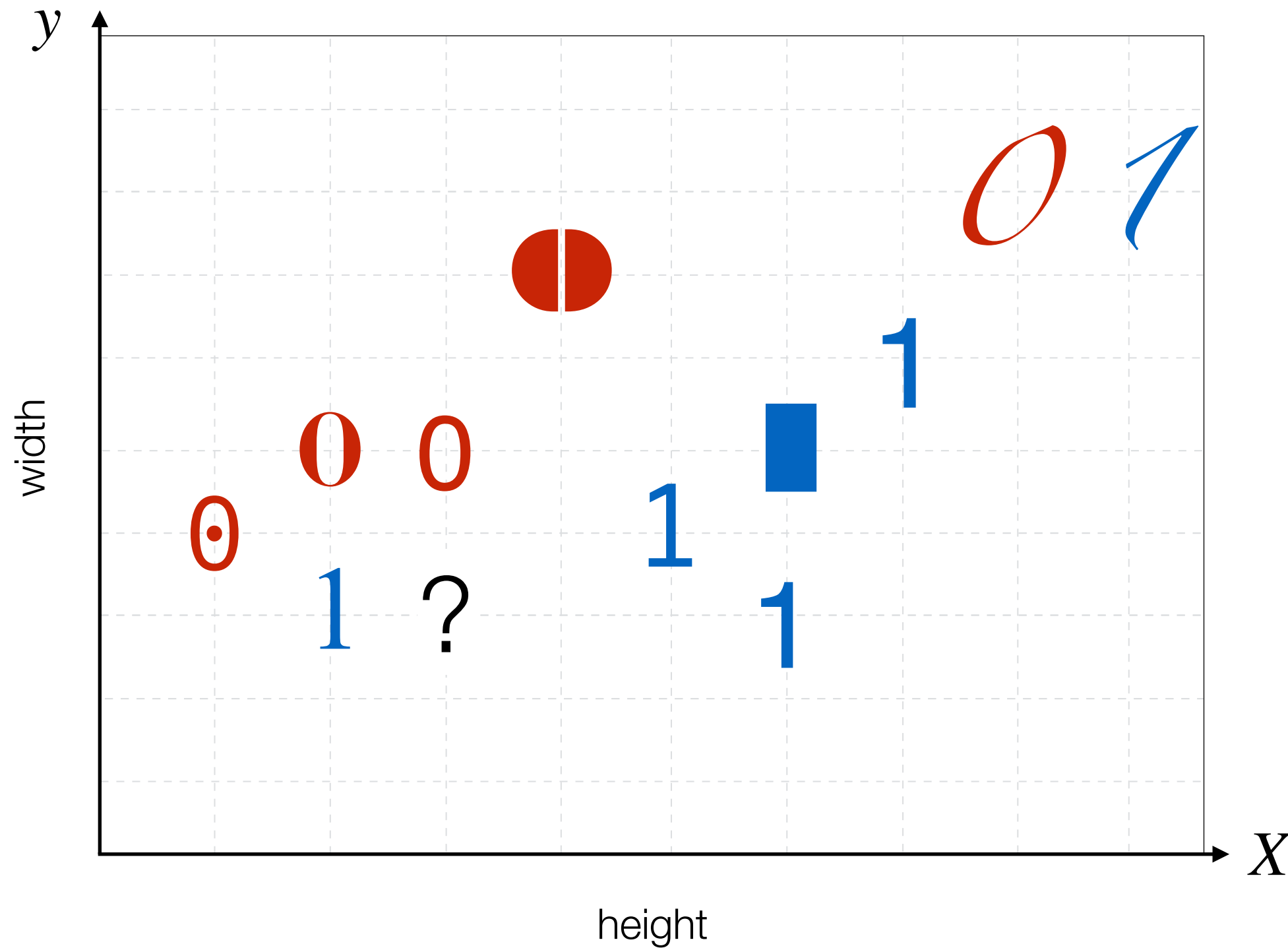
Classification



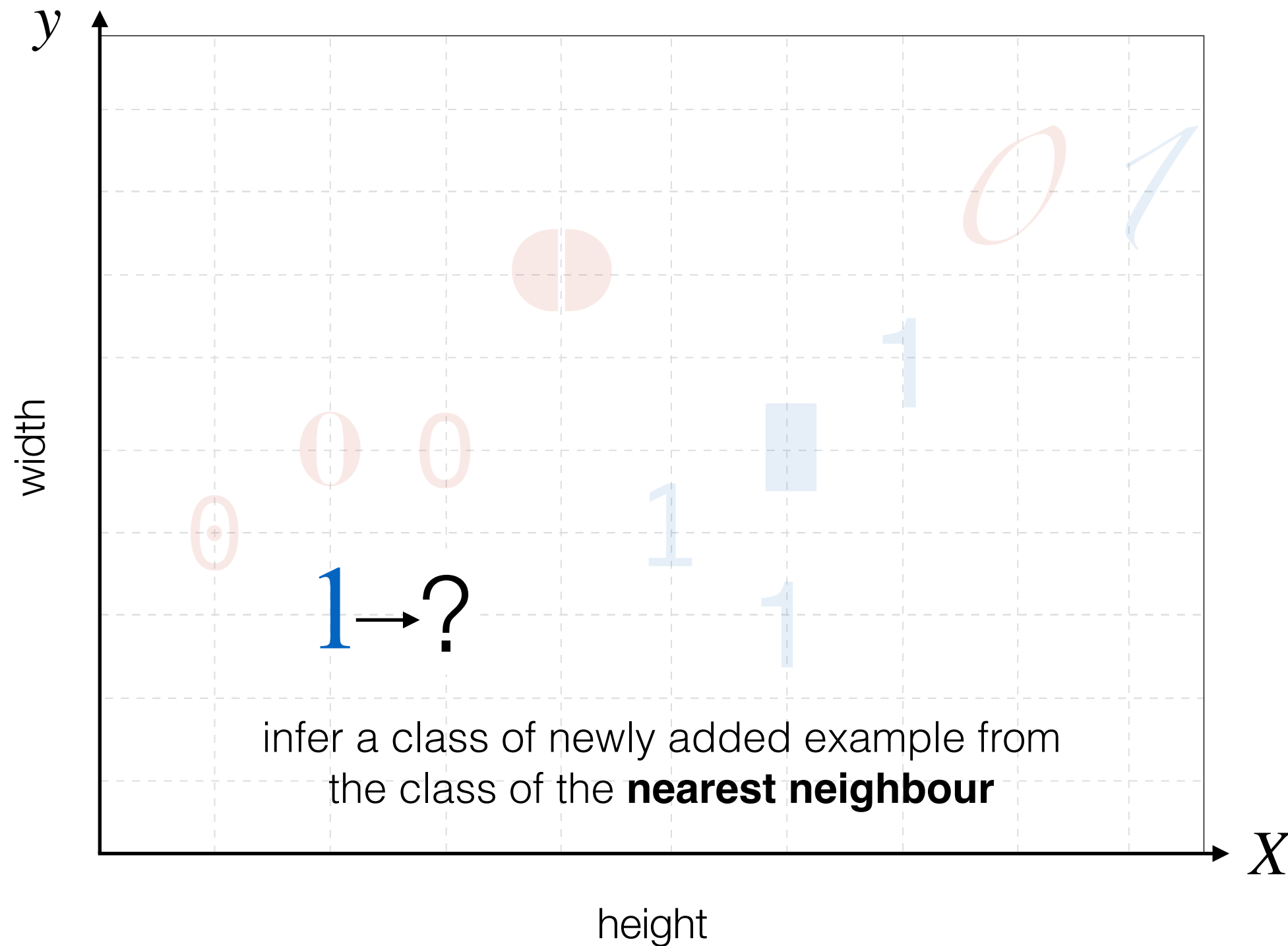
Classification



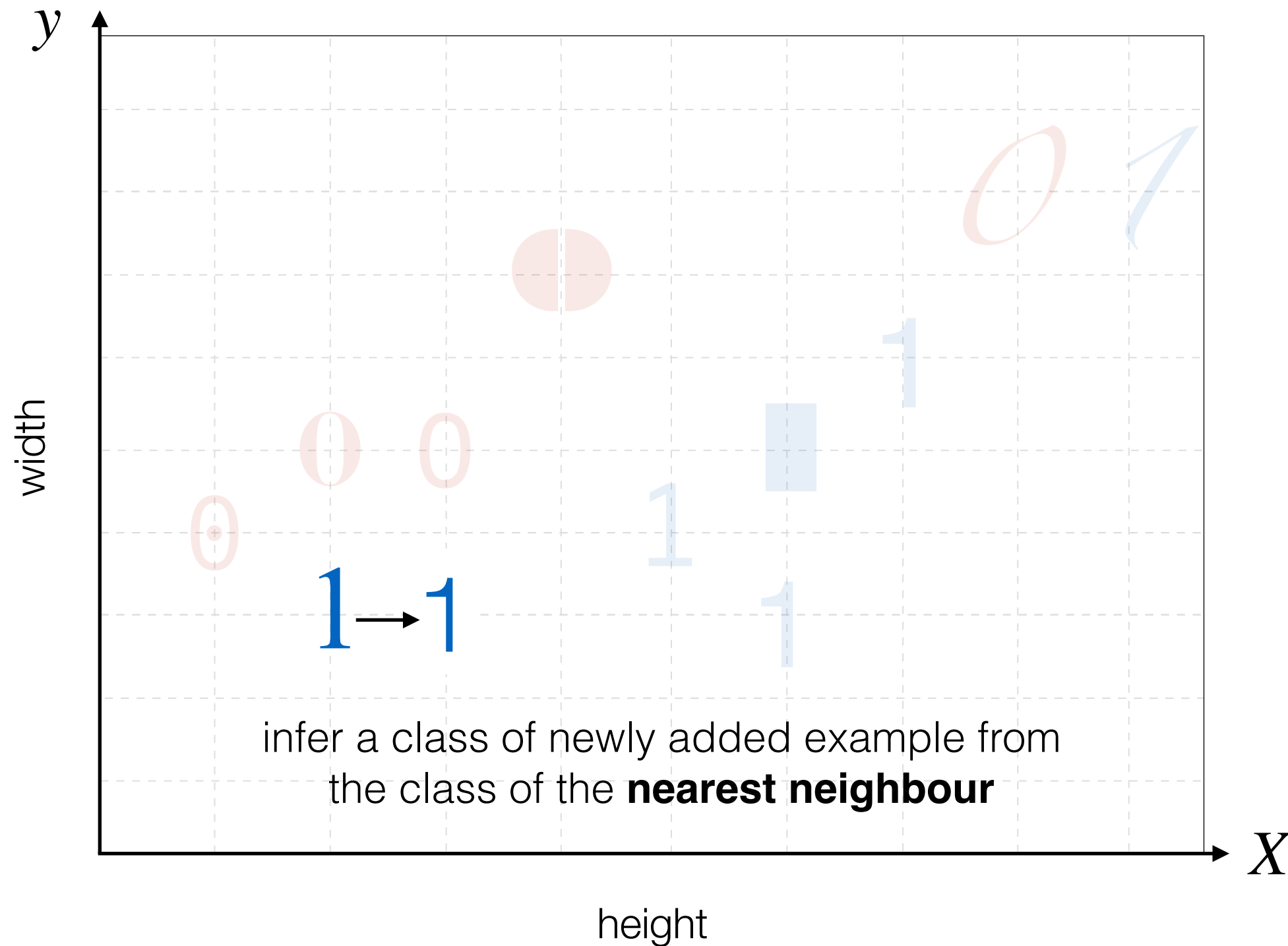
Nearest Neighbour Classifier



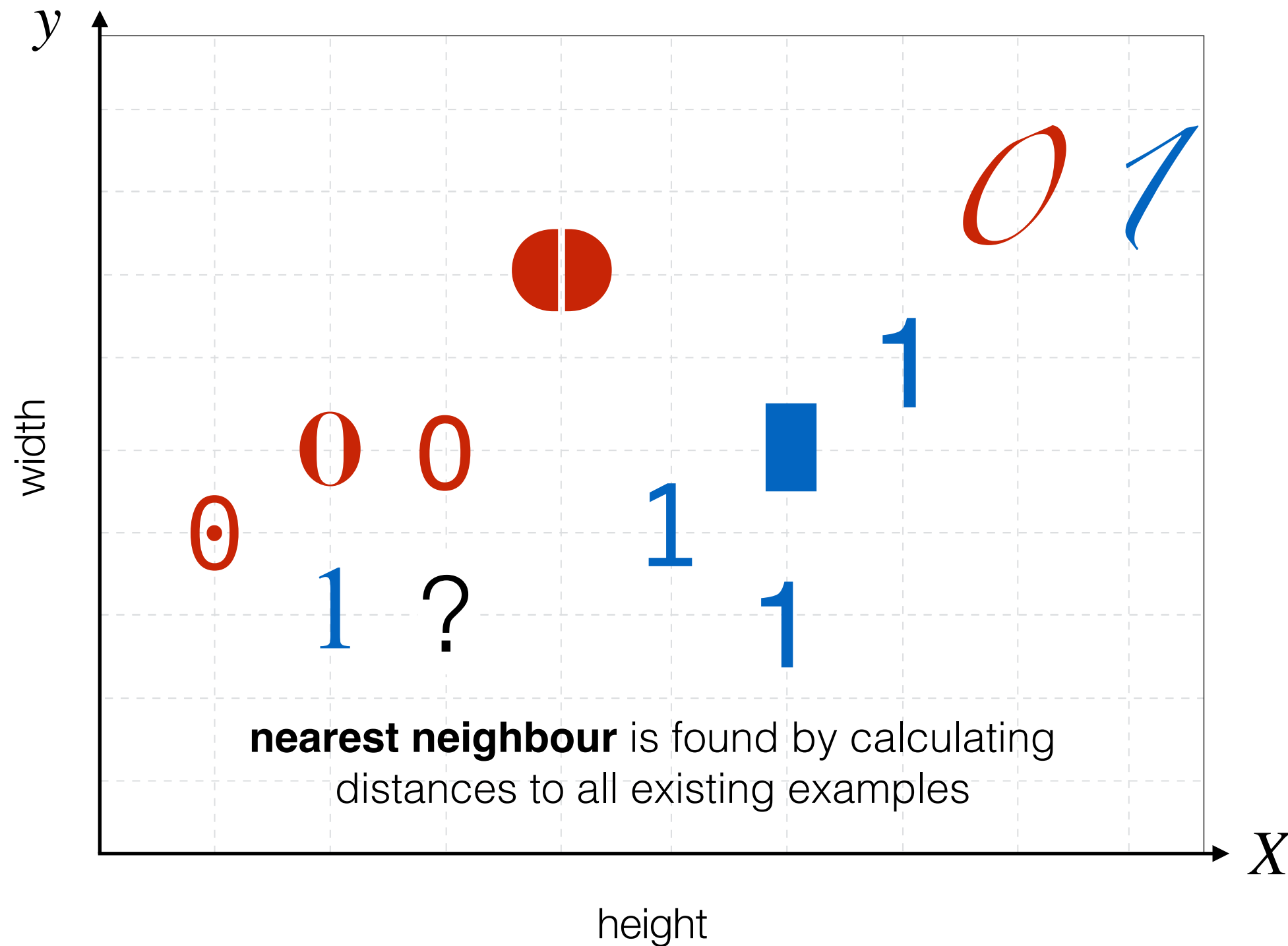
Nearest Neighbour Classifier



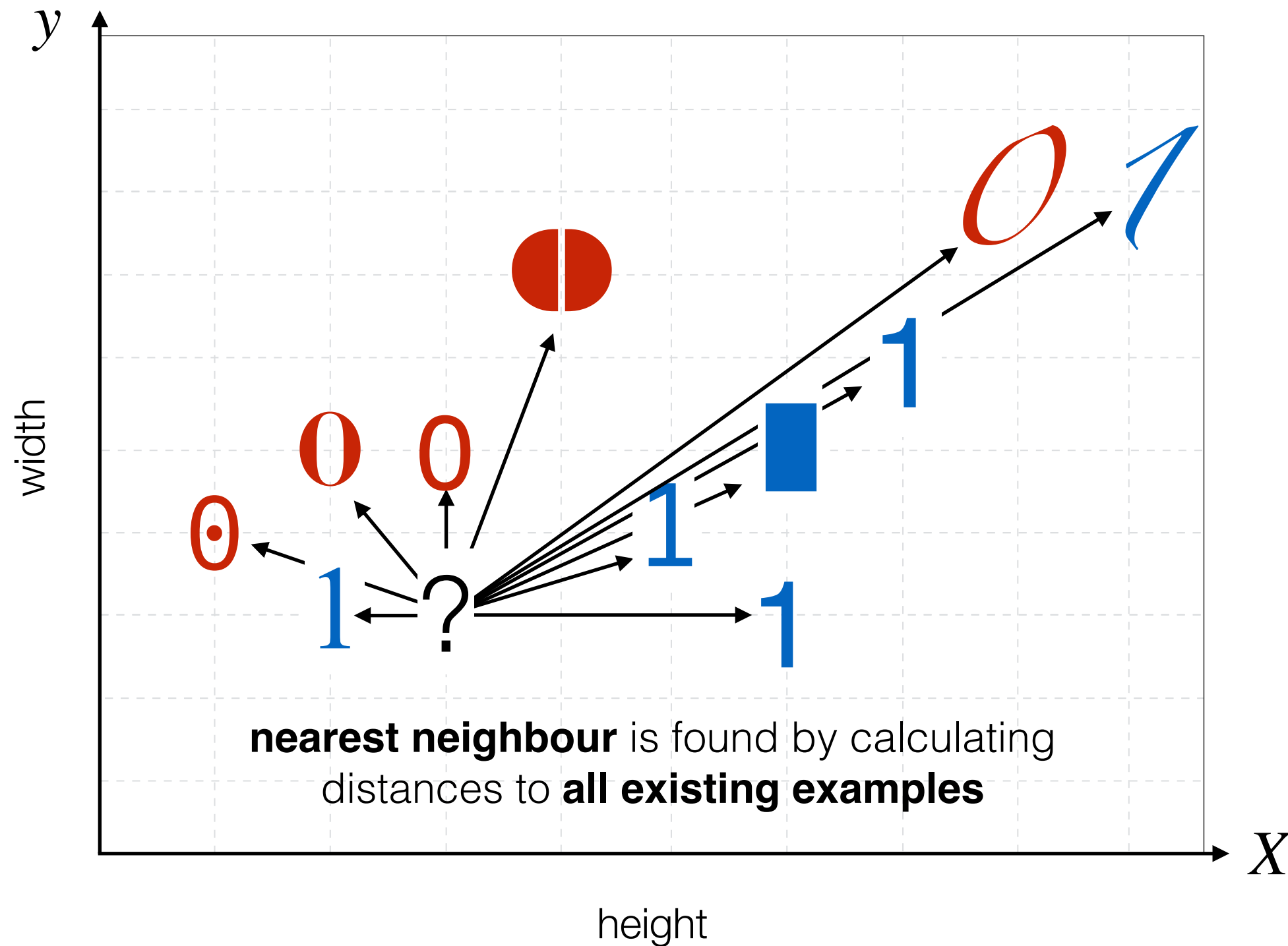
Nearest Neighbour Classifier



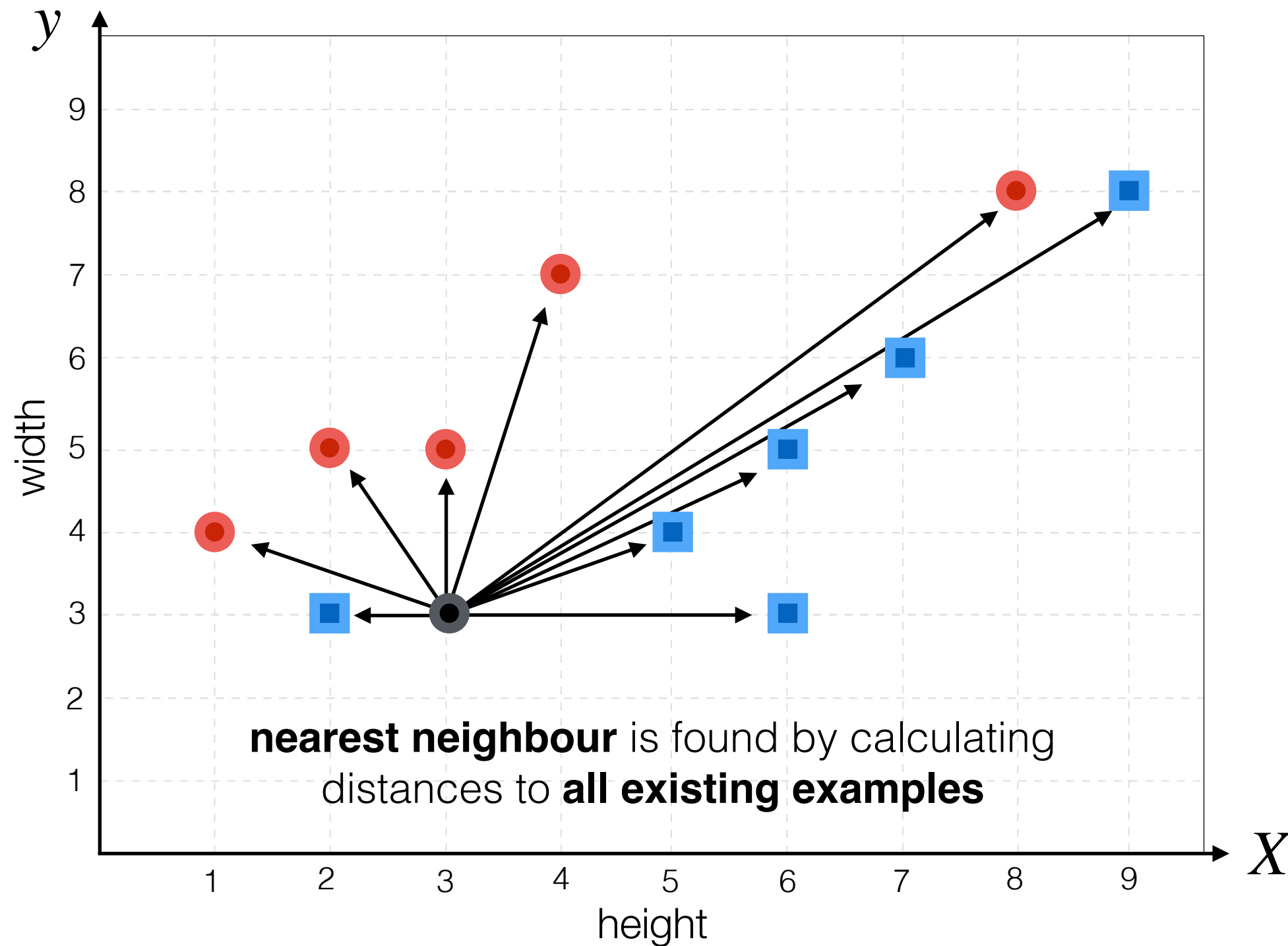
Nearest Neighbour Classifier



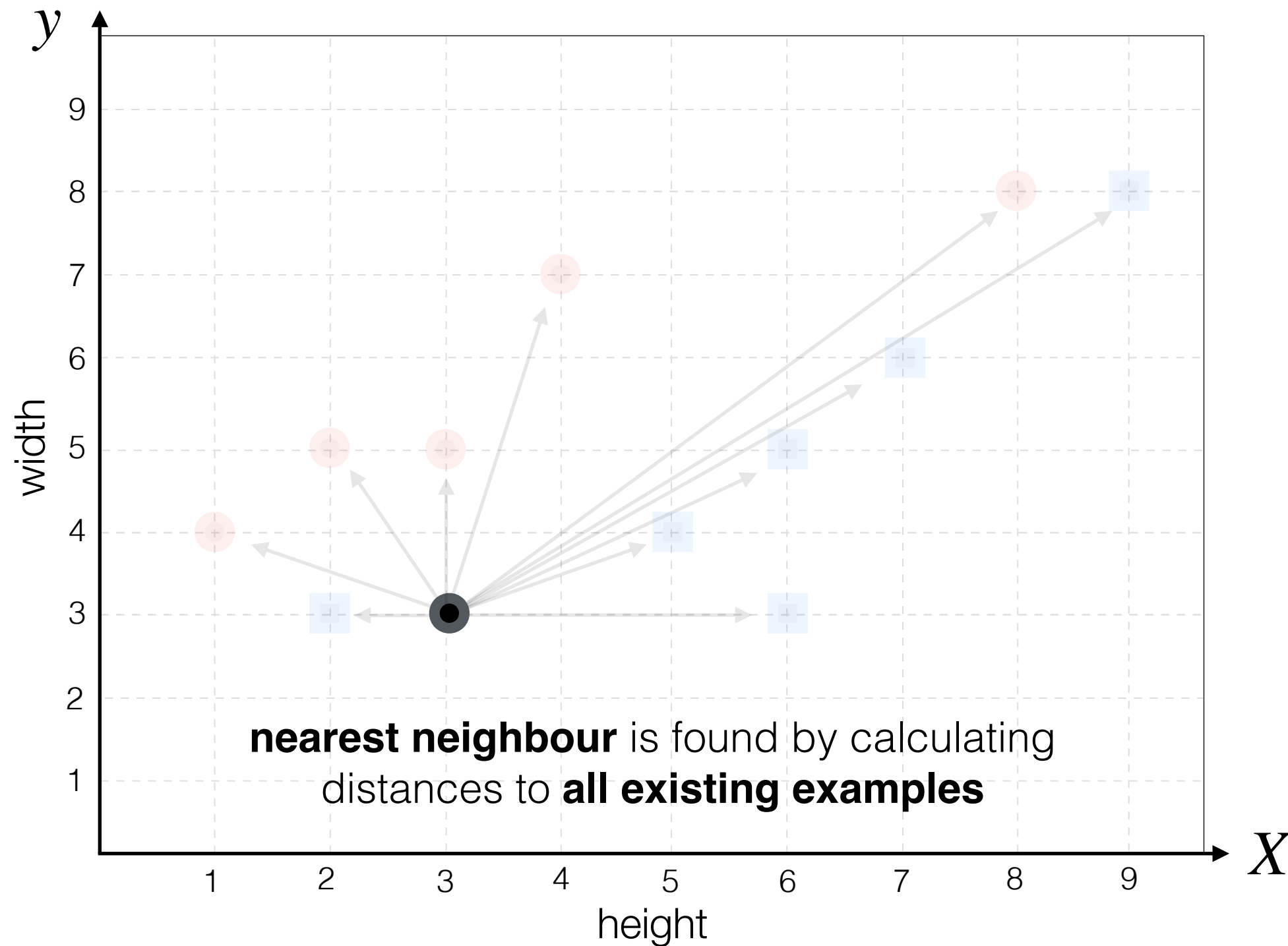
Nearest Neighbour Classifier



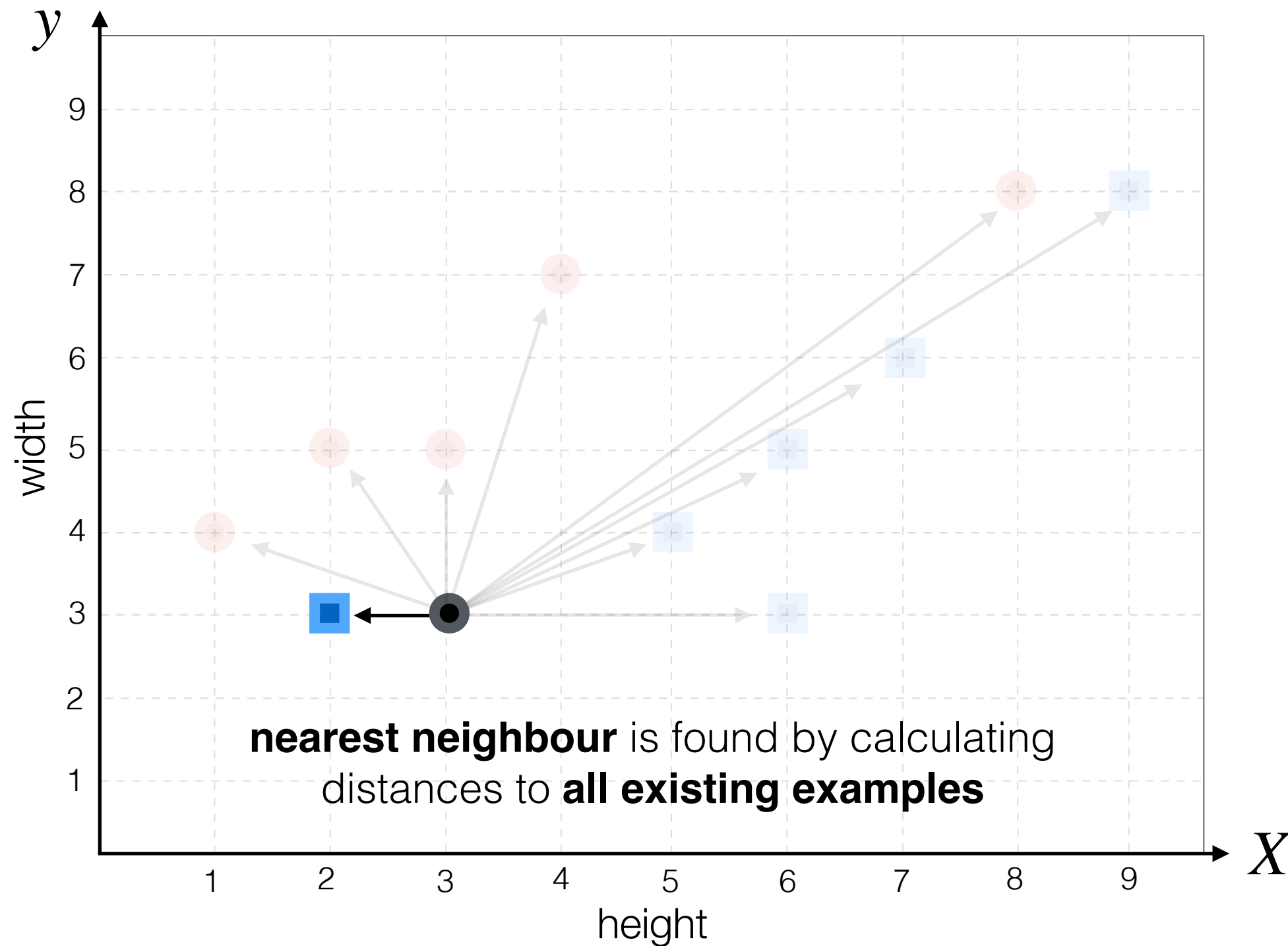
Nearest Neighbour Classifier



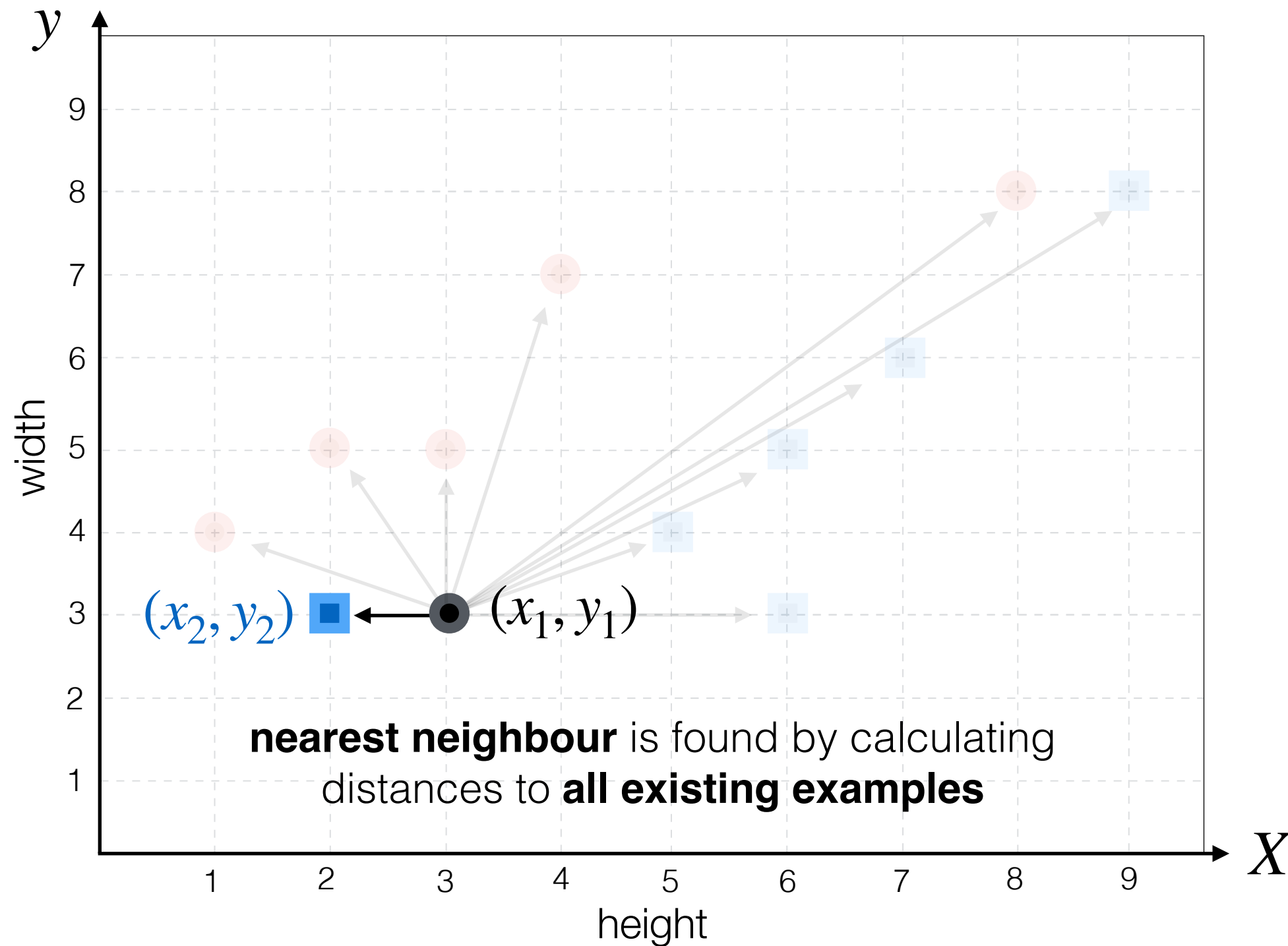
Nearest Neighbour Classifier



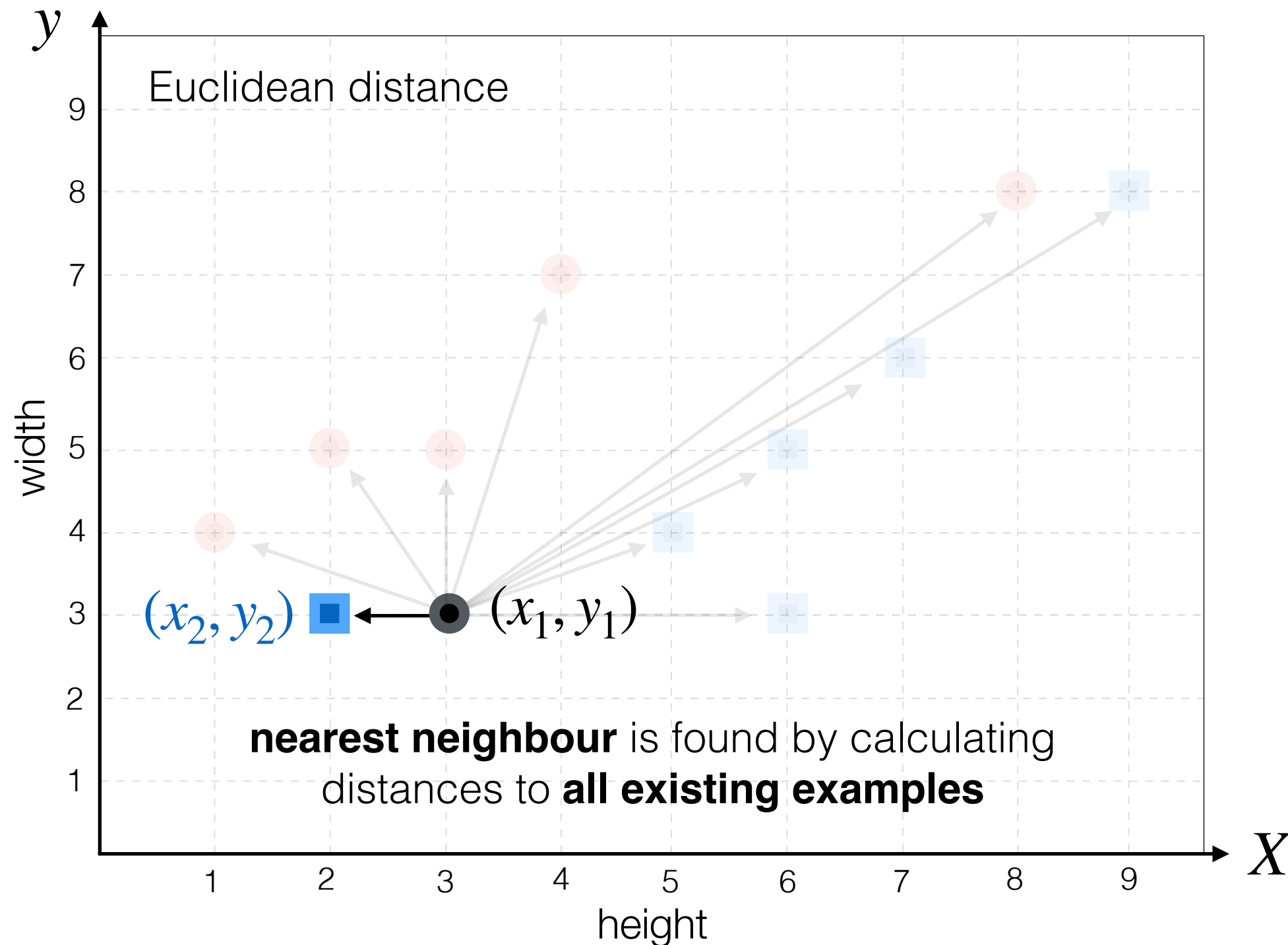
Nearest Neighbour Classifier



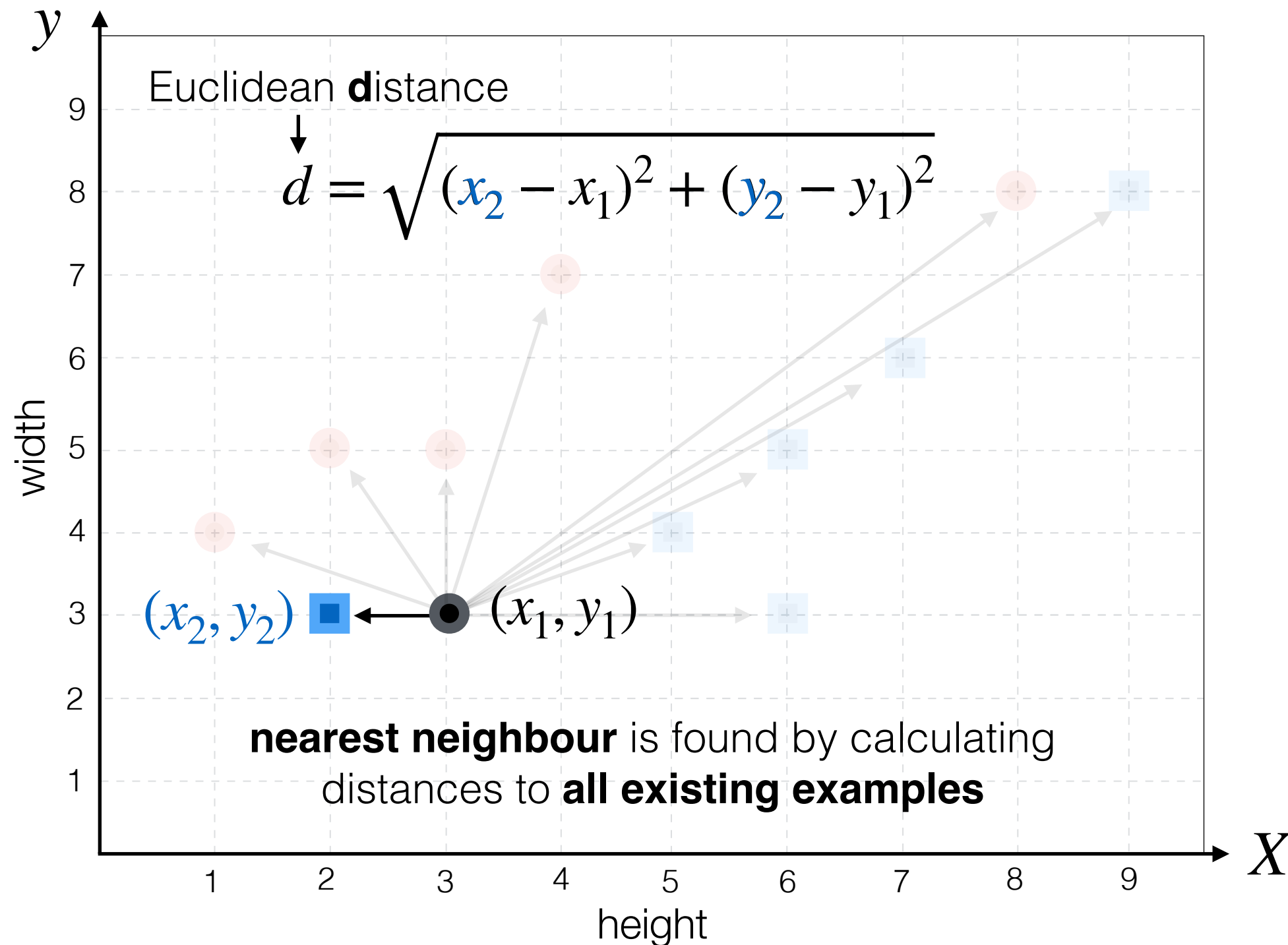
Nearest Neighbour Classifier



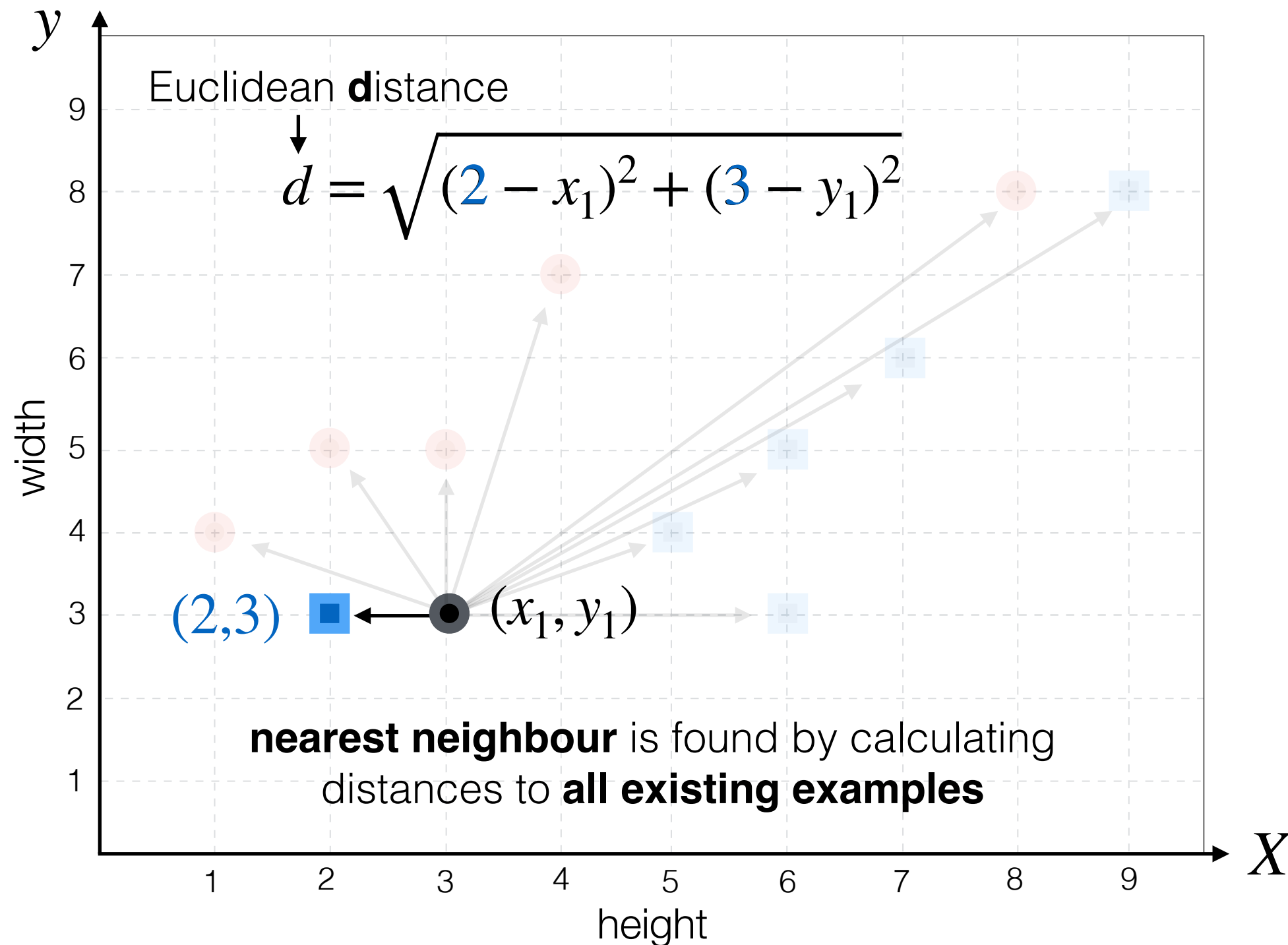
Nearest Neighbour Classifier



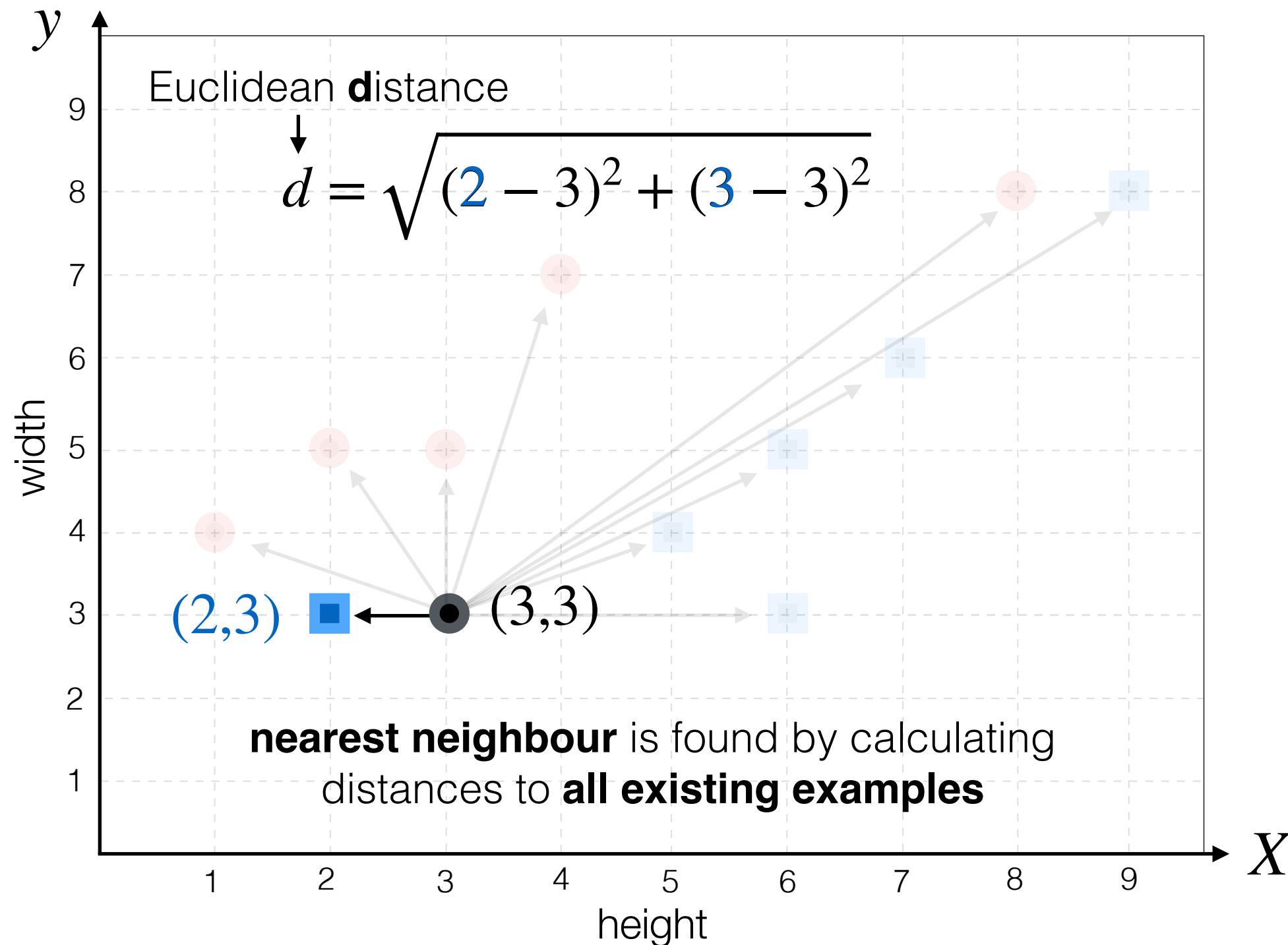
Nearest Neighbour Classifier



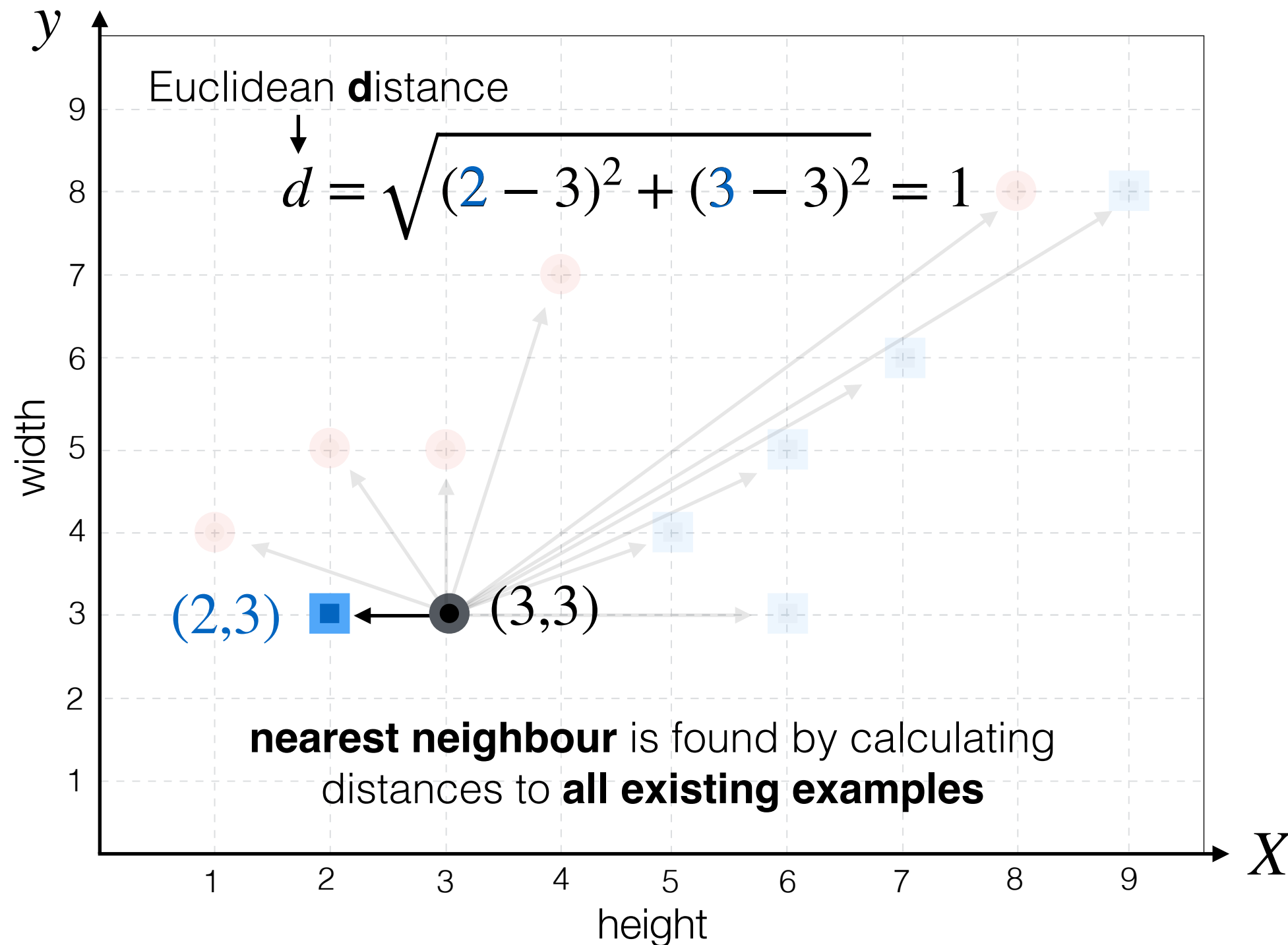
Nearest Neighbour Classifier



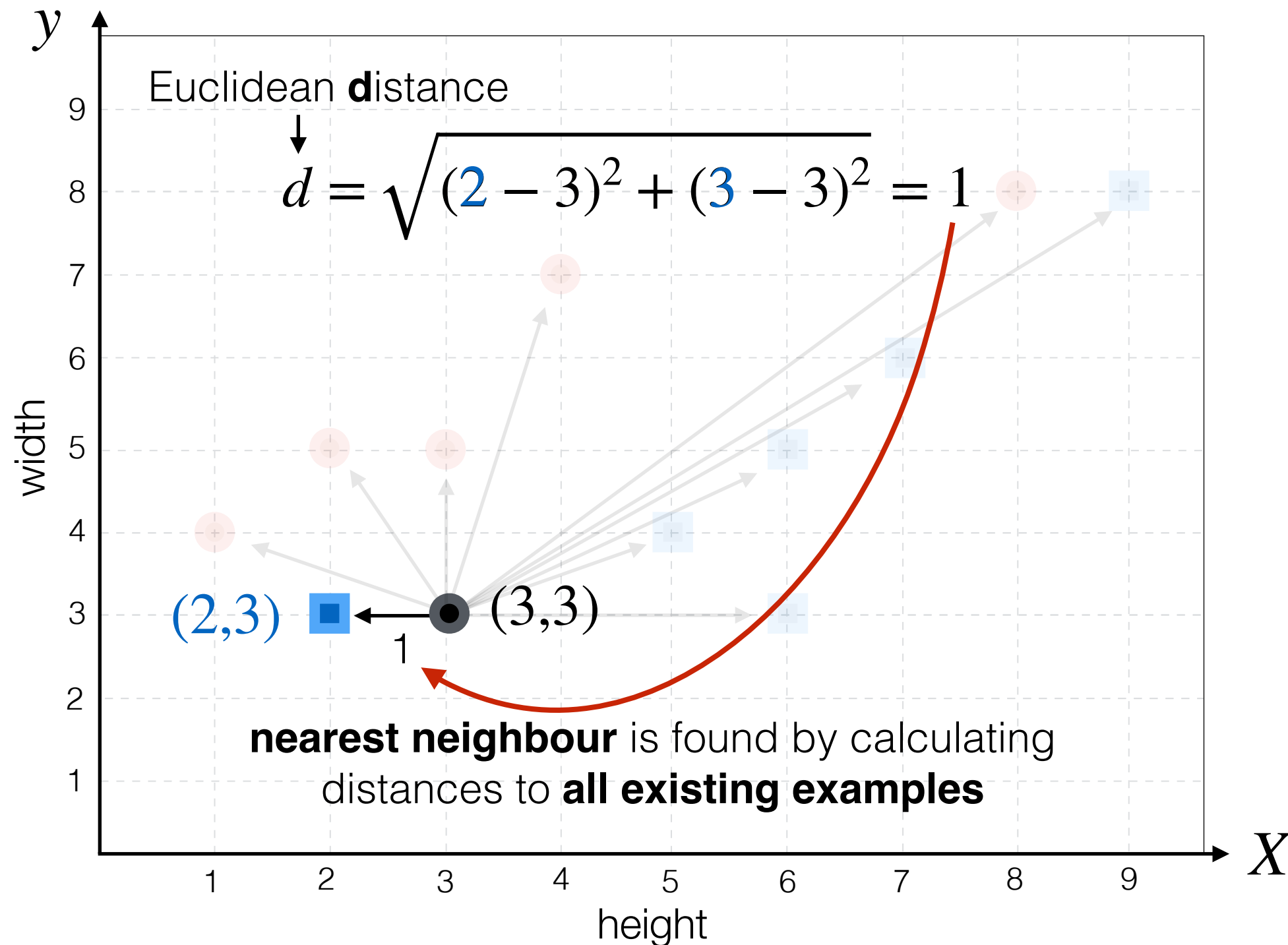
Nearest Neighbour Classifier



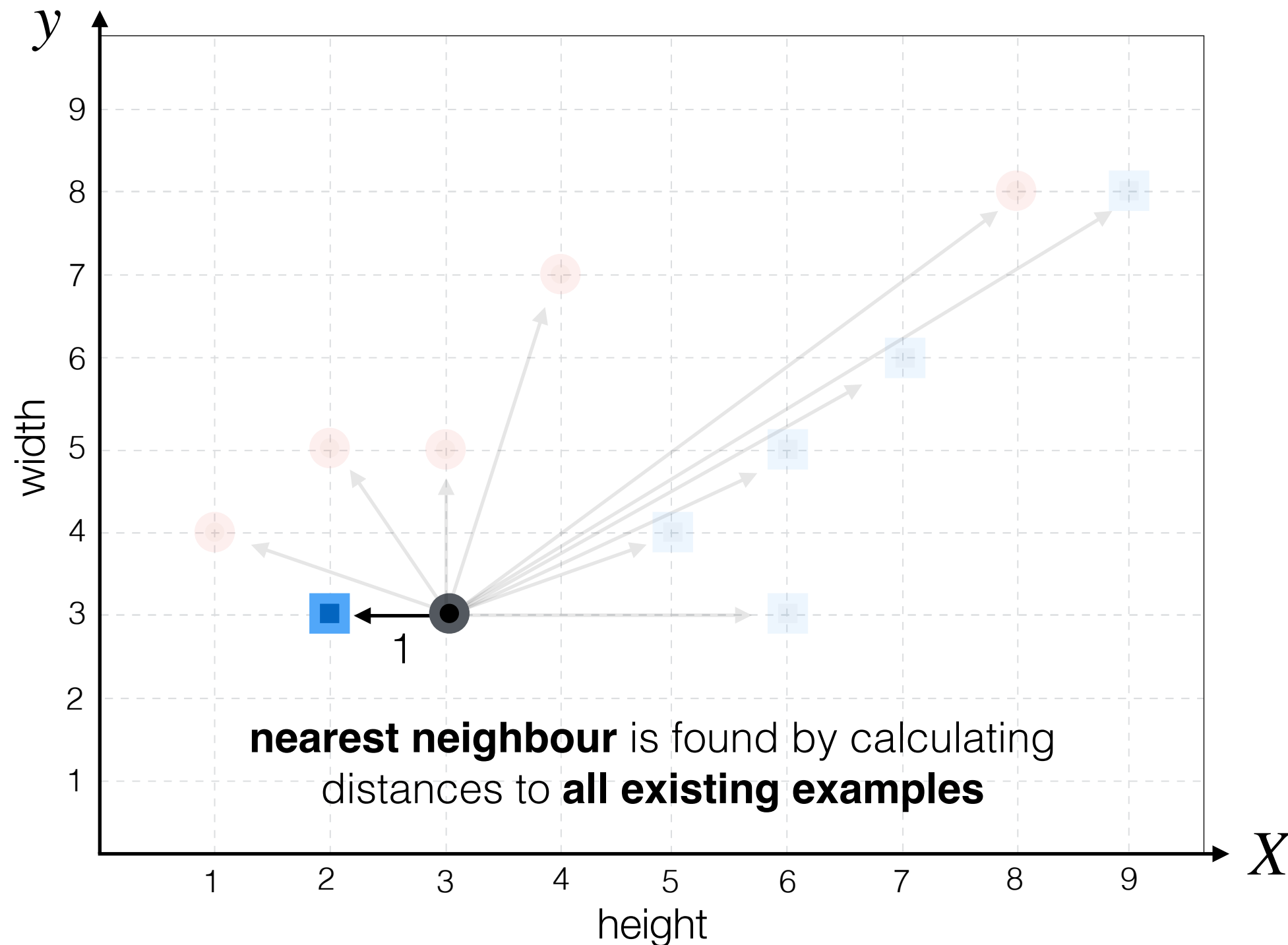
Nearest Neighbour Classifier



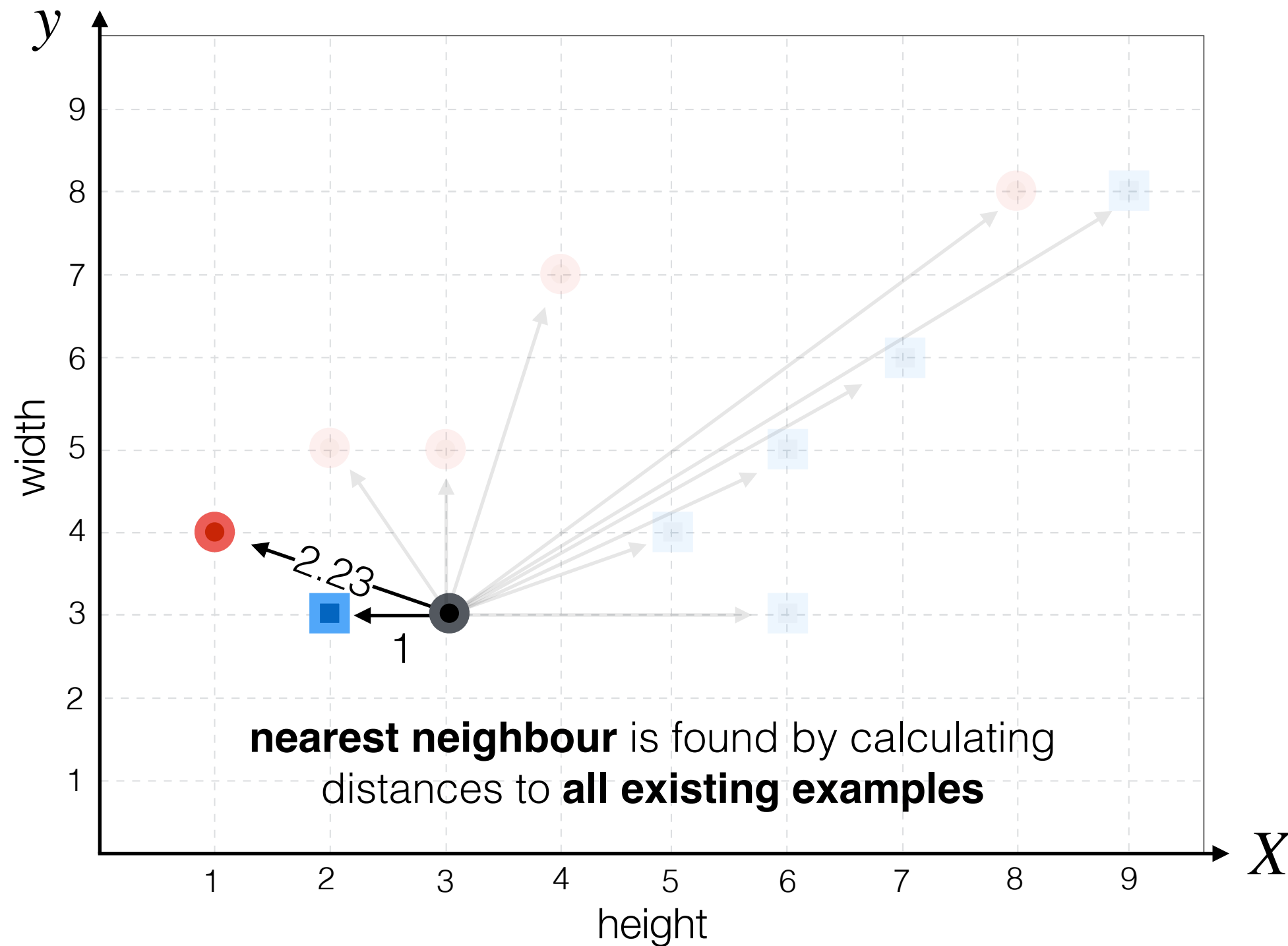
Nearest Neighbour Classifier



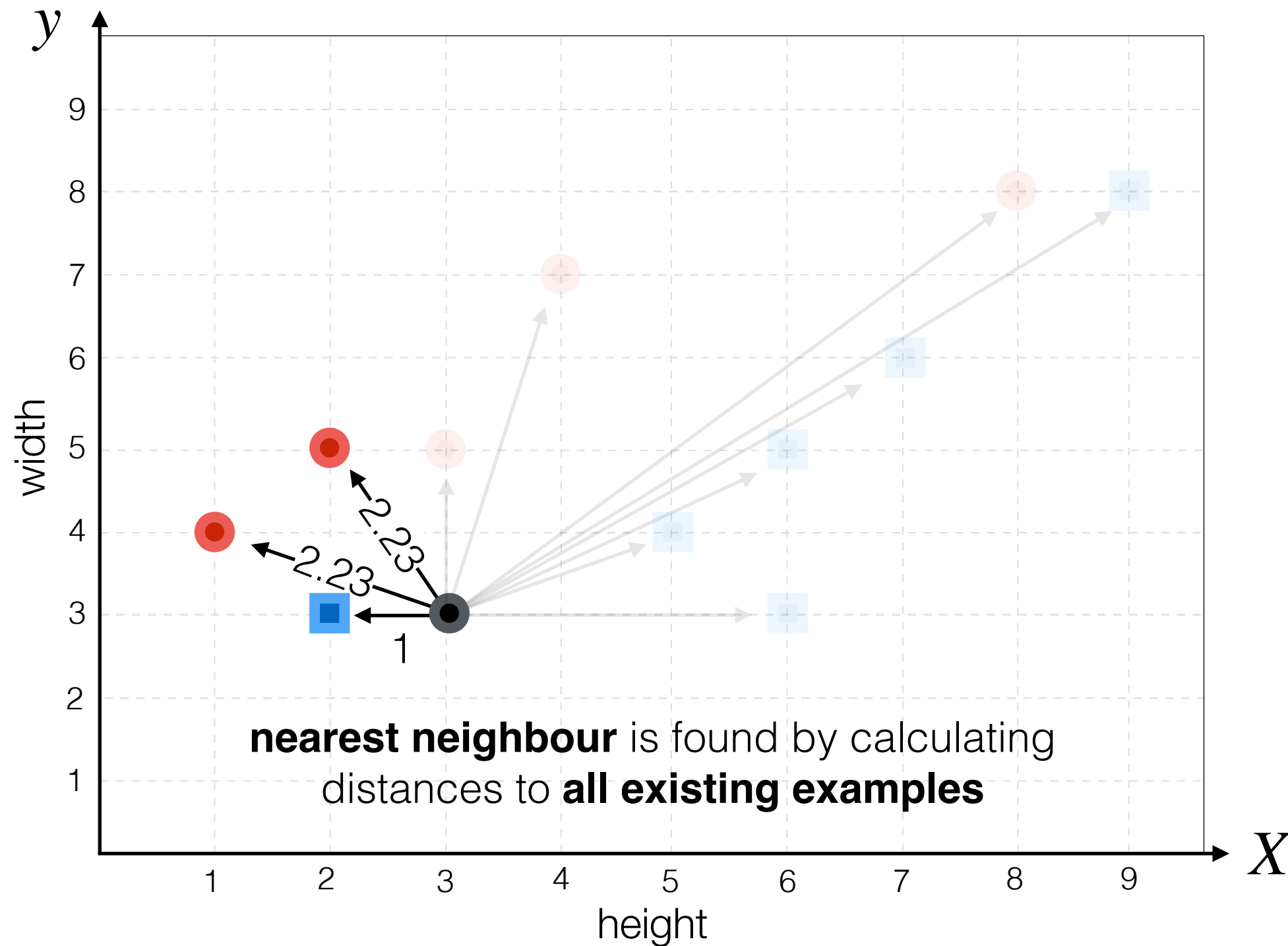
Nearest Neighbour Classifier



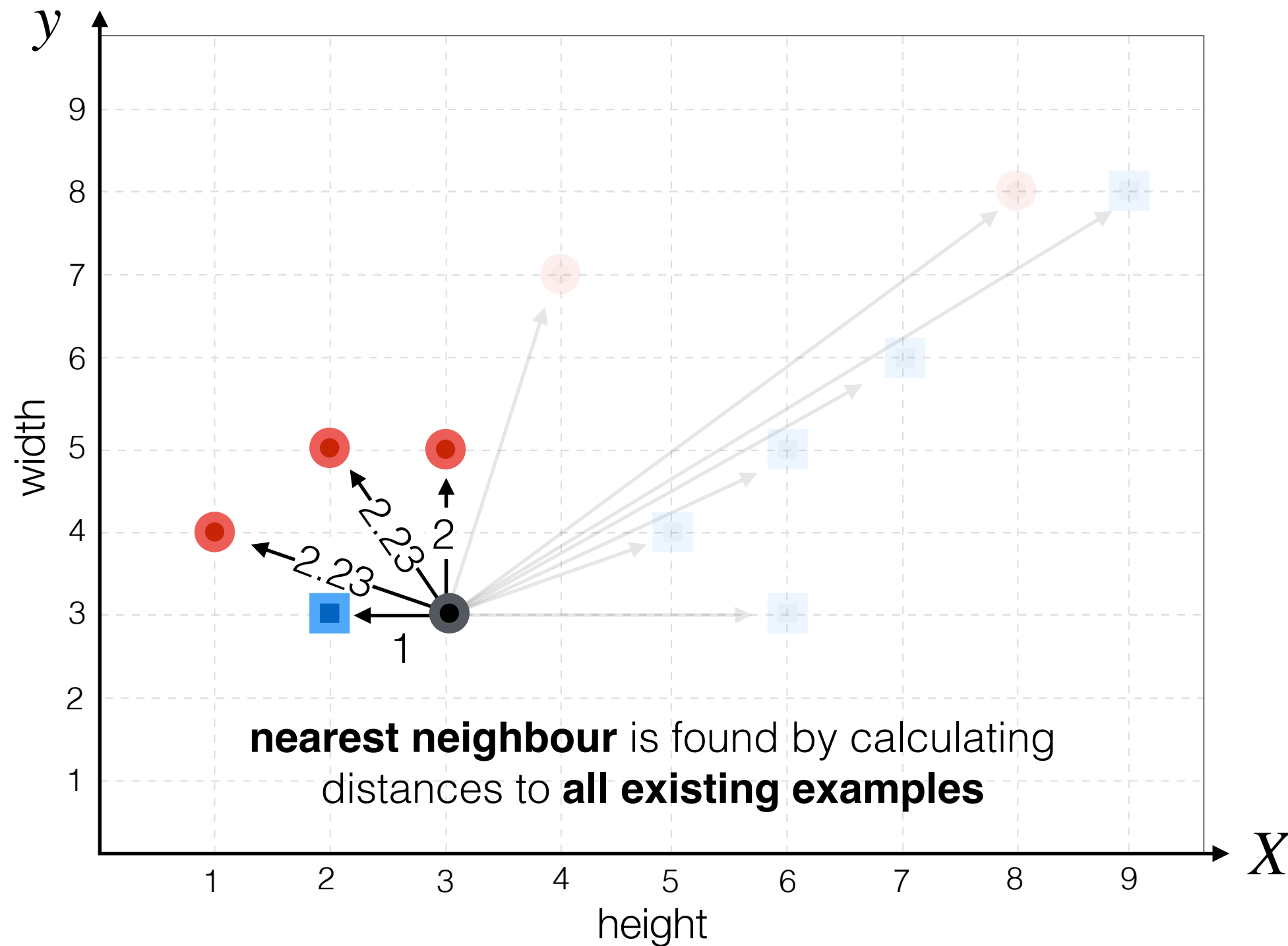
Nearest Neighbour Classifier



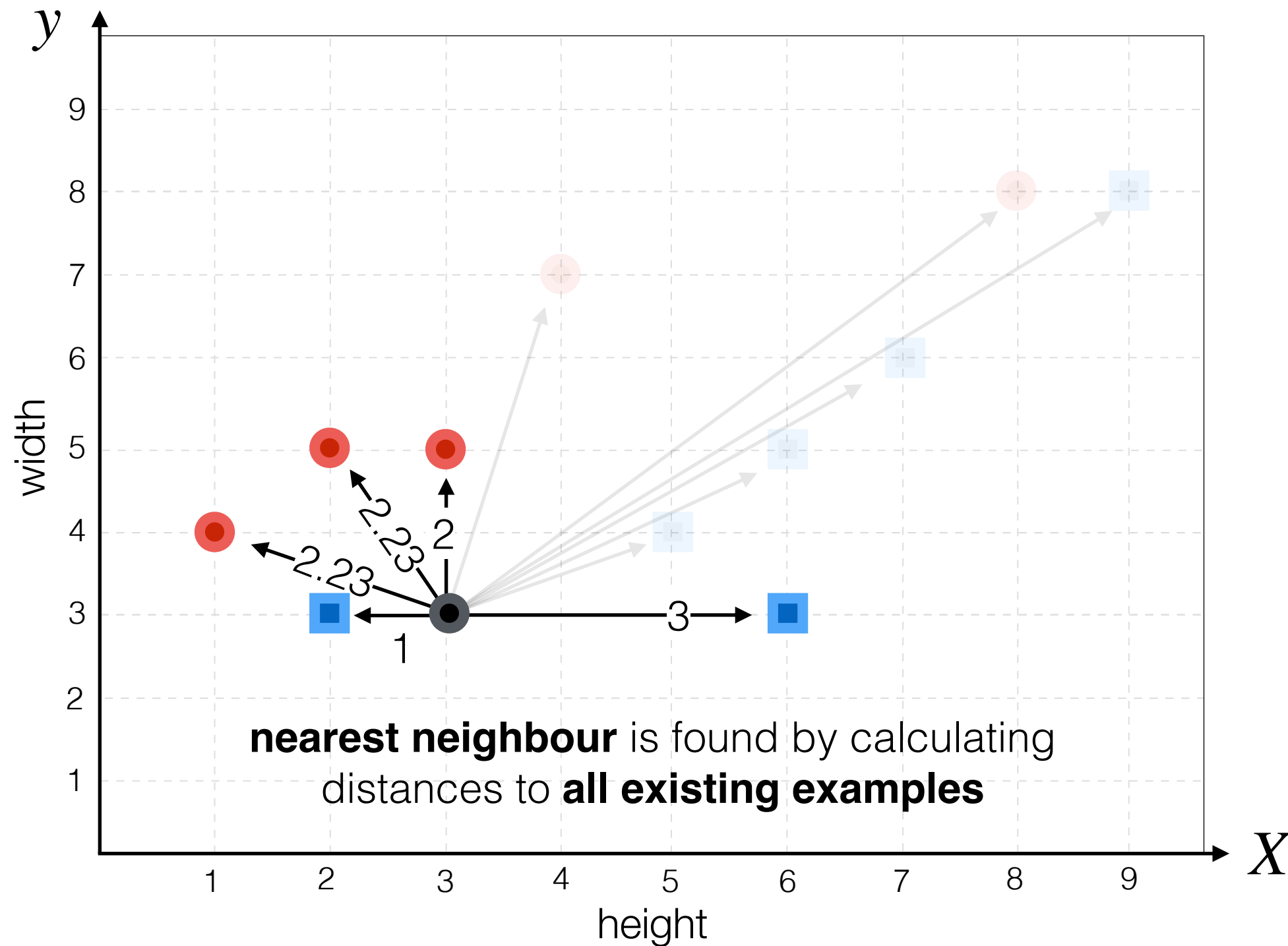
Nearest Neighbour Classifier



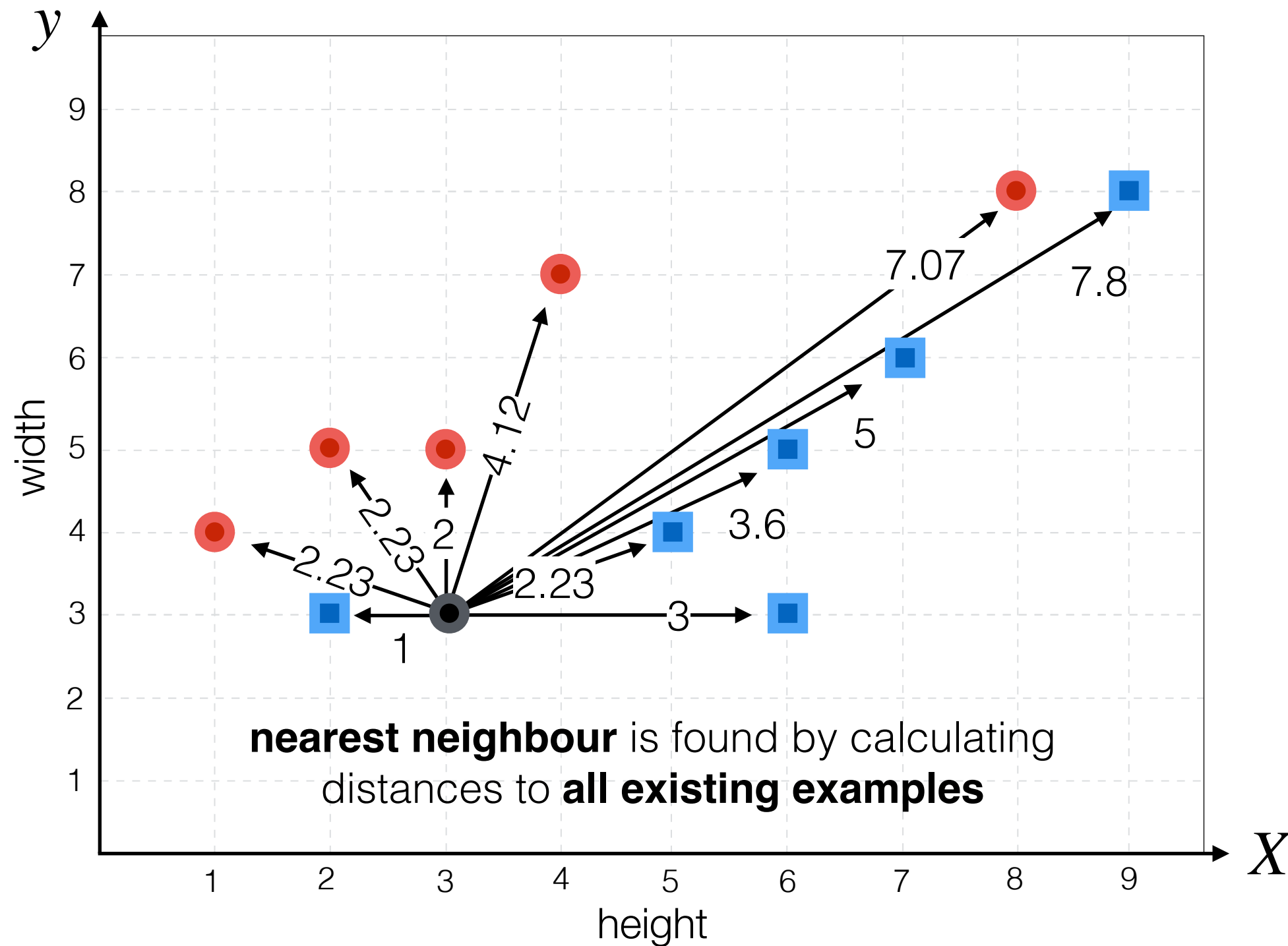
Nearest Neighbour Classifier



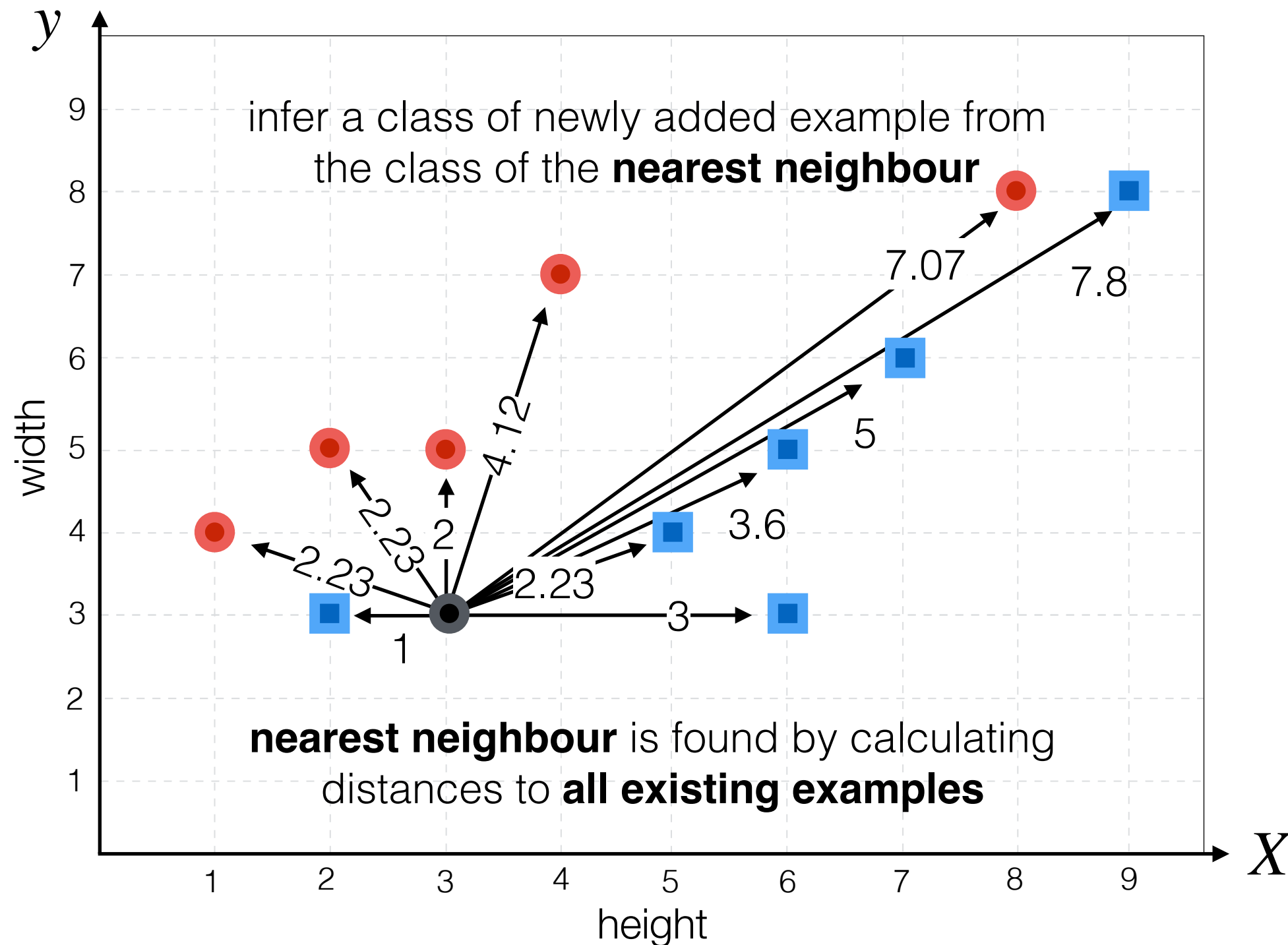
Nearest Neighbour Classifier



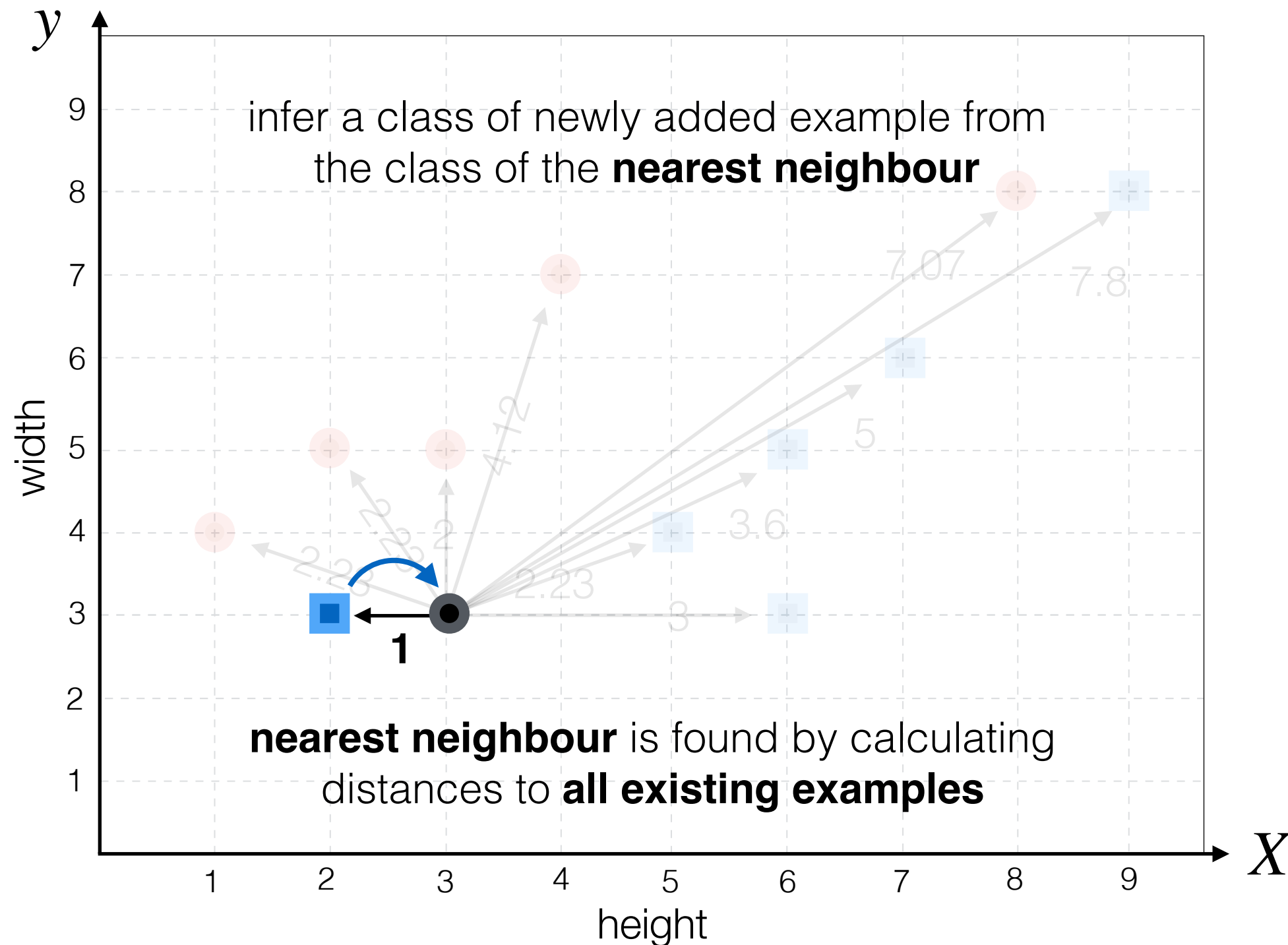
Nearest Neighbour Classifier



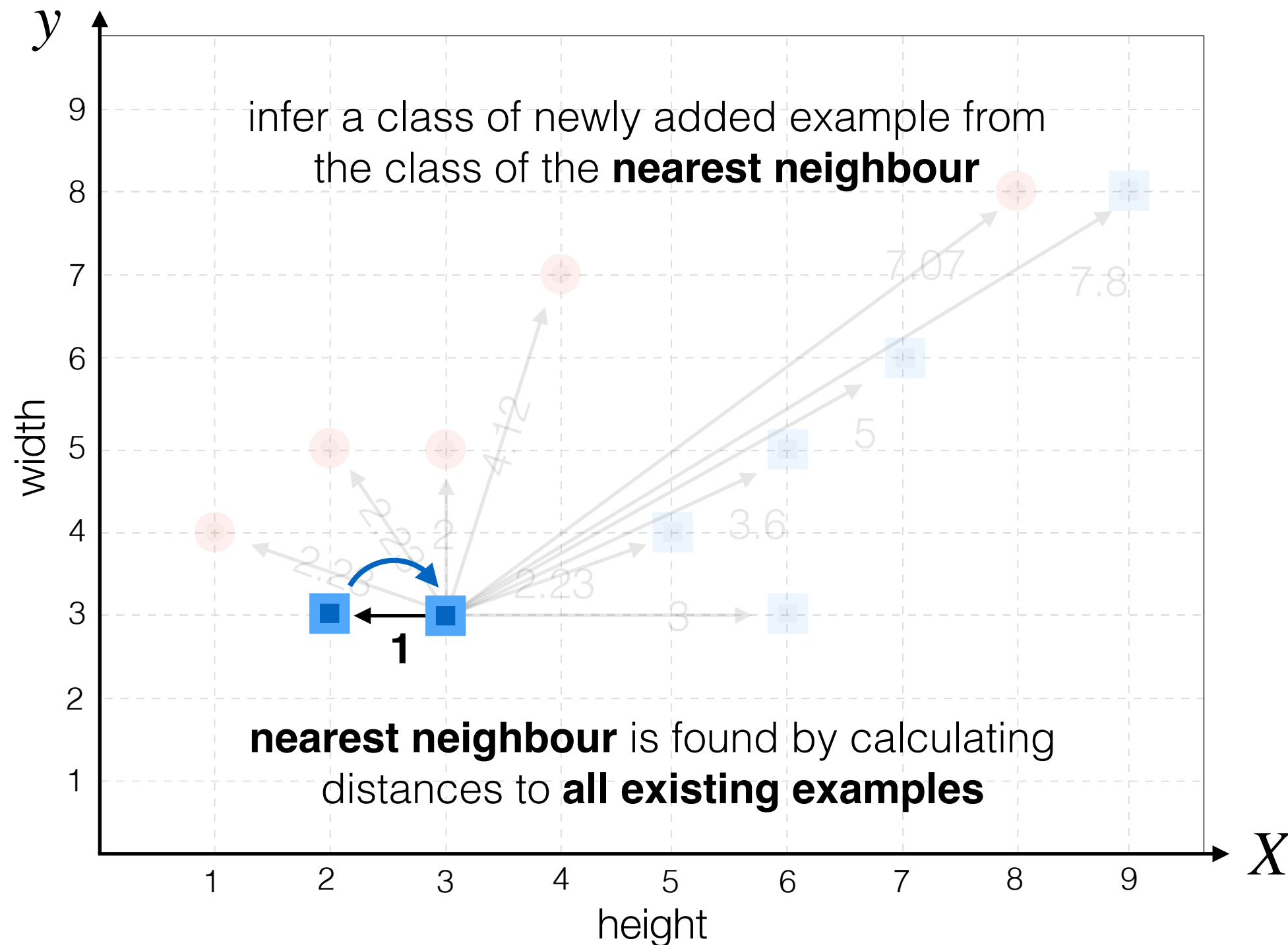
Nearest Neighbour Classifier



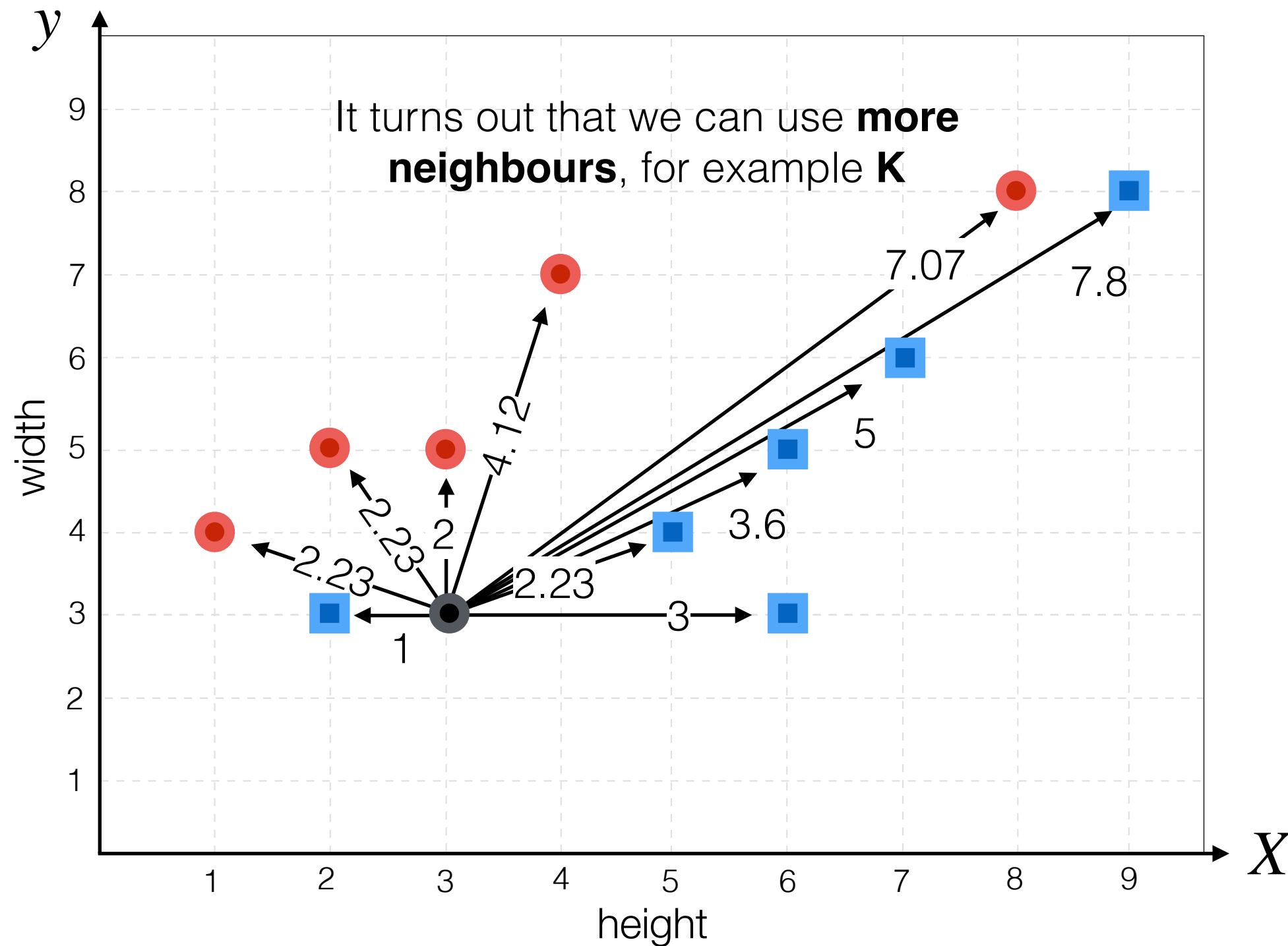
Nearest Neighbour Classifier



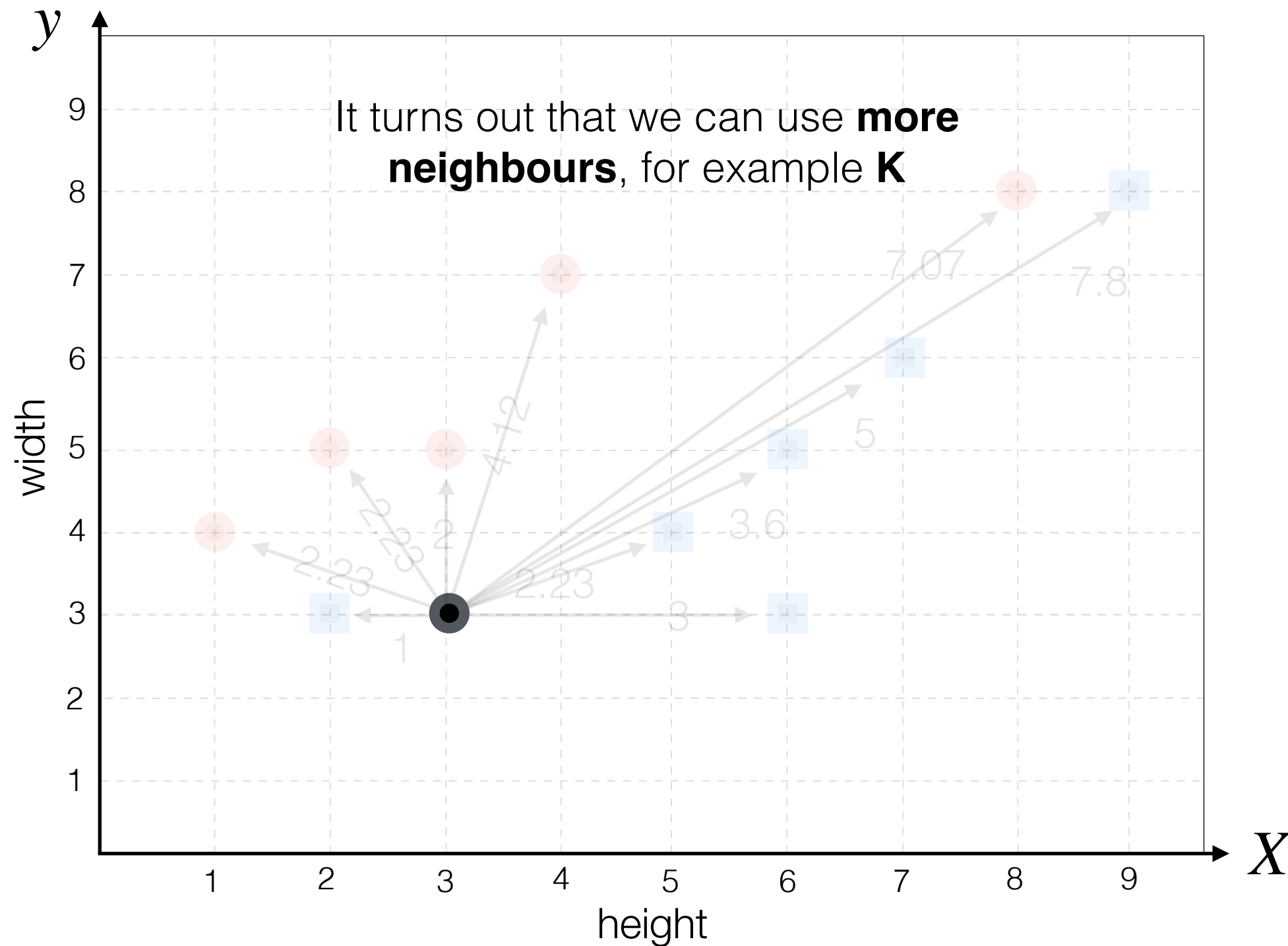
Nearest Neighbour Classifier



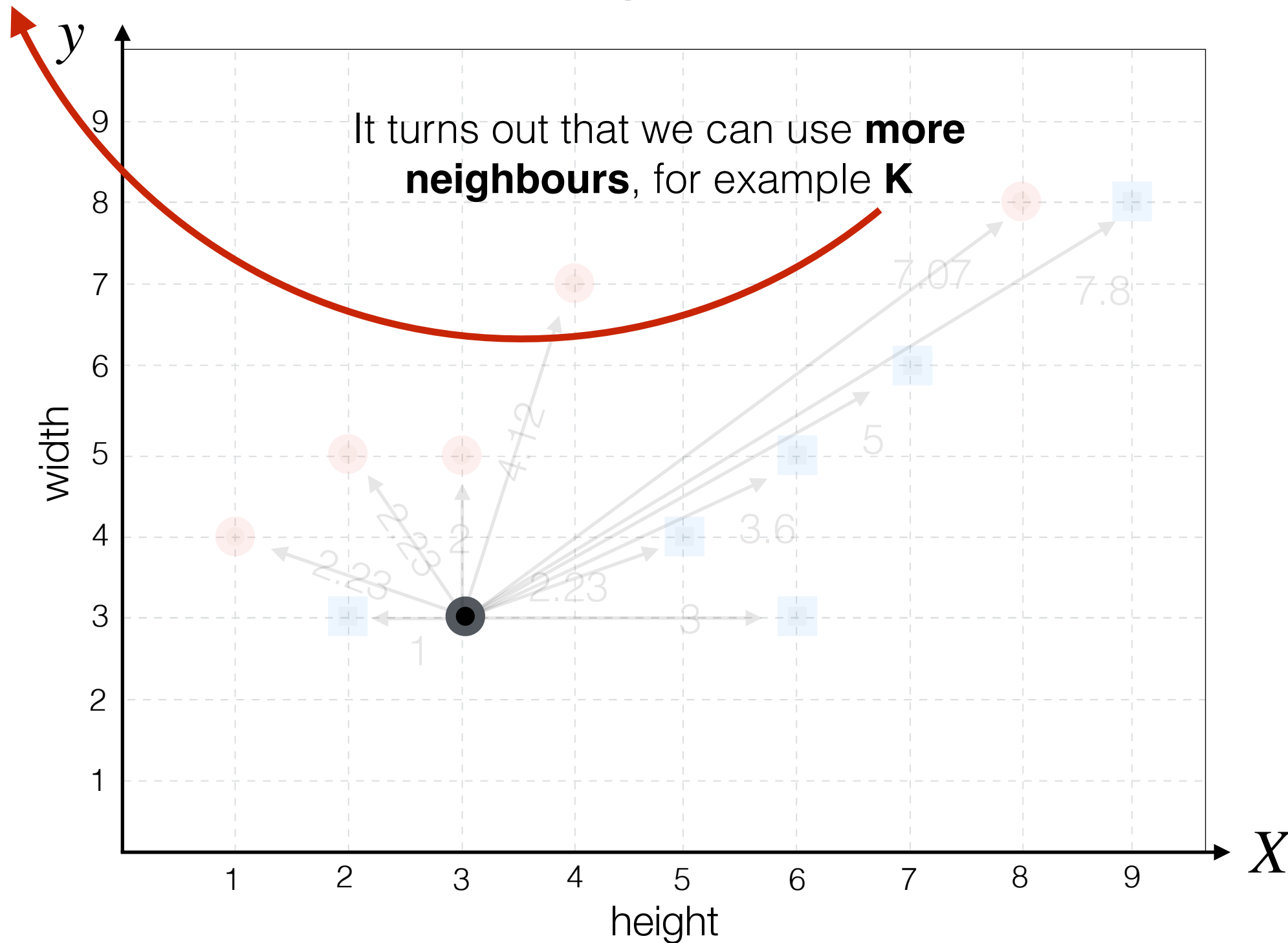
Nearest Neighbour Classifier



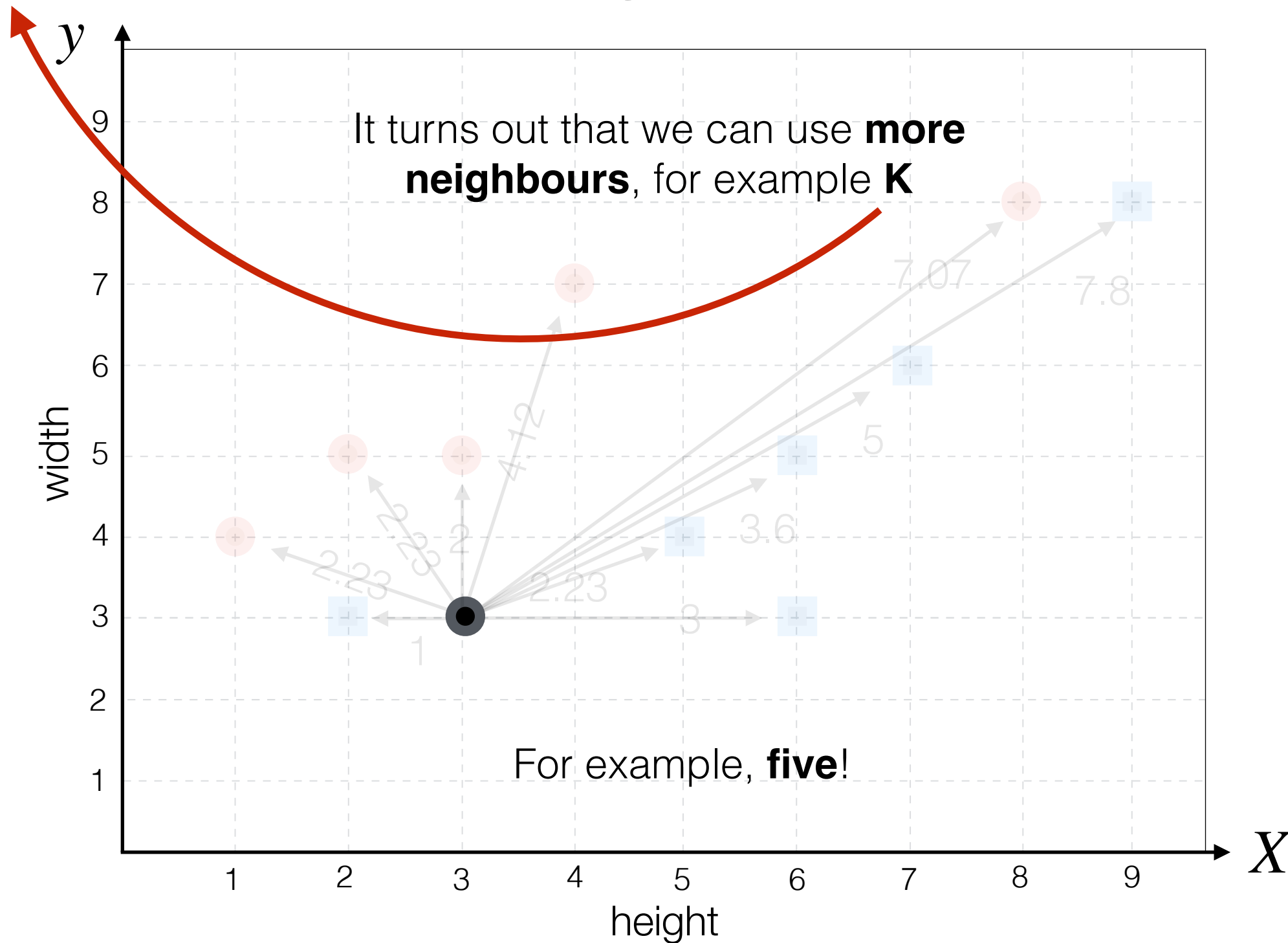
Nearest Neighbour Classifier



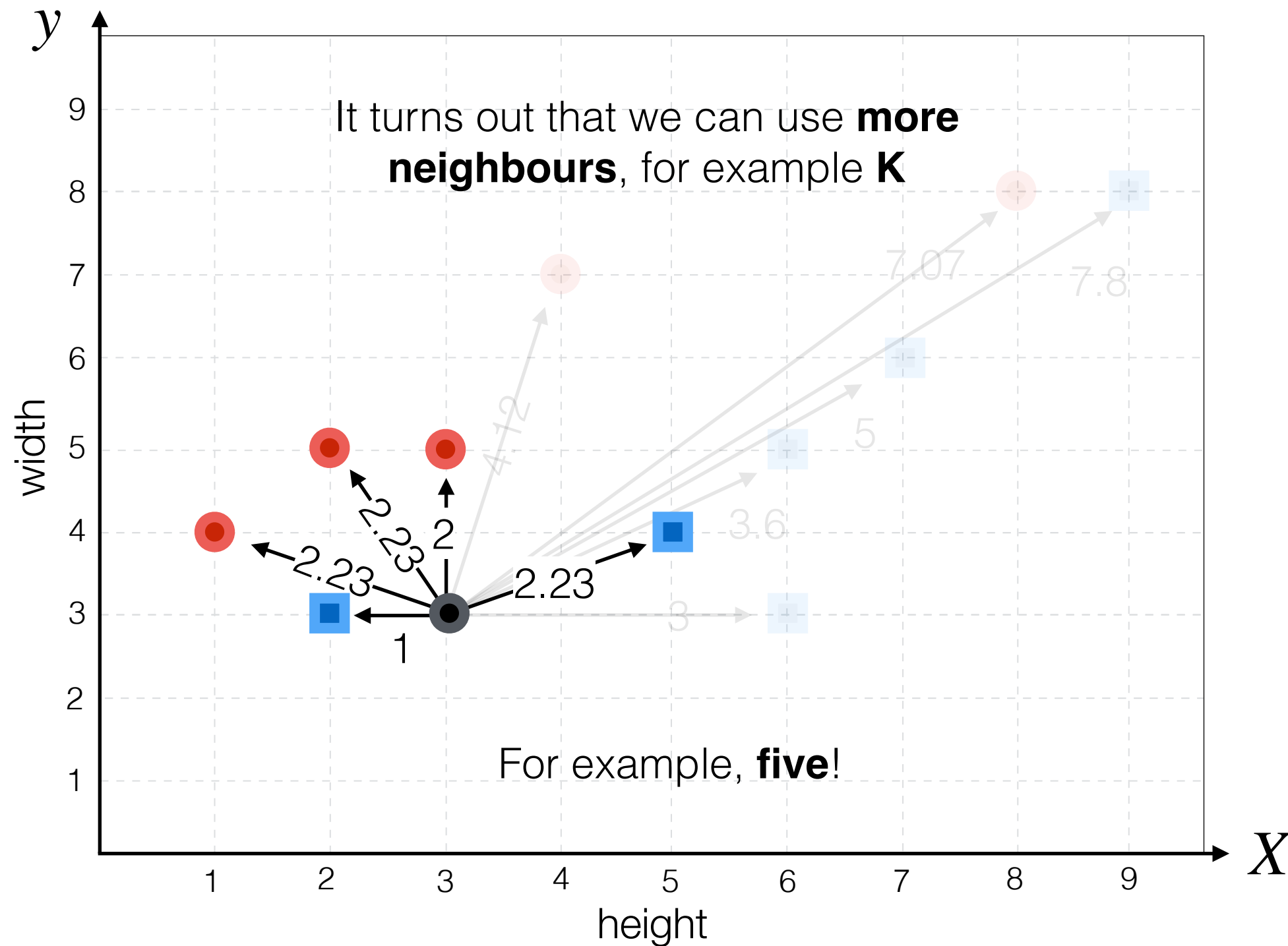
K-Nearest Neighbour Classifier



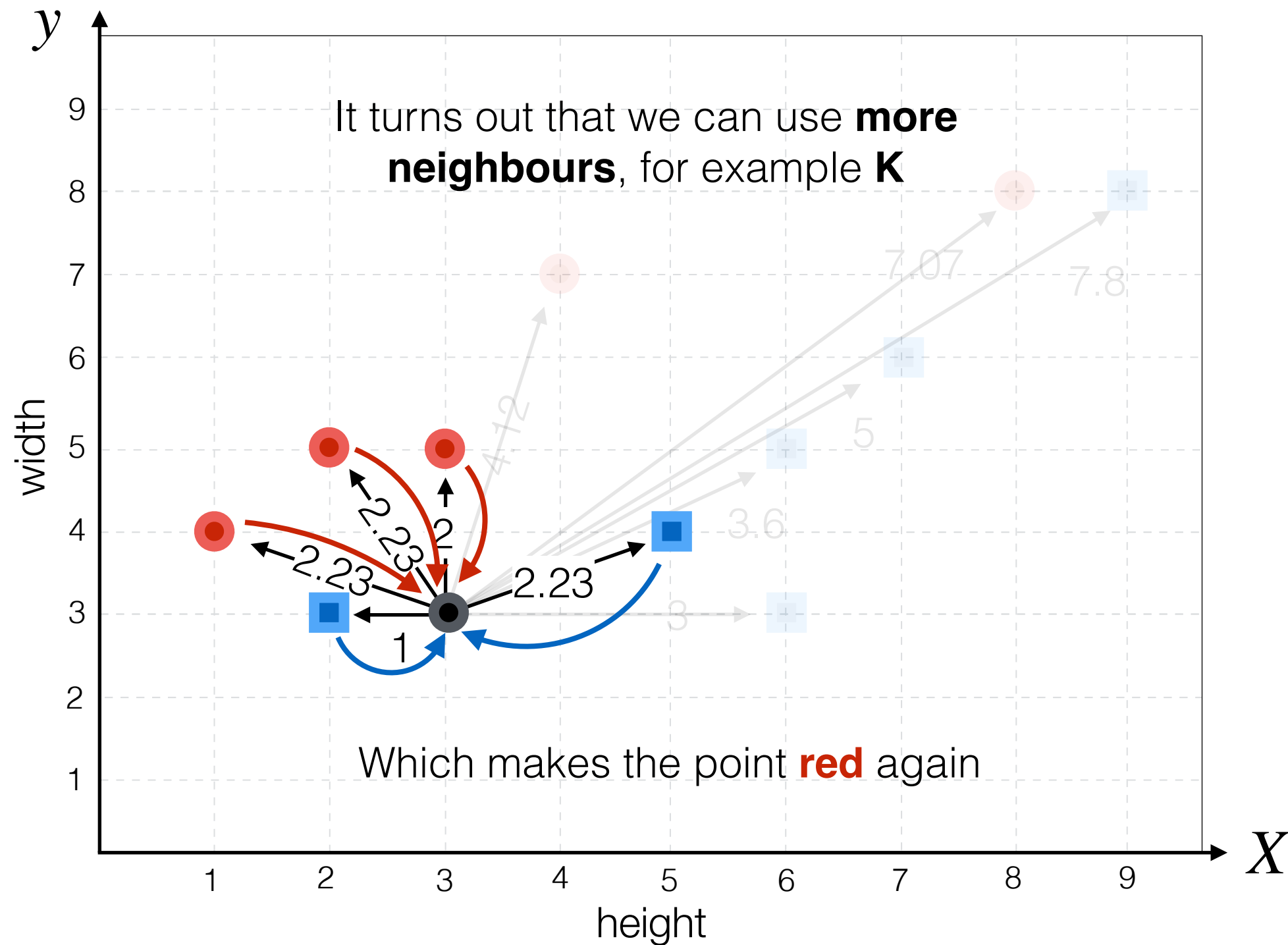
K-Nearest Neighbour Classifier



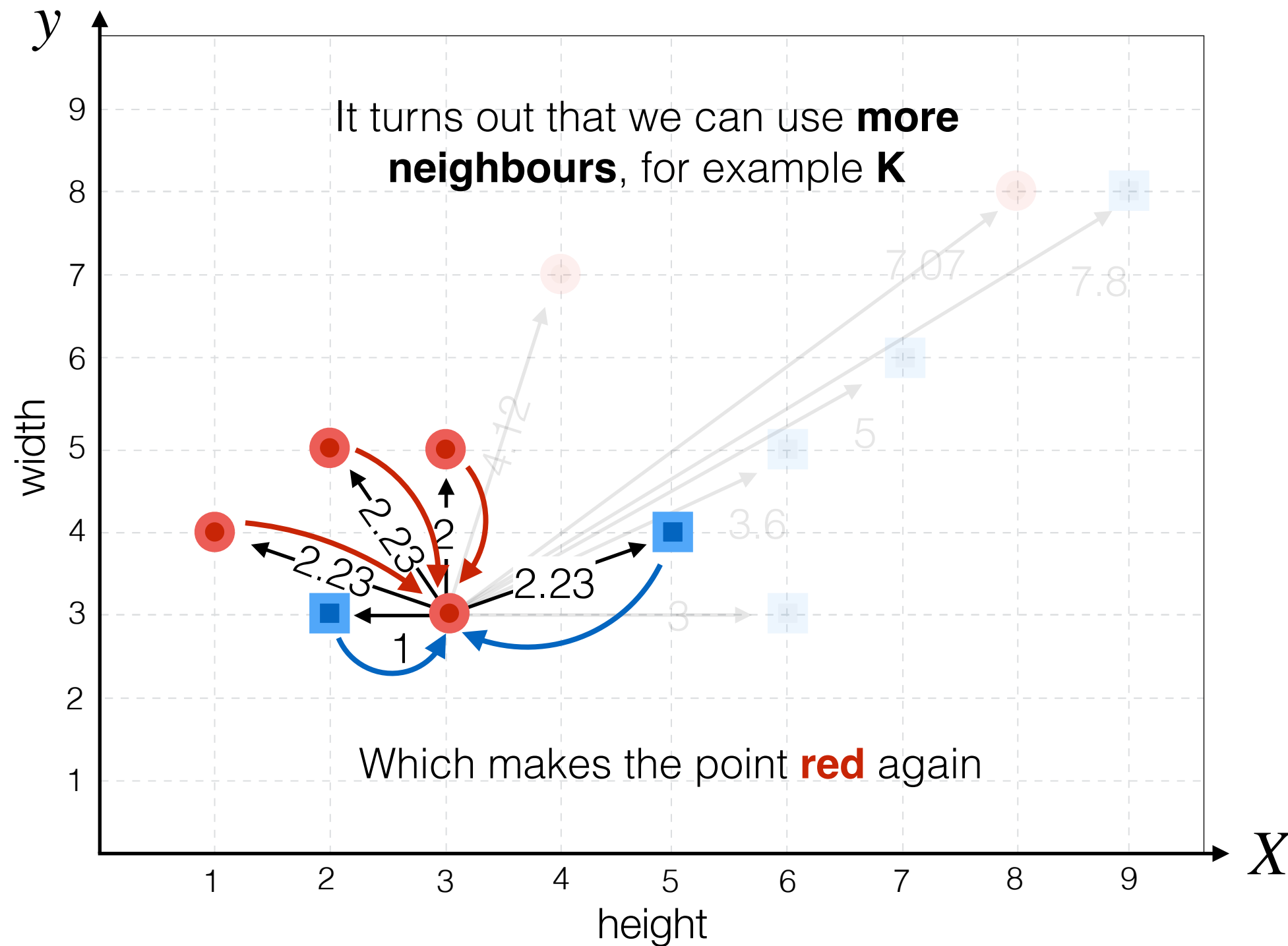
K-Nearest Neighbour Classifier



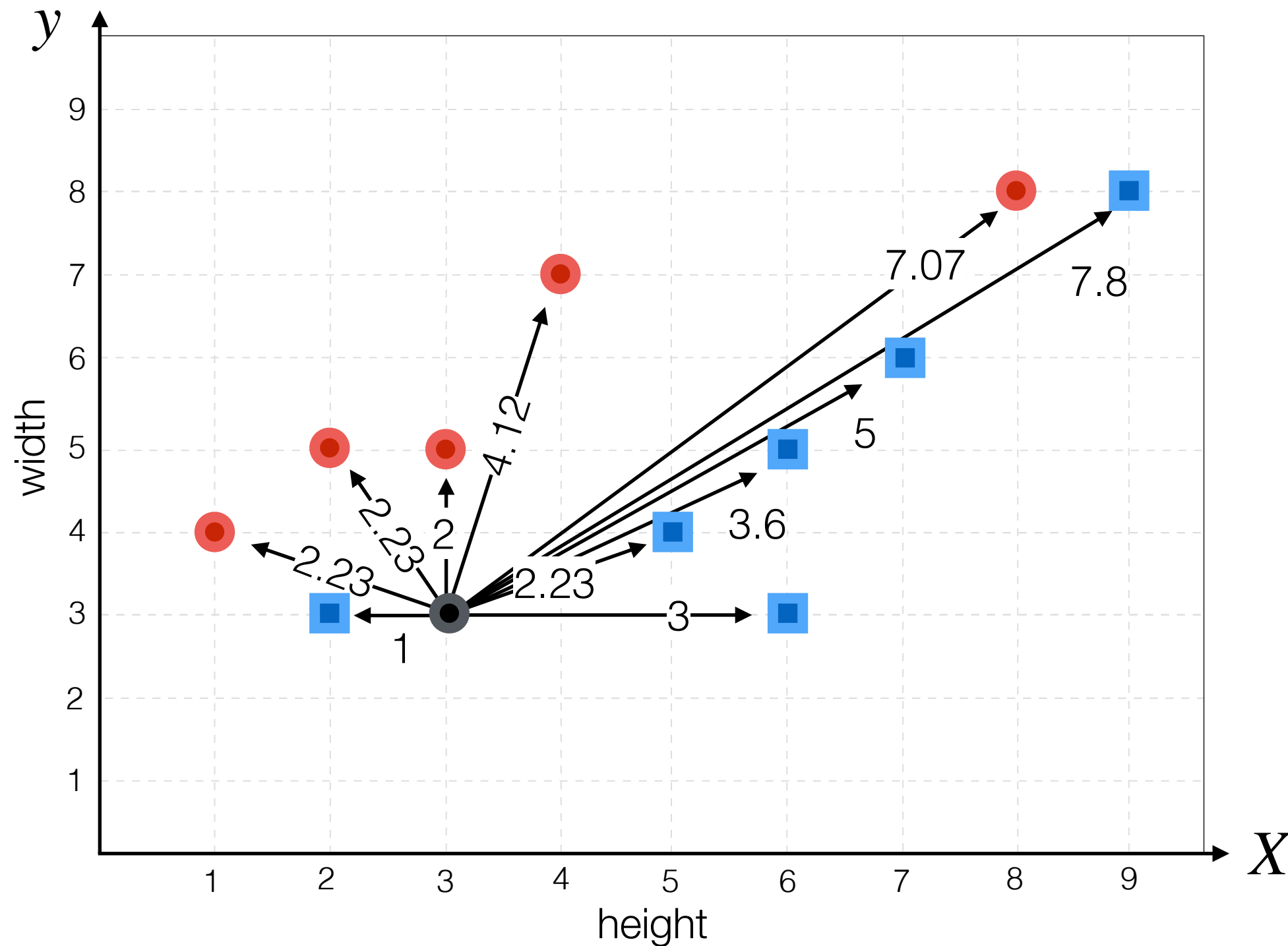
K-Nearest Neighbour Classifier



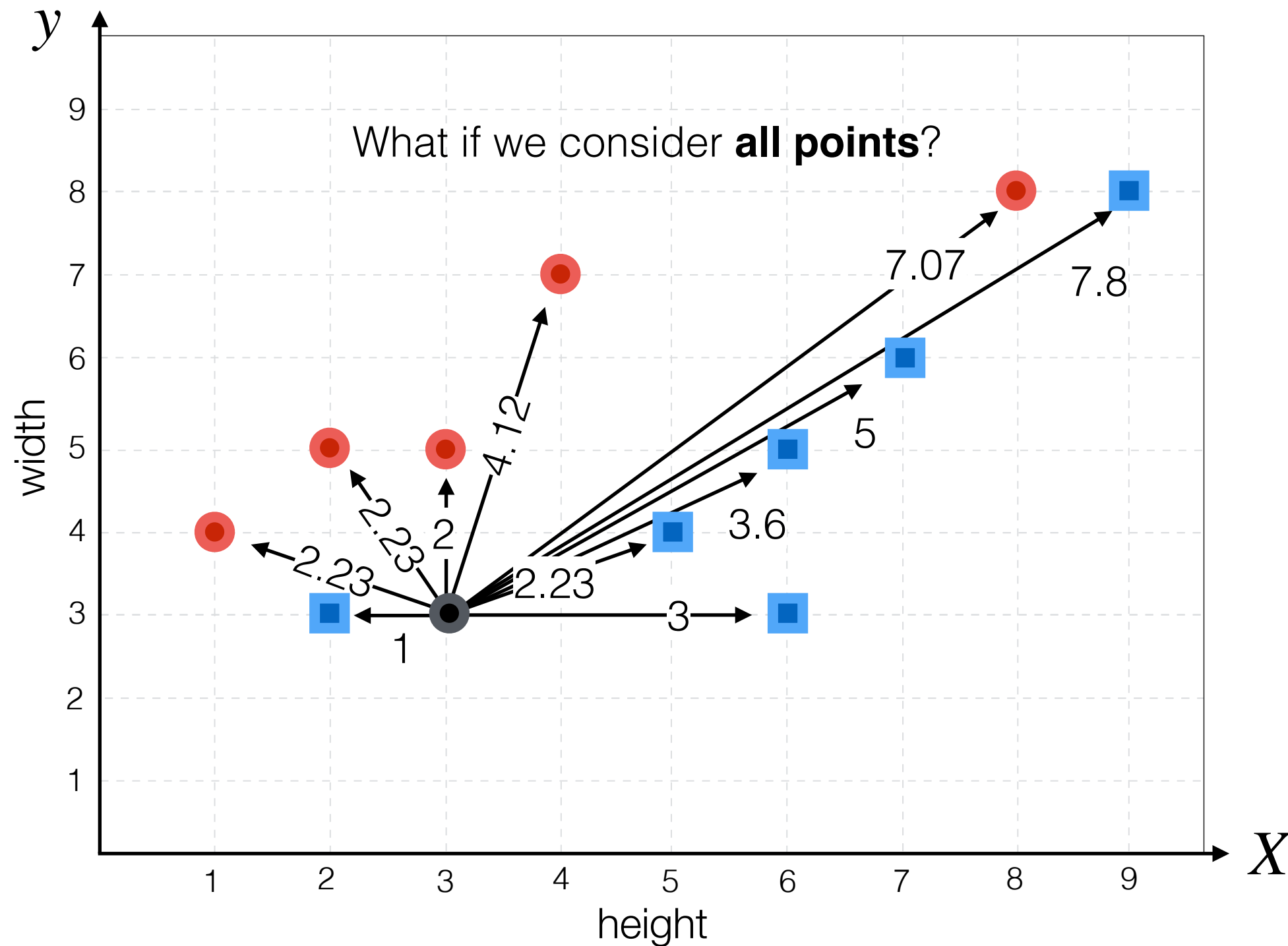
K-Nearest Neighbour Classifier



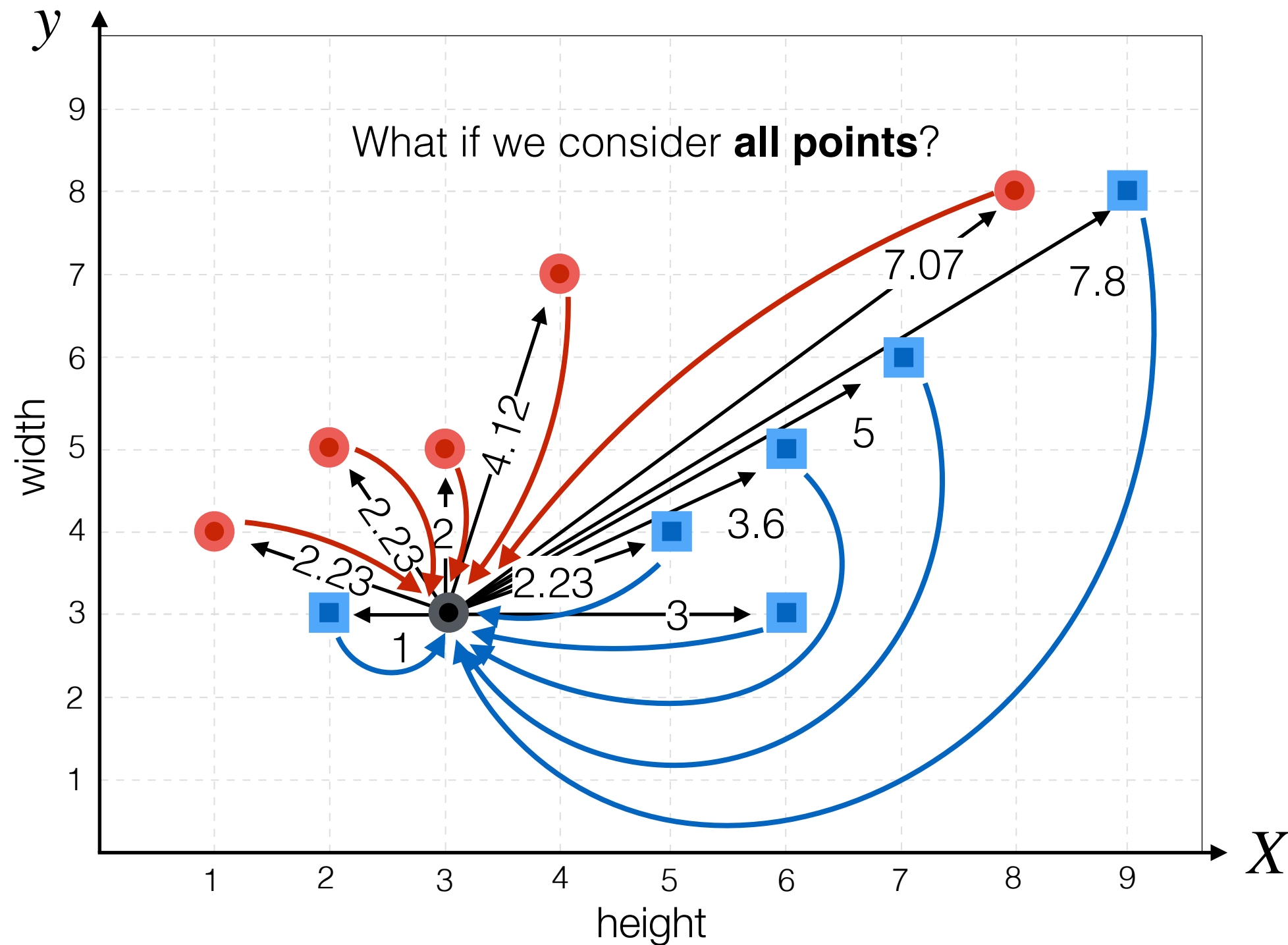
Nearest Neighbour Classifier



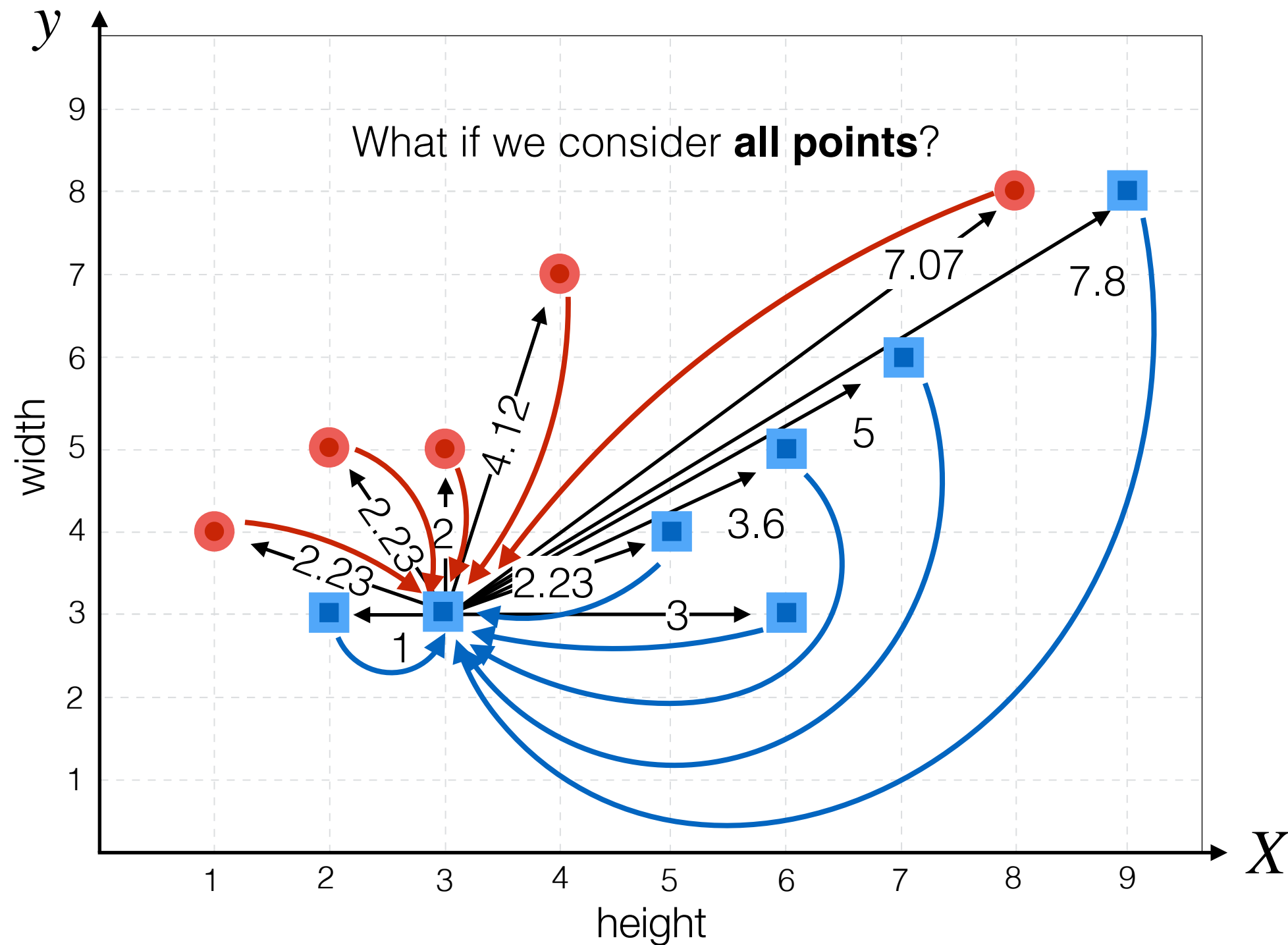
Nearest Neighbour Classifier



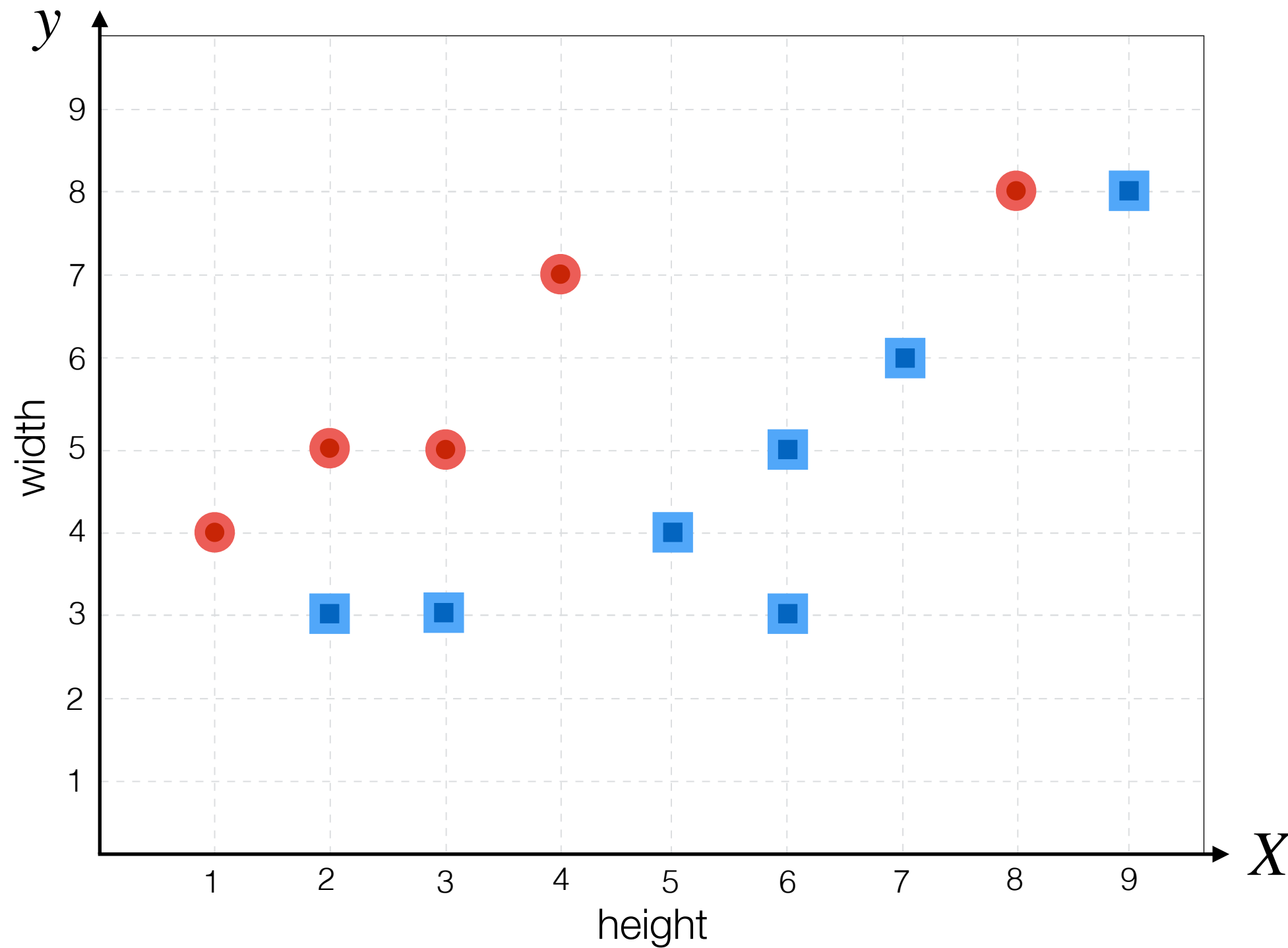
Nearest Neighbour Classifier



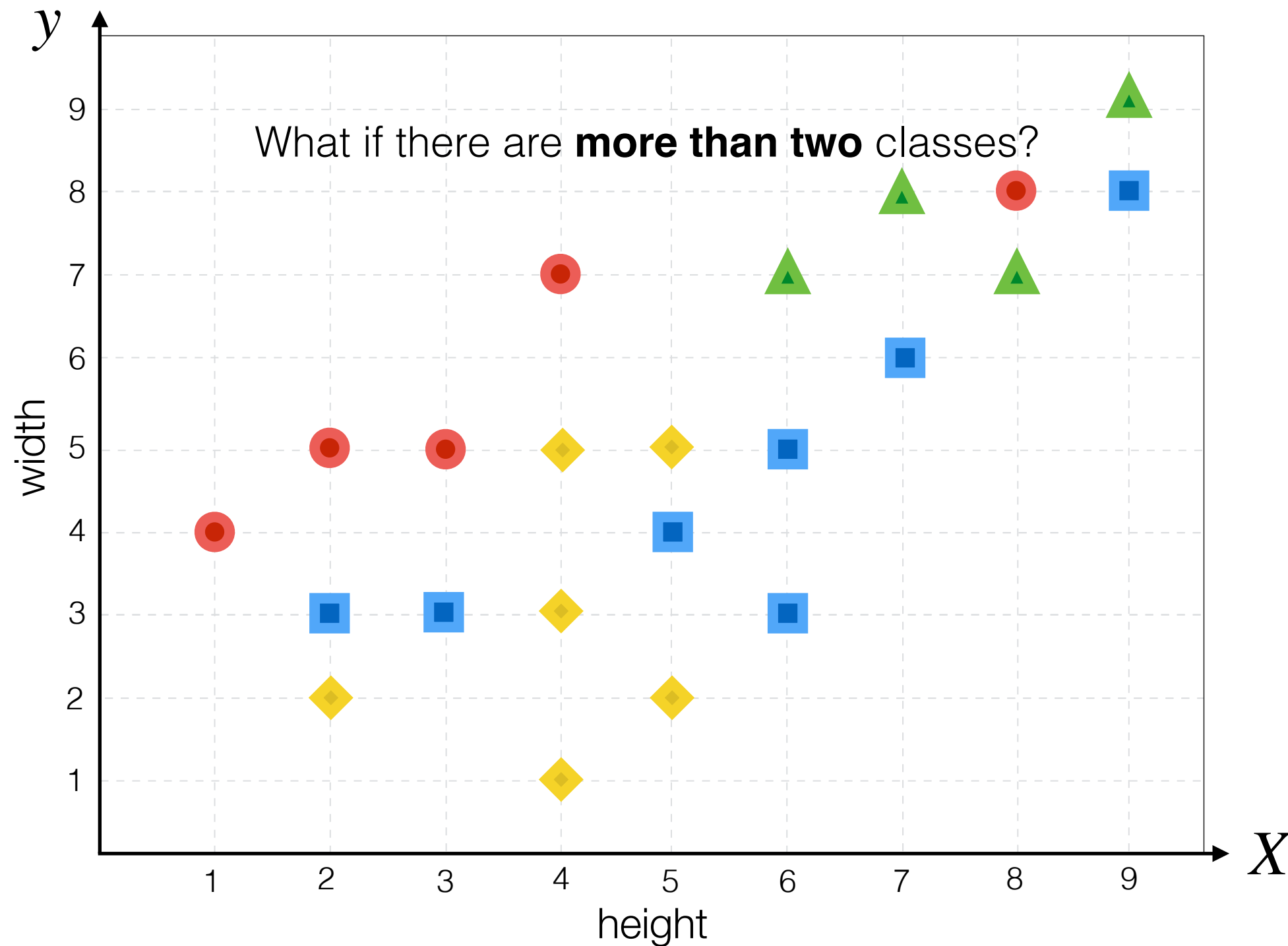
Nearest Neighbour Classifier



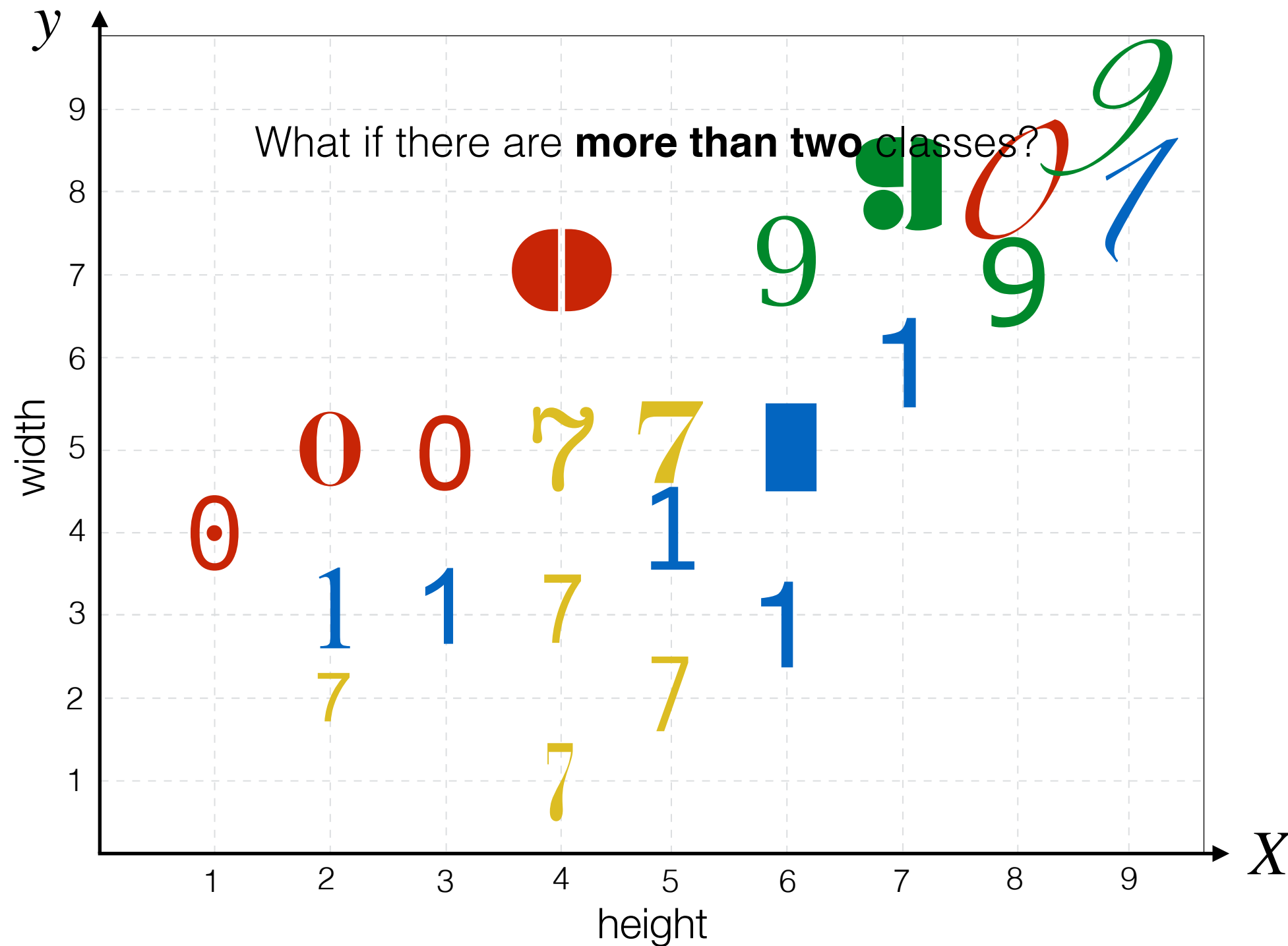
Nearest Neighbour Classifier



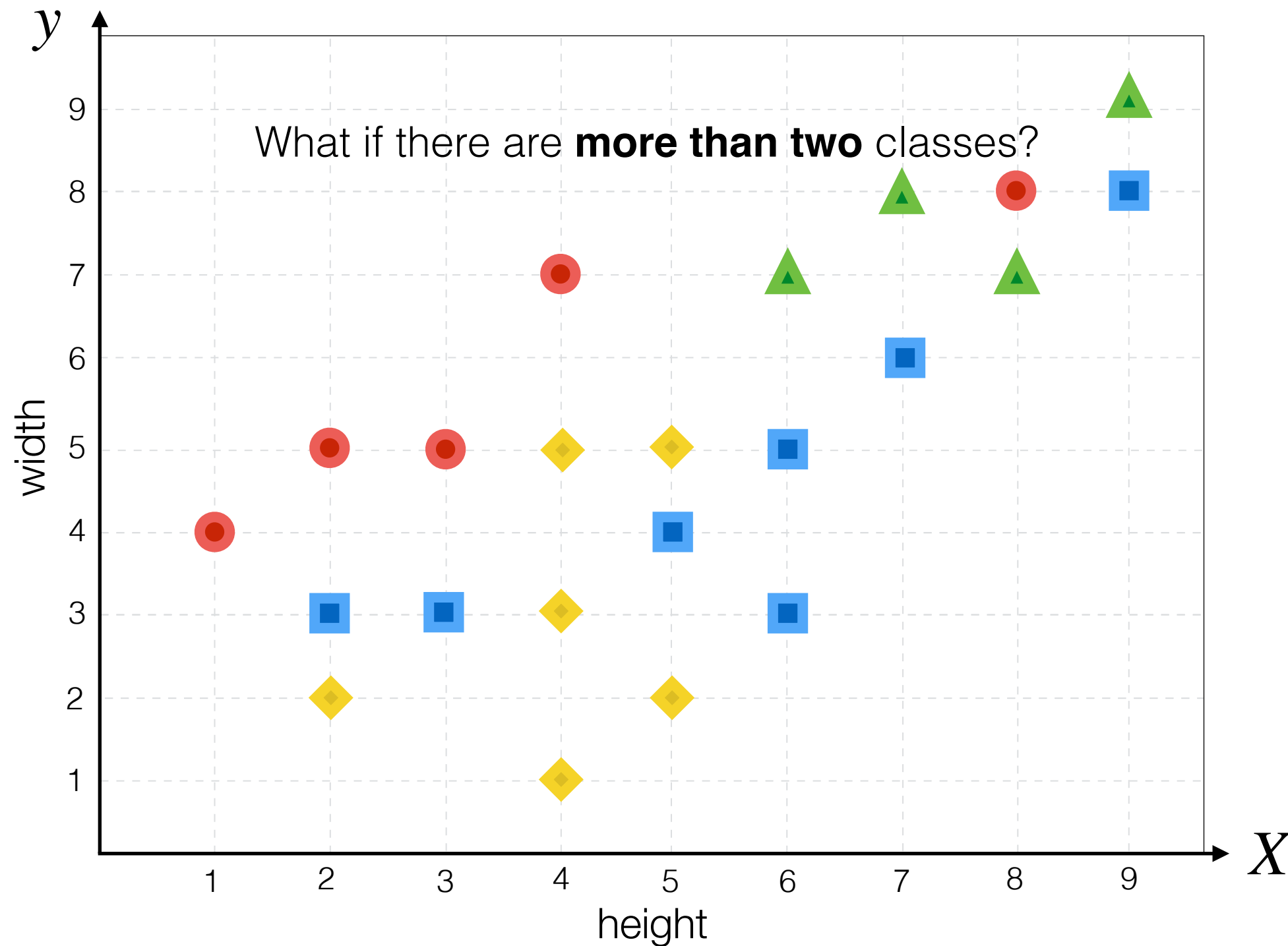
K-Nearest Neighbour Classifier



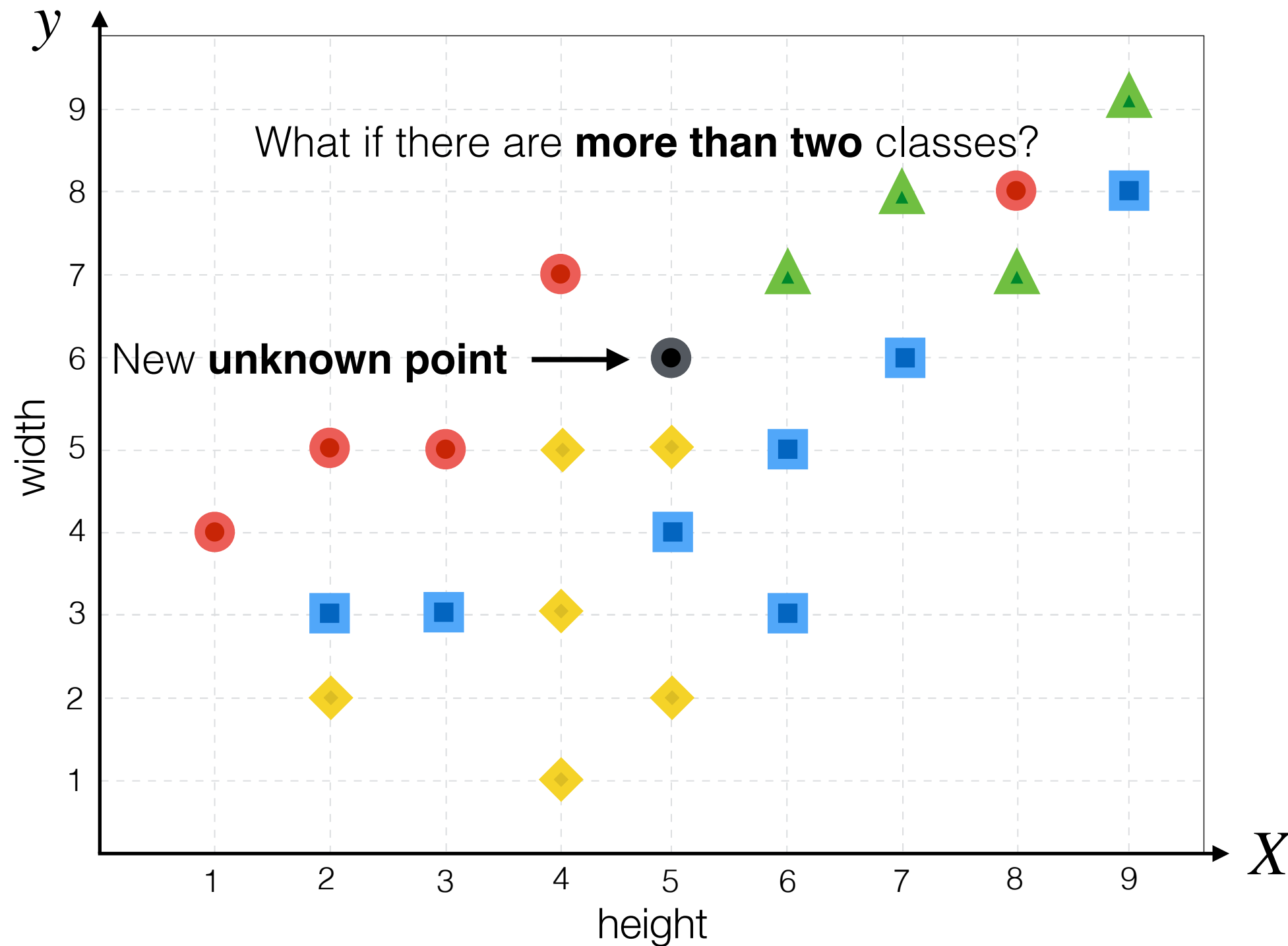
K-Nearest Neighbour Classifier



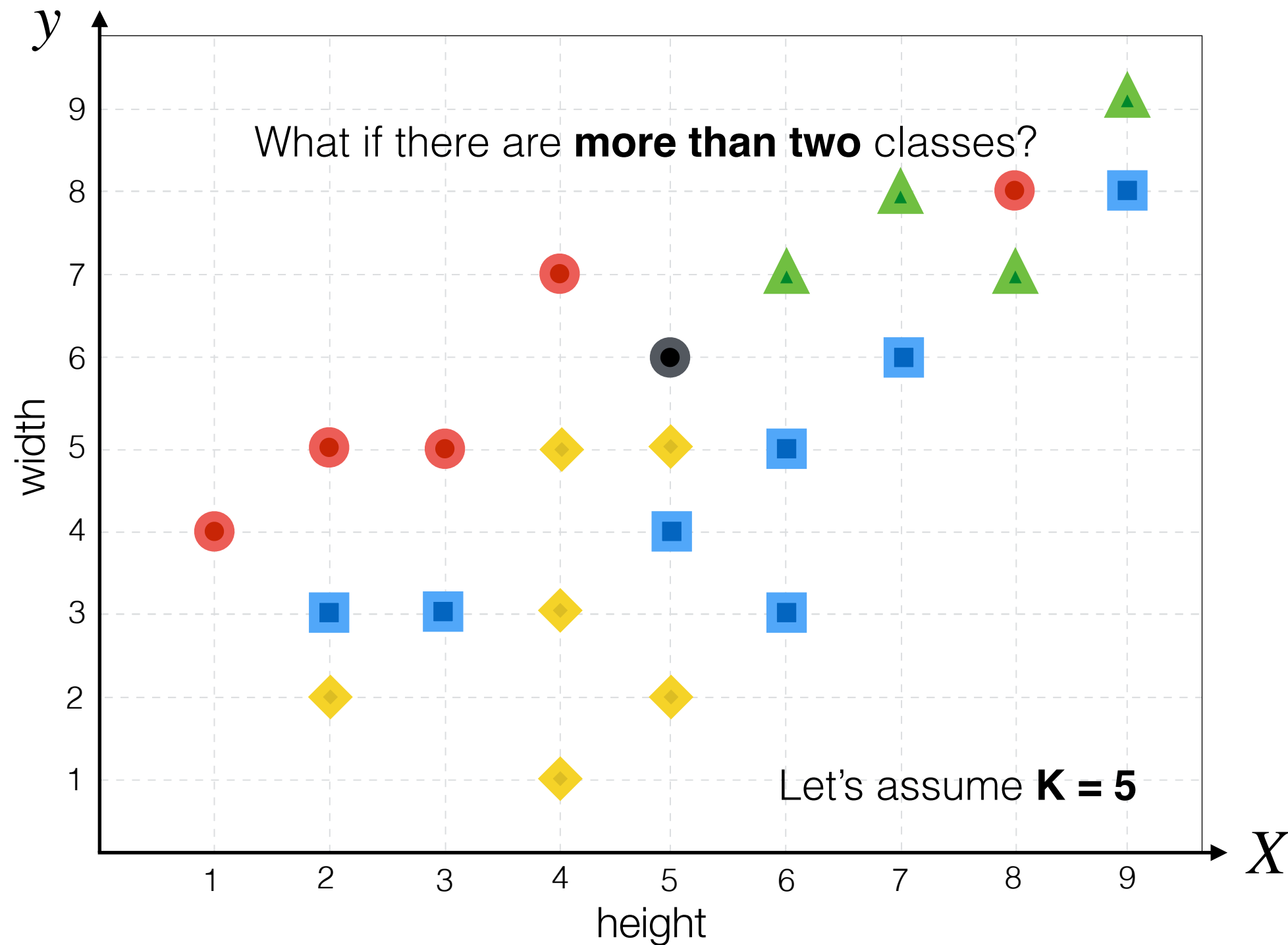
K-Nearest Neighbour Classifier



K-Nearest Neighbour Classifier



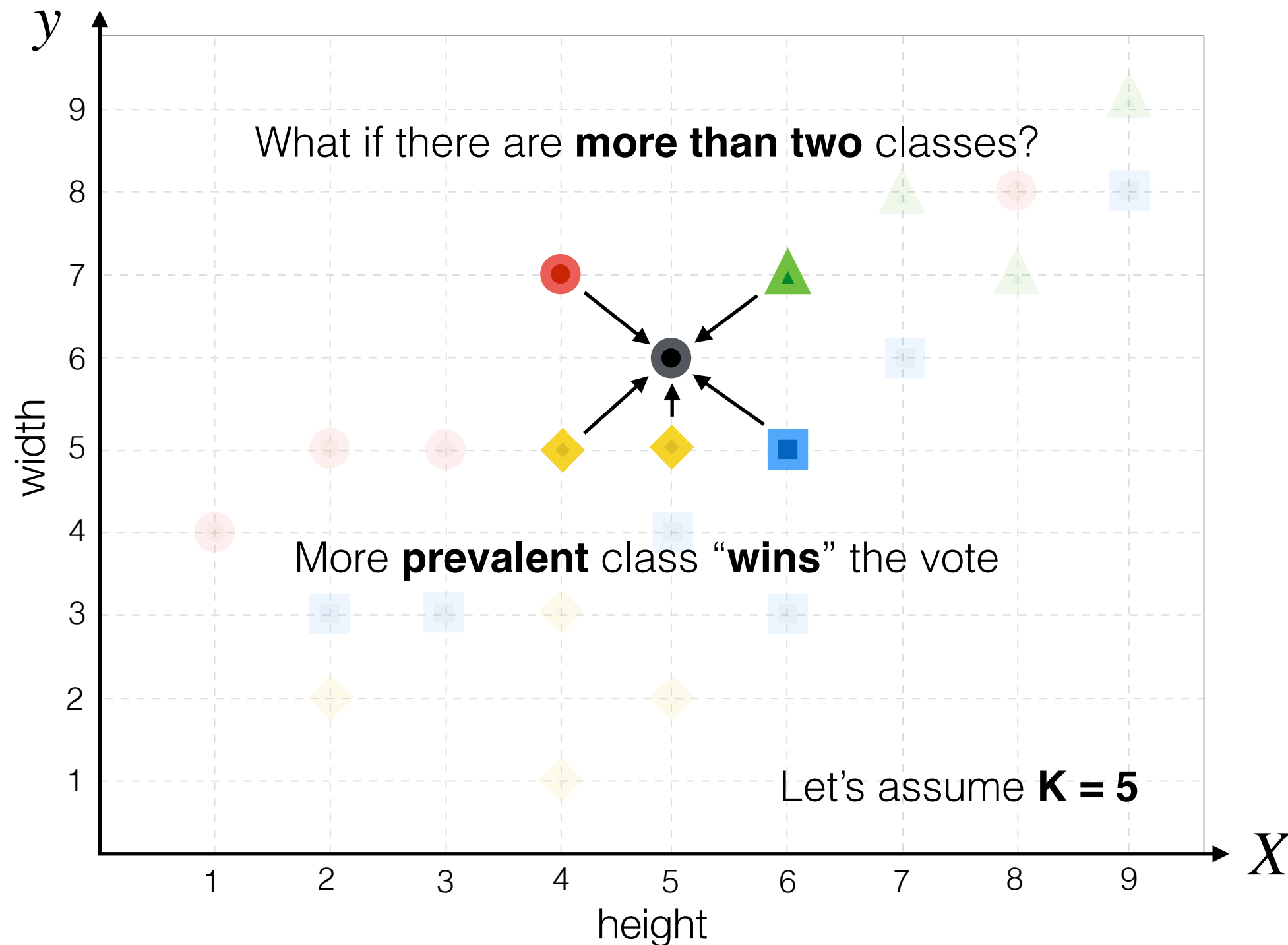
K-Nearest Neighbour Classifier



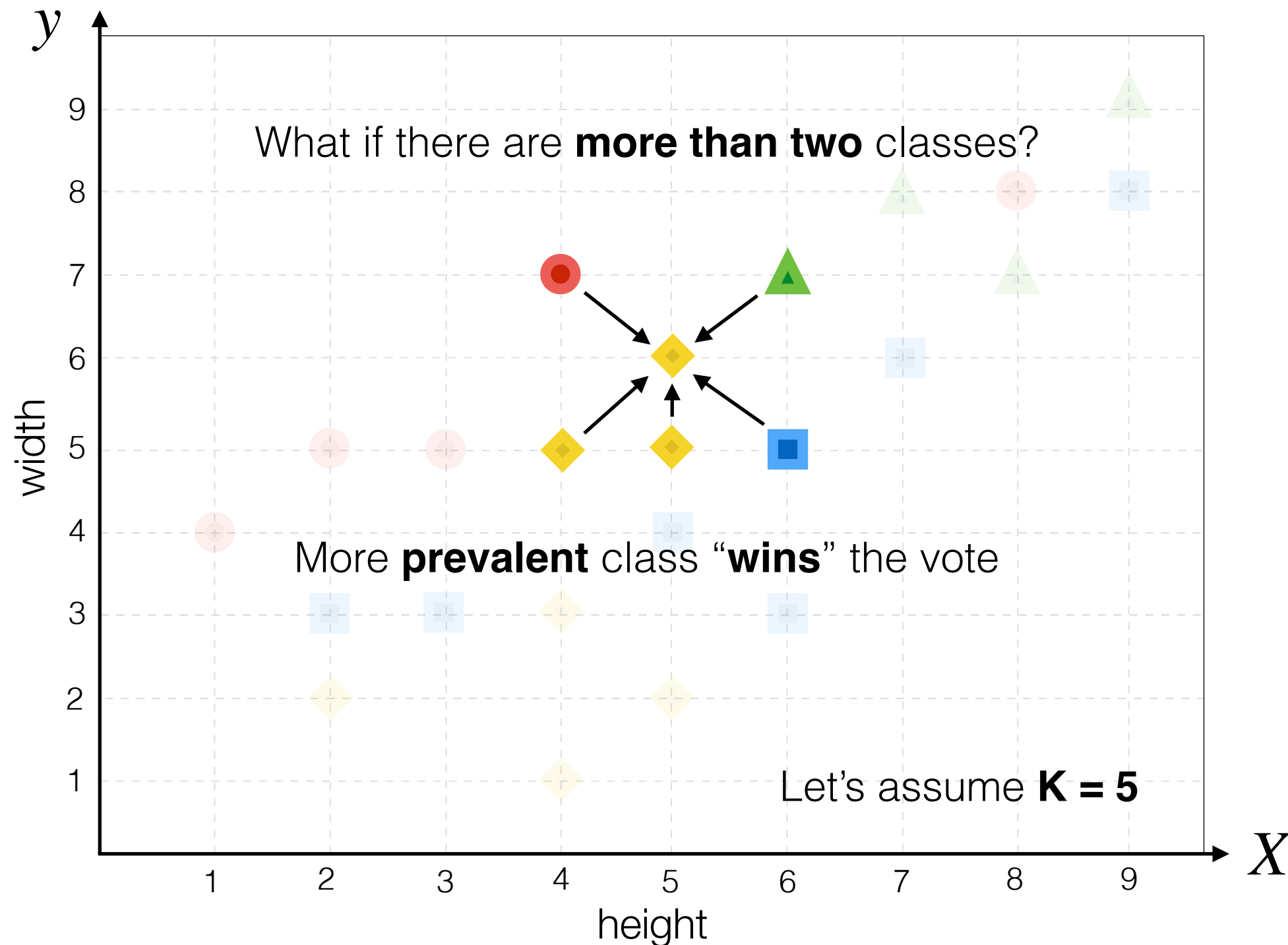
K-Nearest Neighbour Classifier



K-Nearest Neighbour Classifier



K-Nearest Neighbour Classifier



0 $\longleftrightarrow$? $\longleftrightarrow$ 1

Classification

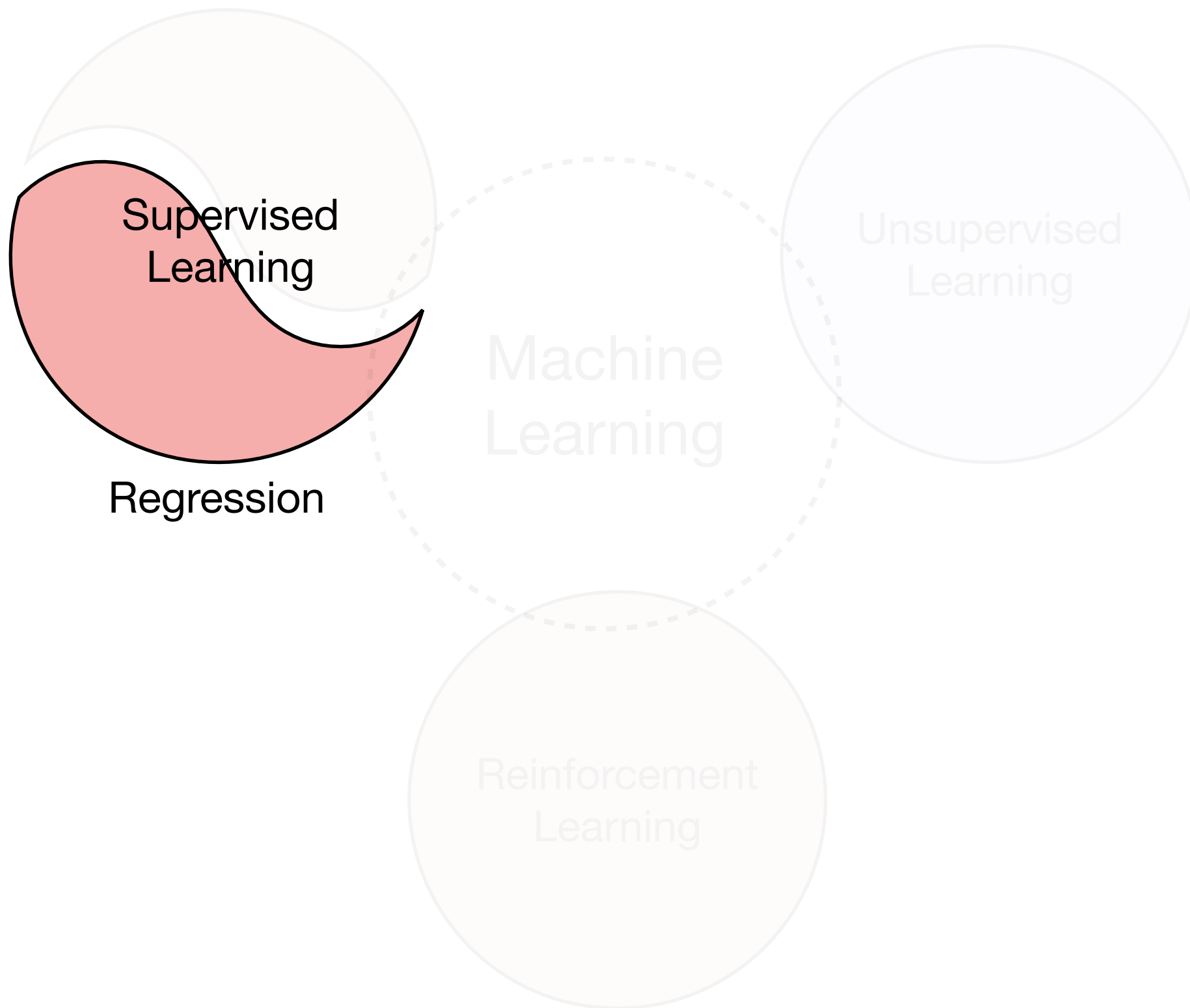
Supervised
Learning

Regression

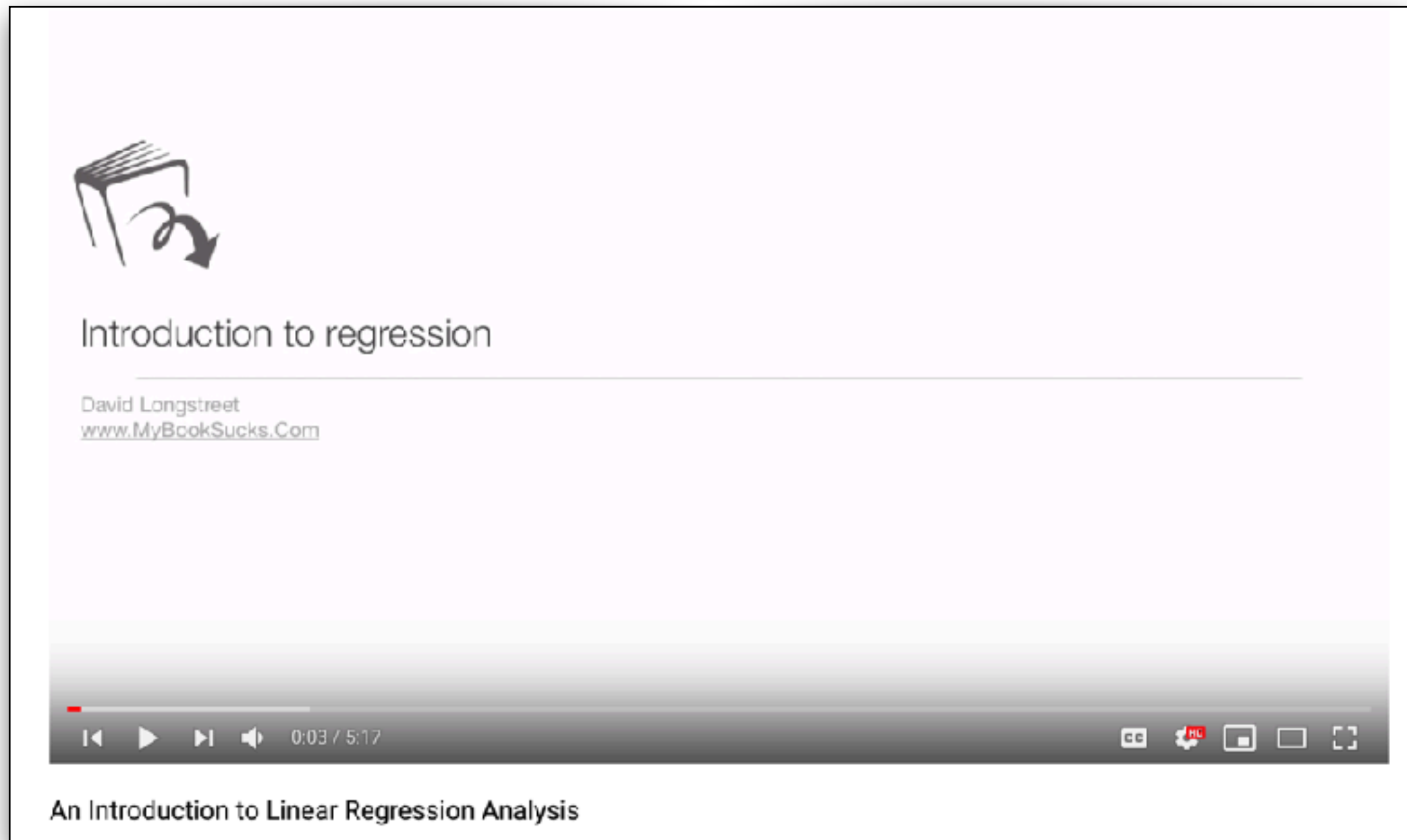
Machine
Learning

Unsupervised
Learning

Reinforcement
Learning

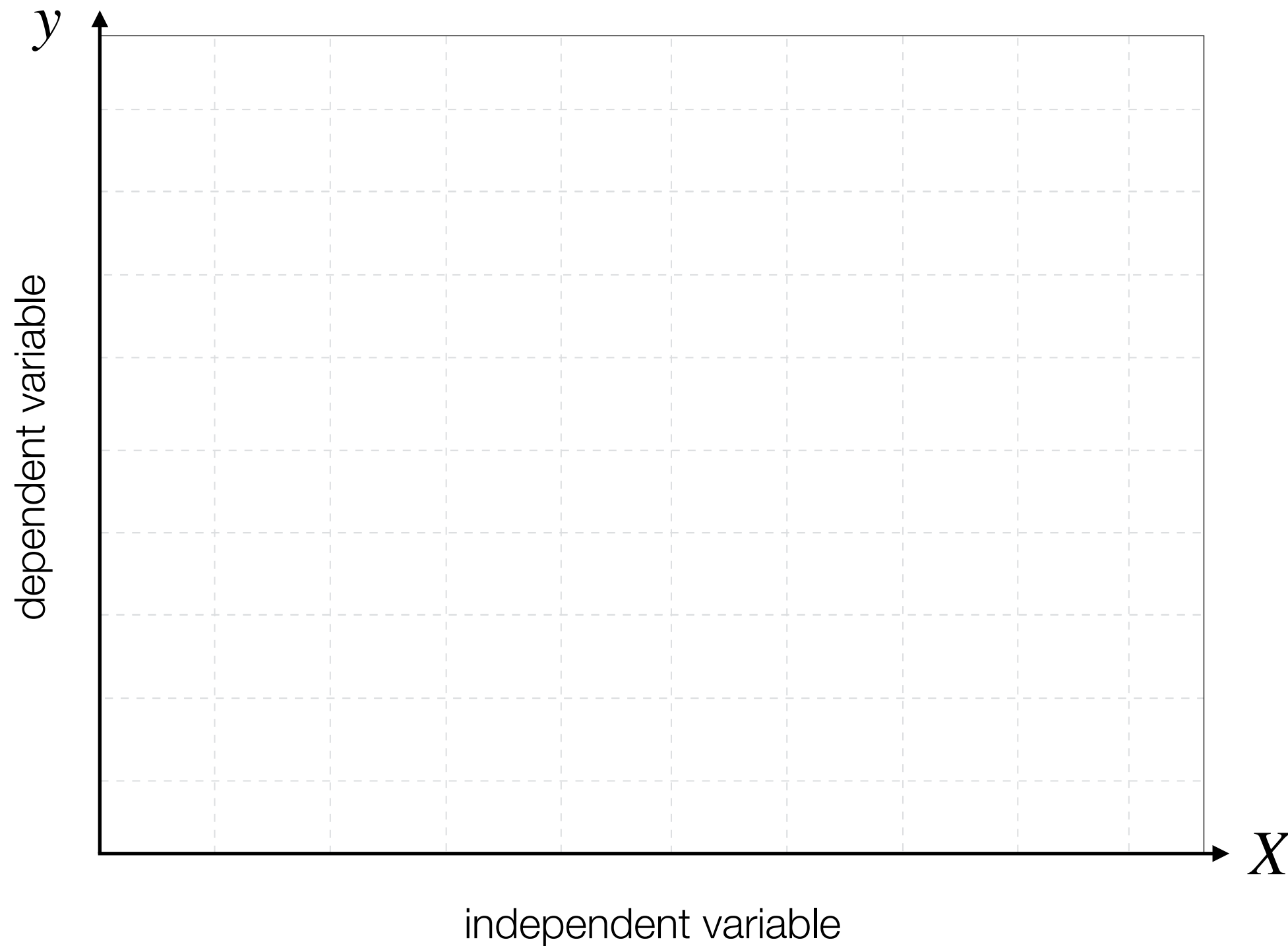


The following slides are inspired by
“An Introduction to Linear Regression Analysis” video

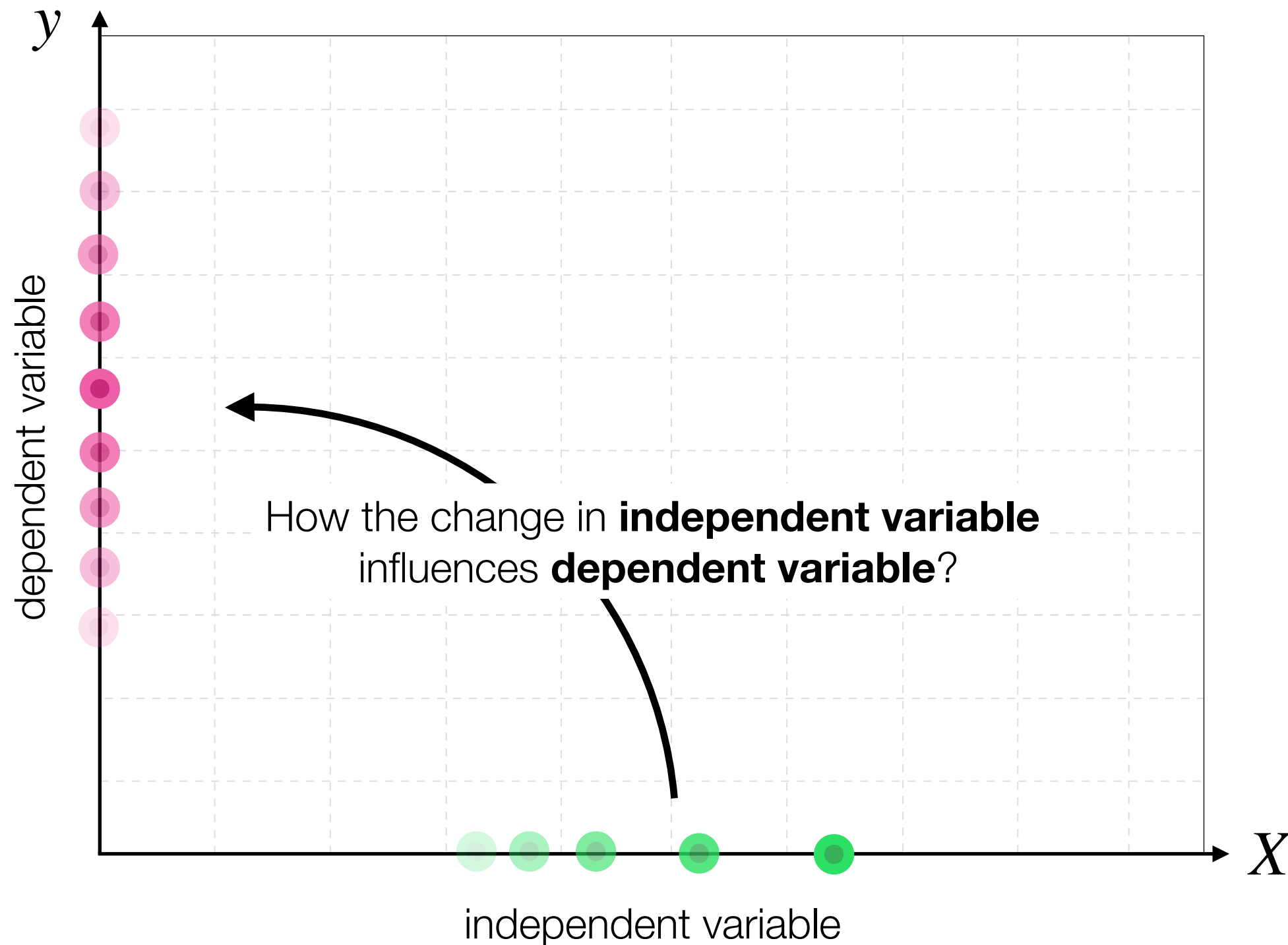


<https://youtu.be/zPG4NjIkCjc>

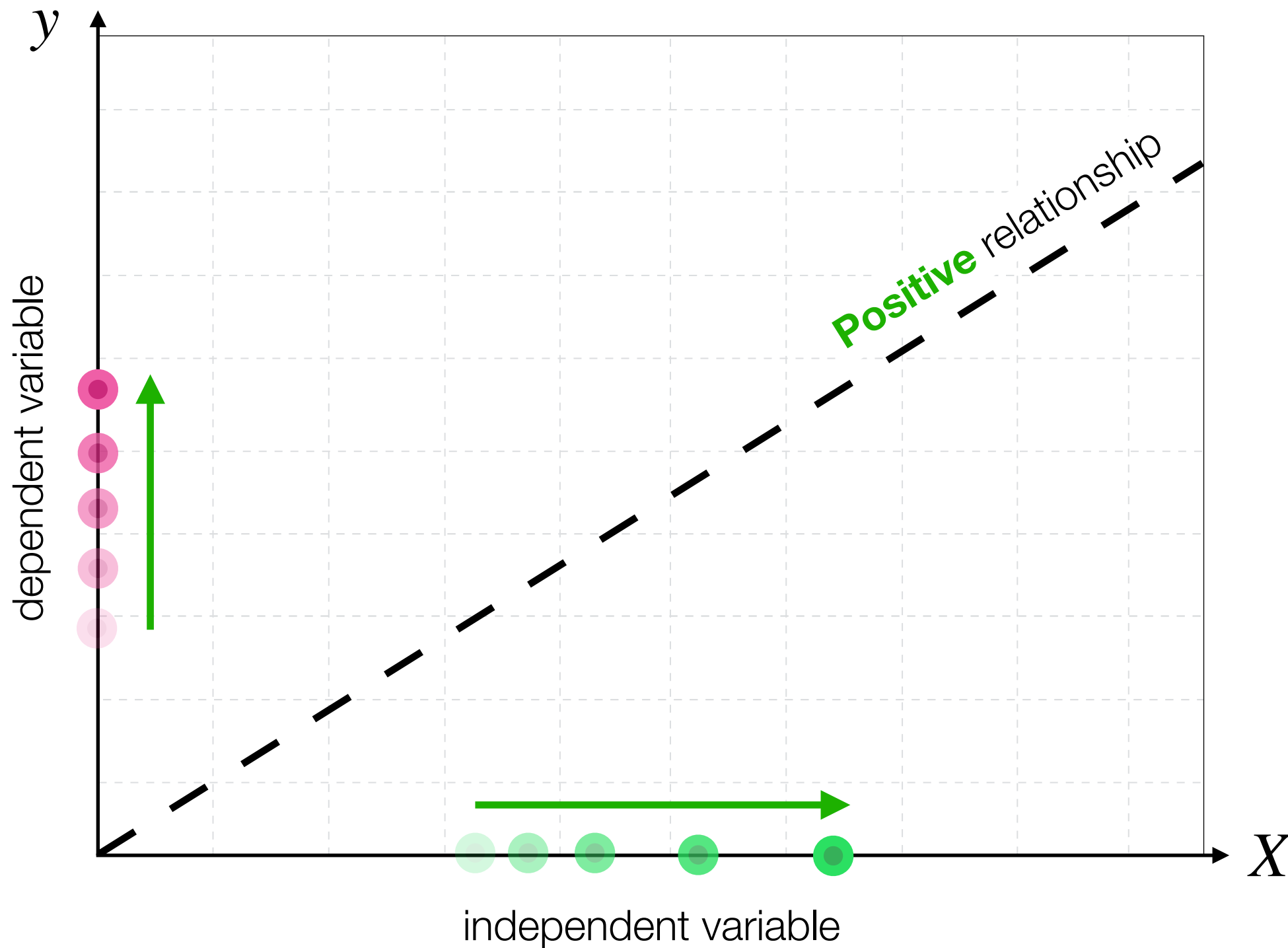
Linear Regression



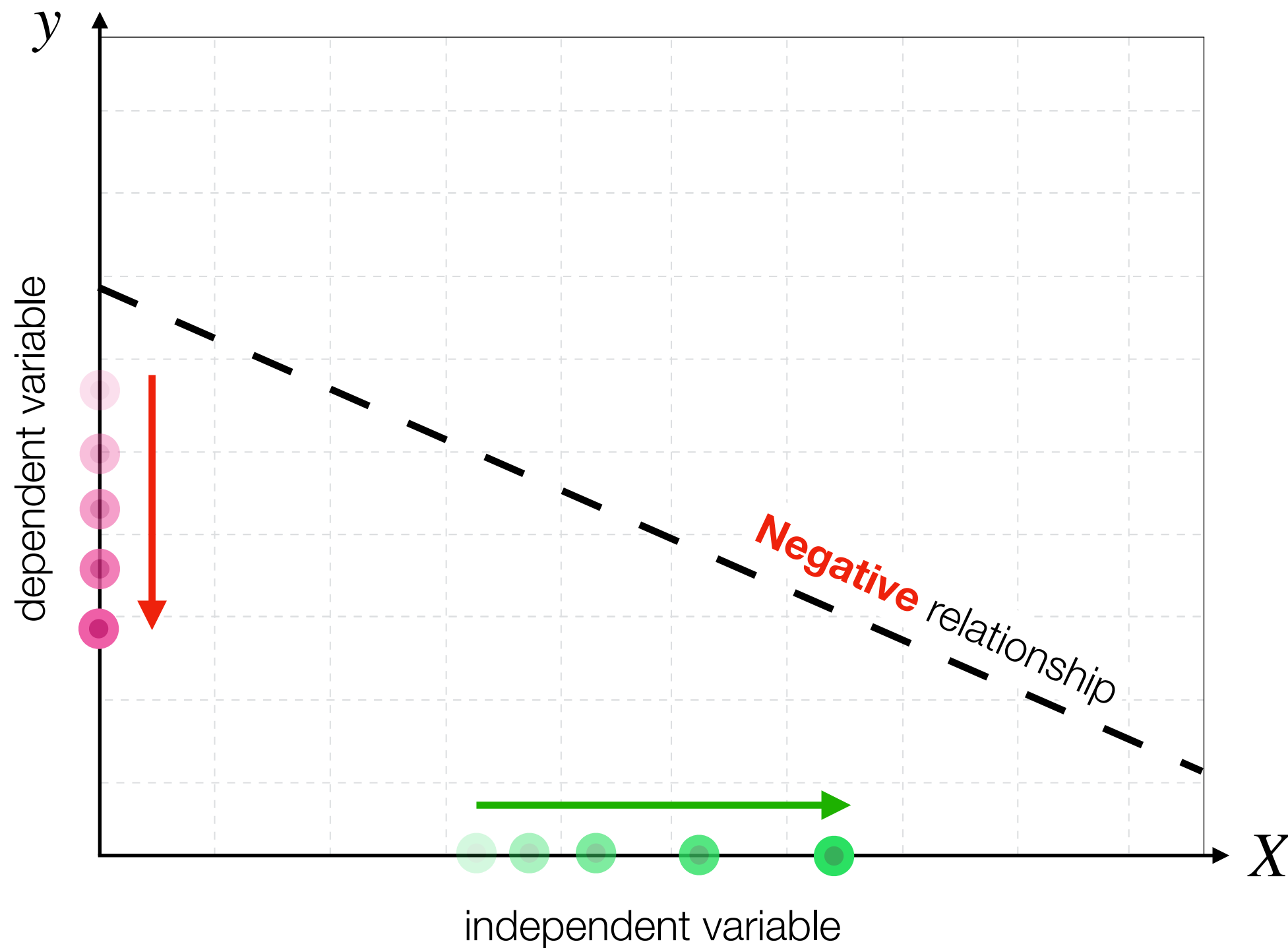
Linear Regression



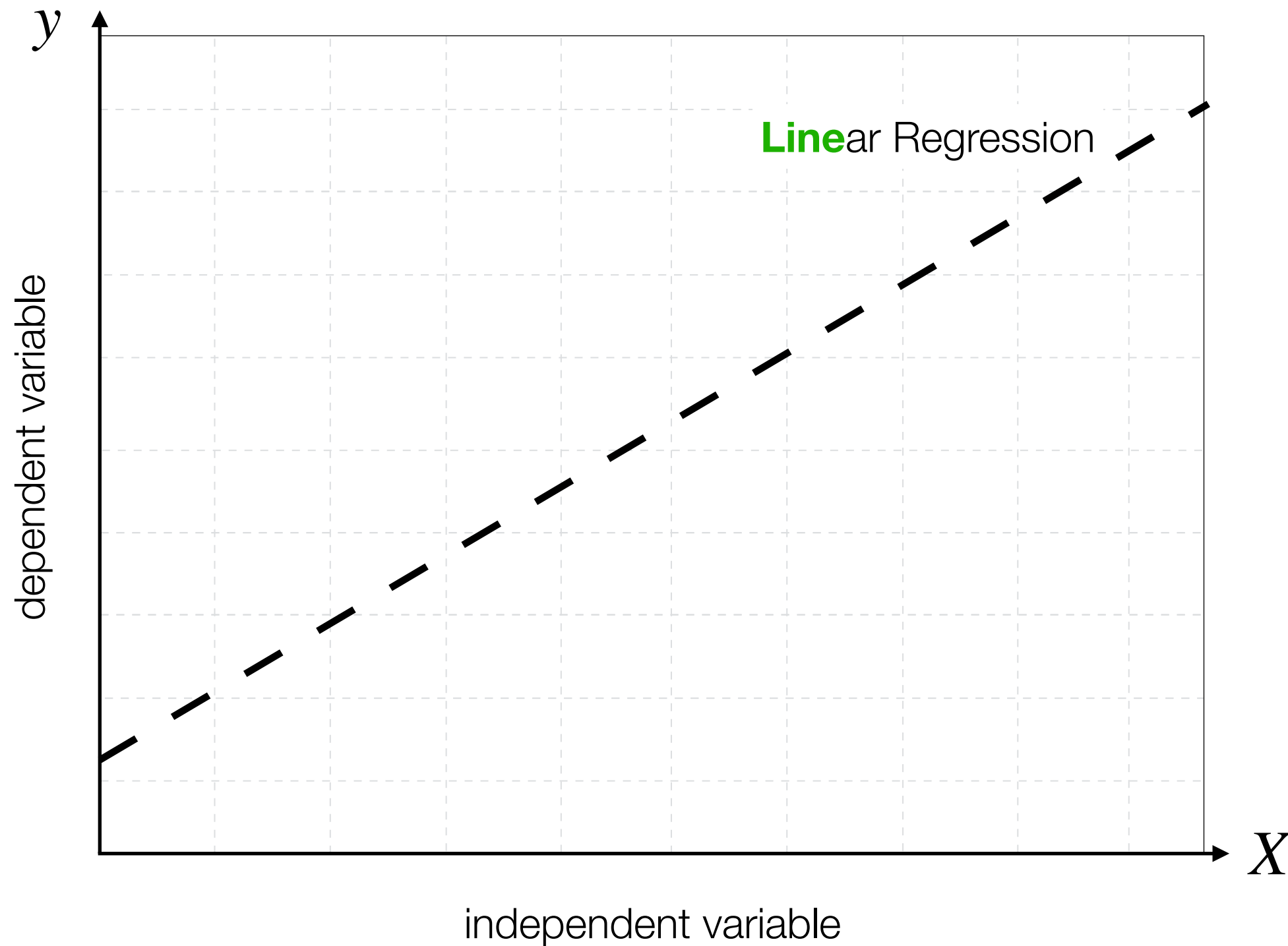
Linear Regression



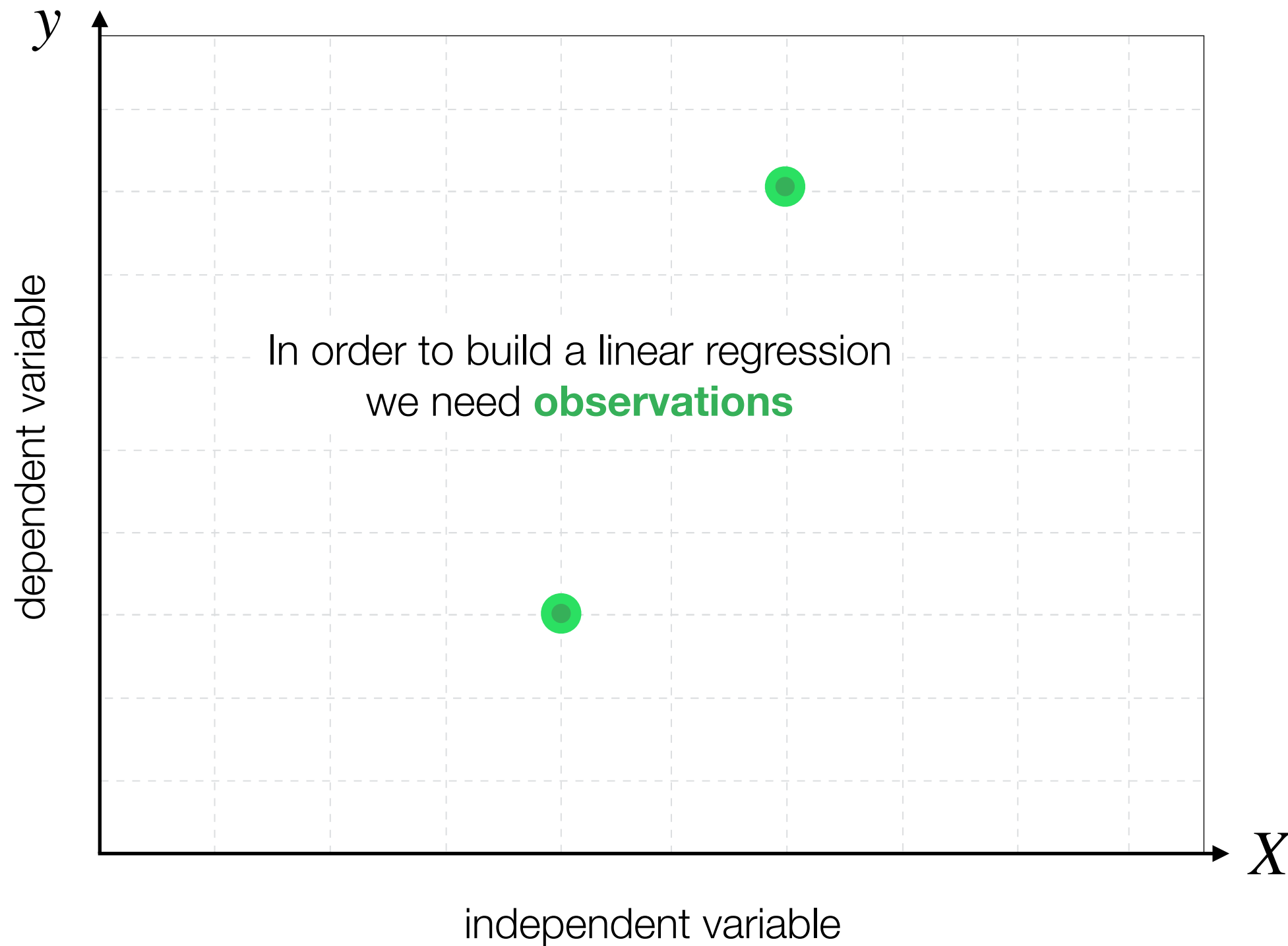
Linear Regression



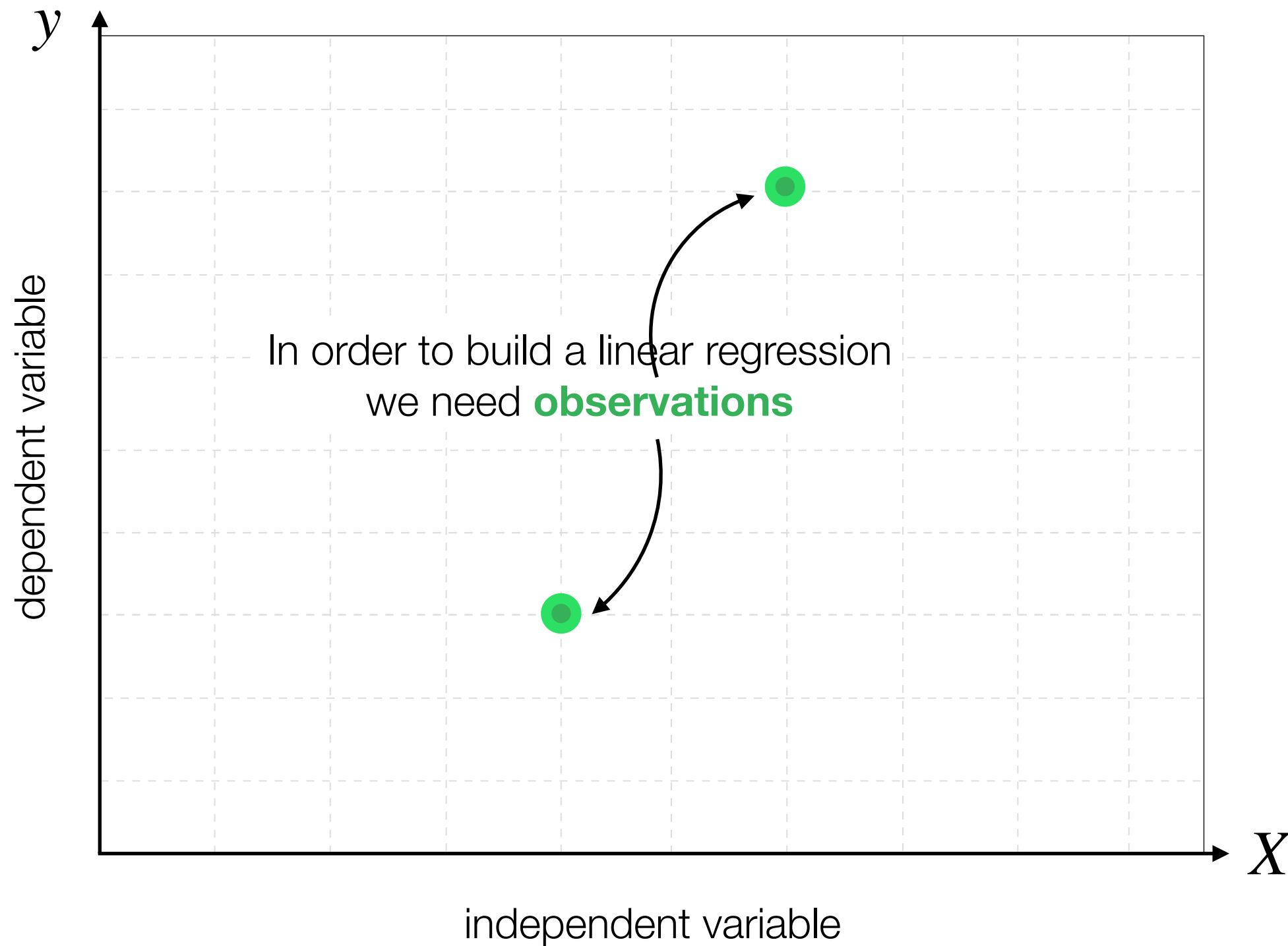
Linear Regression



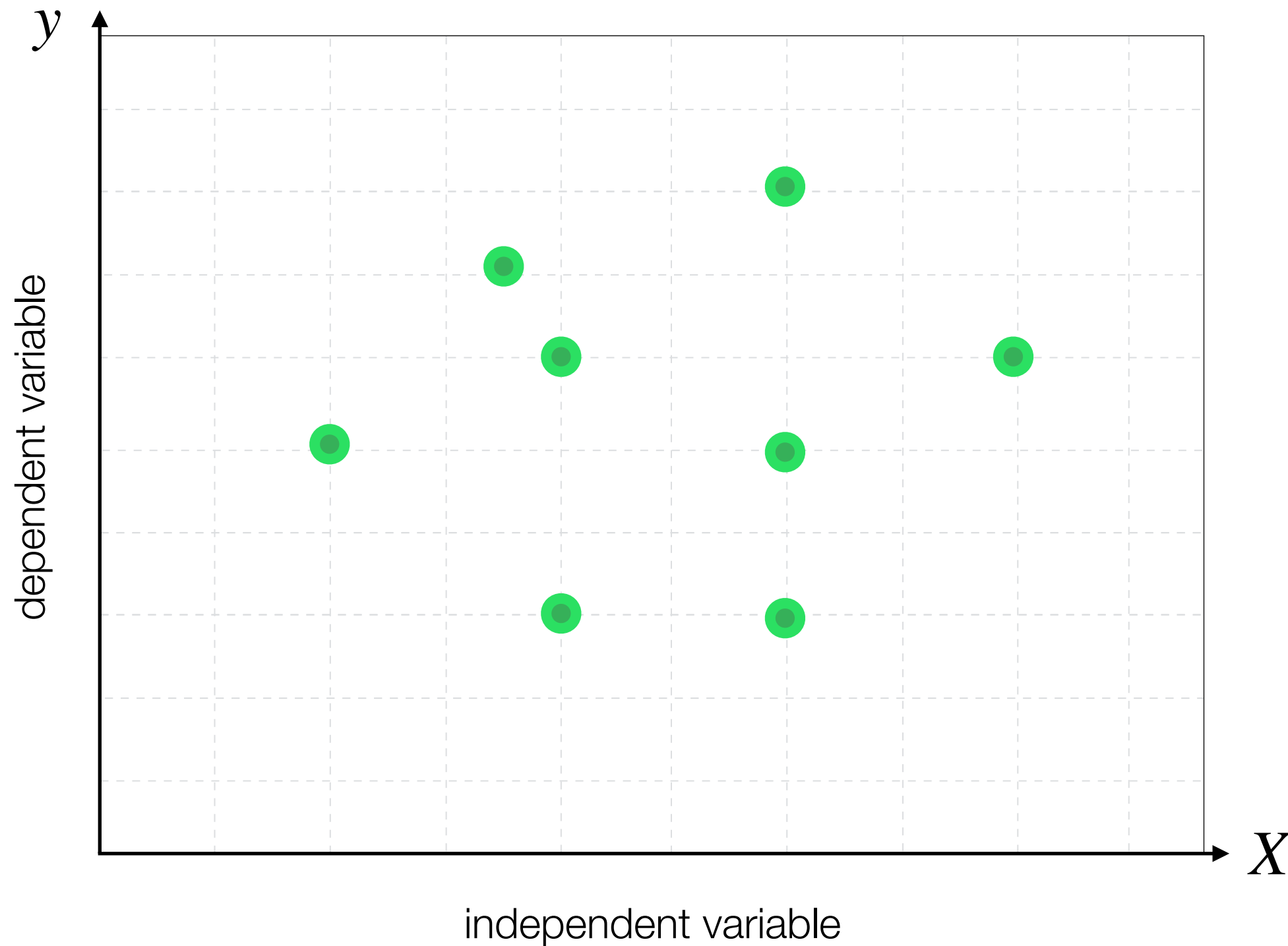
Linear Regression



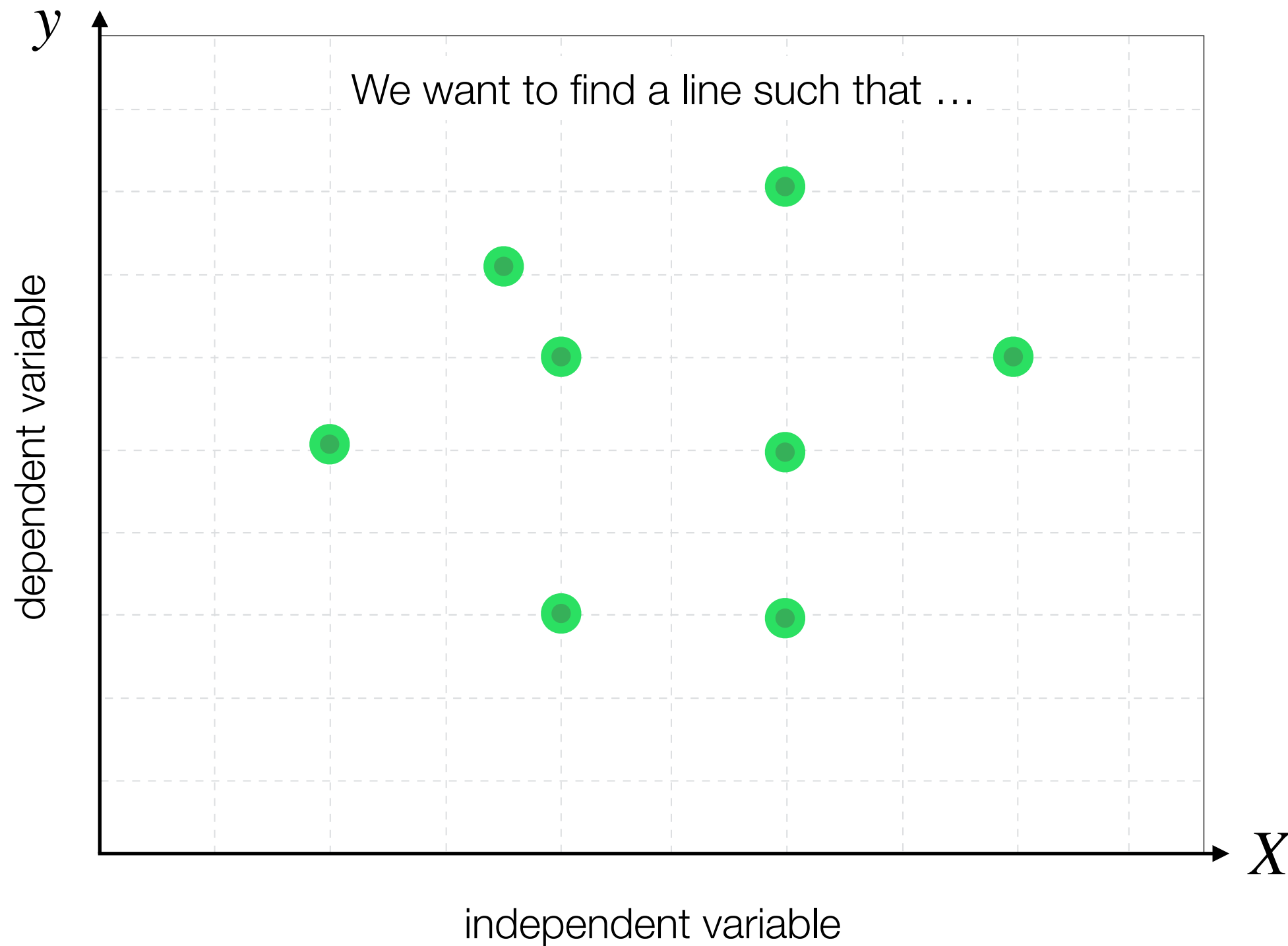
Linear Regression



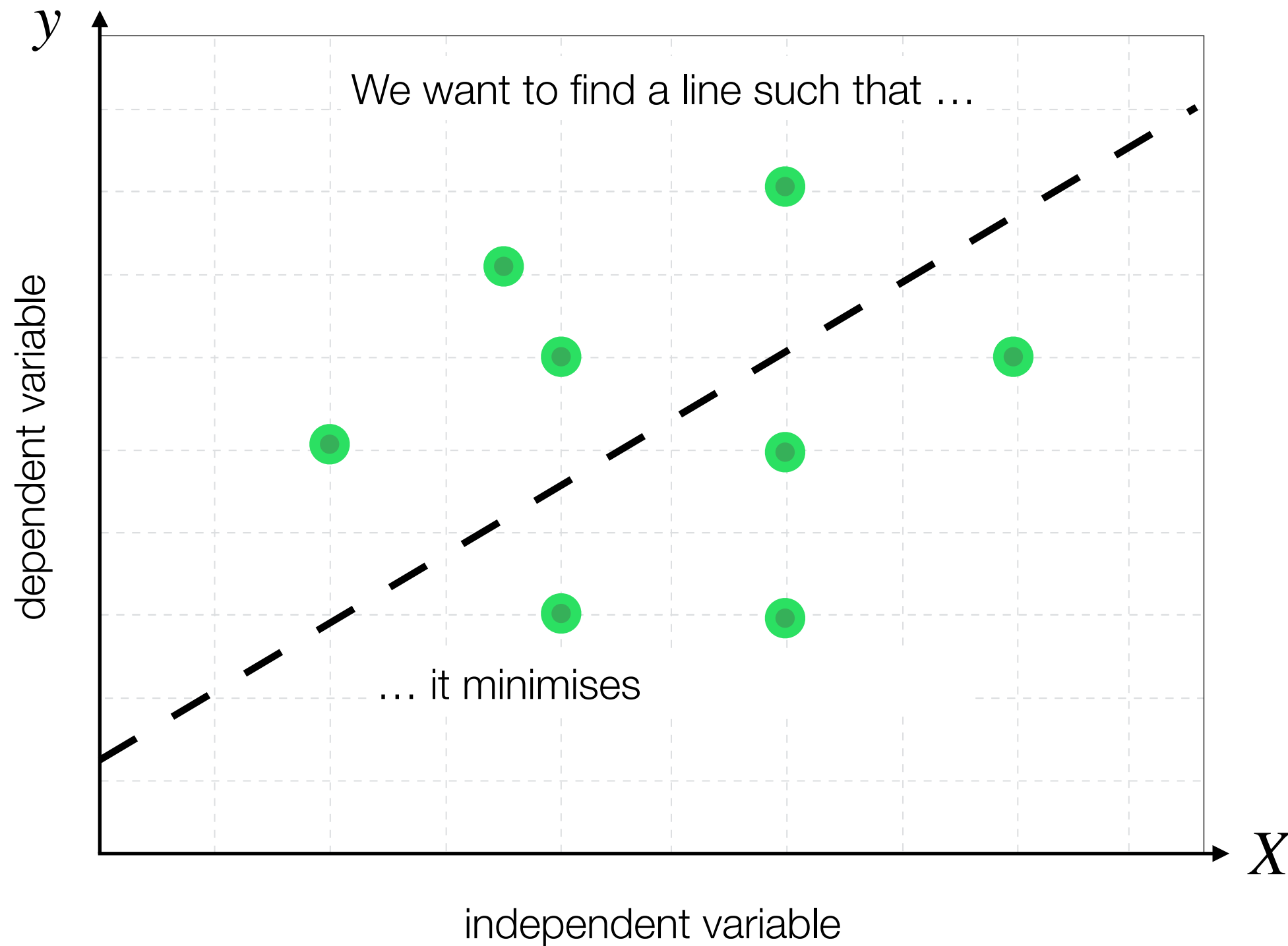
Linear Regression



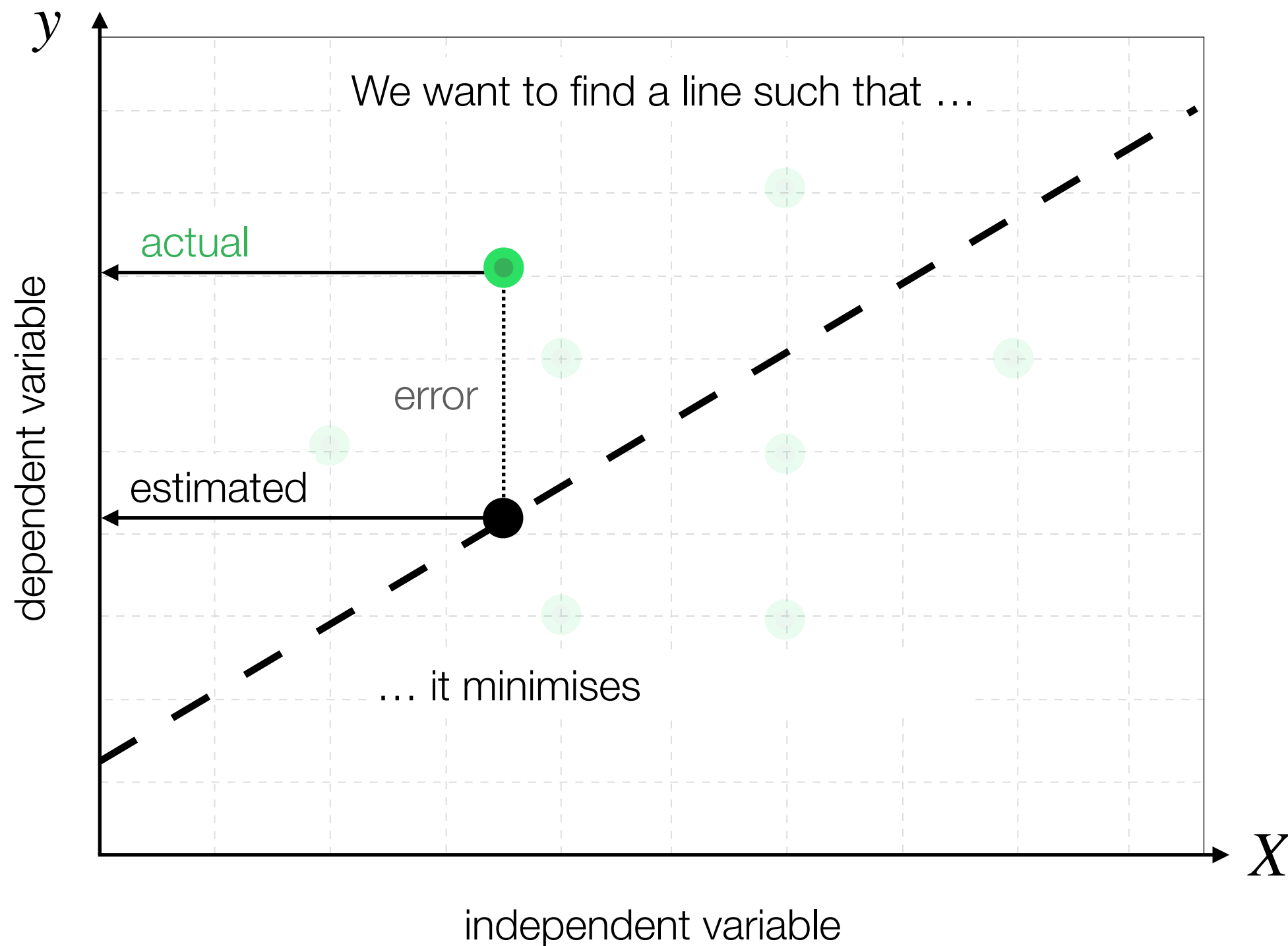
Linear Regression



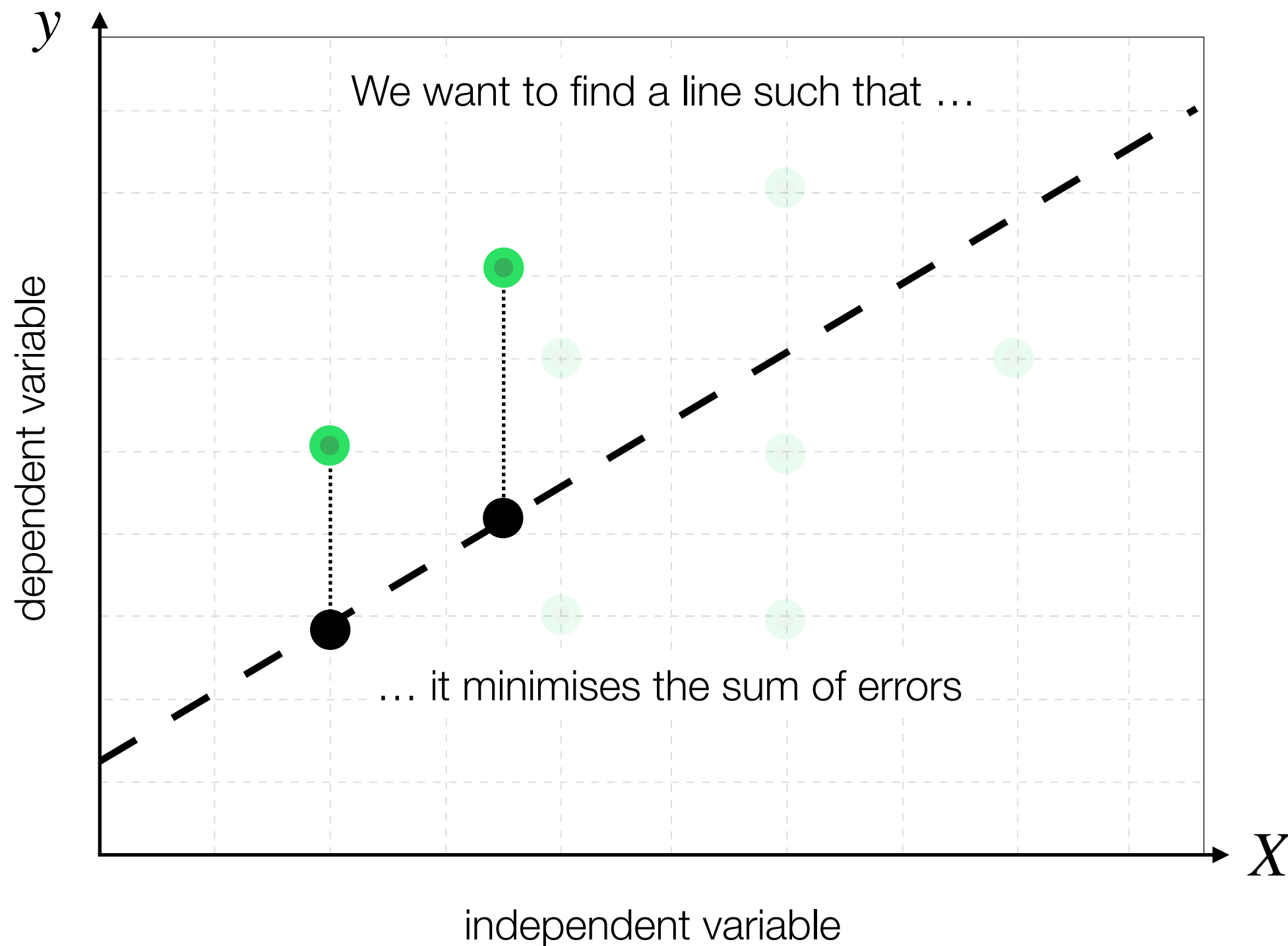
Linear Regression



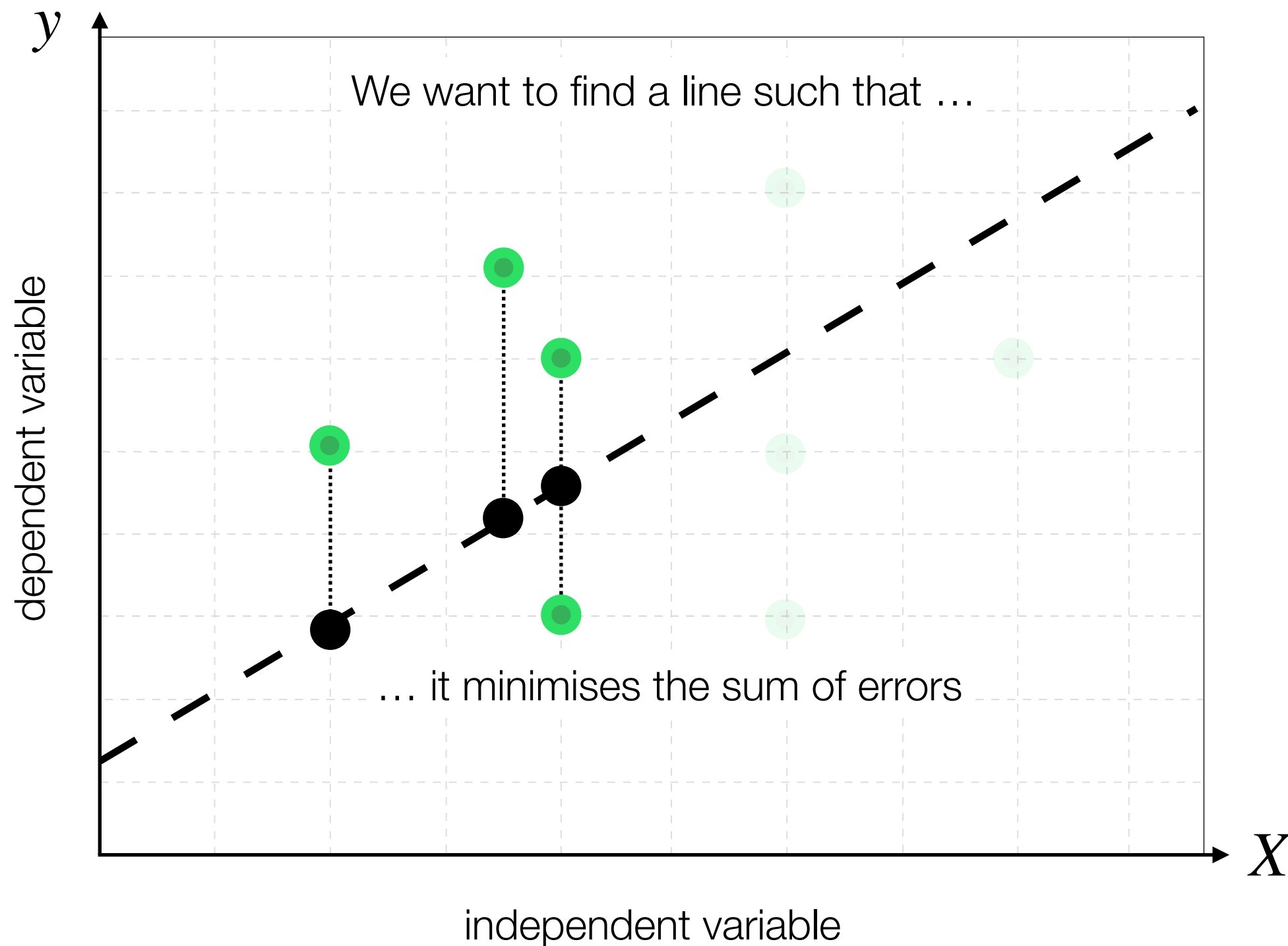
Linear Regression



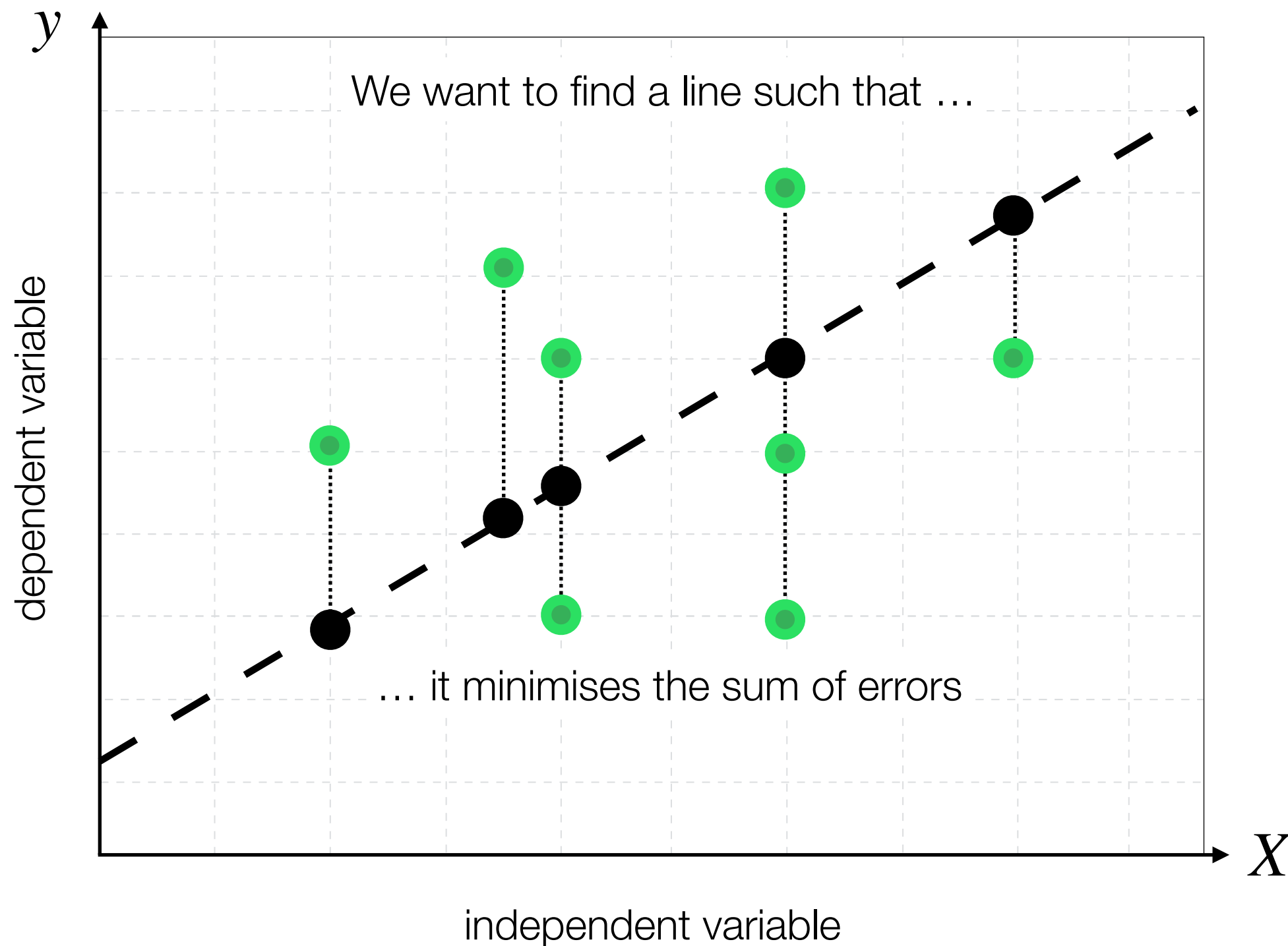
Linear Regression



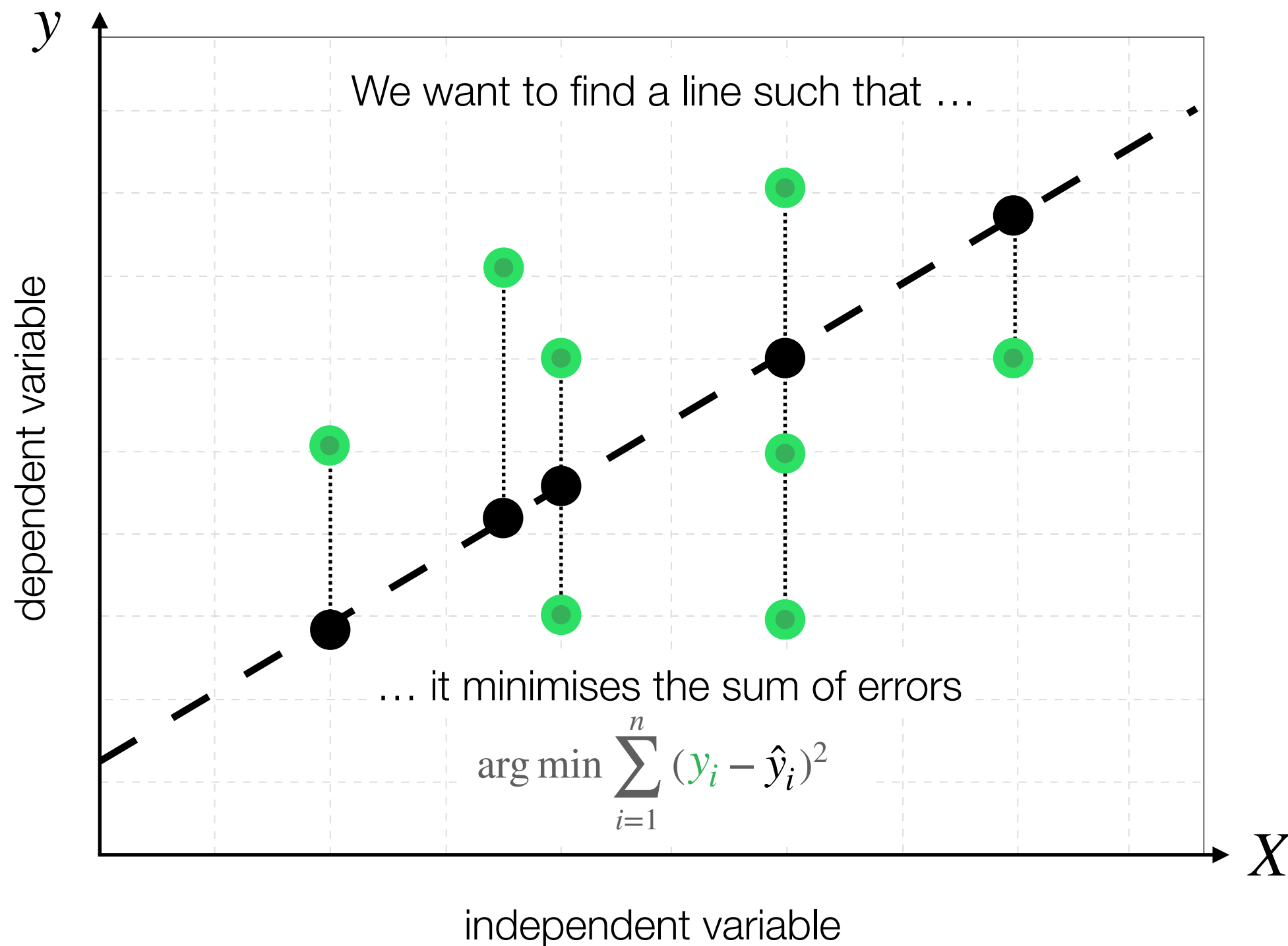
Linear Regression



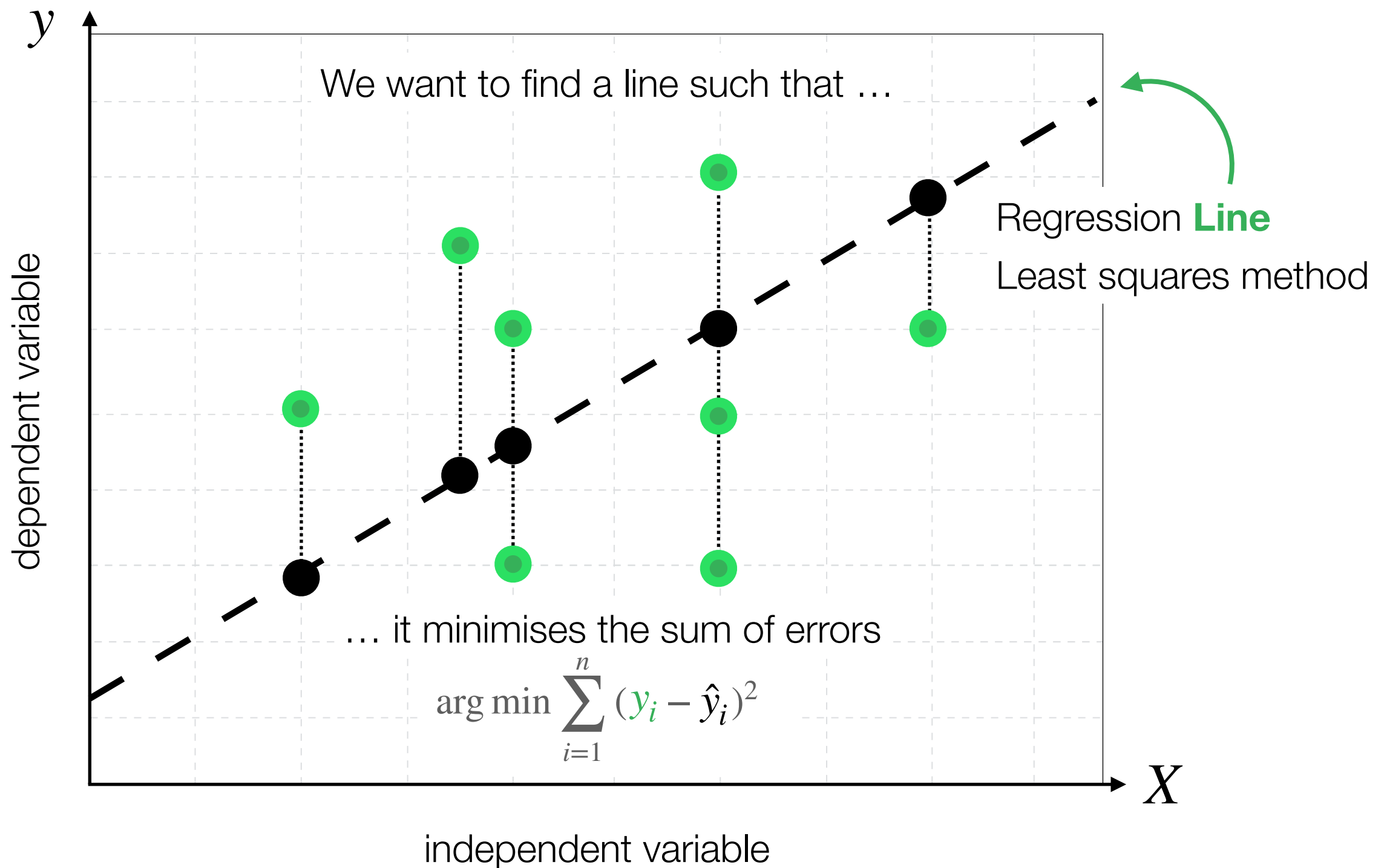
Linear Regression



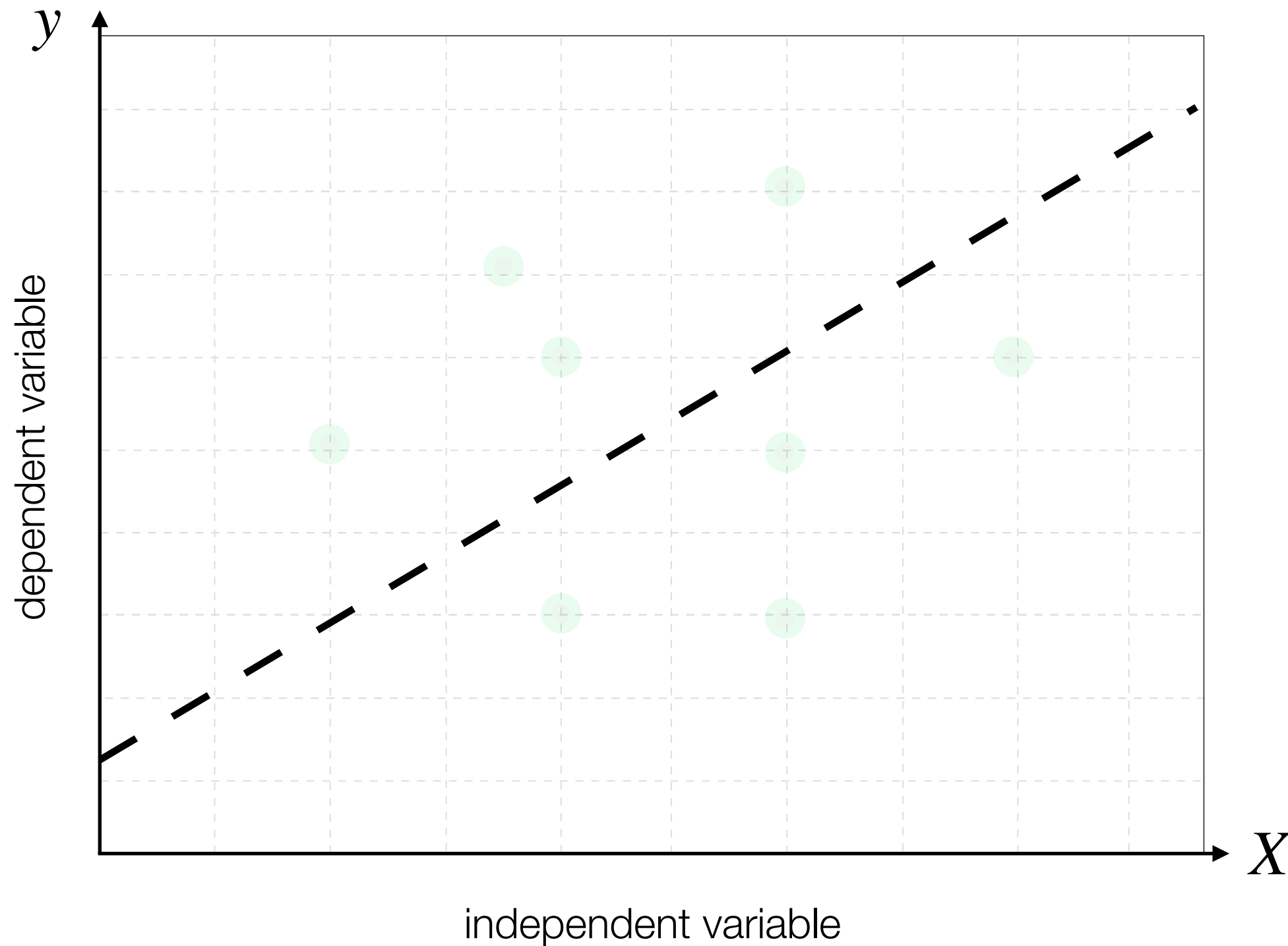
Linear Regression



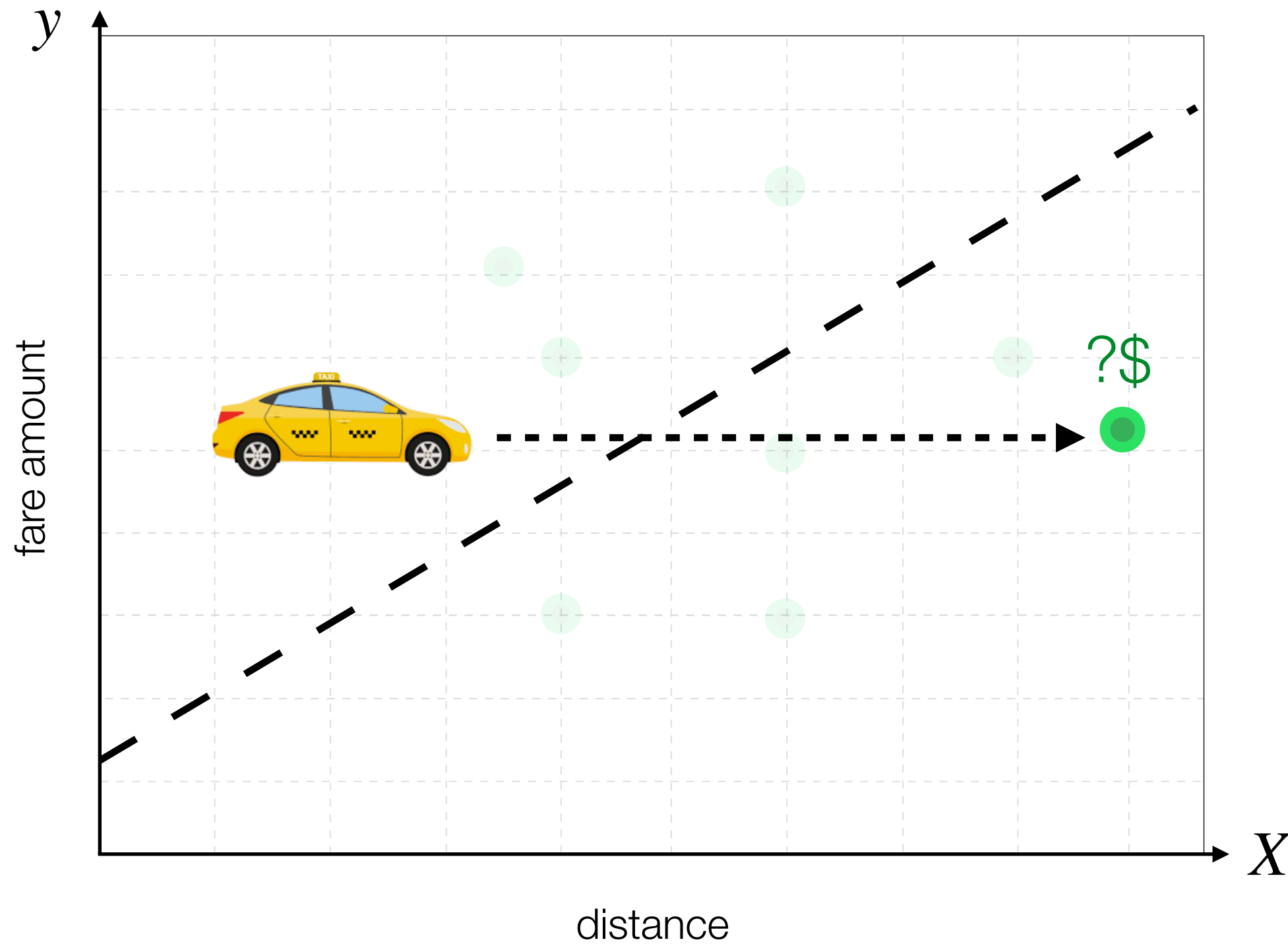
Linear Regression



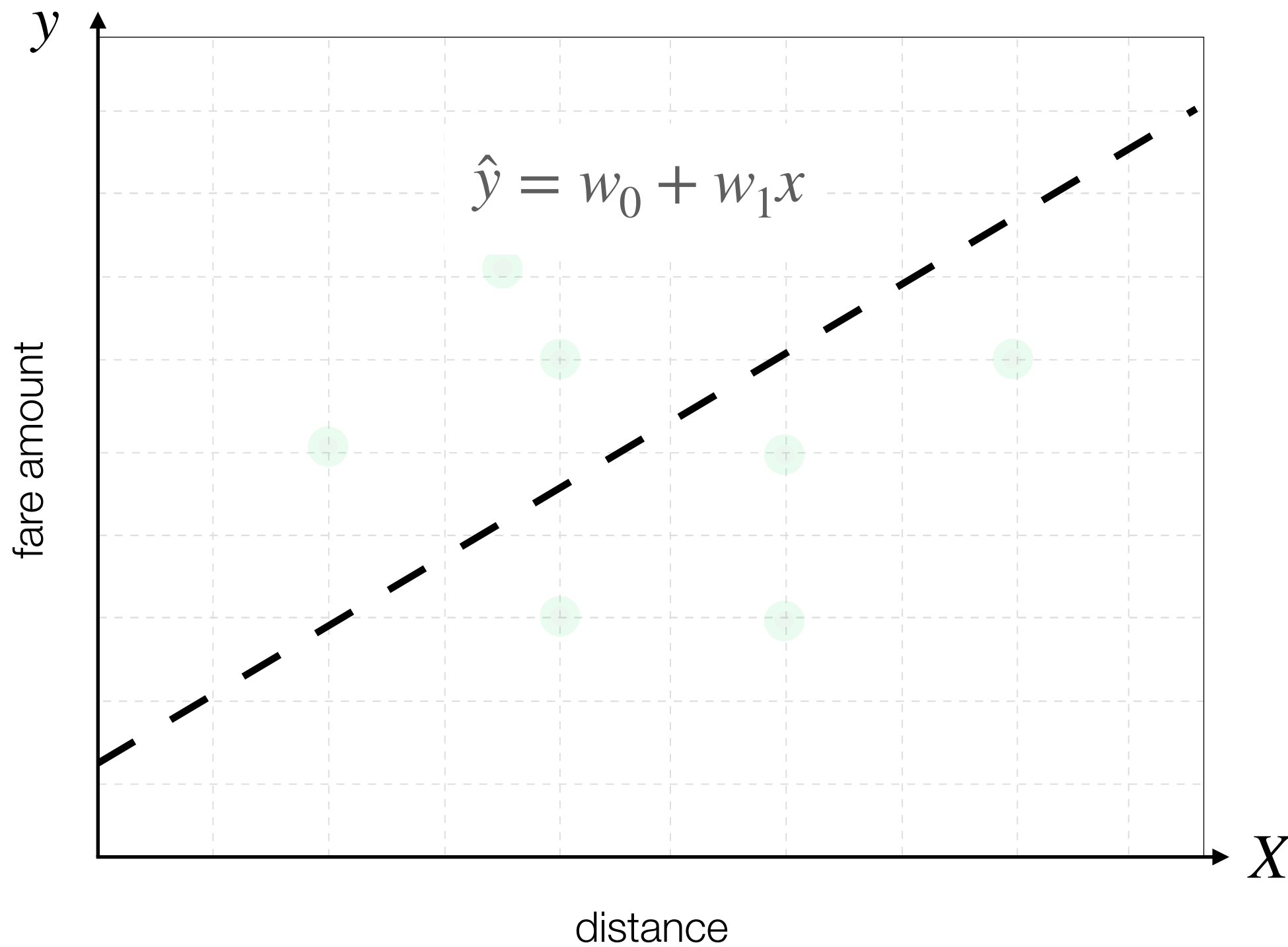
Linear Regression



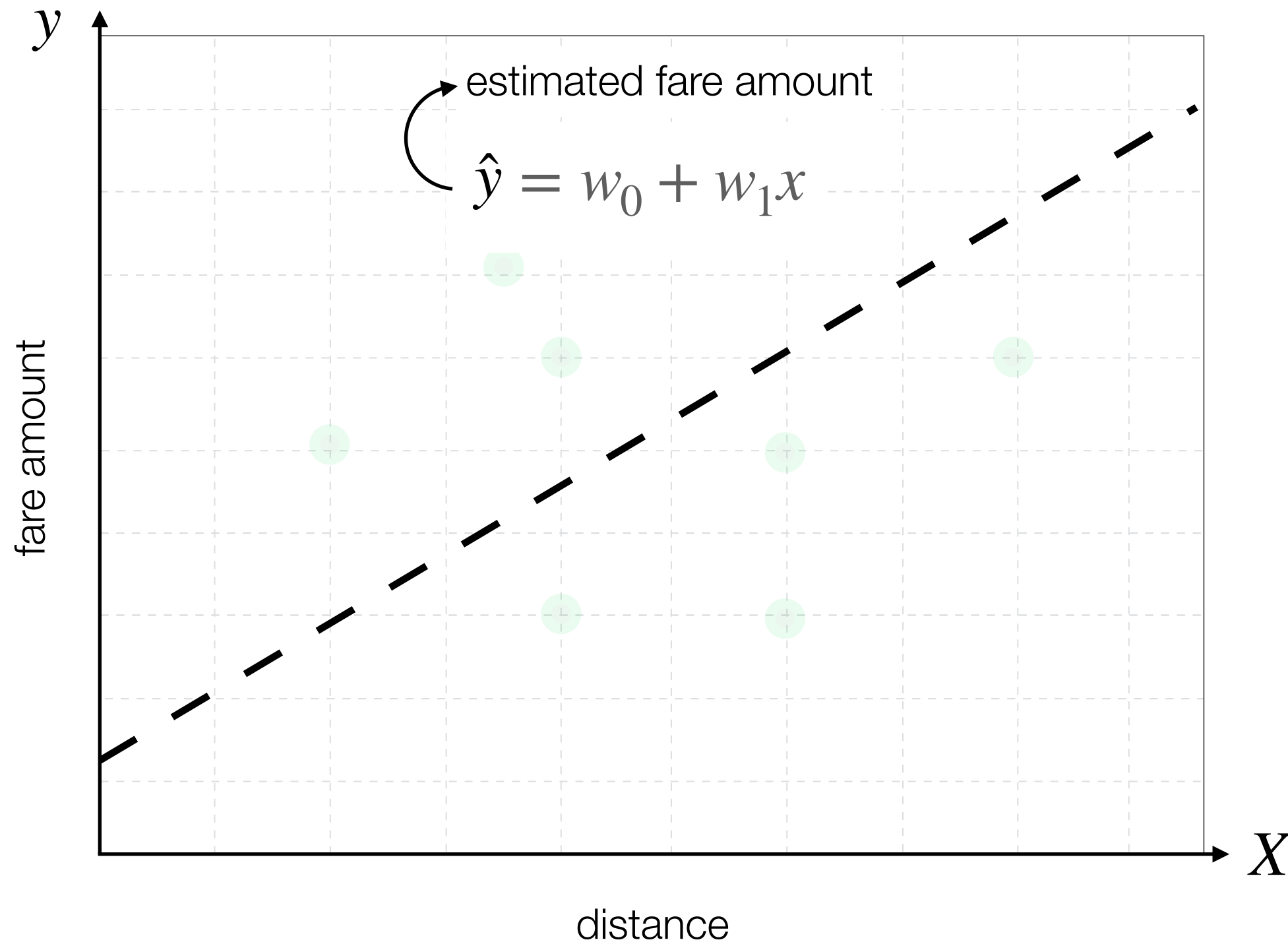
Linear Regression



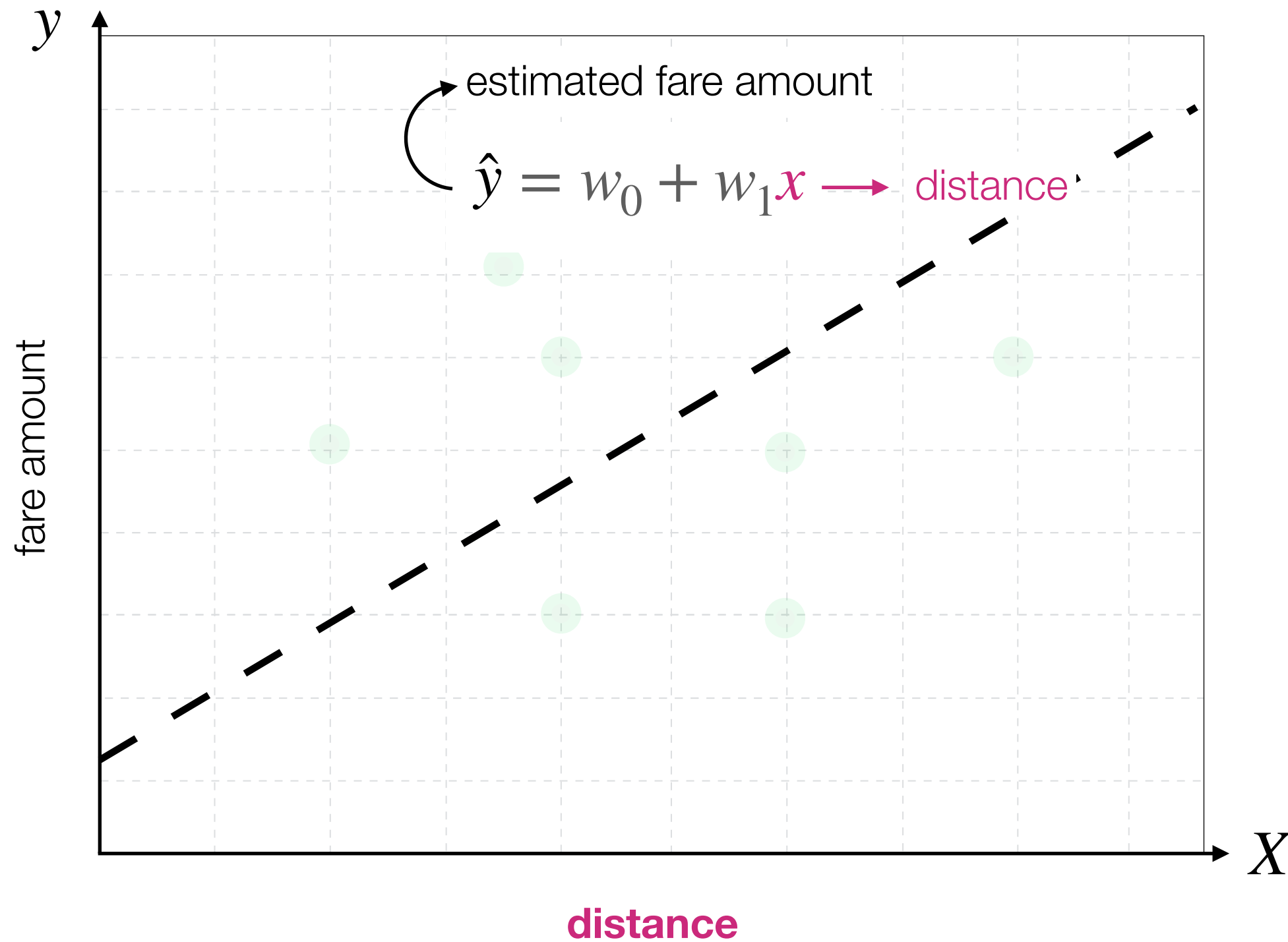
Linear Regression



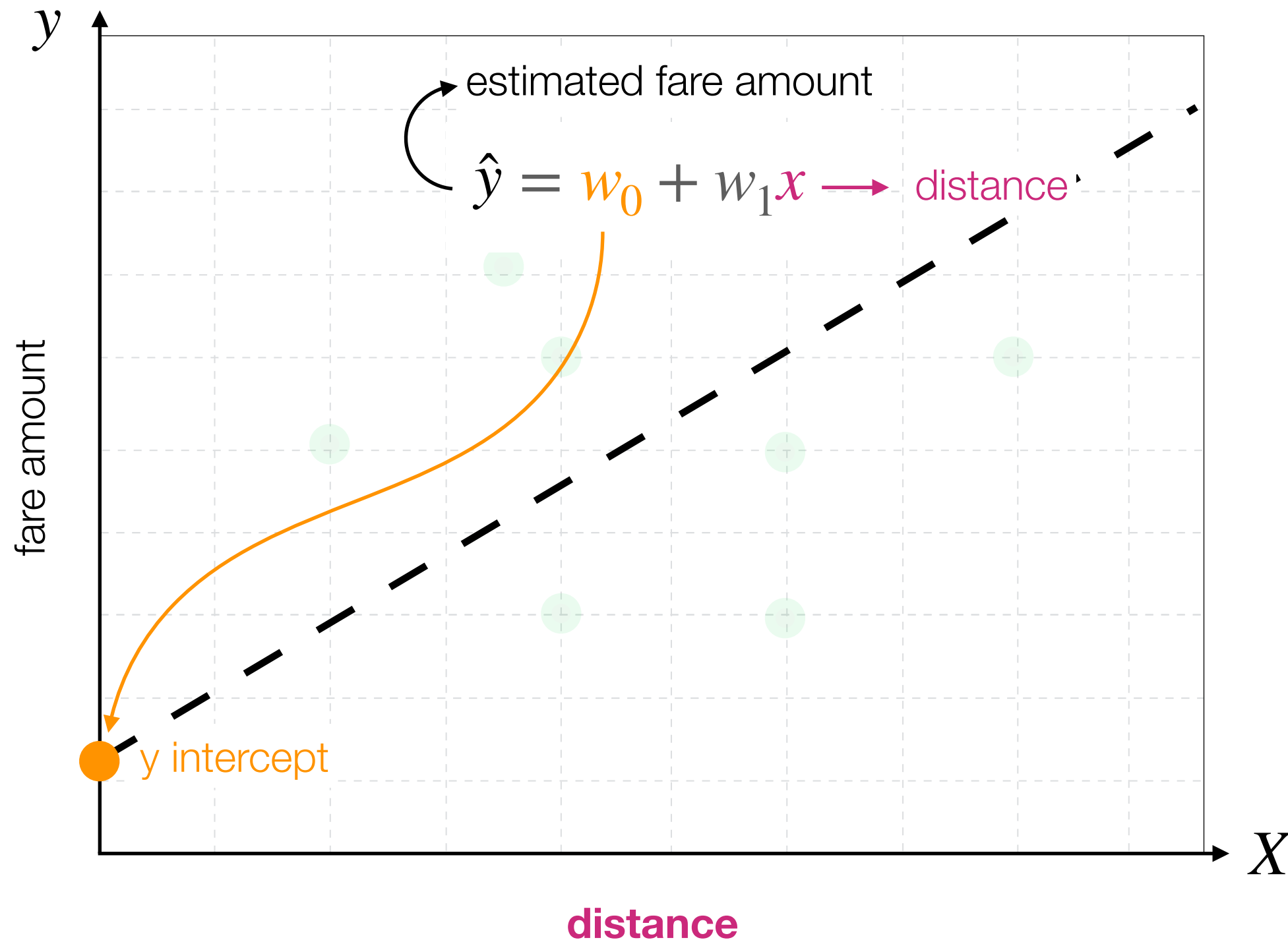
Linear Regression



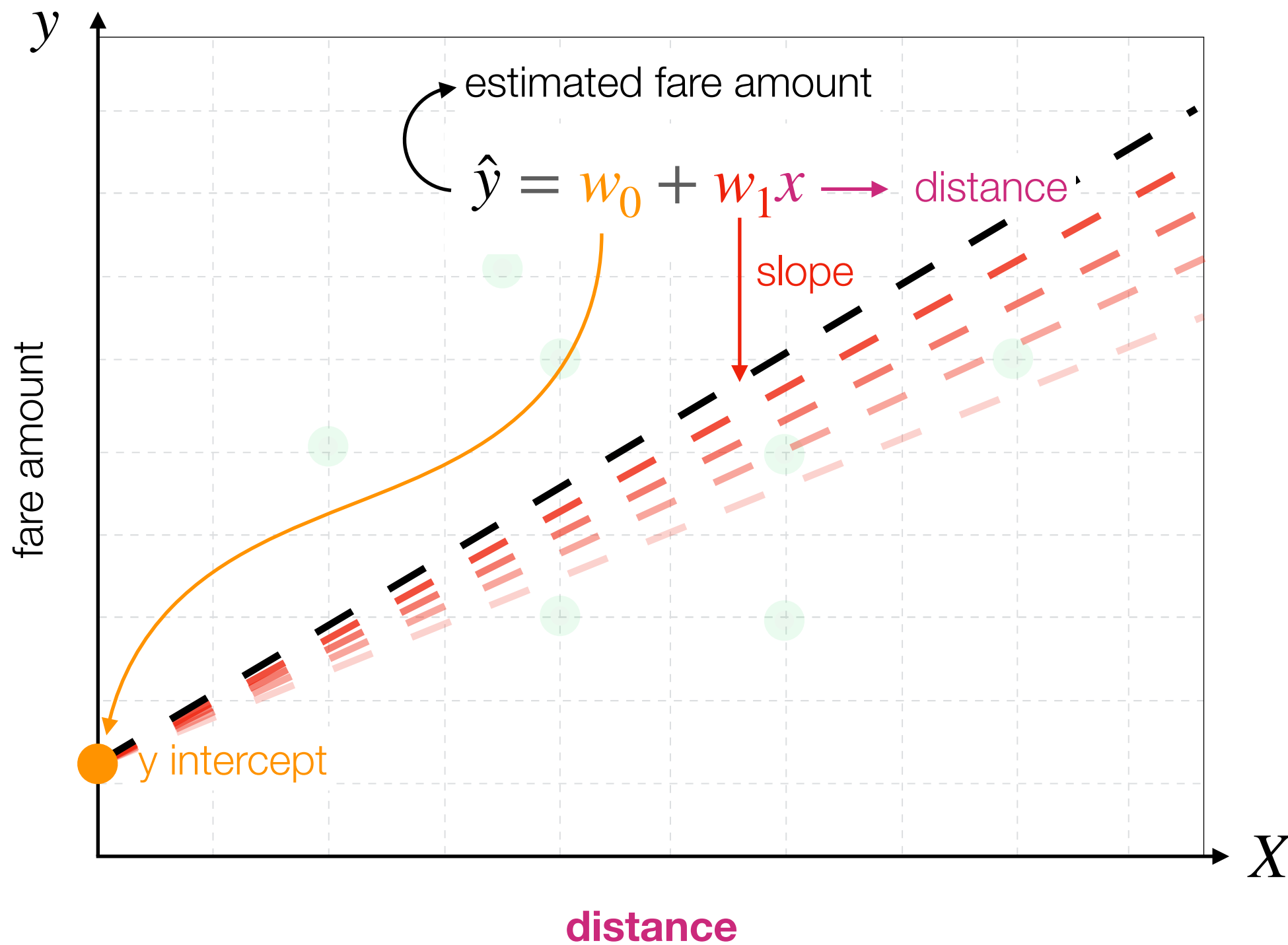
Linear Regression



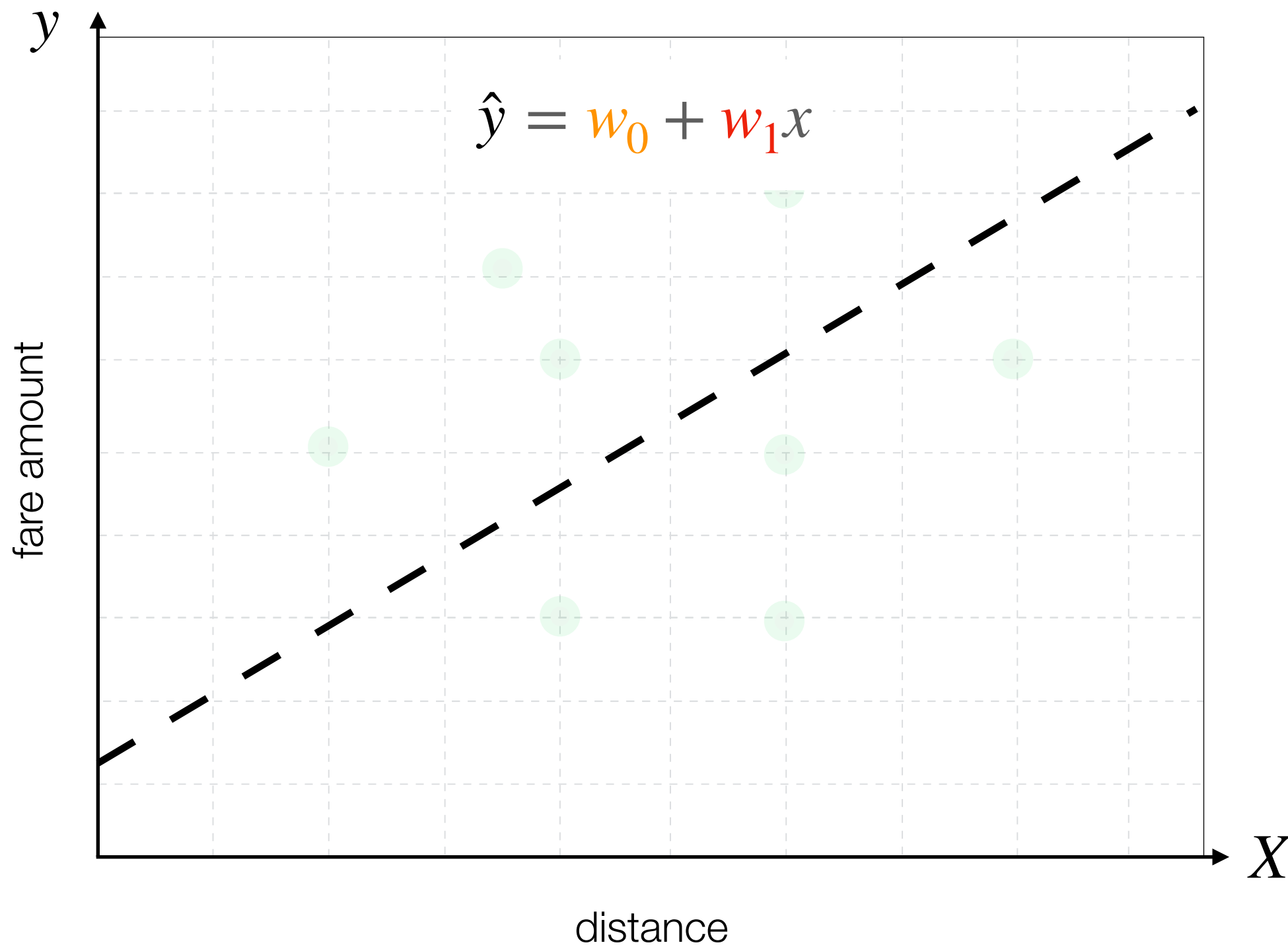
Linear Regression



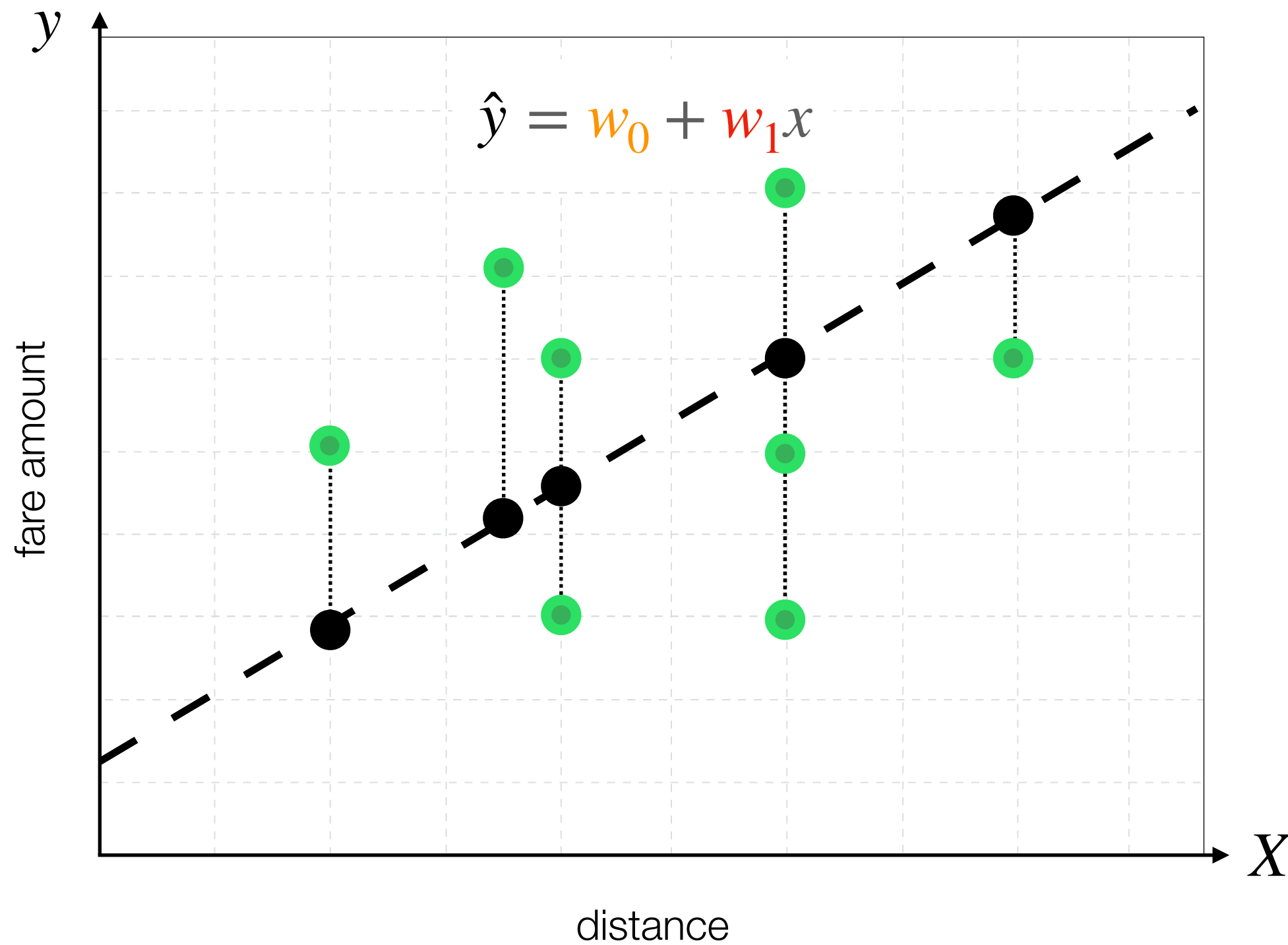
Linear Regression



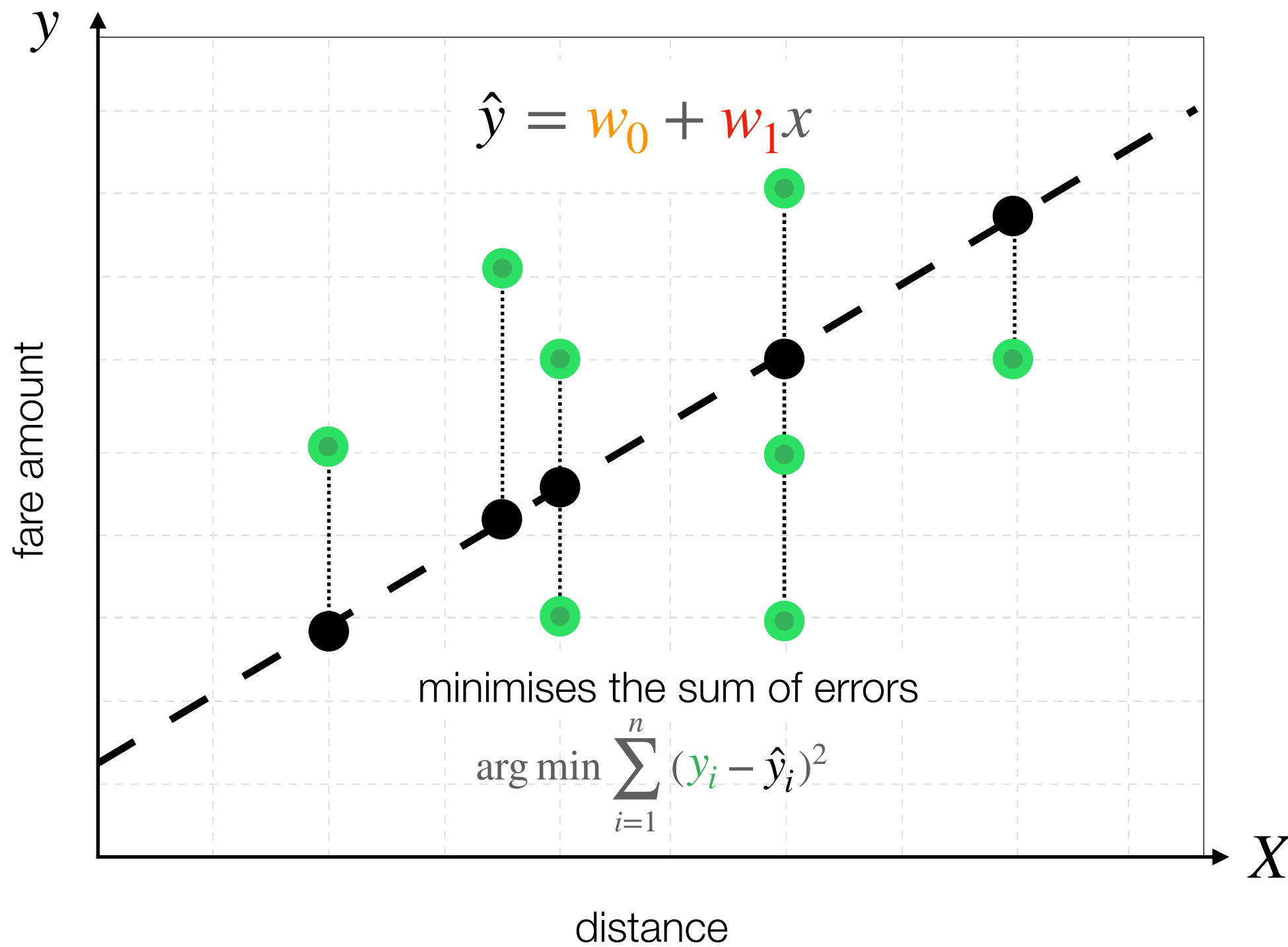
Linear Regression



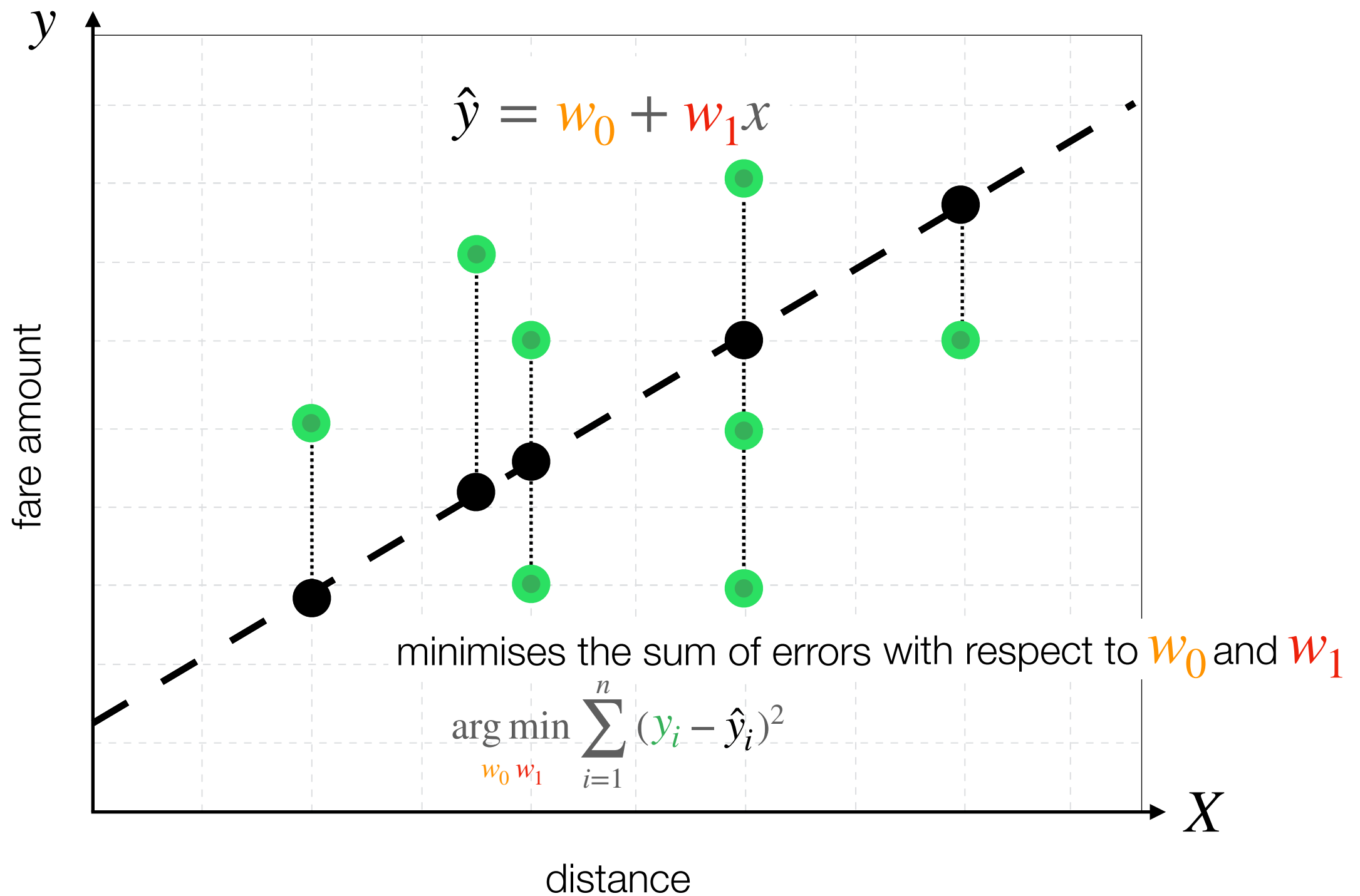
Linear Regression



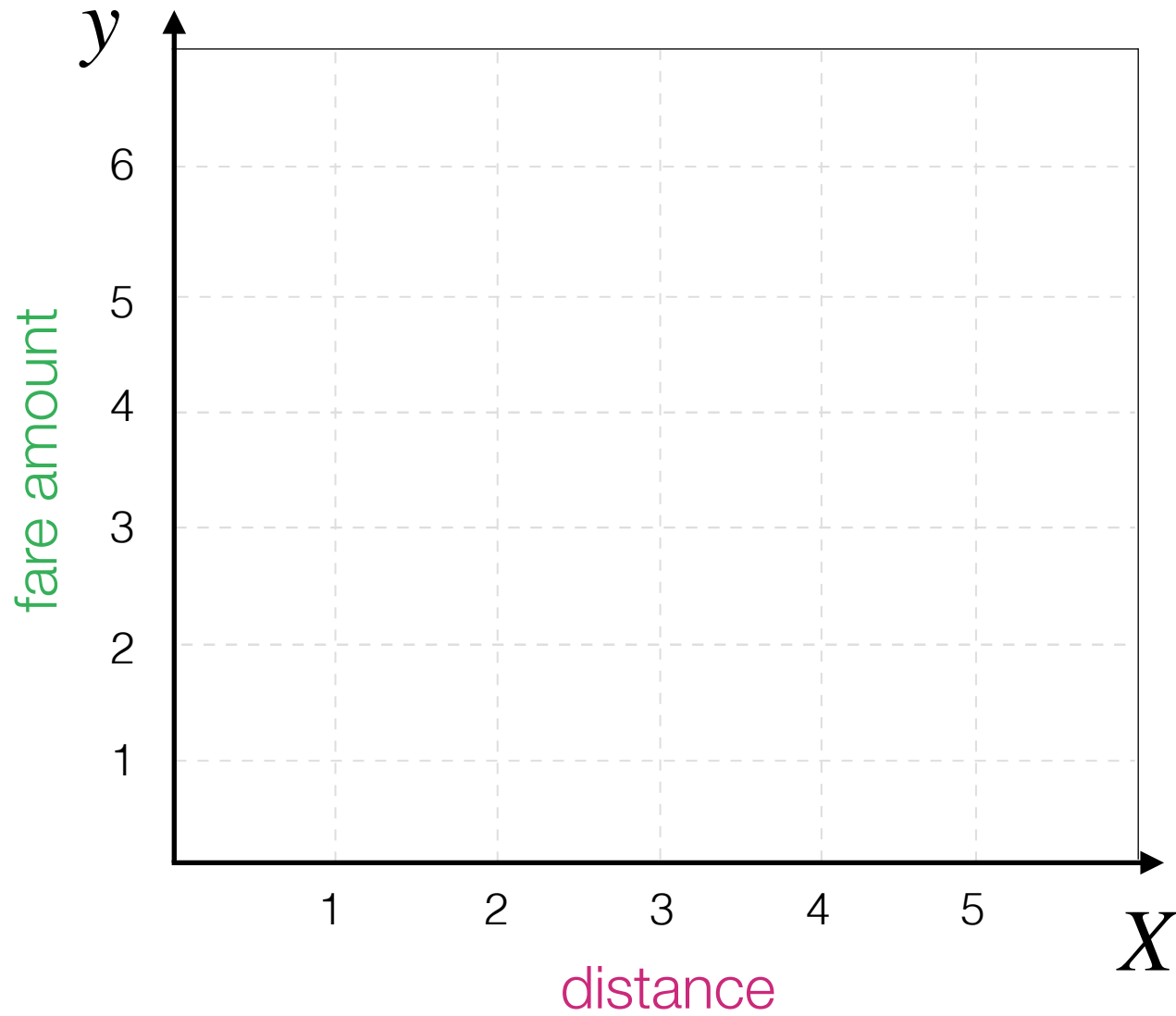
Linear Regression



Linear Regression

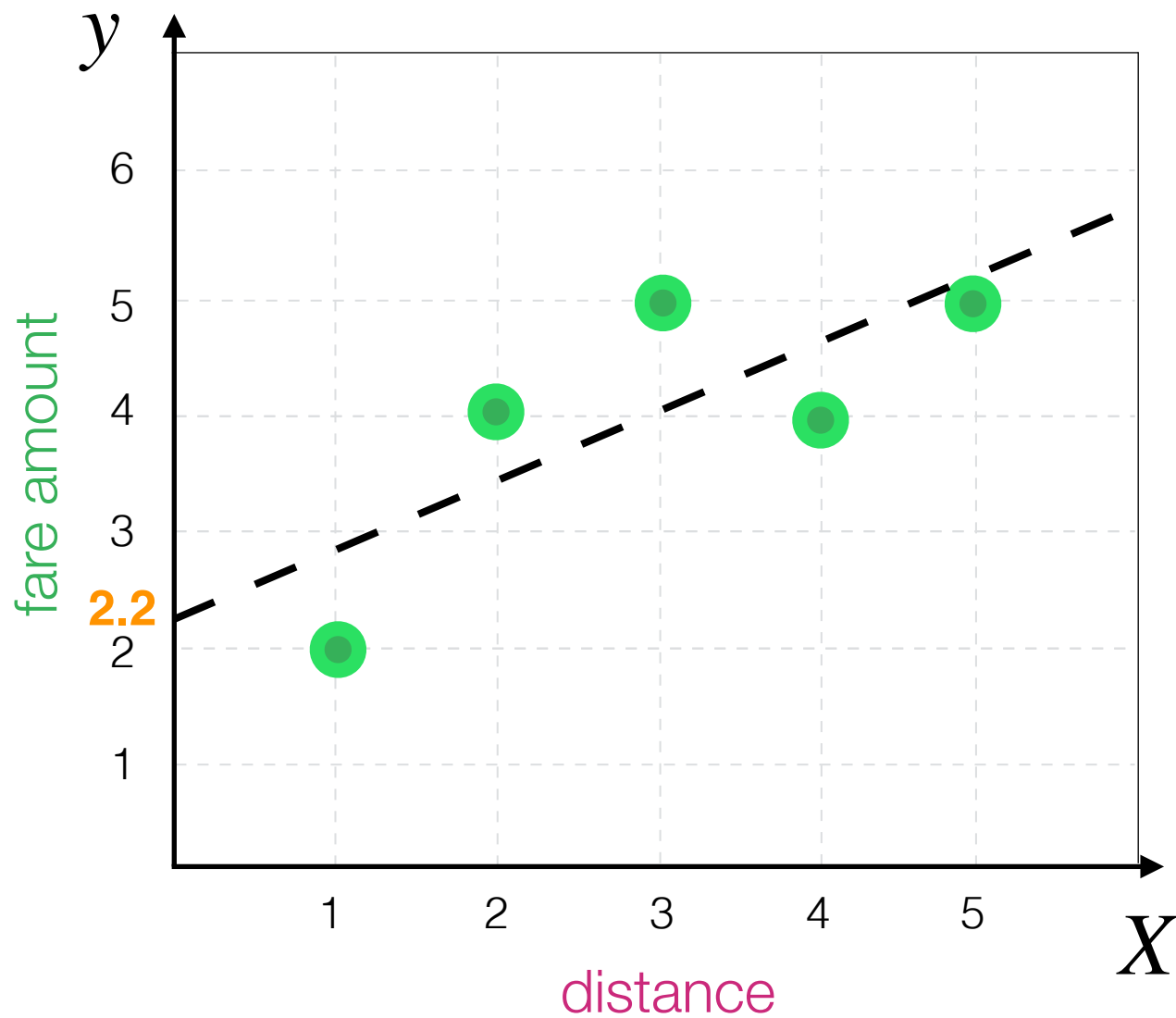


Linear Regression (example)



Linear Regression (example)

$$\hat{y} = 2.2 + .6x$$



x	y	x - $\bar{x}$	y - $\bar{y}$	(x - $\bar{x}$) ²	(x - $\bar{x}$)(y - $\bar{y}$)
1	2	-2	-2	4	4
2	4	-1	0	1	0
3	5	0	1	0	0
4	4	1	0	1	0
5	5	2	1	4	2

$$\bar{x} = 3 \quad \bar{y} = 4$$

$$10$$

$$6$$

$$w_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{6}{10} = .6$$

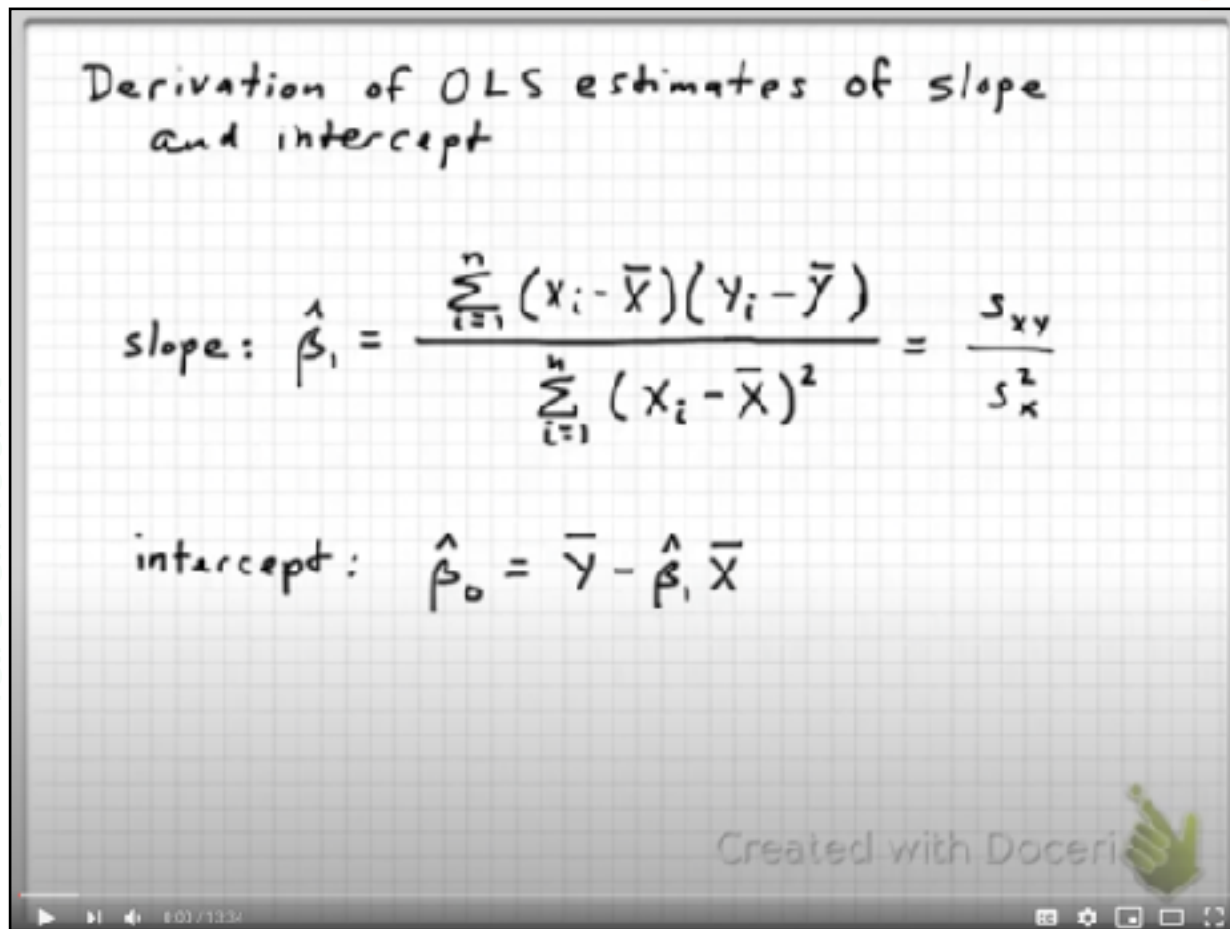
$$4 = w_0 + .6 * 3$$

$$w_0 = 2.2$$

Linear Regression (example)

$$\hat{y} = w_0 + w_1 x$$

Check out the video for how the formulas are derived!



Derivation of OLS estimates of slope and intercept

slope: $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{s_{xy}}{s_x^2}$

intercept: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Created with Doceri

$$w_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

$$w_0 = \bar{y} - w_1 \bar{x}$$

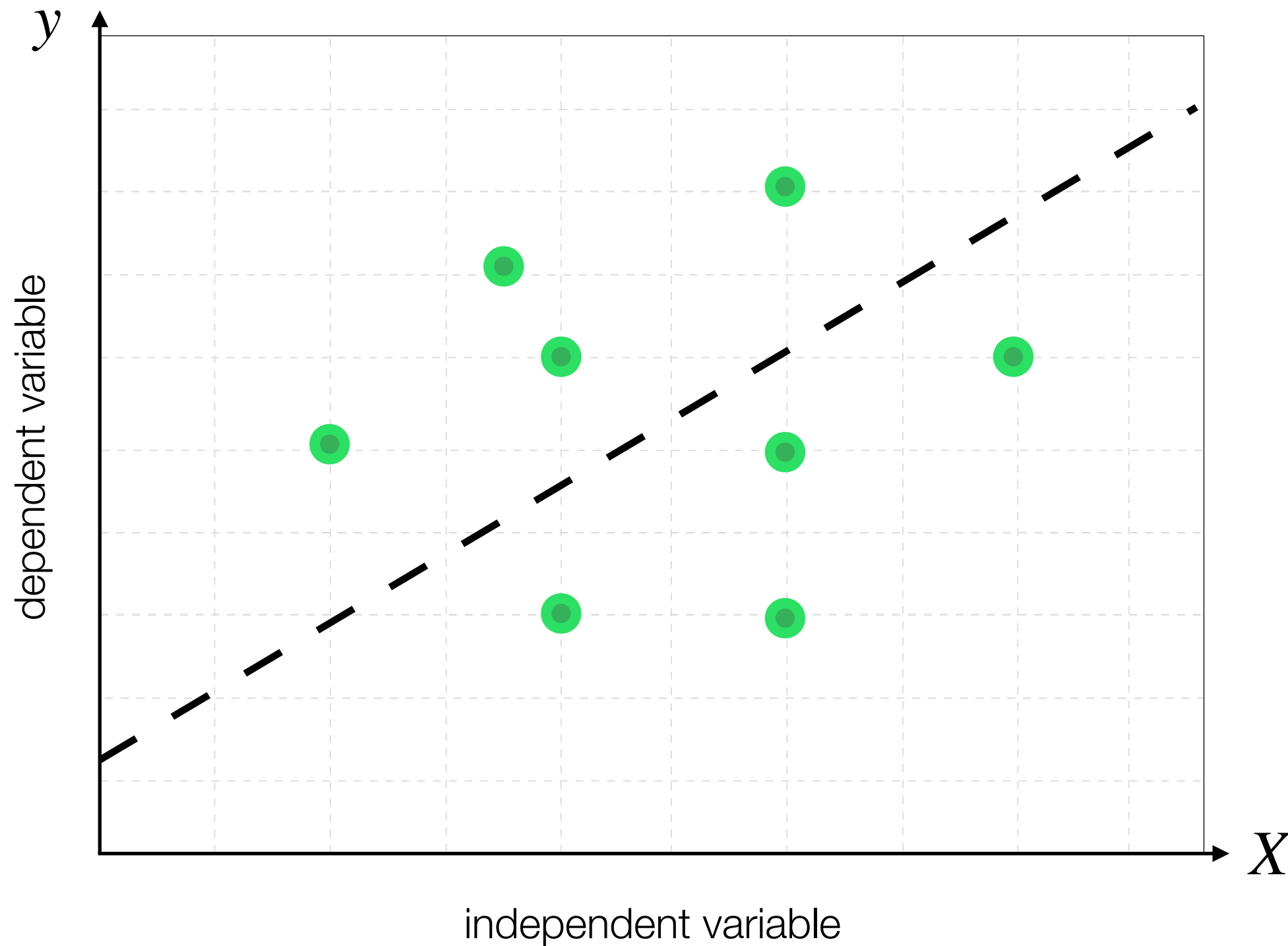
<https://youtu.be/jqoHefilf9U>

$$\bar{x} = 3 \quad \bar{y} = 4$$

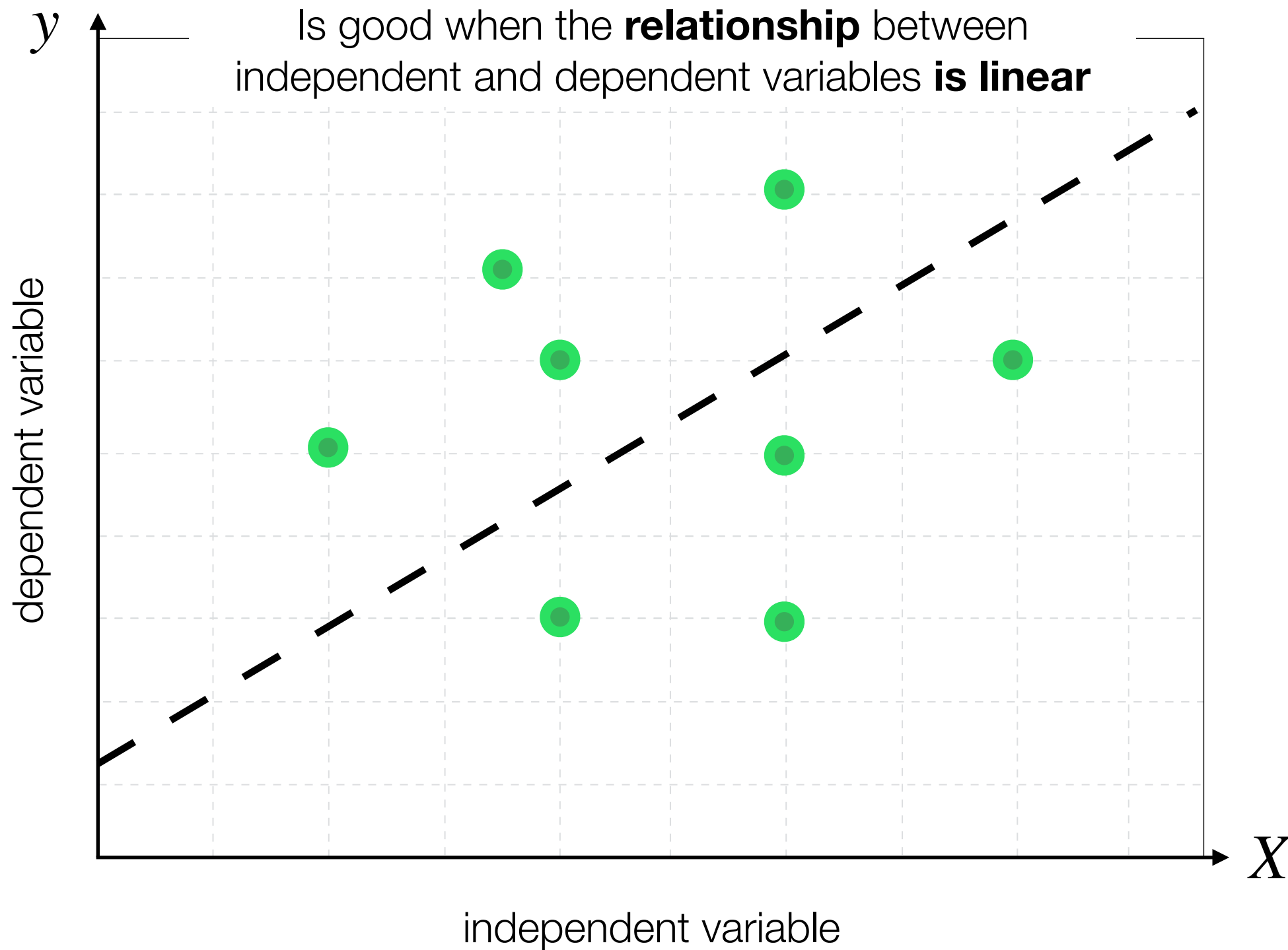
$$10$$

$$6$$

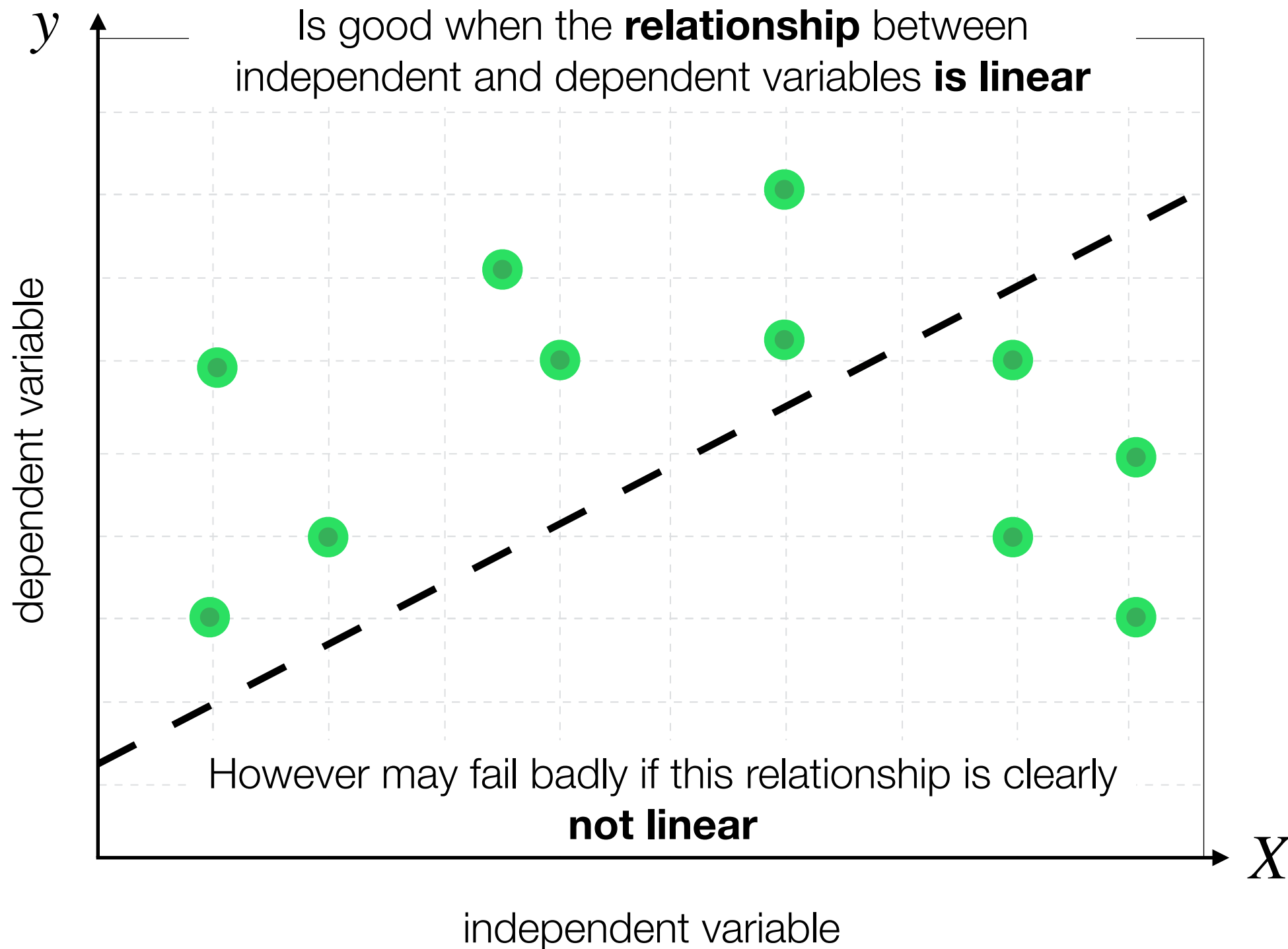
Linear Regression



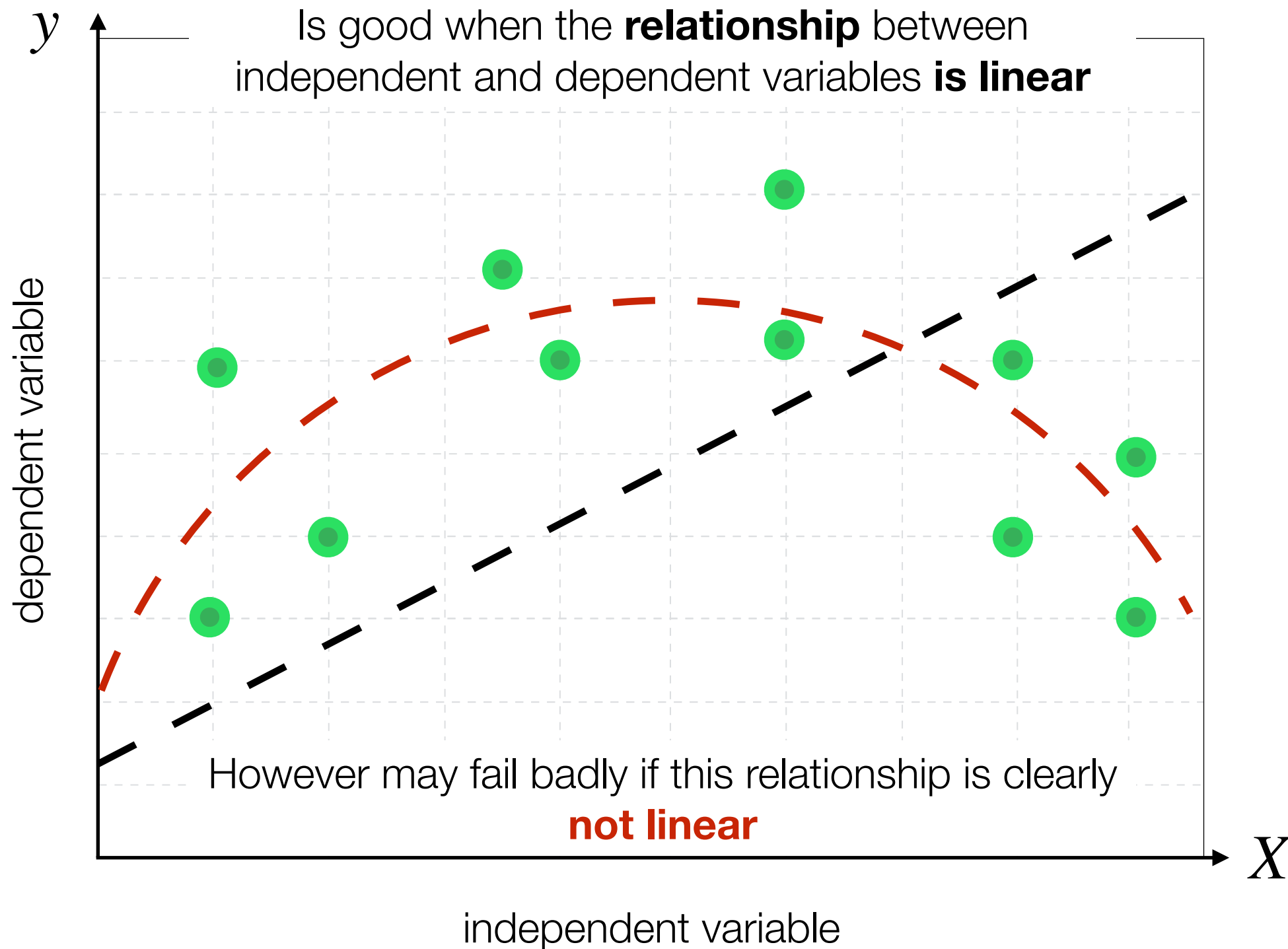
Linear Regression



Linear Regression

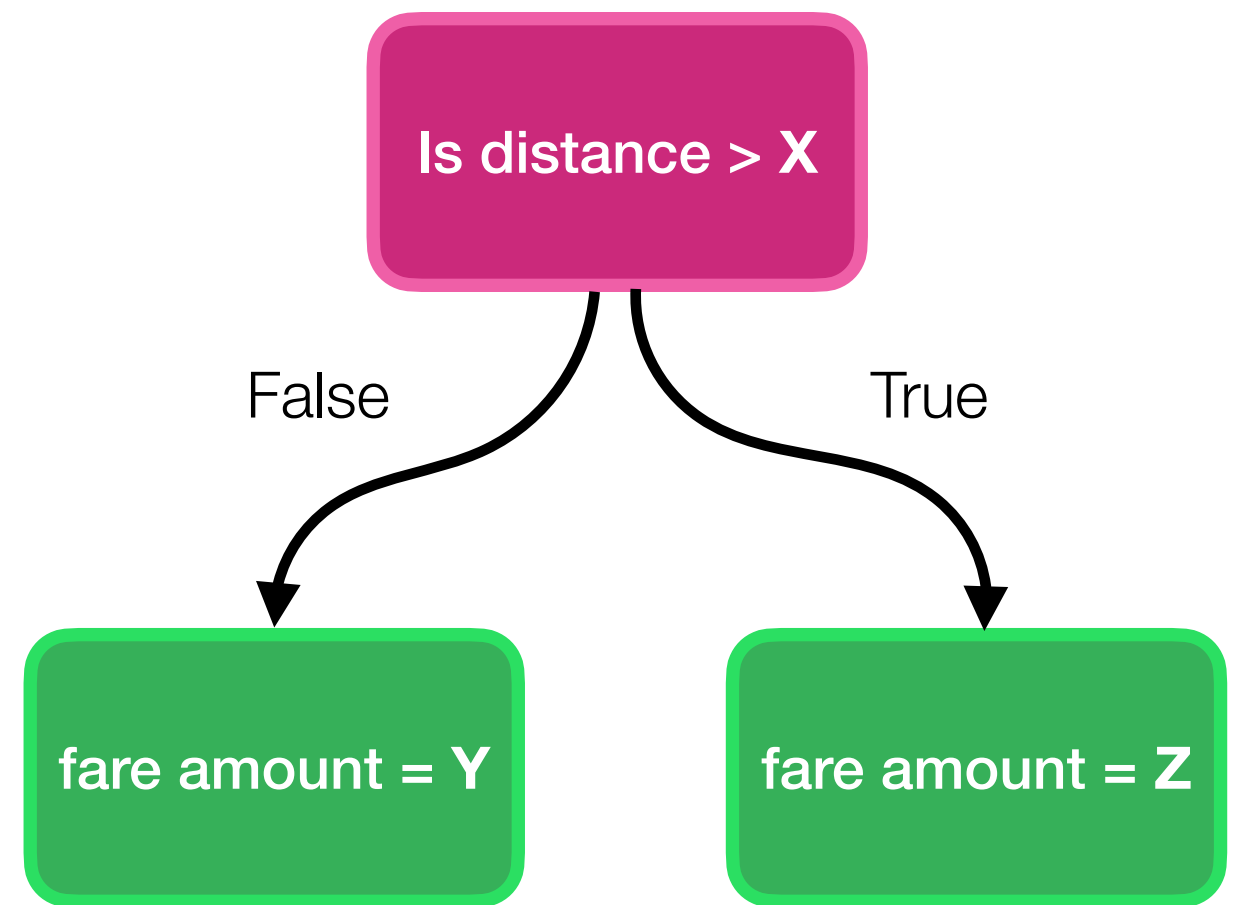
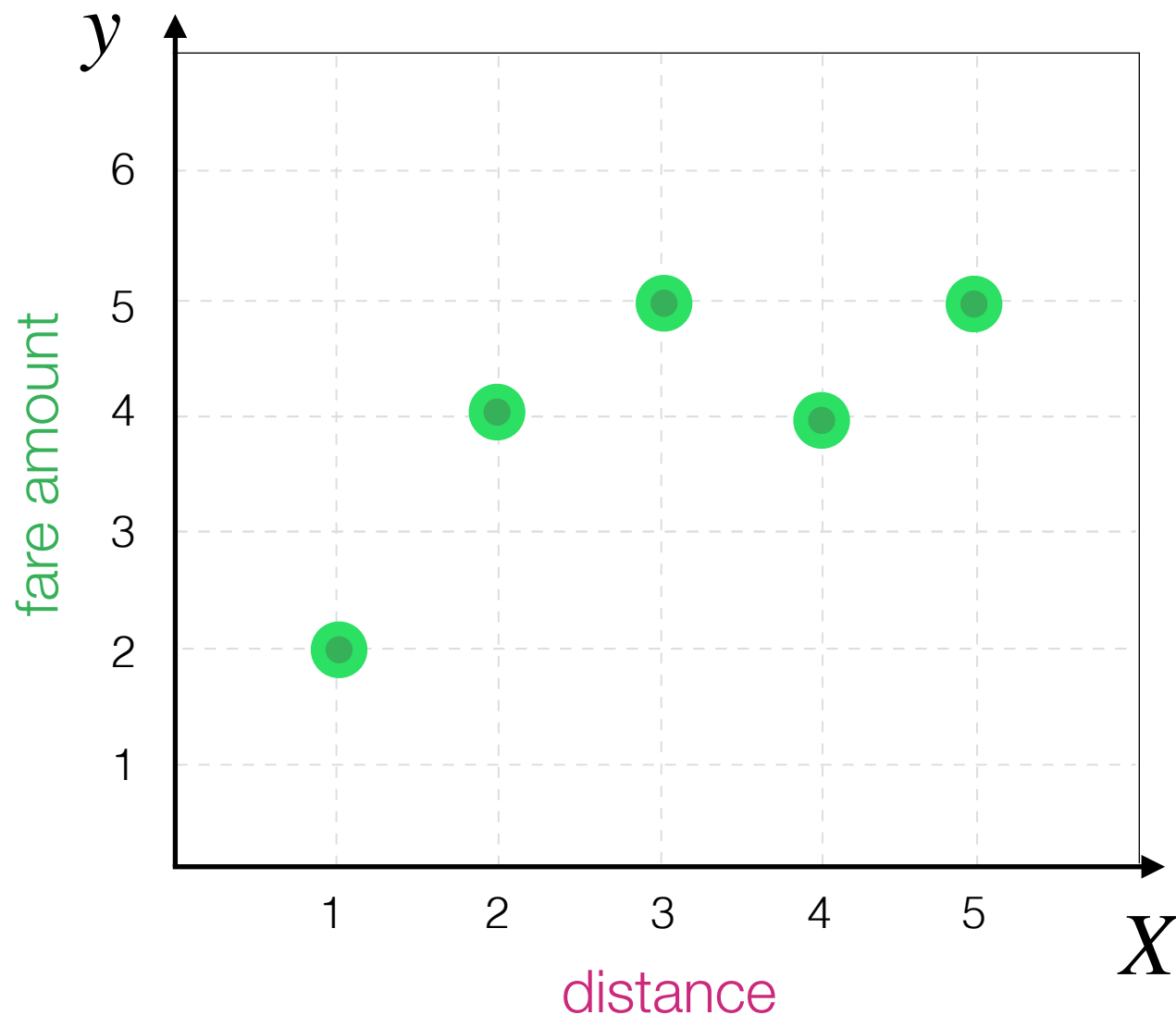


Linear Regression



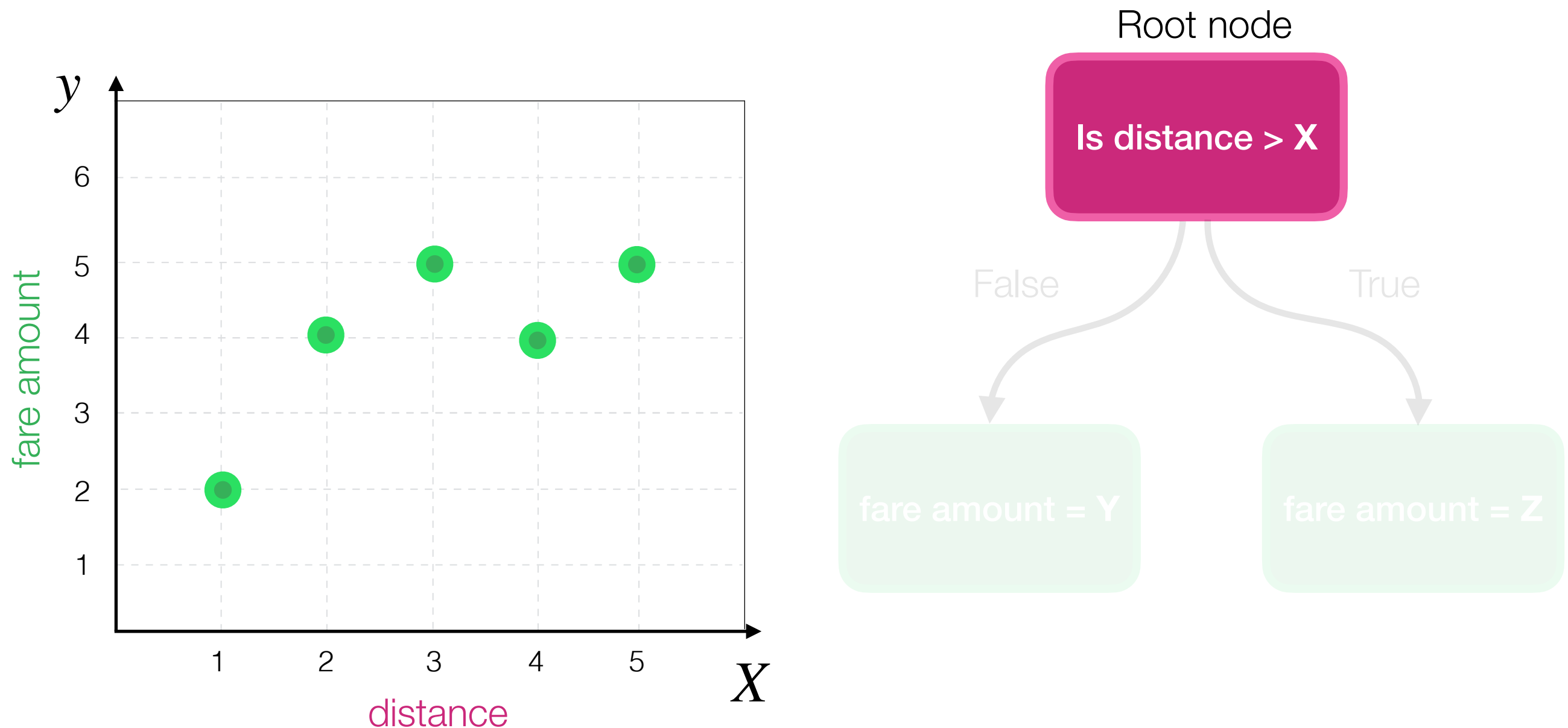
Decision Tree Algorithm

By asking a simple **question** about value of **independent variable** it tries to predict a value of **dependent variable**



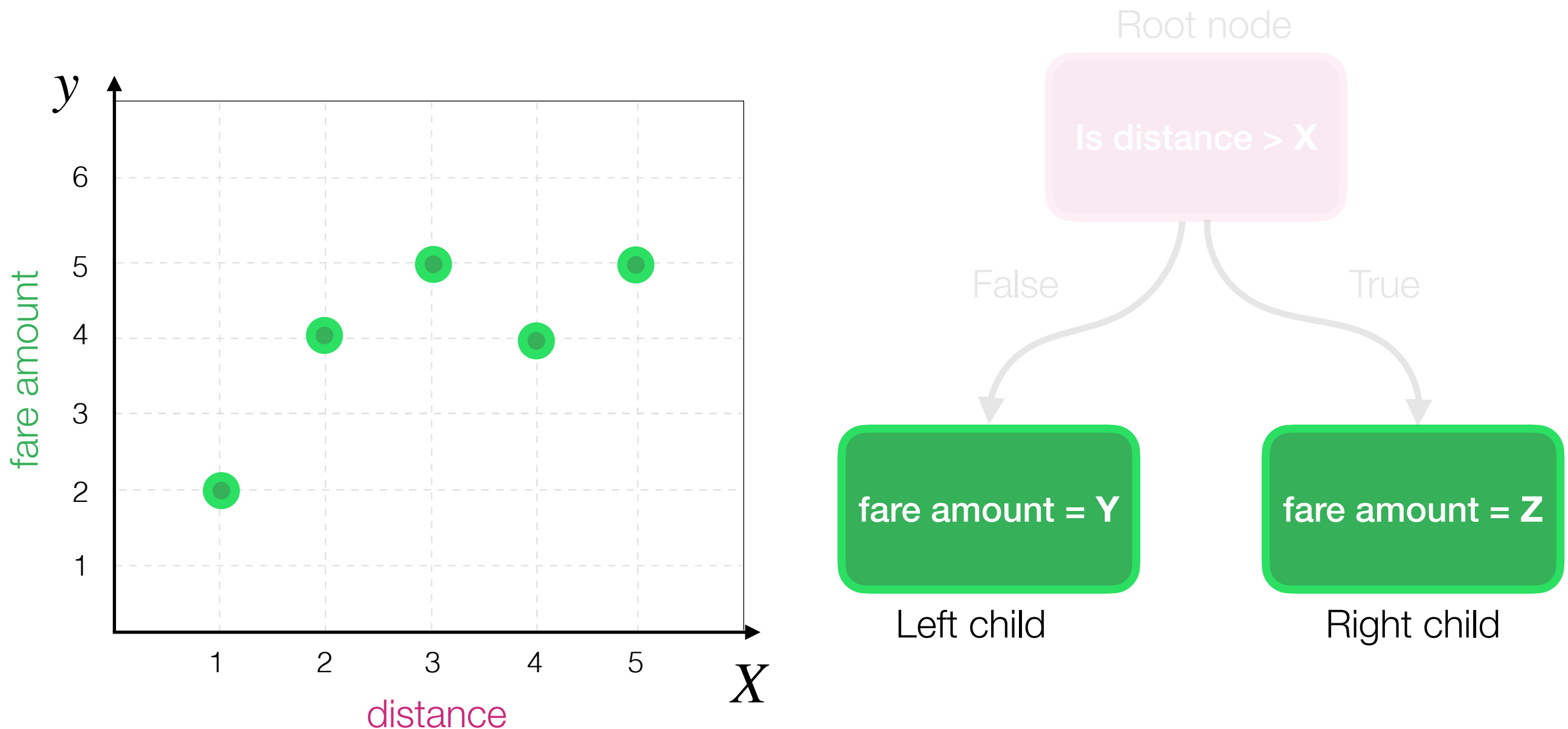
Decision Tree Algorithm

By asking a simple **question** about value of **independent variable** it tries to predict a value of **dependent variable**



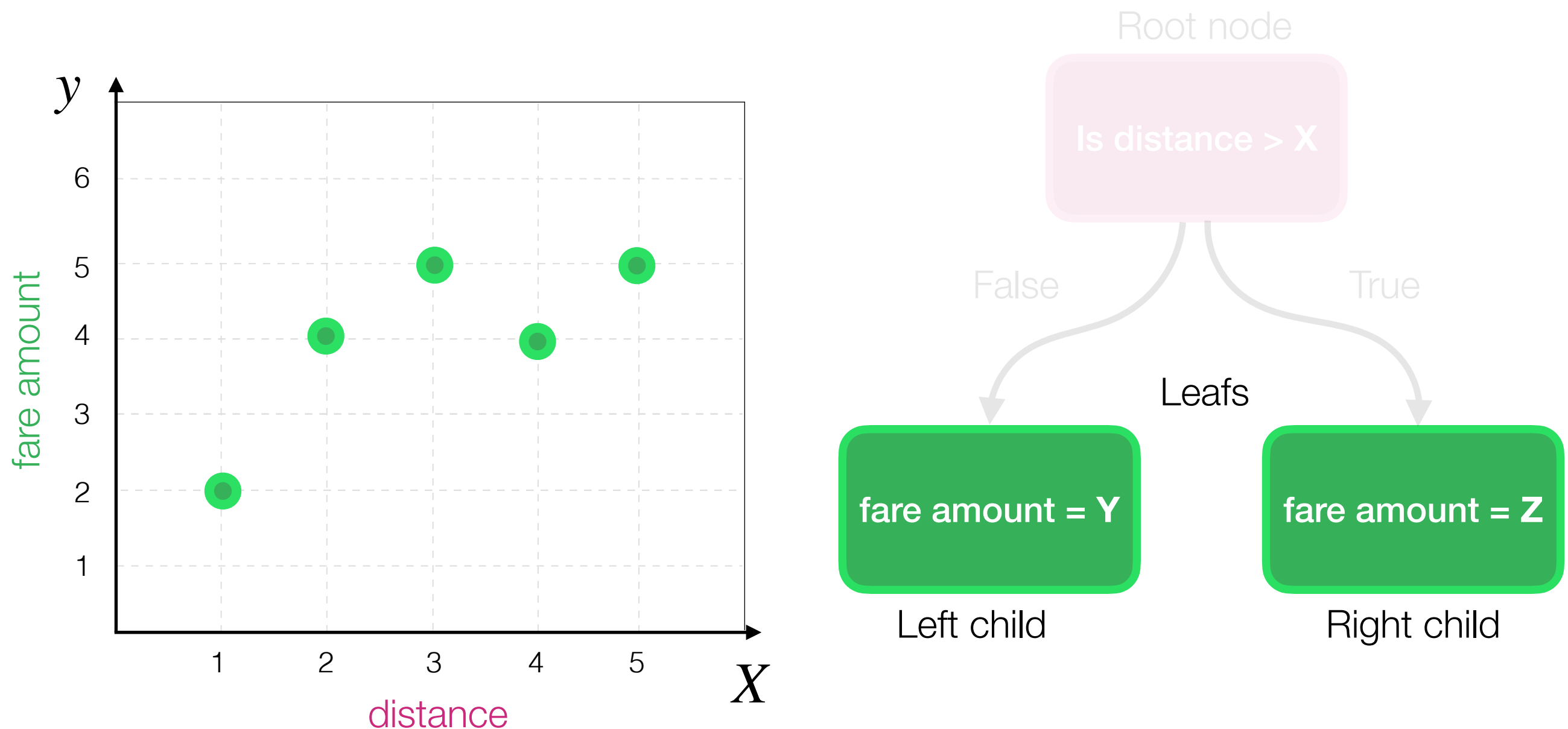
Decision Tree Algorithm

By asking a simple **question** about value of **independent variable** it tries to predict a value of **dependent variable**



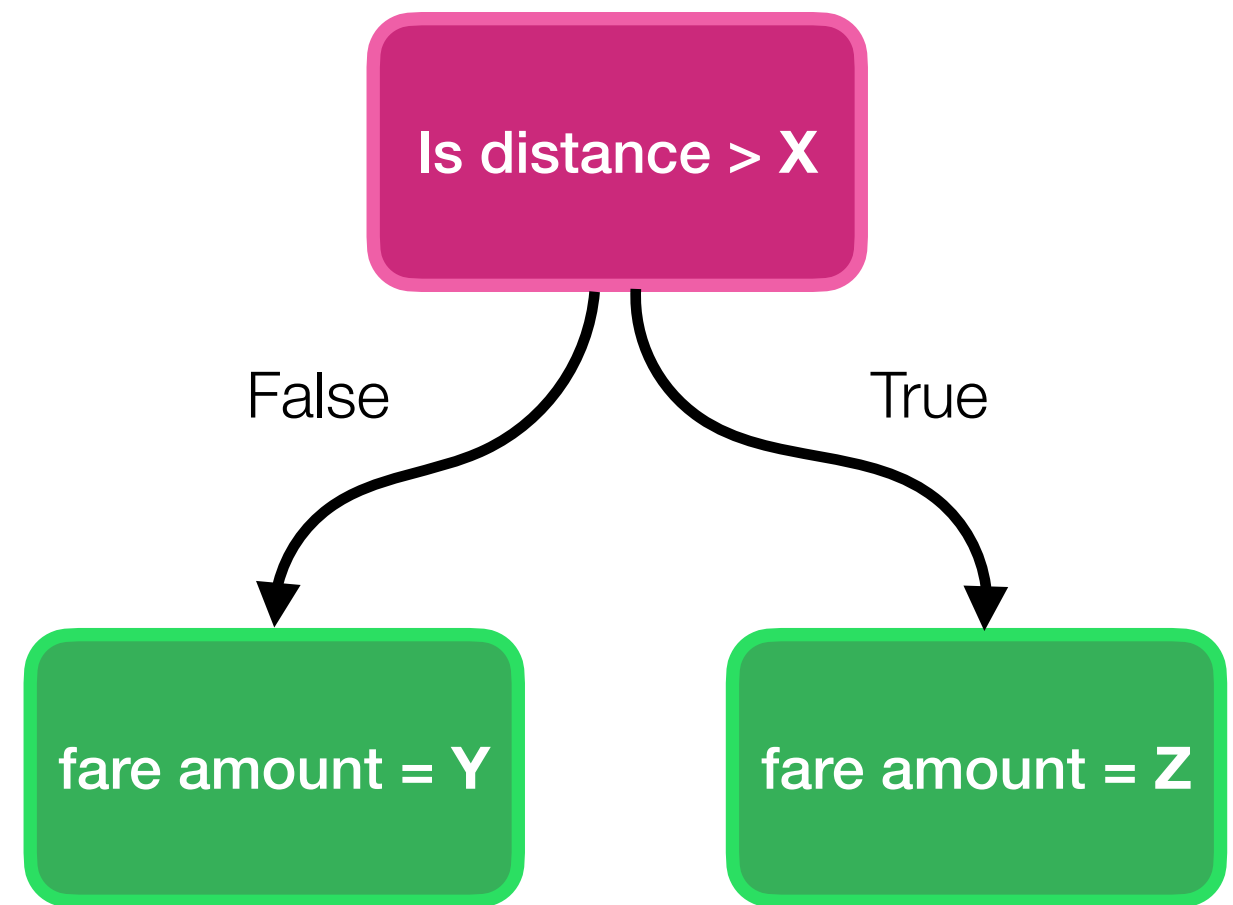
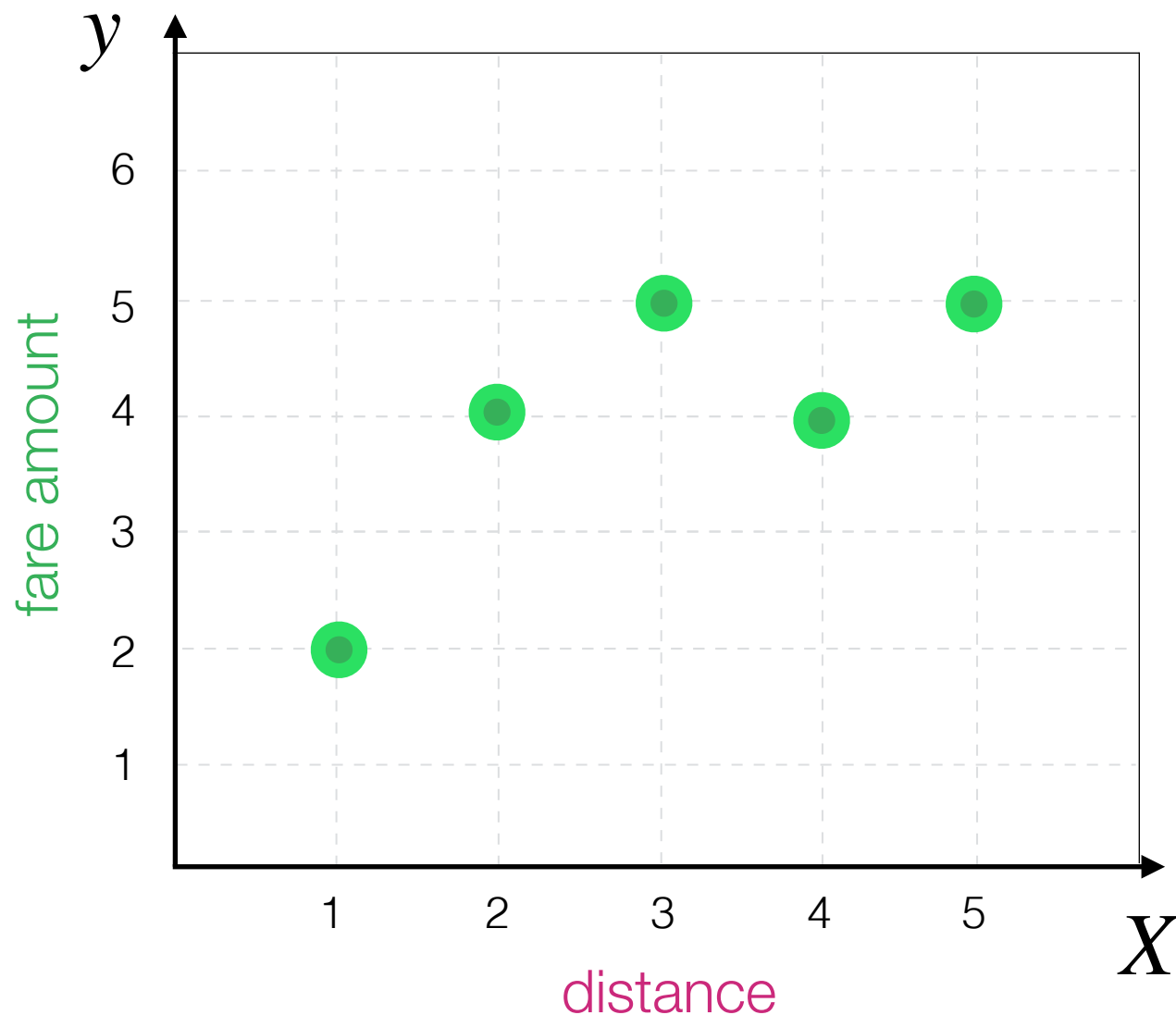
Decision Tree Algorithm

By asking a simple **question** about value of **independent variable** it tries to predict a value of **dependent variable**



Decision Tree Algorithm

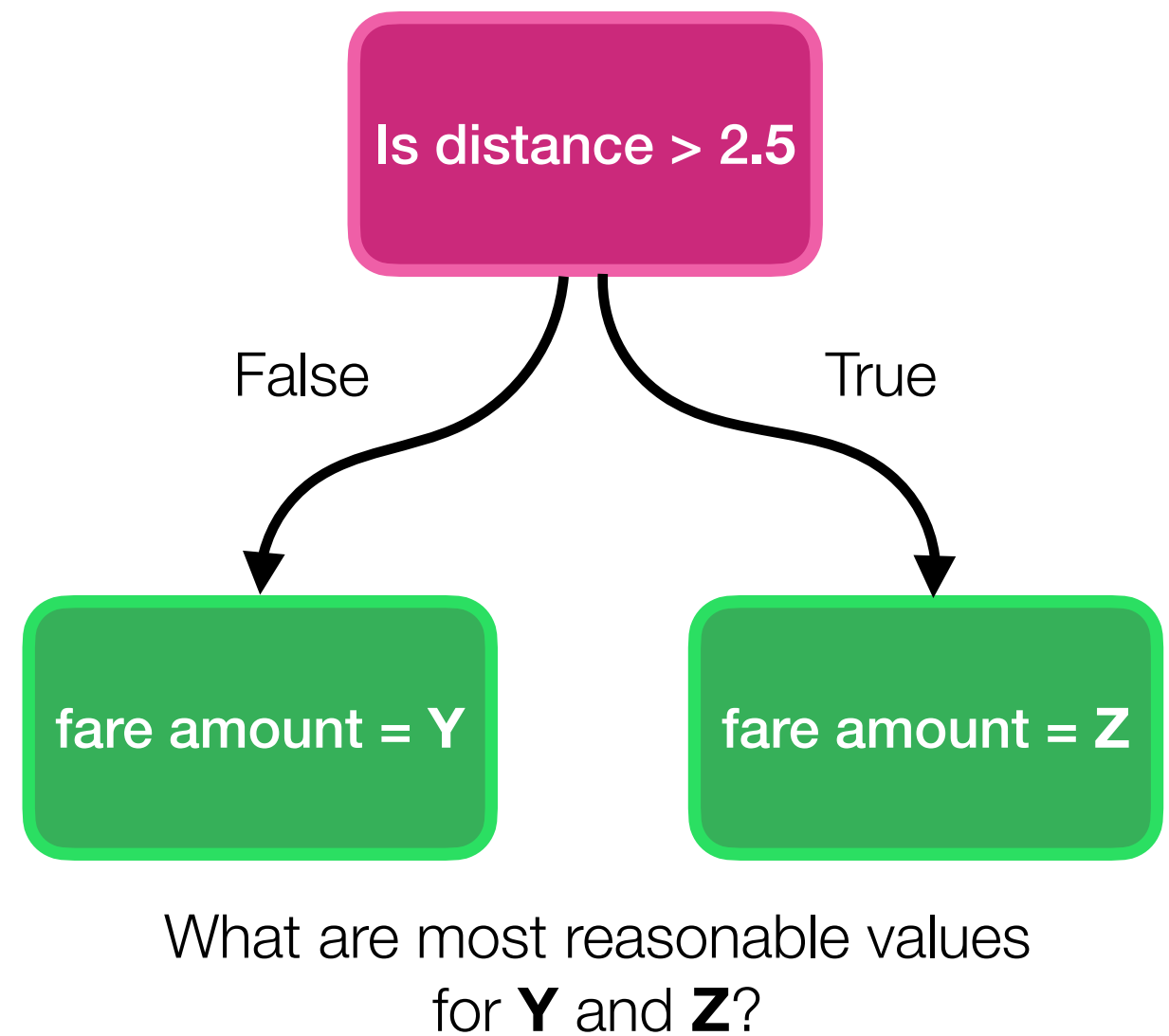
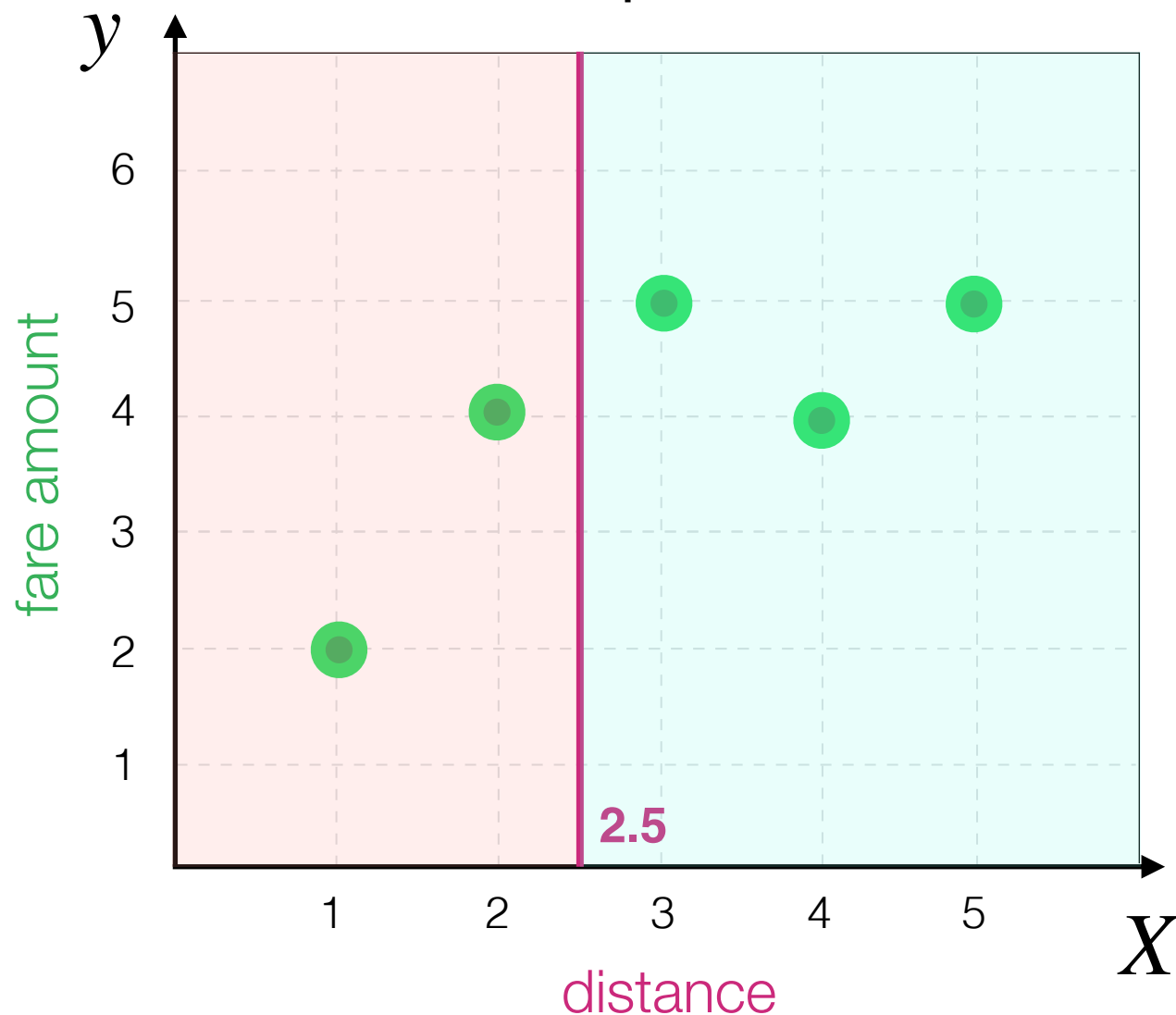
By asking a simple **question** about value of **independent variable** it tries to predict a value of **dependent variable**



Decision Tree Algorithm

Here, **X** may correspond to any **vertical** line.

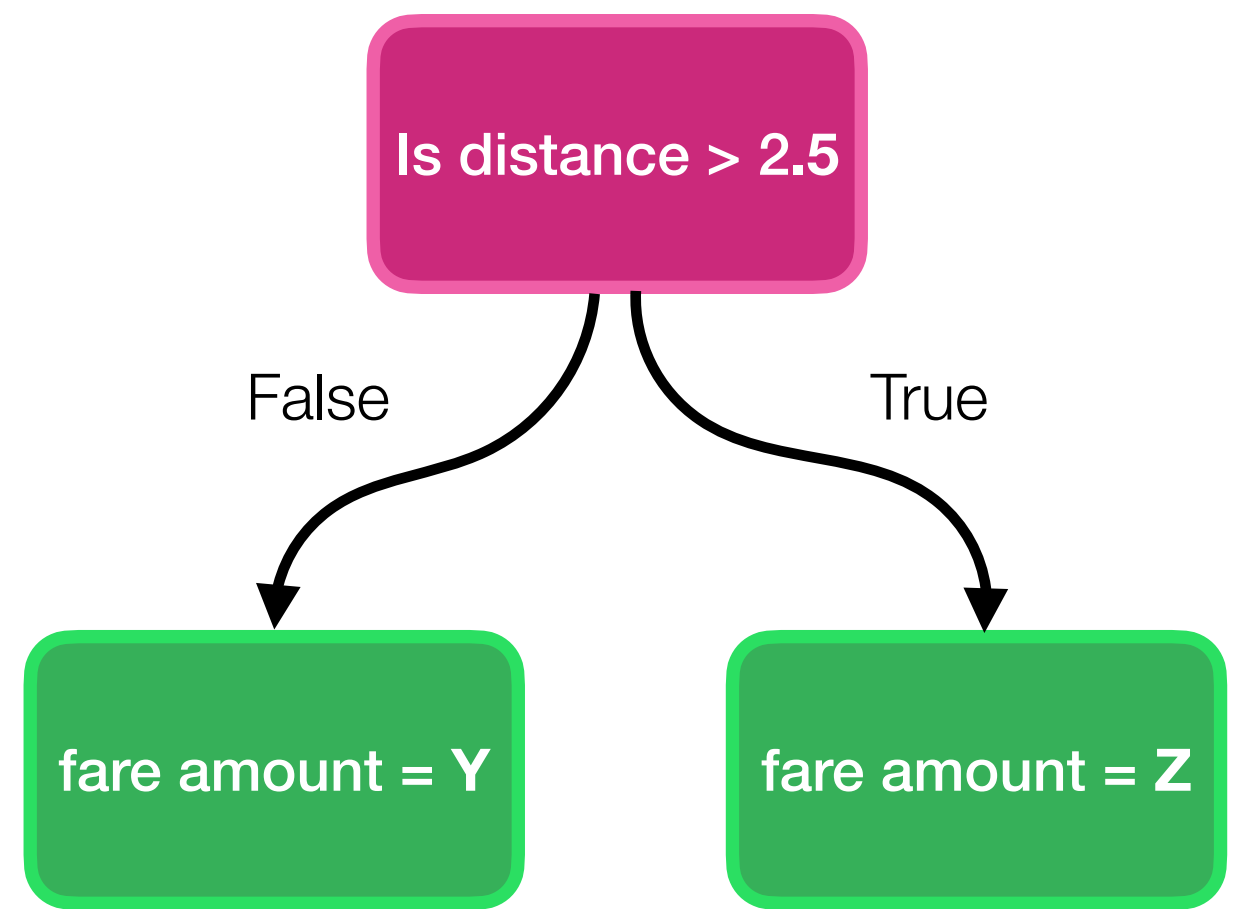
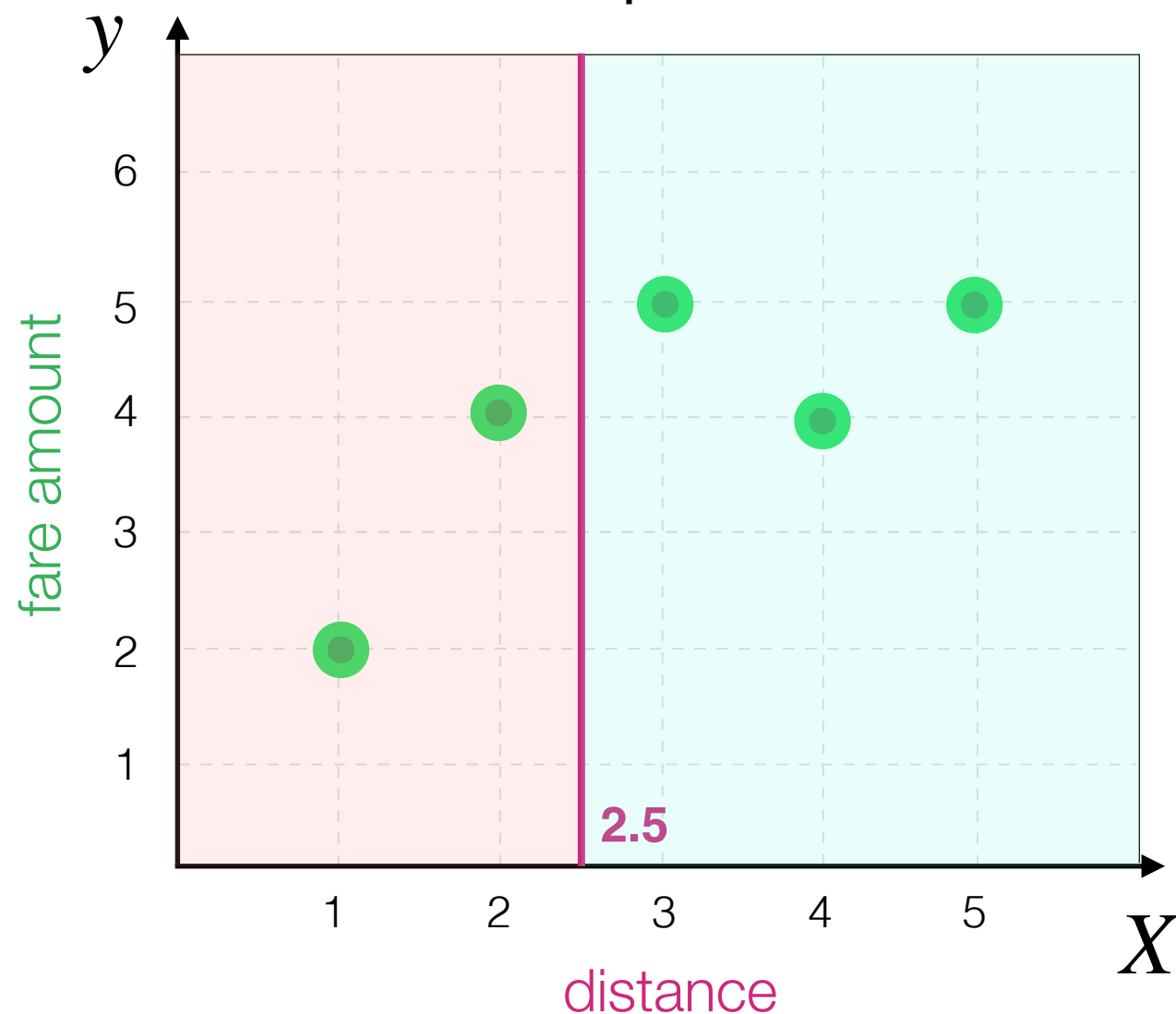
For example if $X = 2.5$:



Decision Tree Algorithm

Here, **X** may correspond to any **vertical** line.

For example if $X = 2.5$:

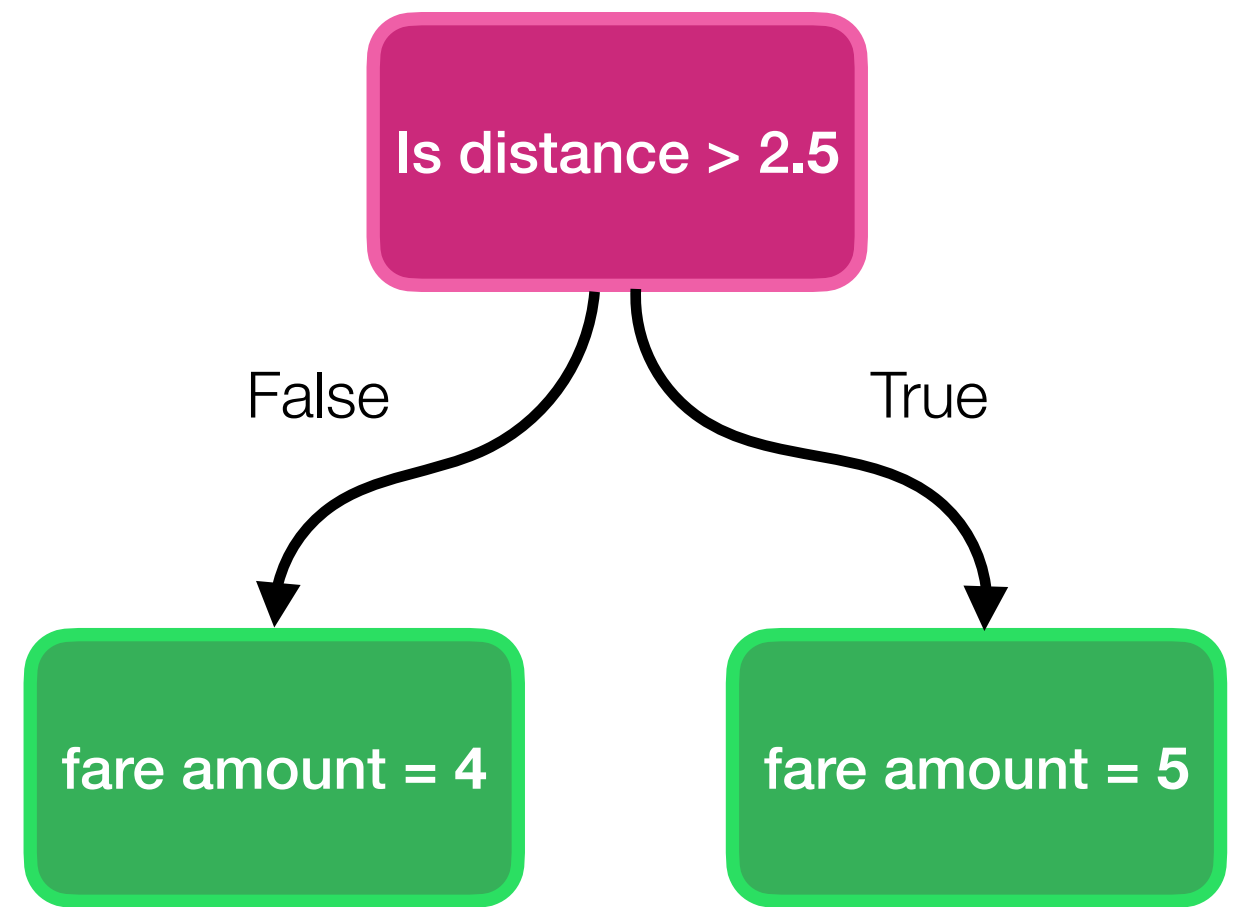
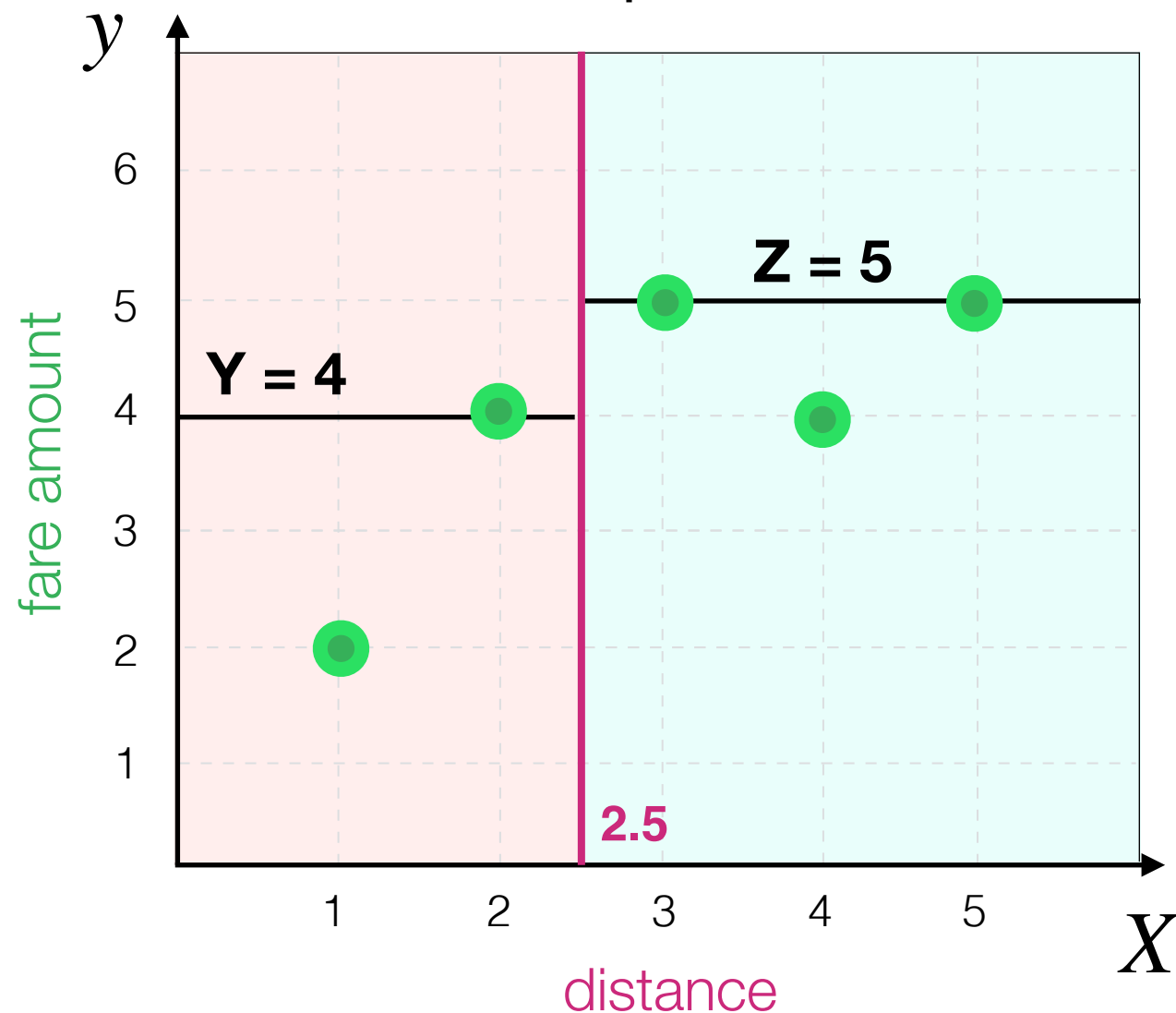


What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

What would be MSE if **Y = 4** and **Z = 5**?

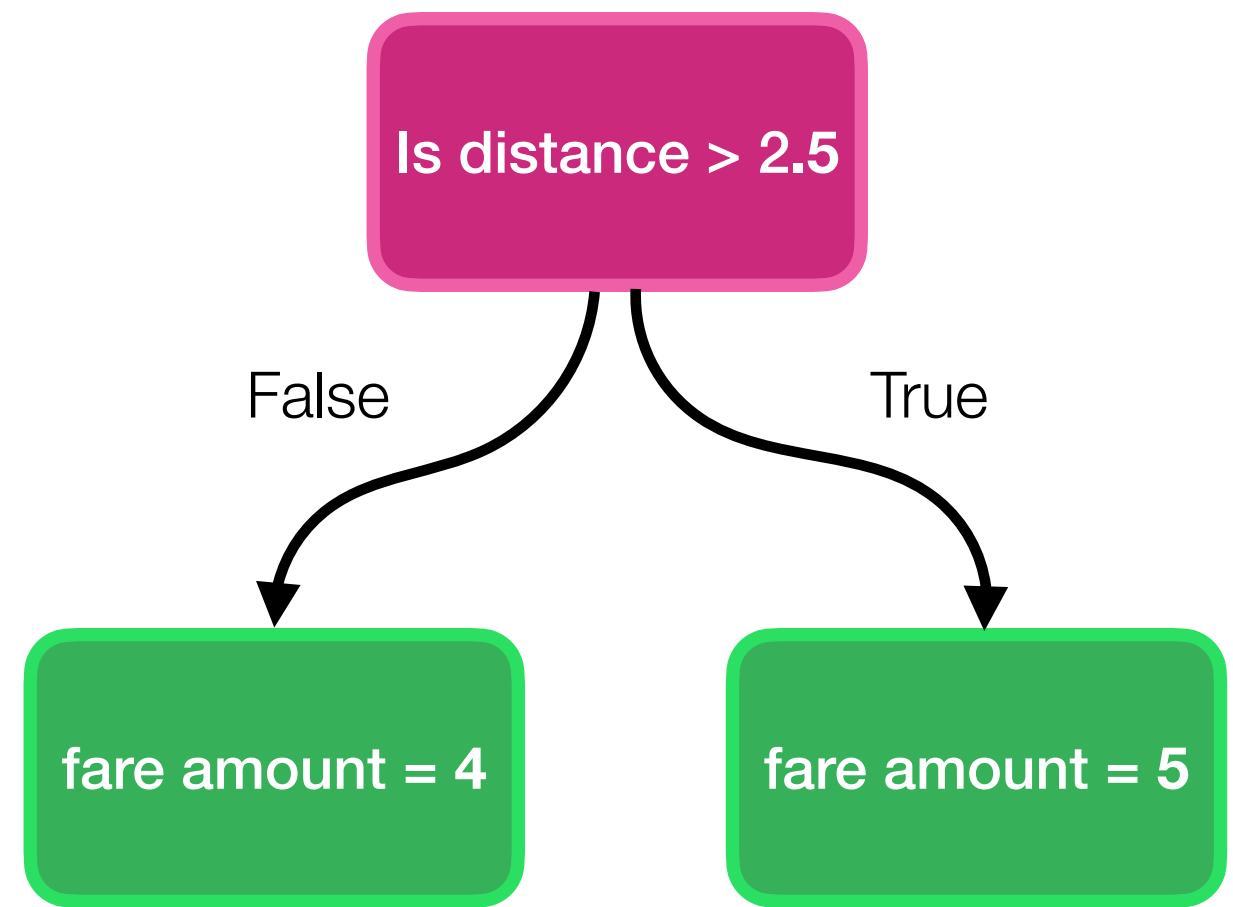
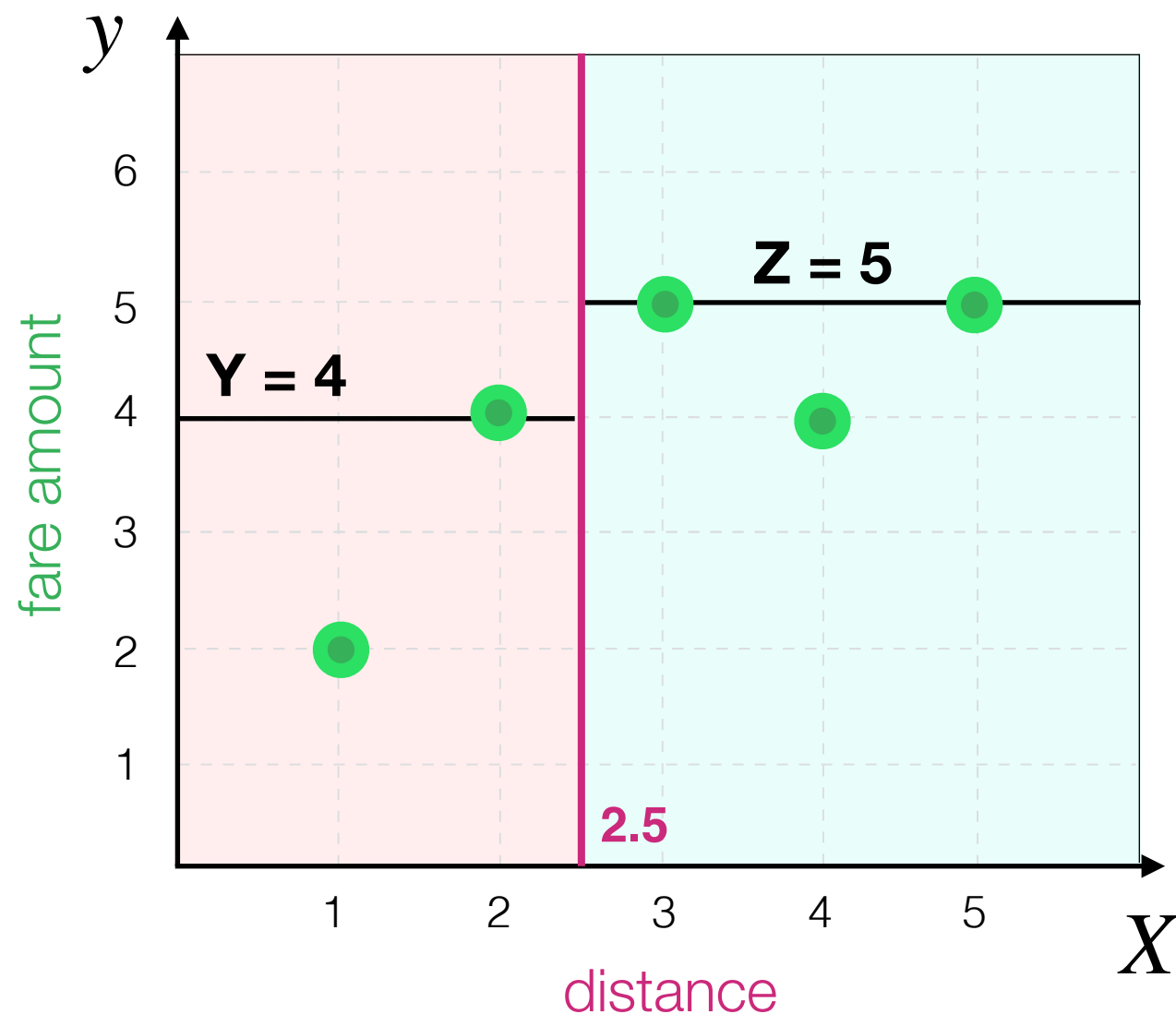
For example if $X = 2.5$:



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

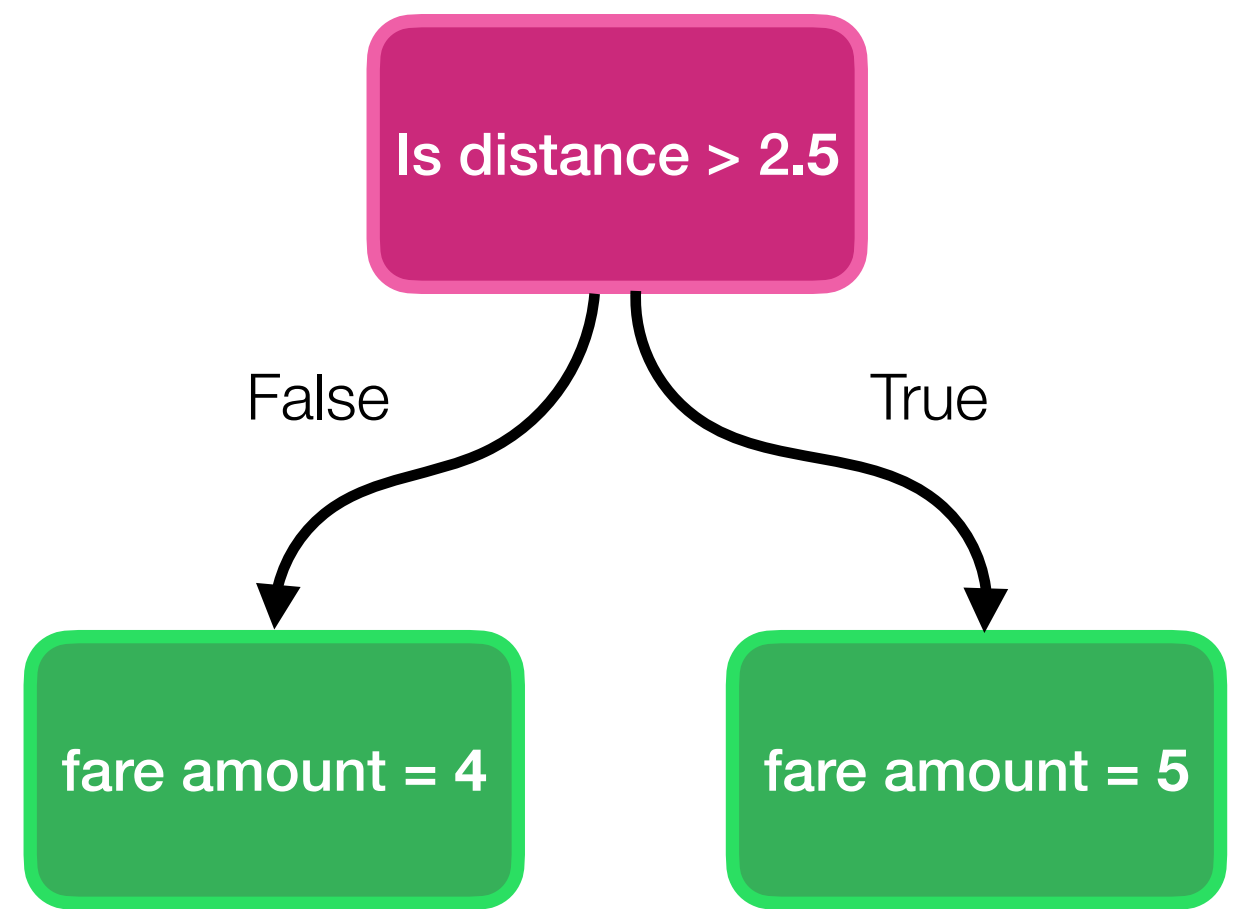
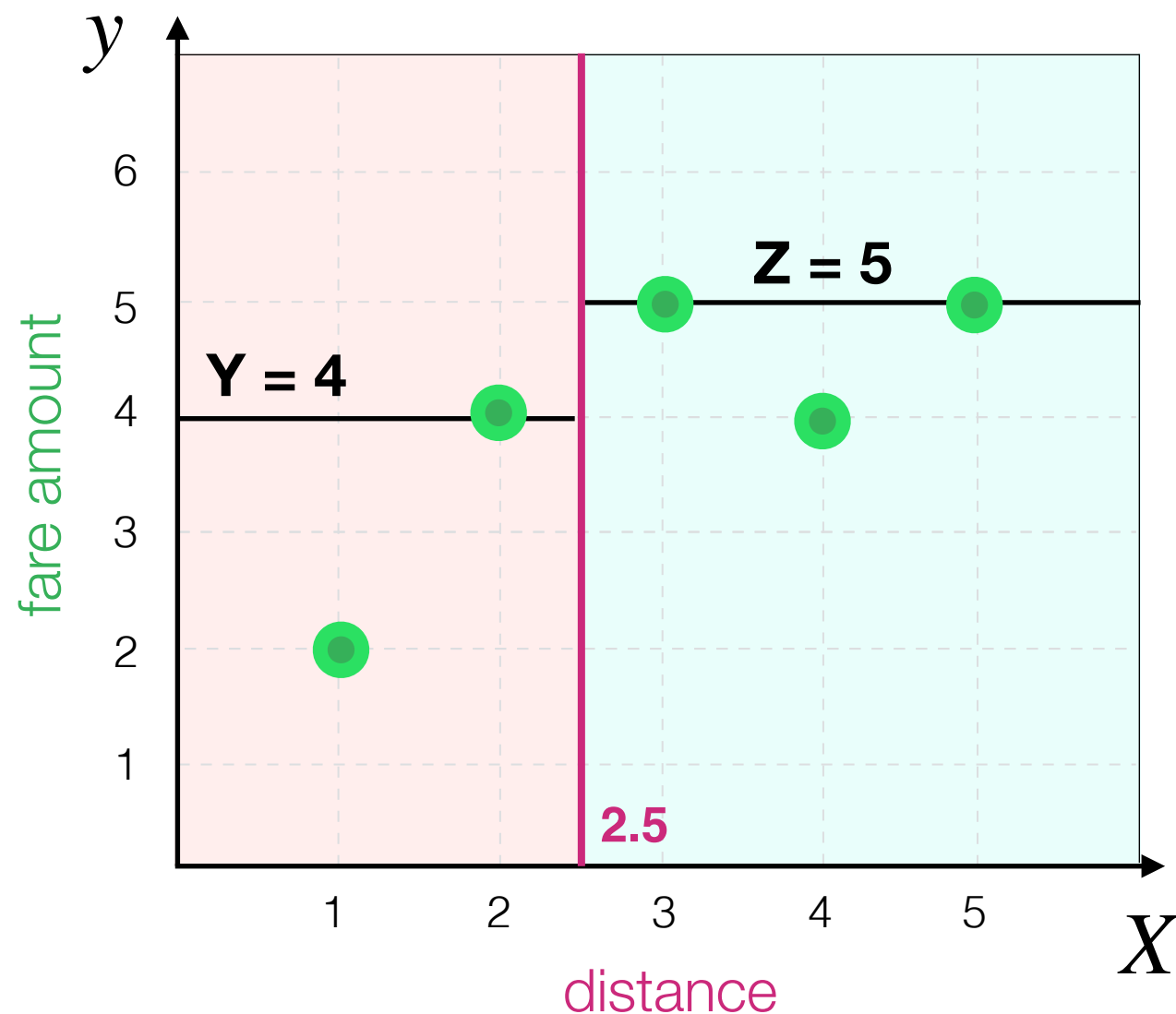
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

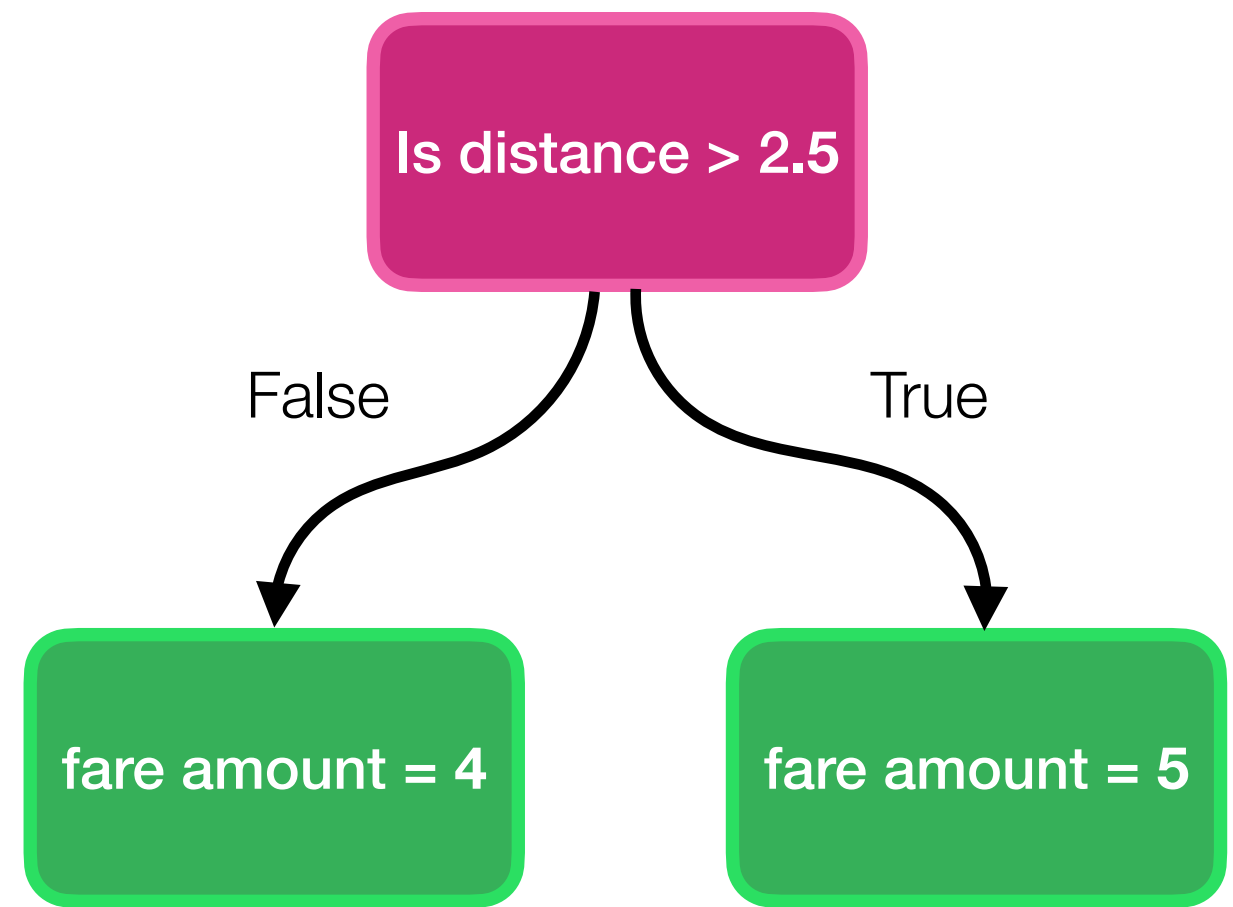
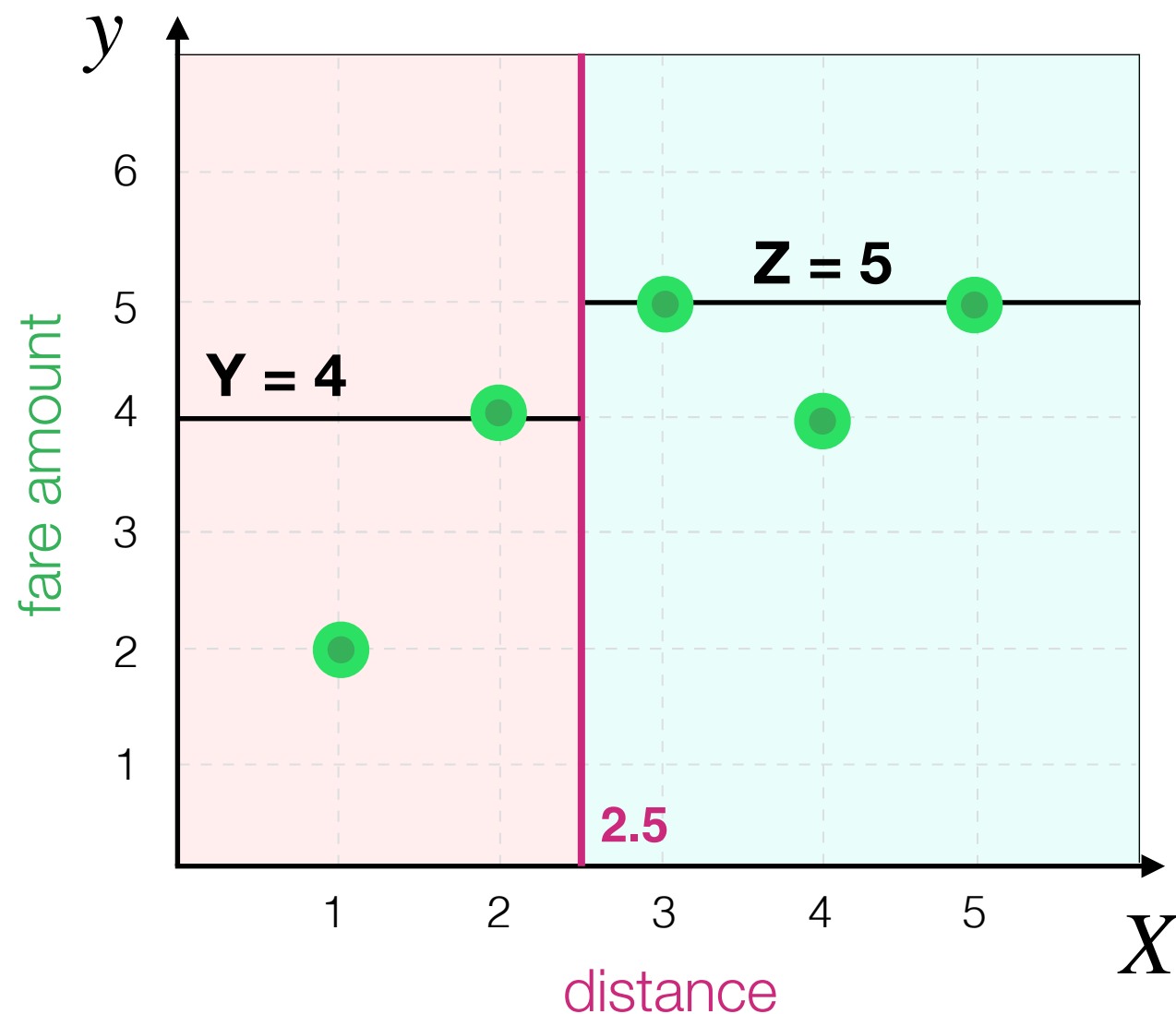
$$MSE = \frac{1}{n} \sum_{i=1}^n (\overset{\text{real value}}{y_i} - \underset{\text{predicted value}}{\hat{y}_i})^2$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

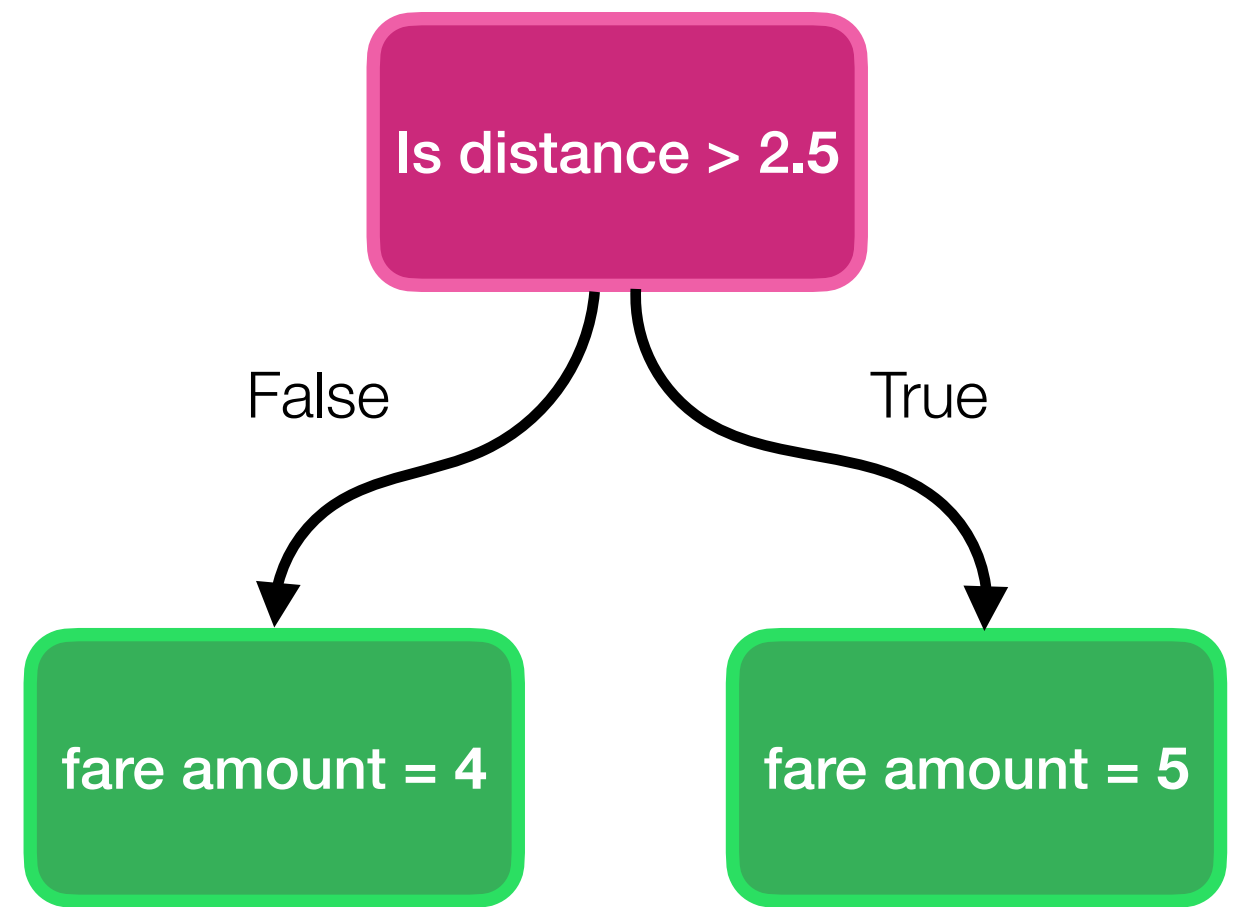
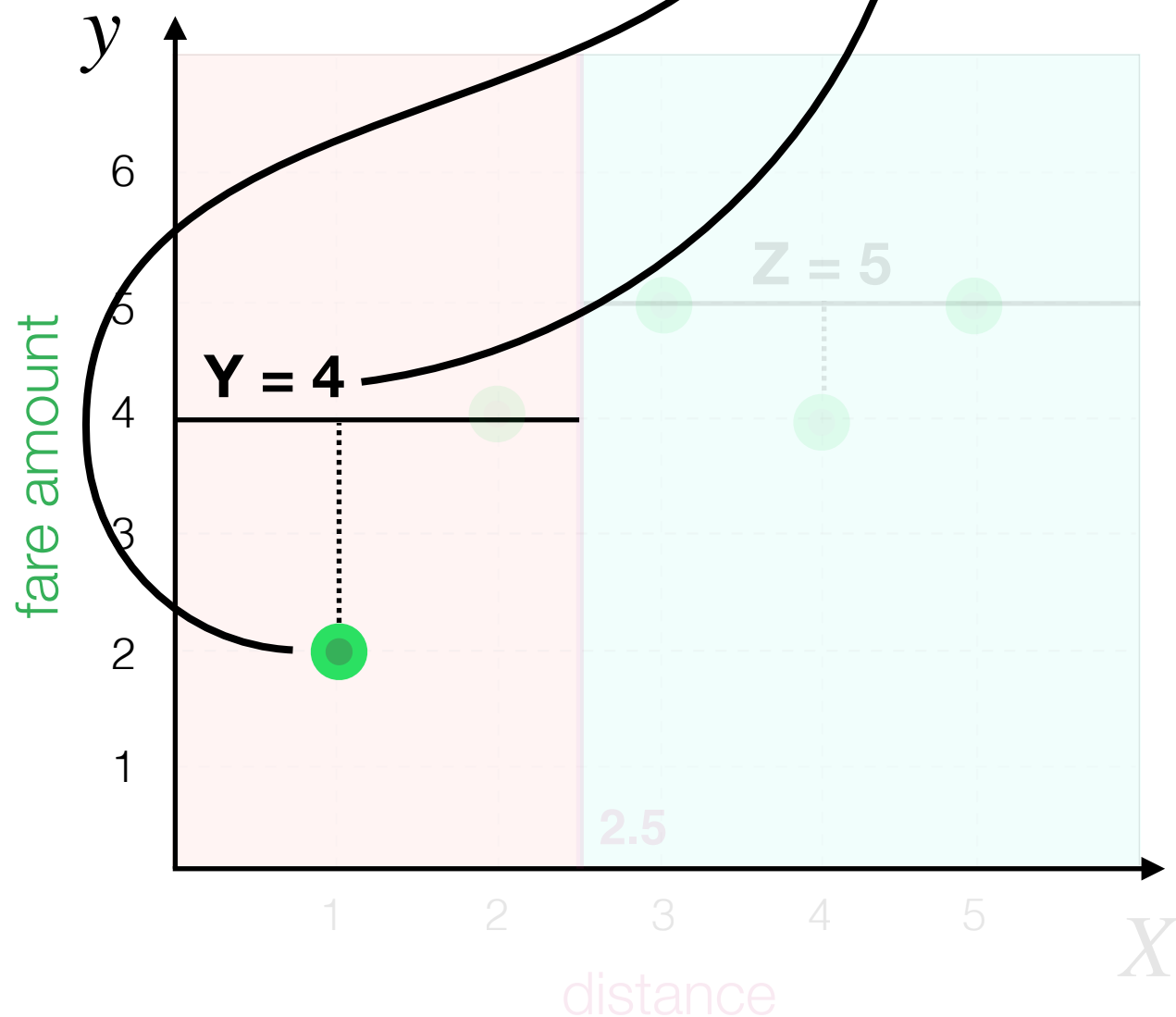
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

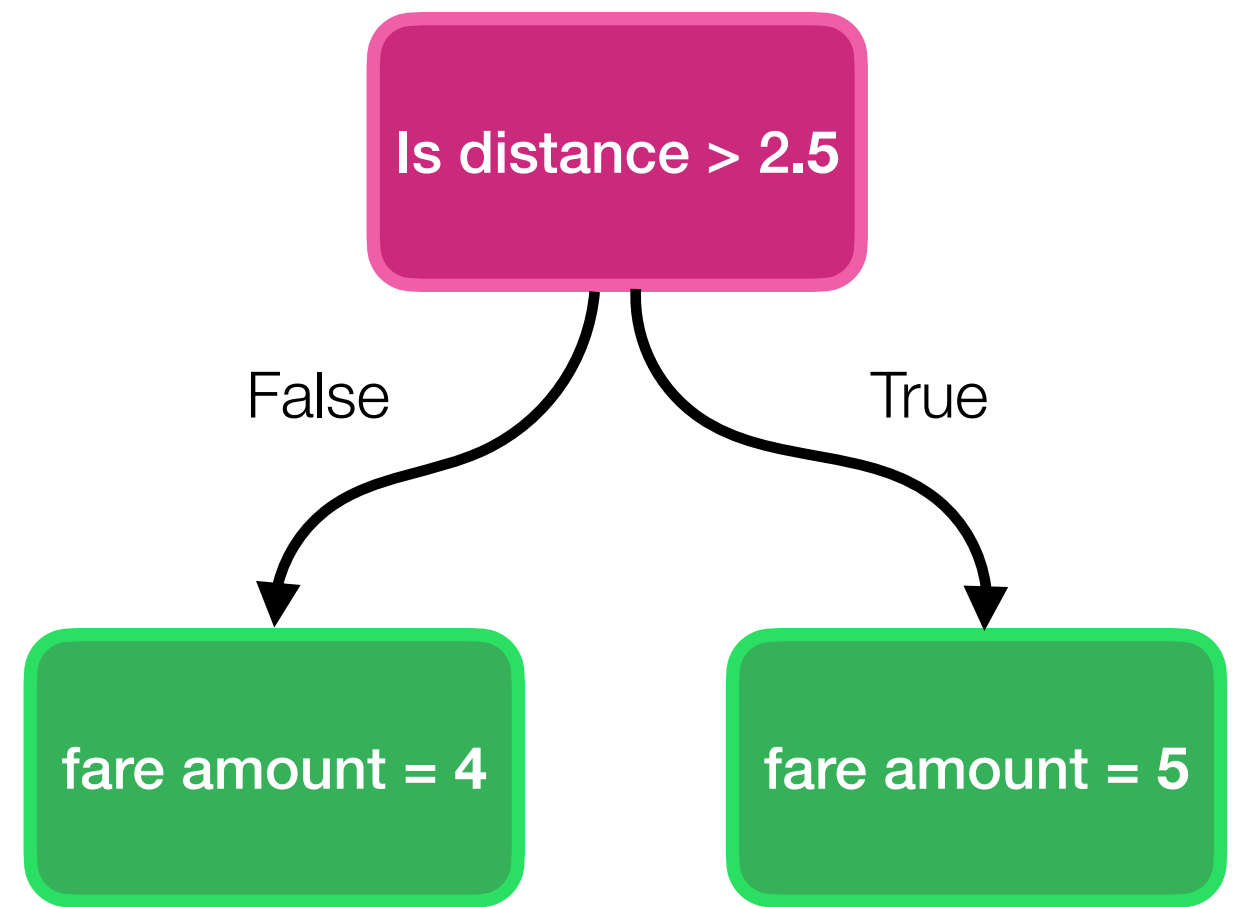
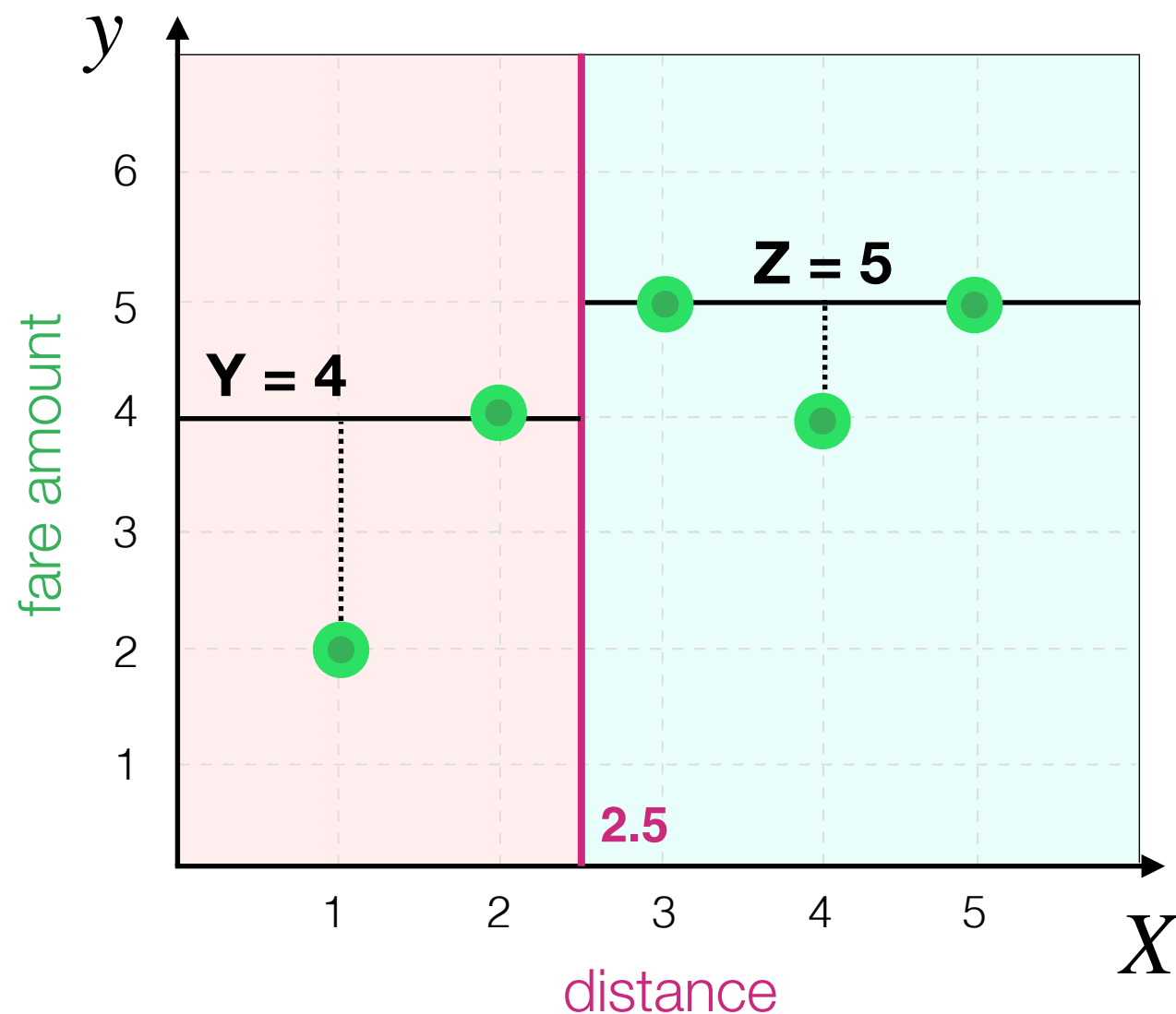
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 4)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

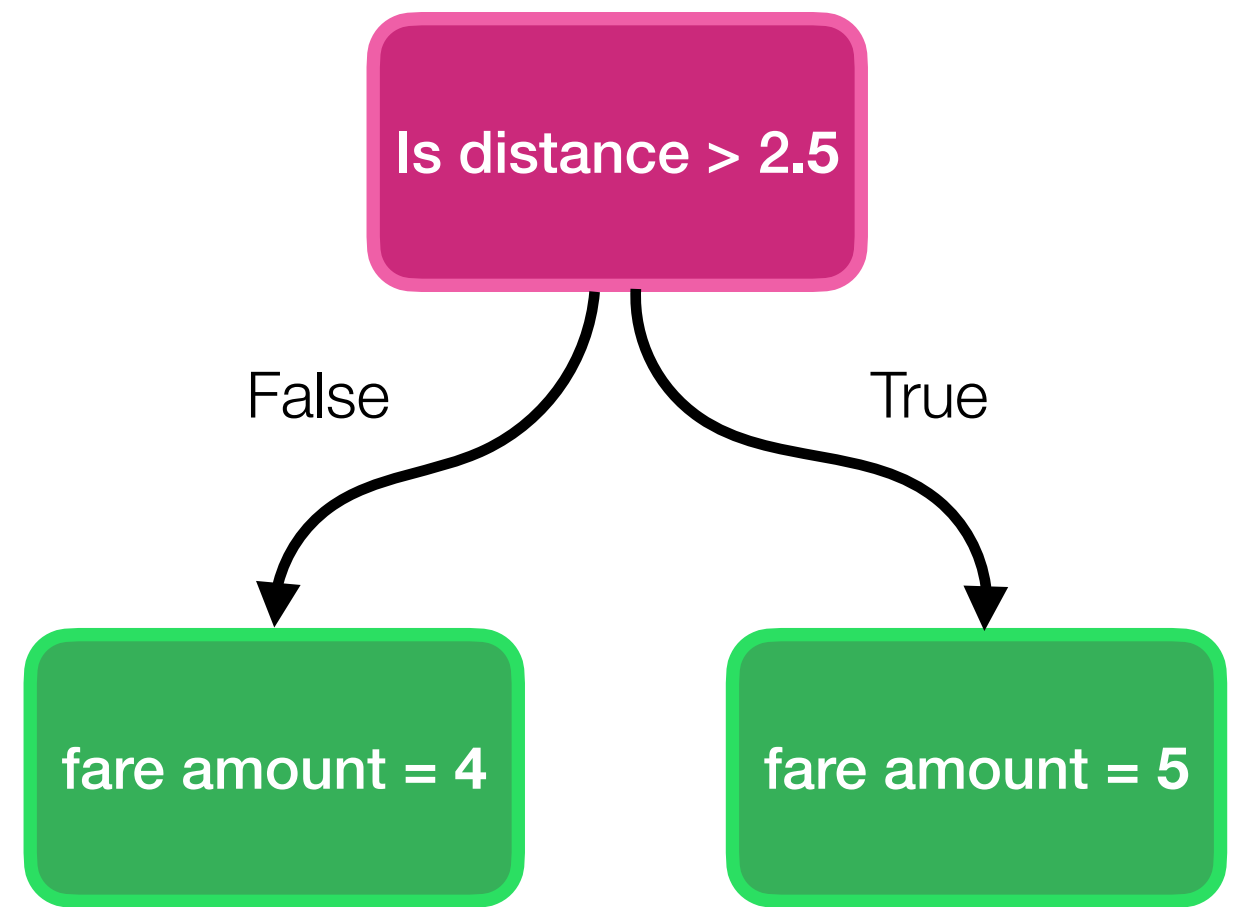
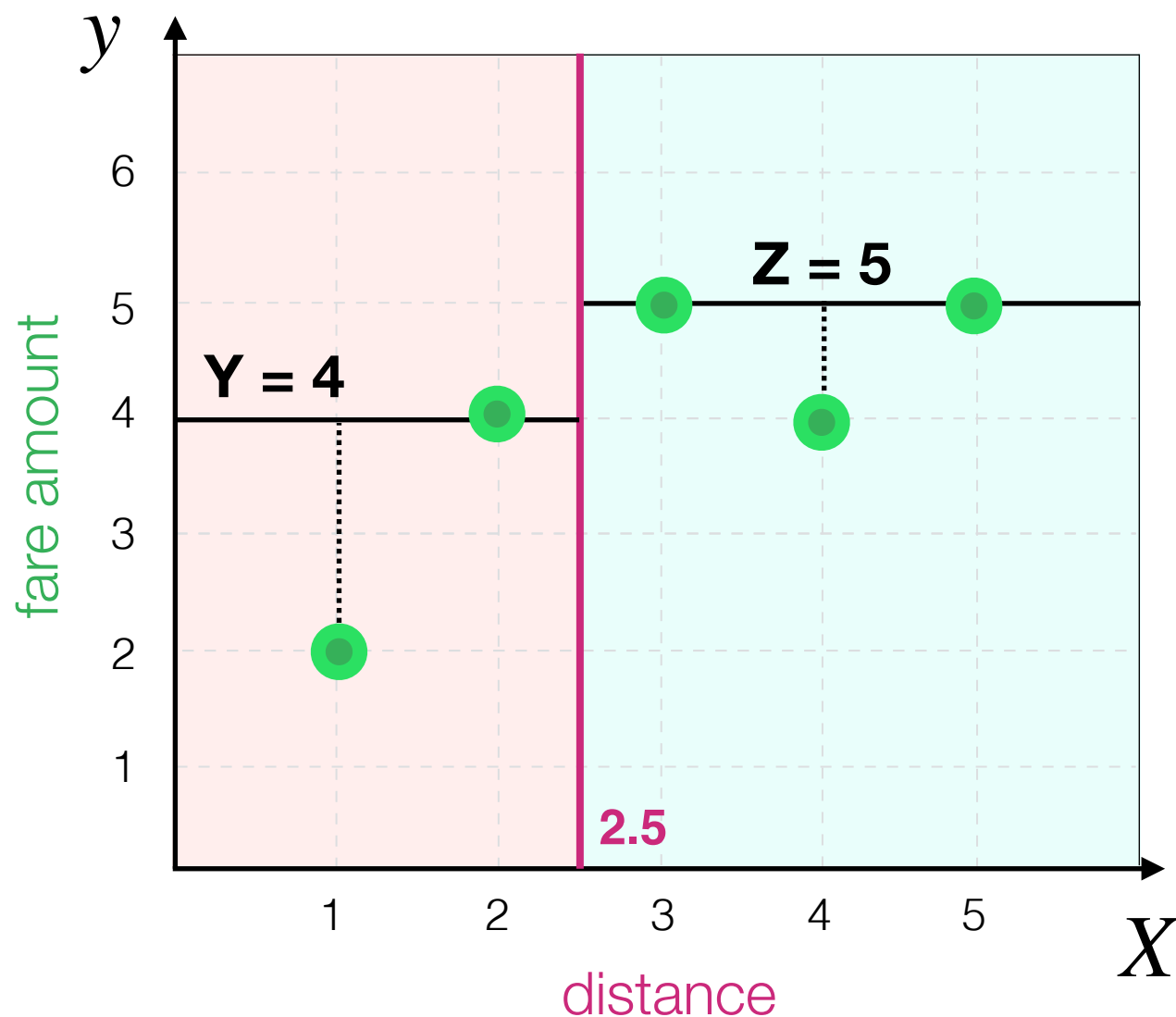
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

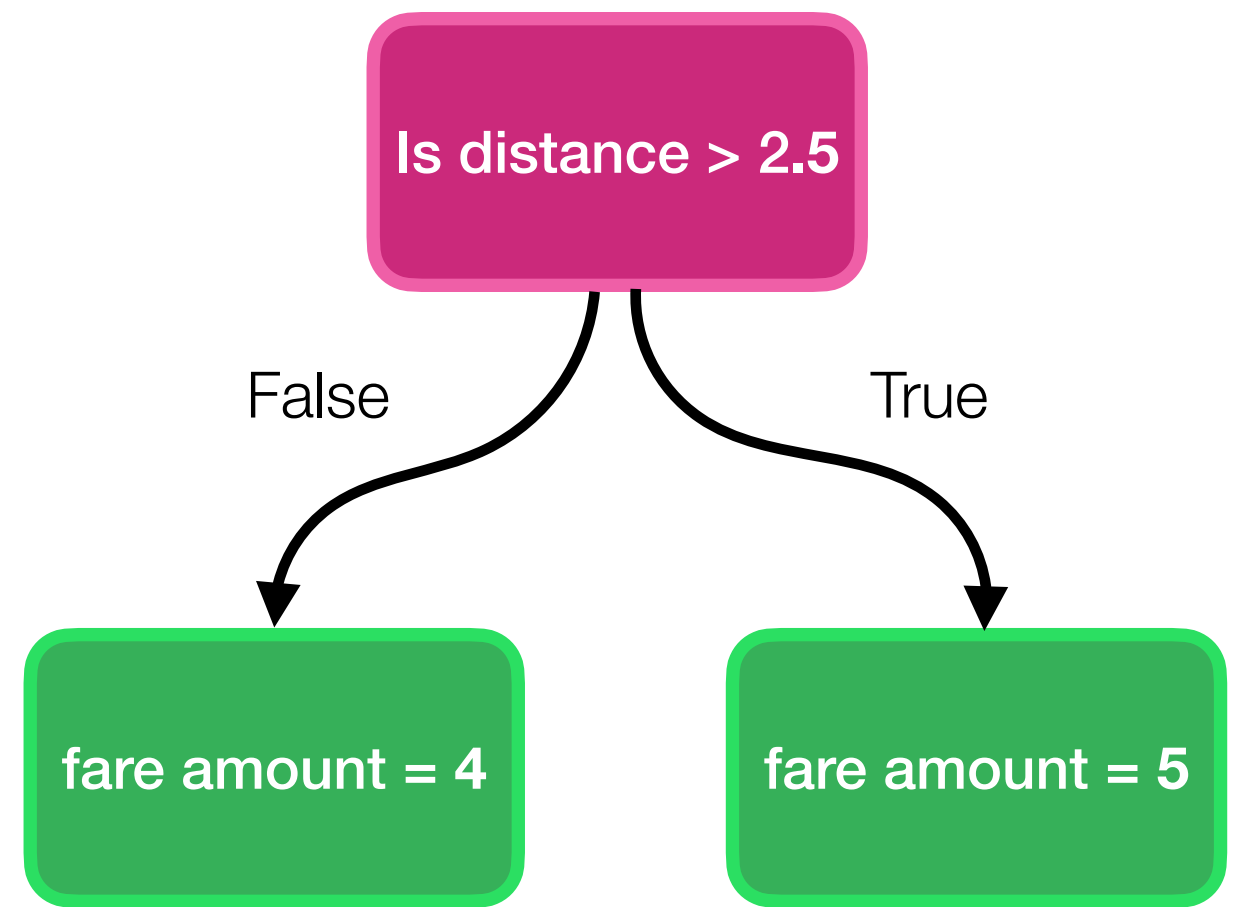
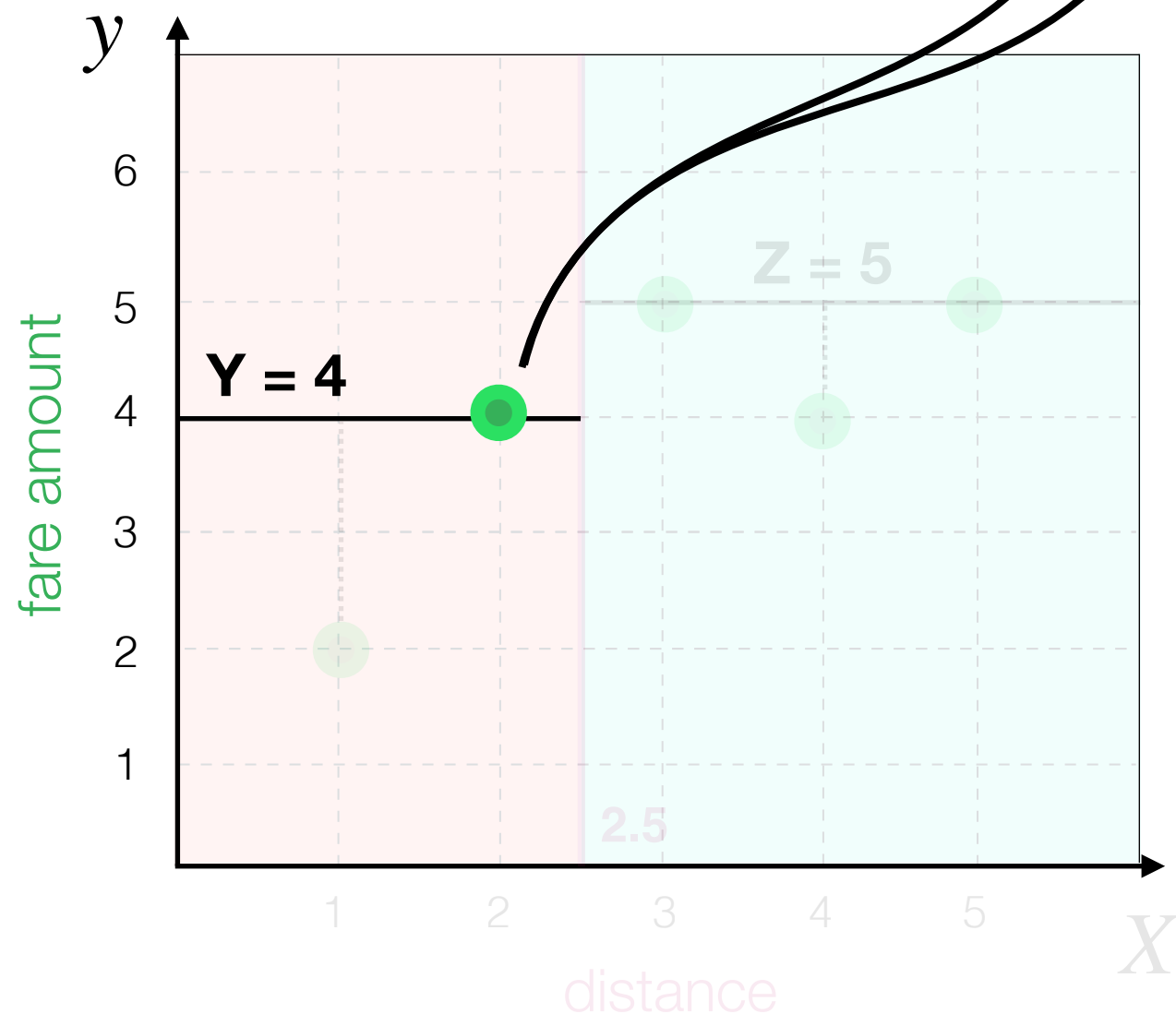
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 4)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

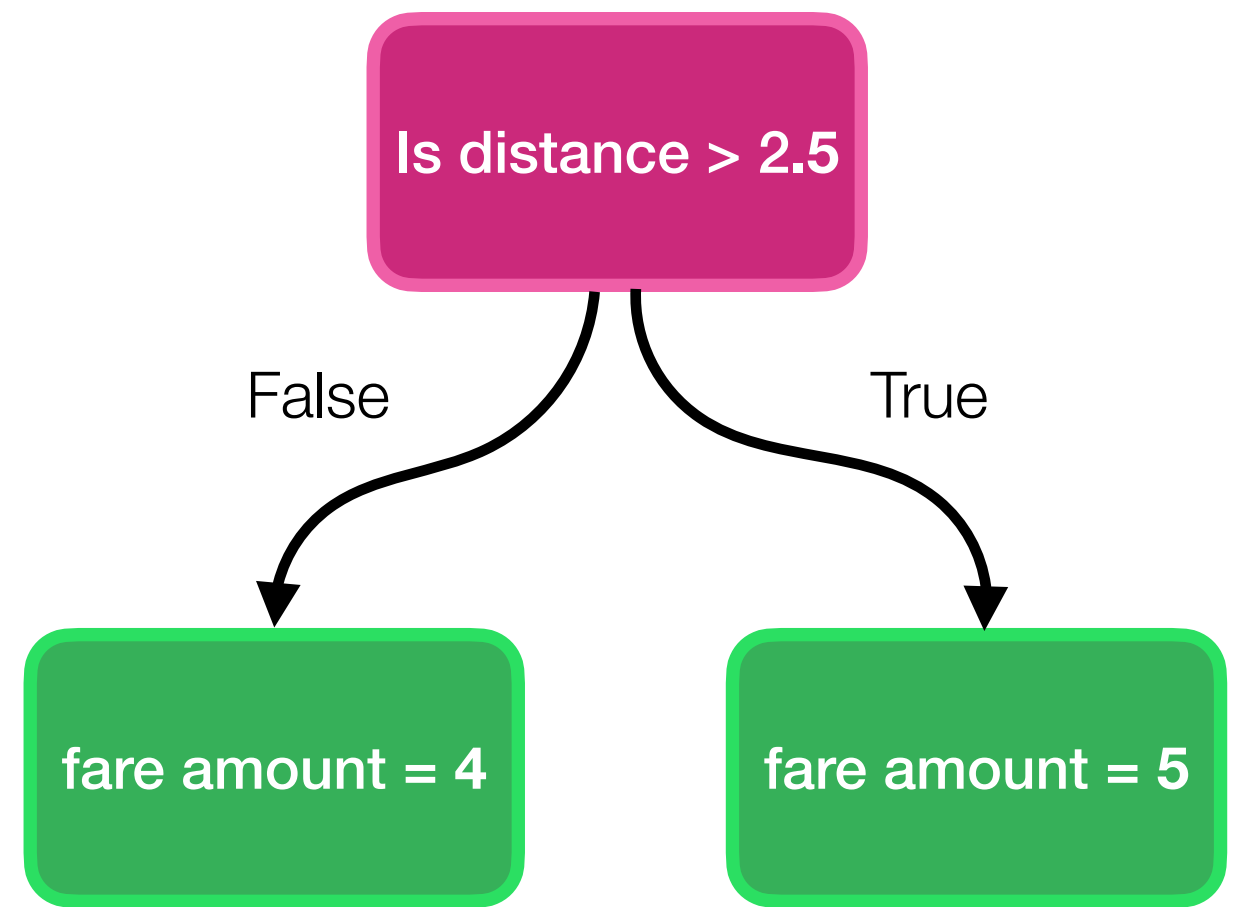
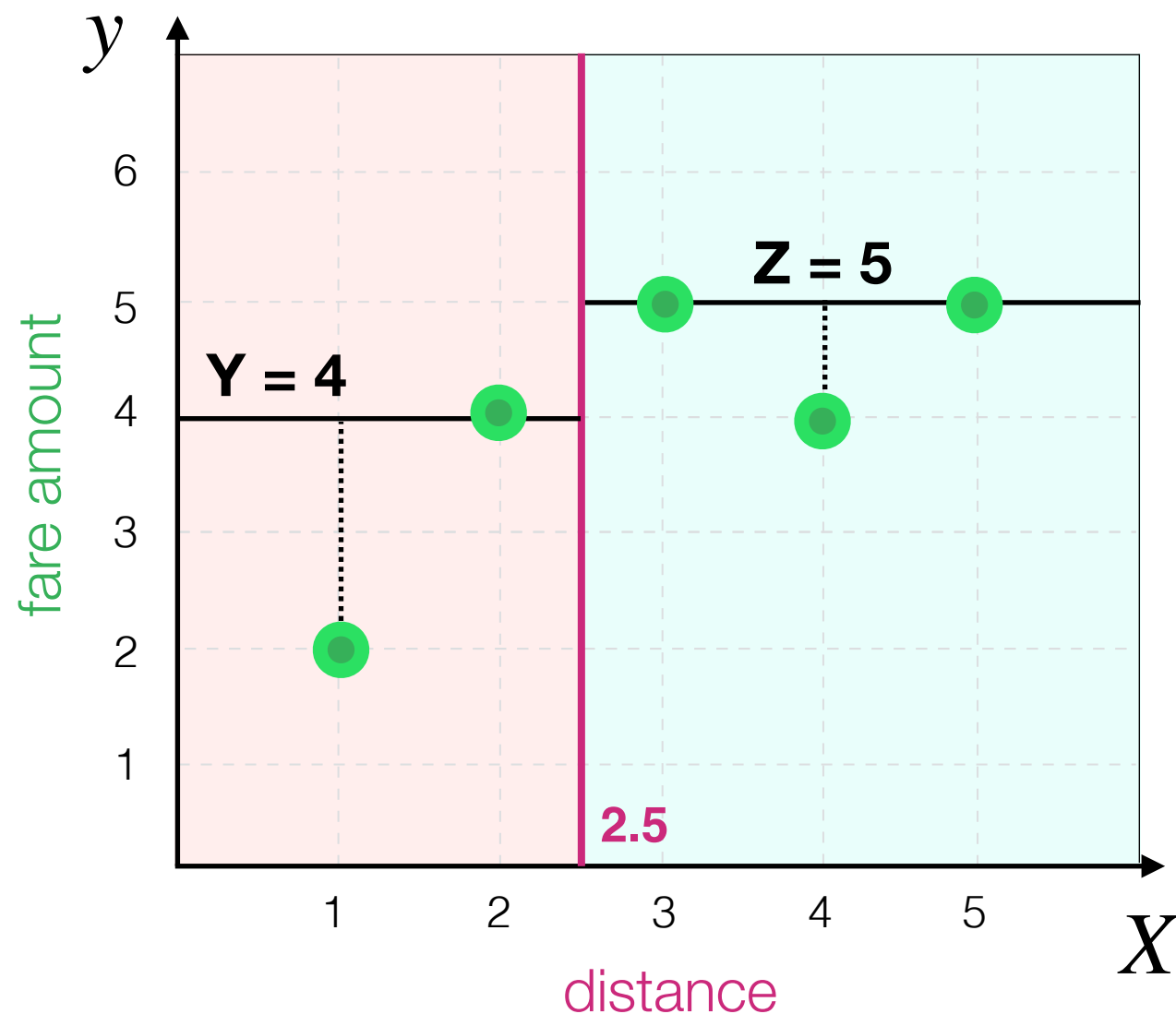
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 4)^2 + (4 - 4)^2 + (y_3 - \hat{y}_3)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

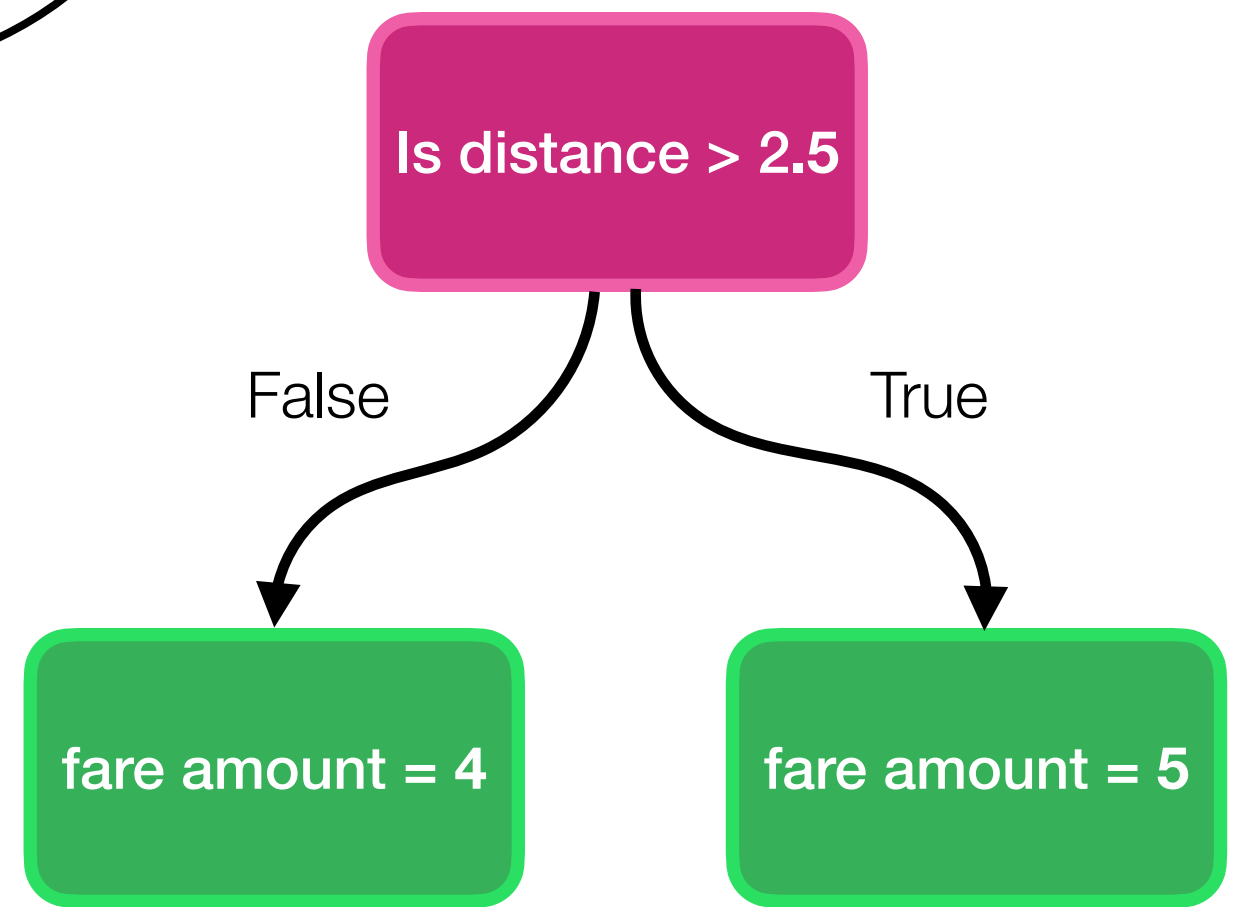
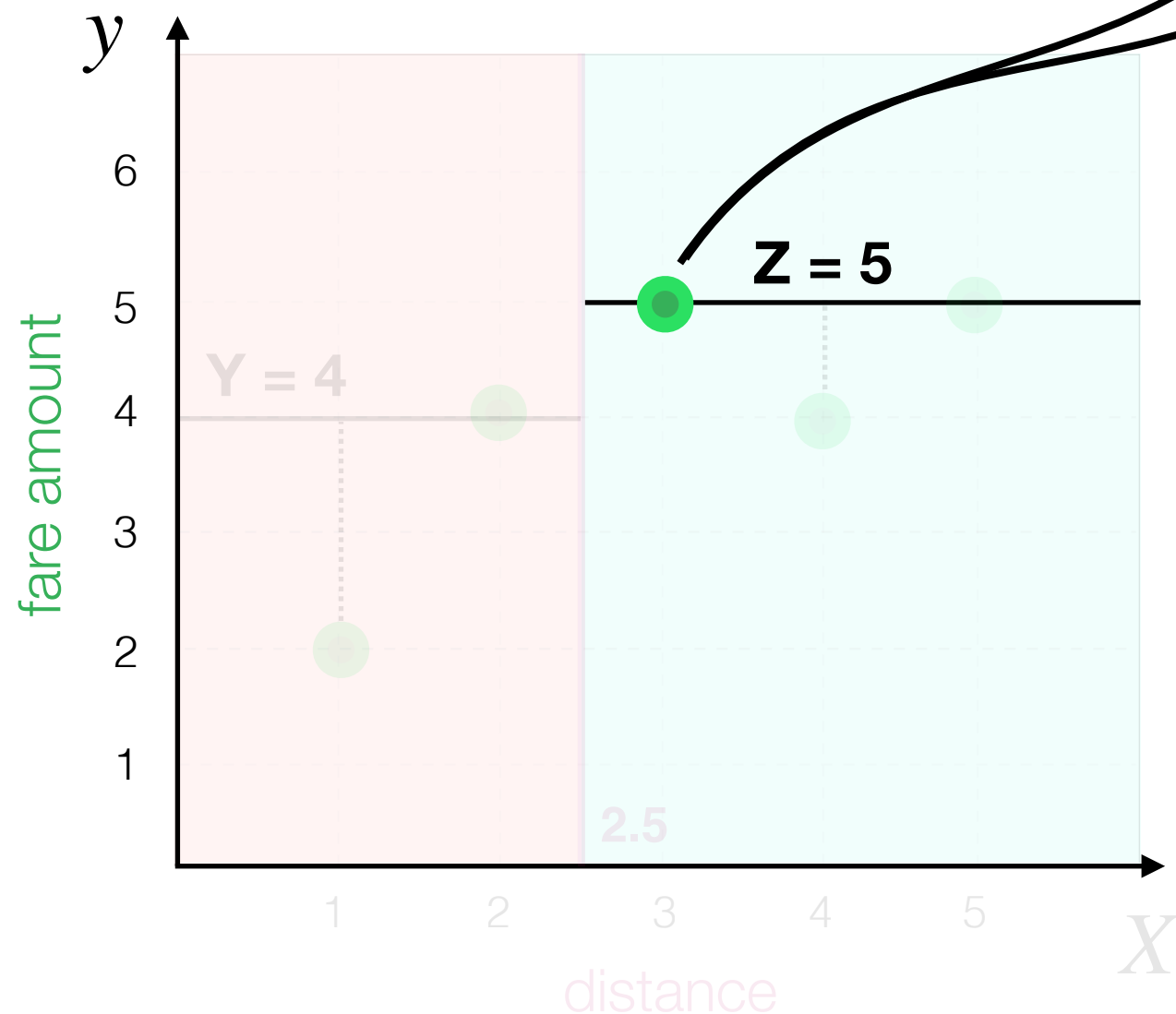
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 4)^2 + (4 - 4)^2 + (y_3 - \hat{y}_3)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

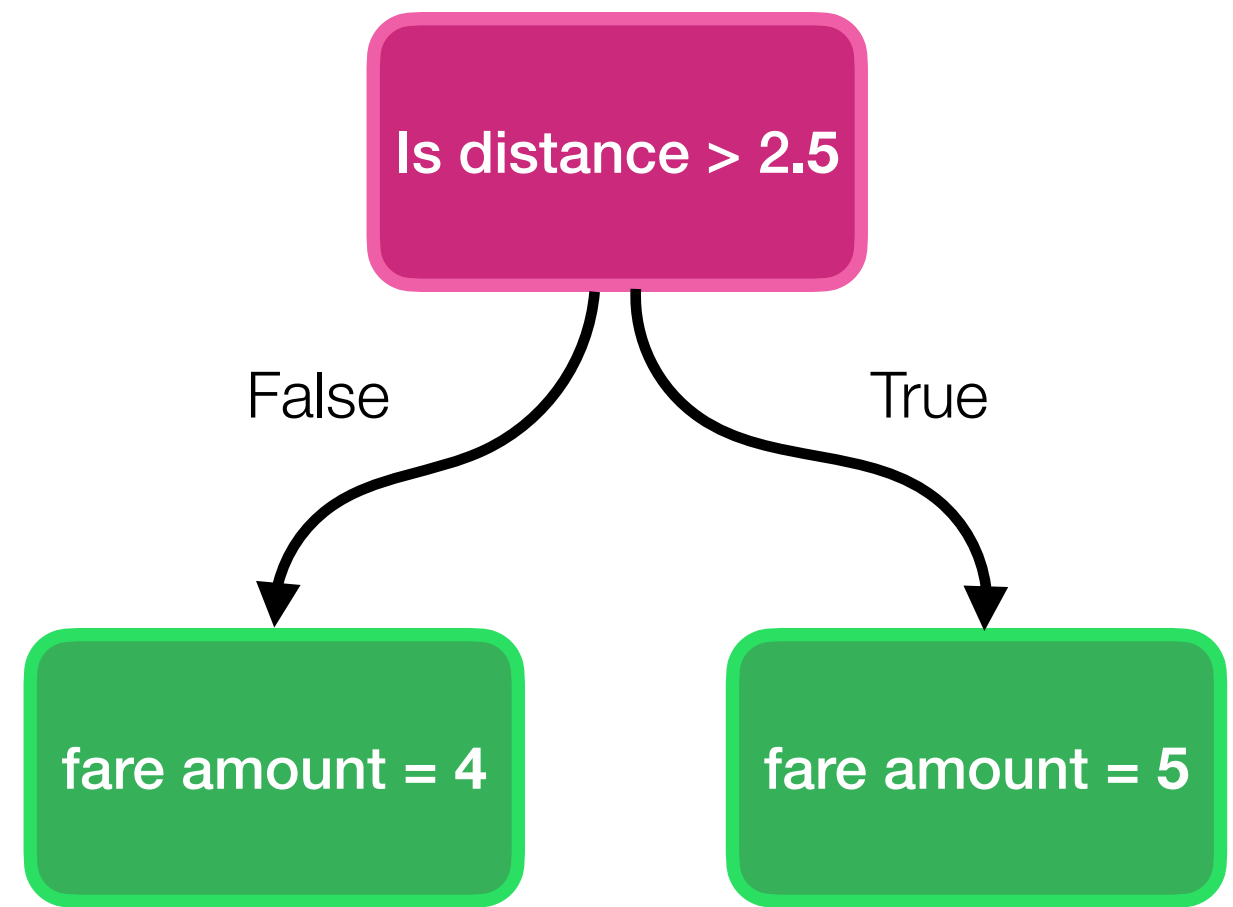
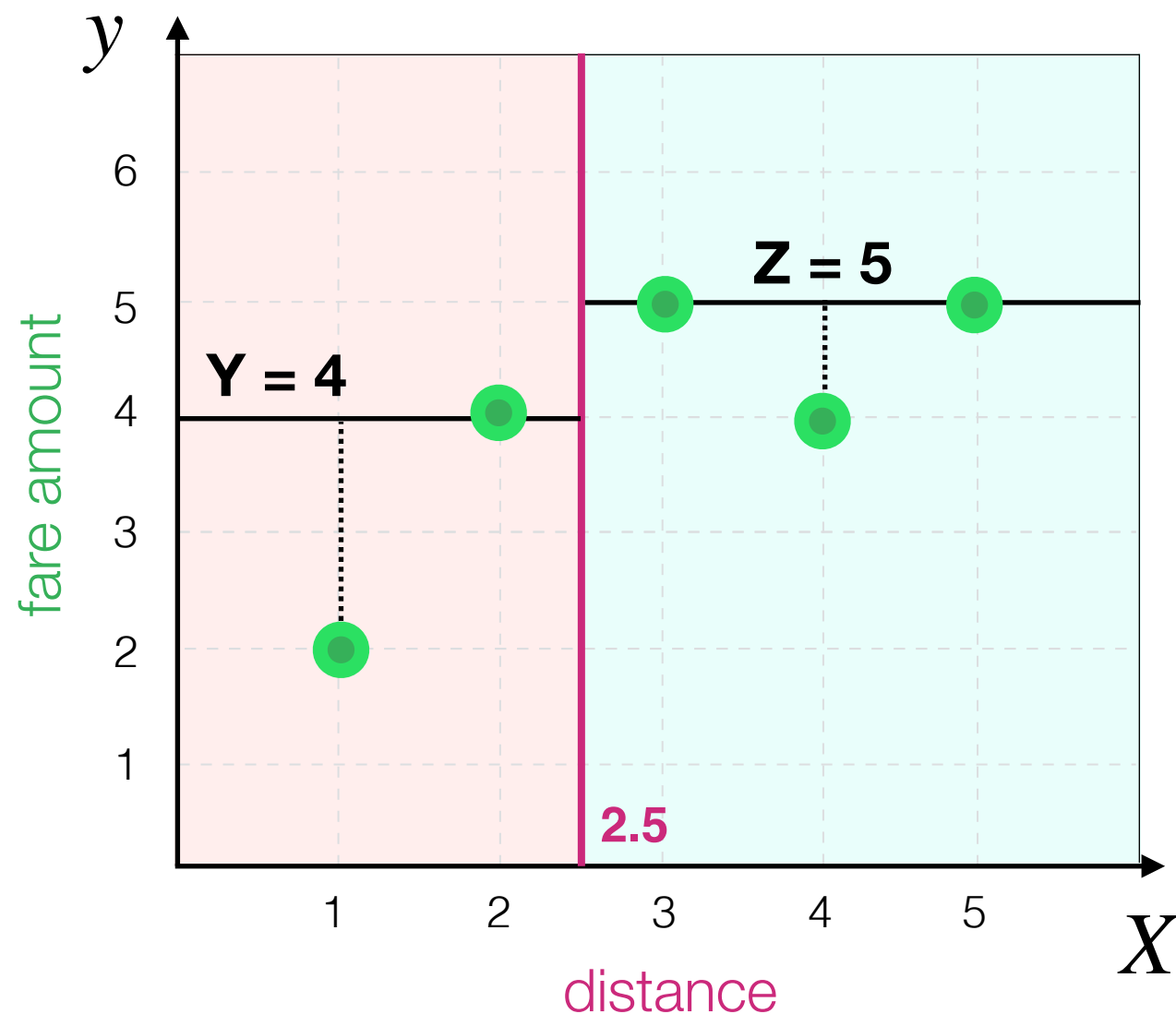
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 4)^2 + (4 - 4)^2 + (5 - 5)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

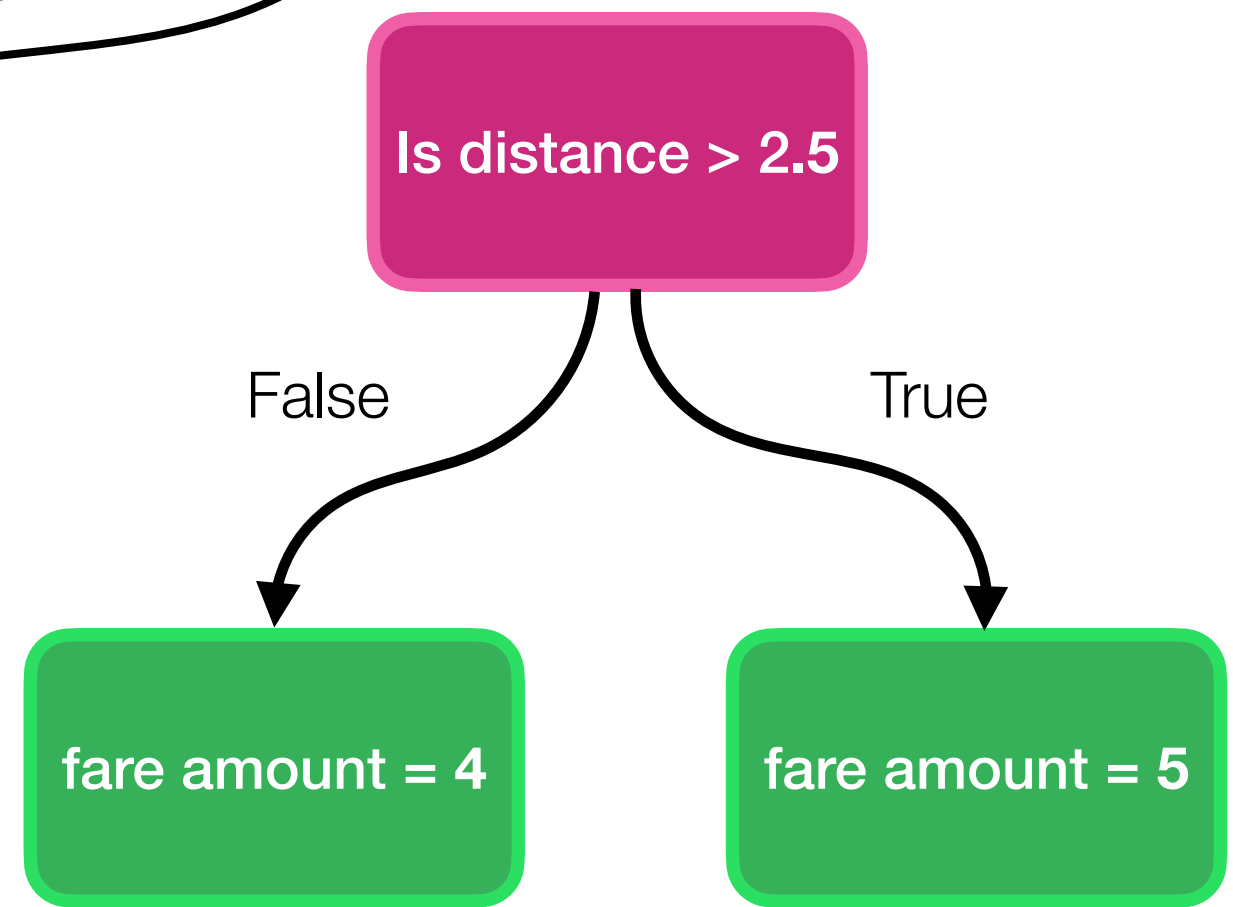
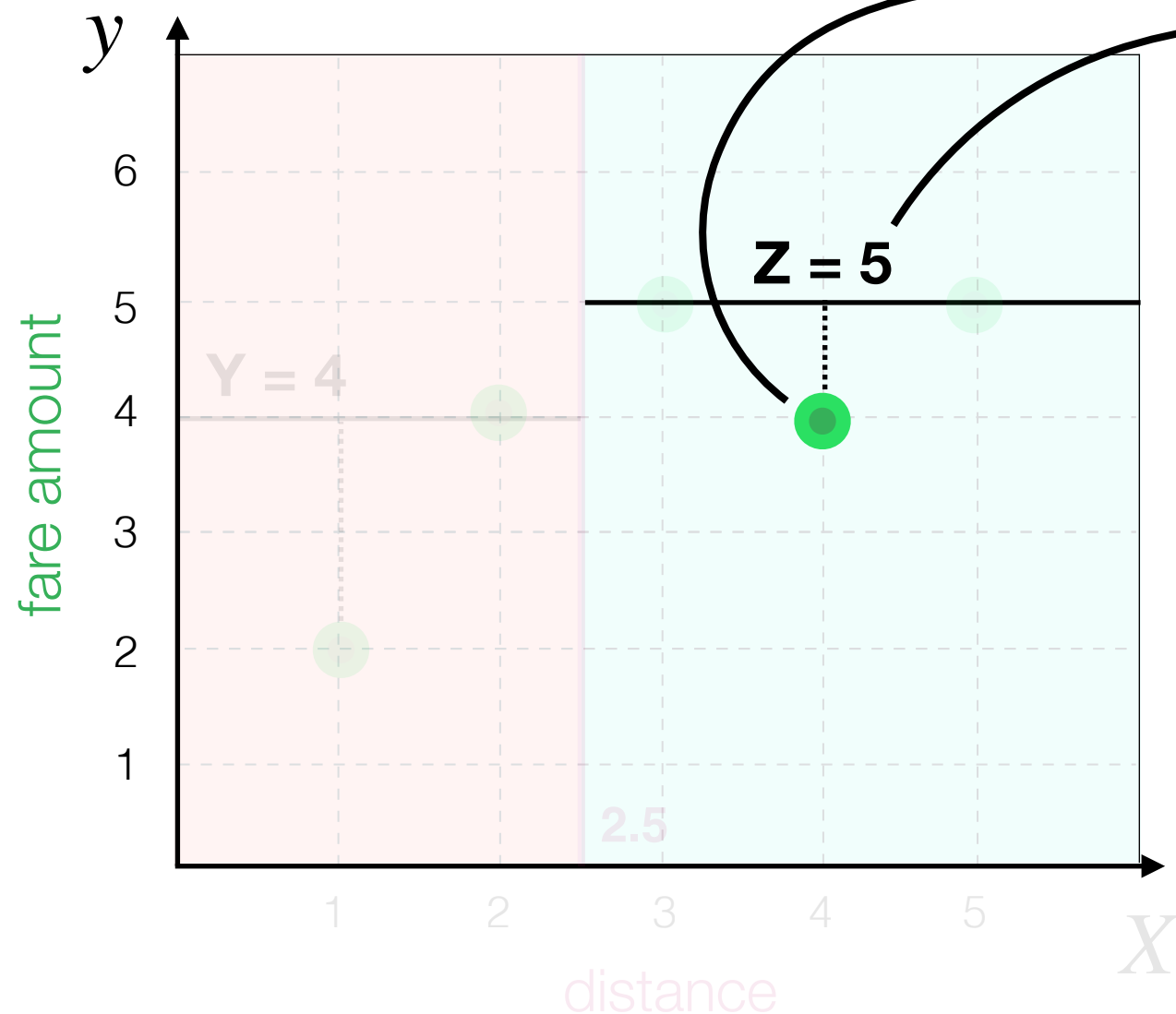
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 4)^2 + (4 - 4)^2 + (5 - 5)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

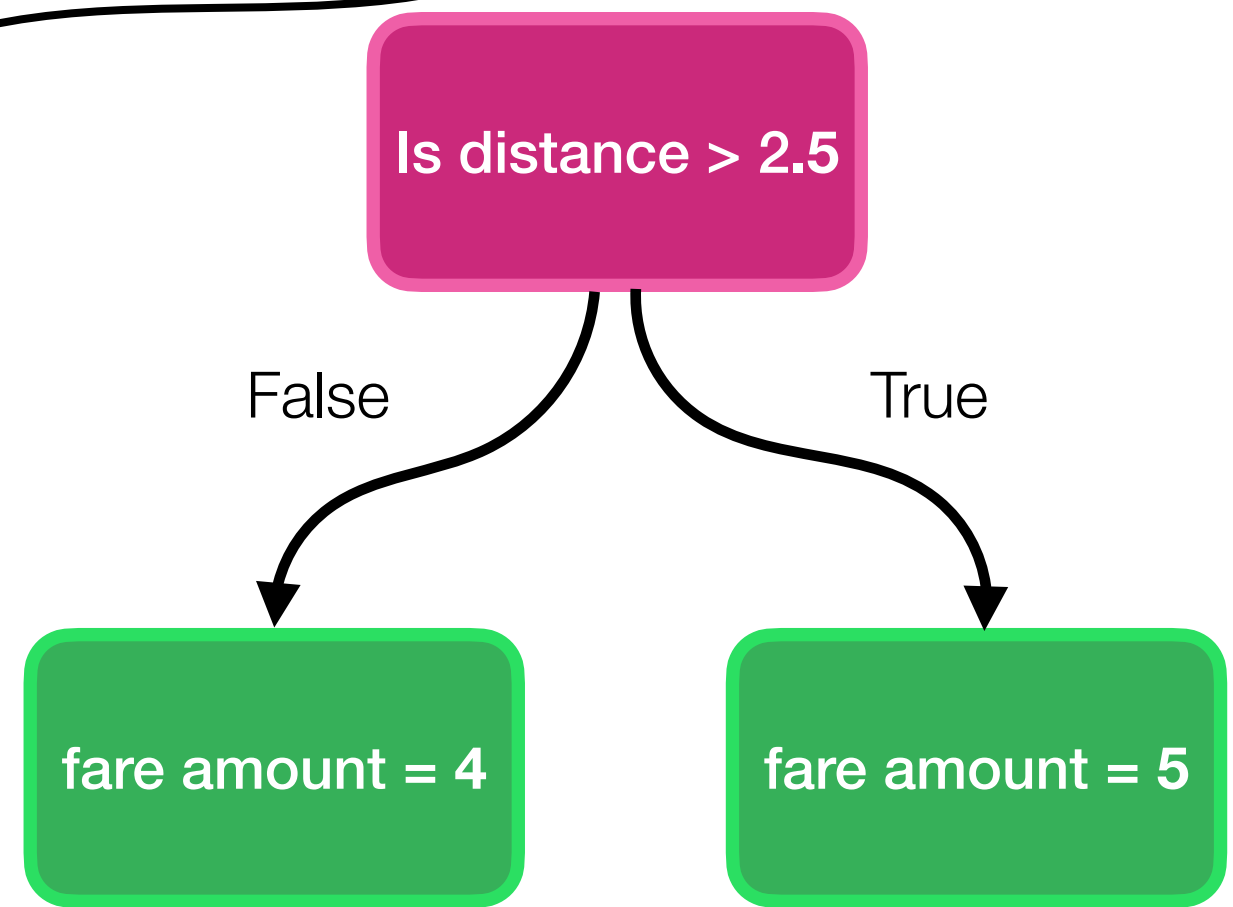
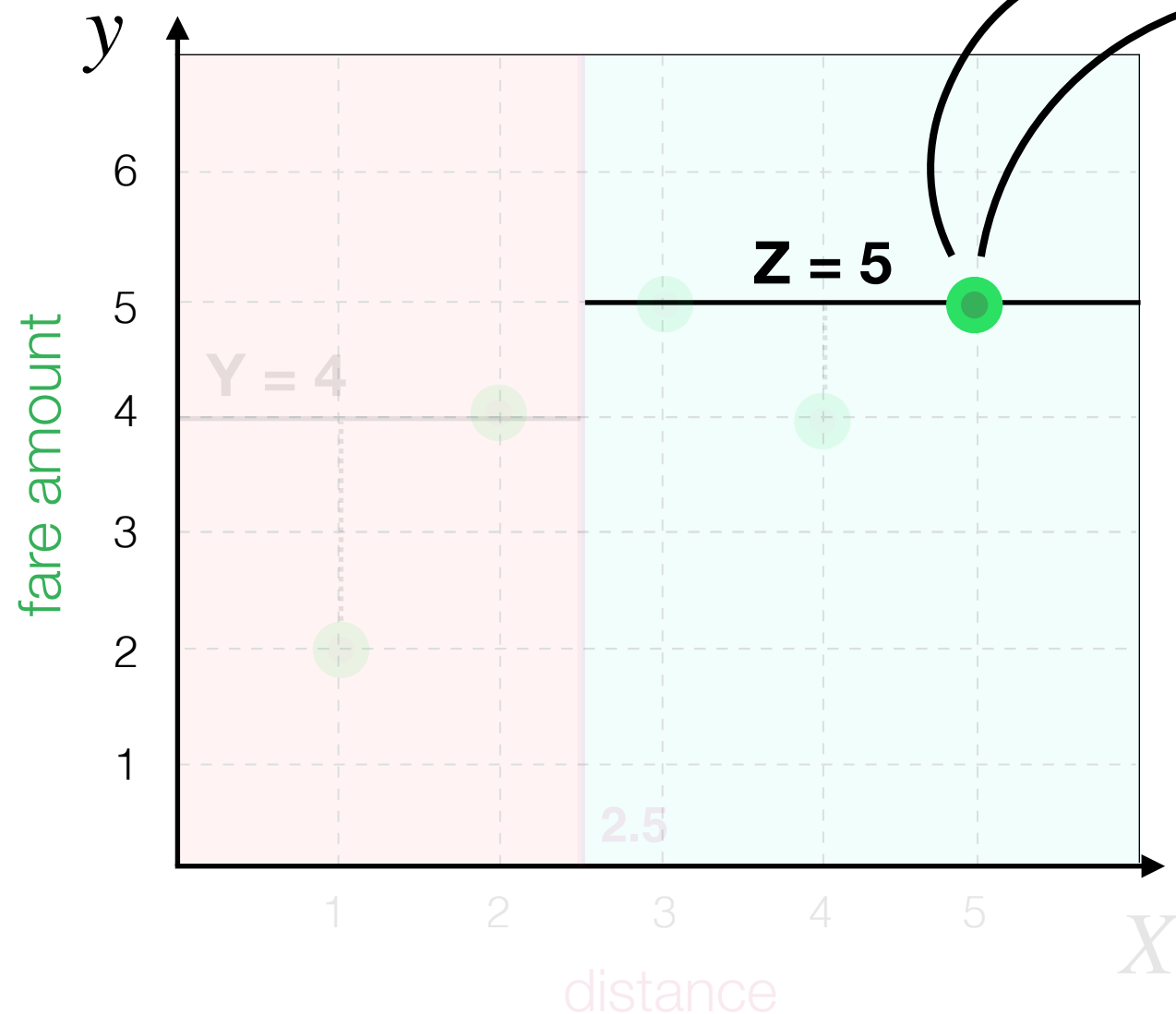
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 4)^2 + (4 - 4)^2 + (5 - 5)^2 + (4 - 5)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

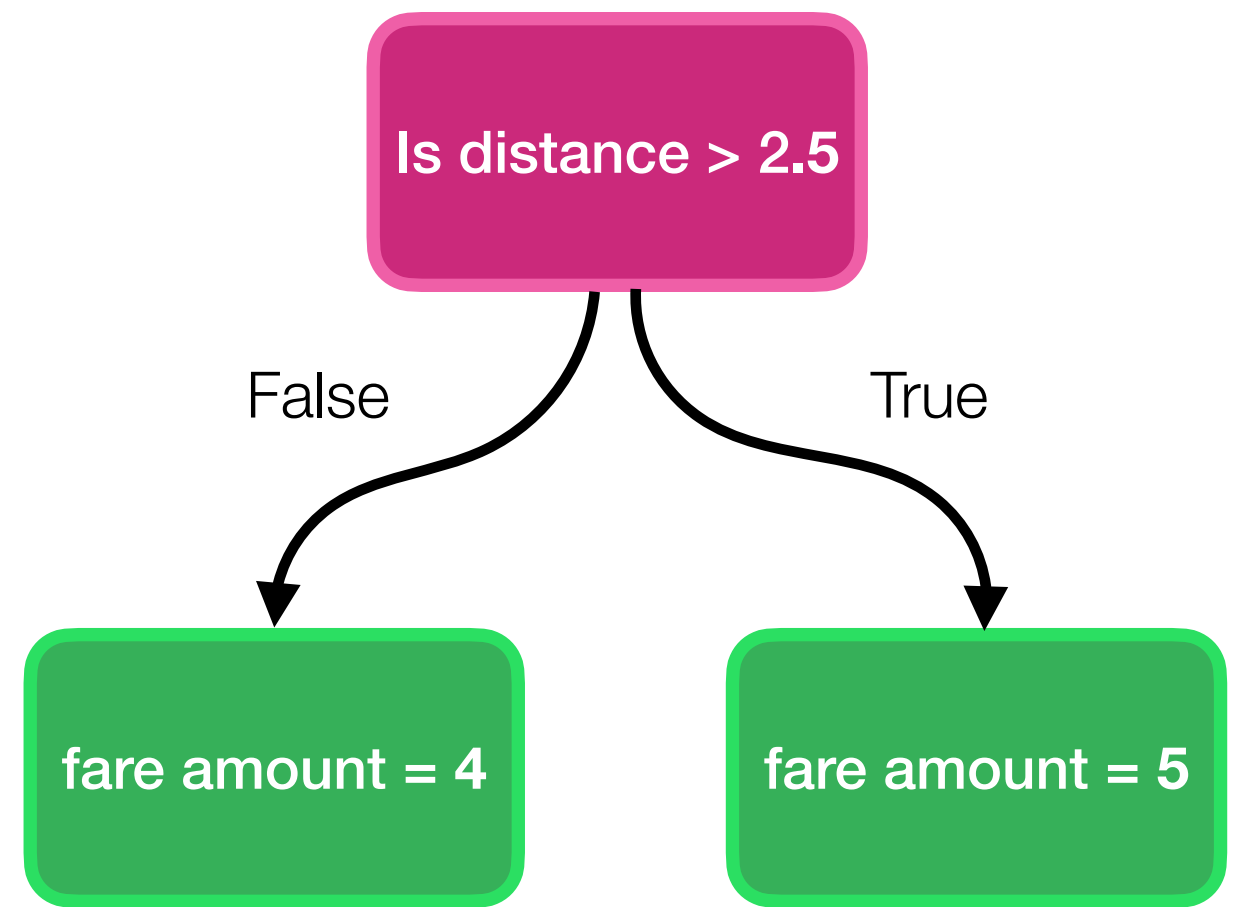
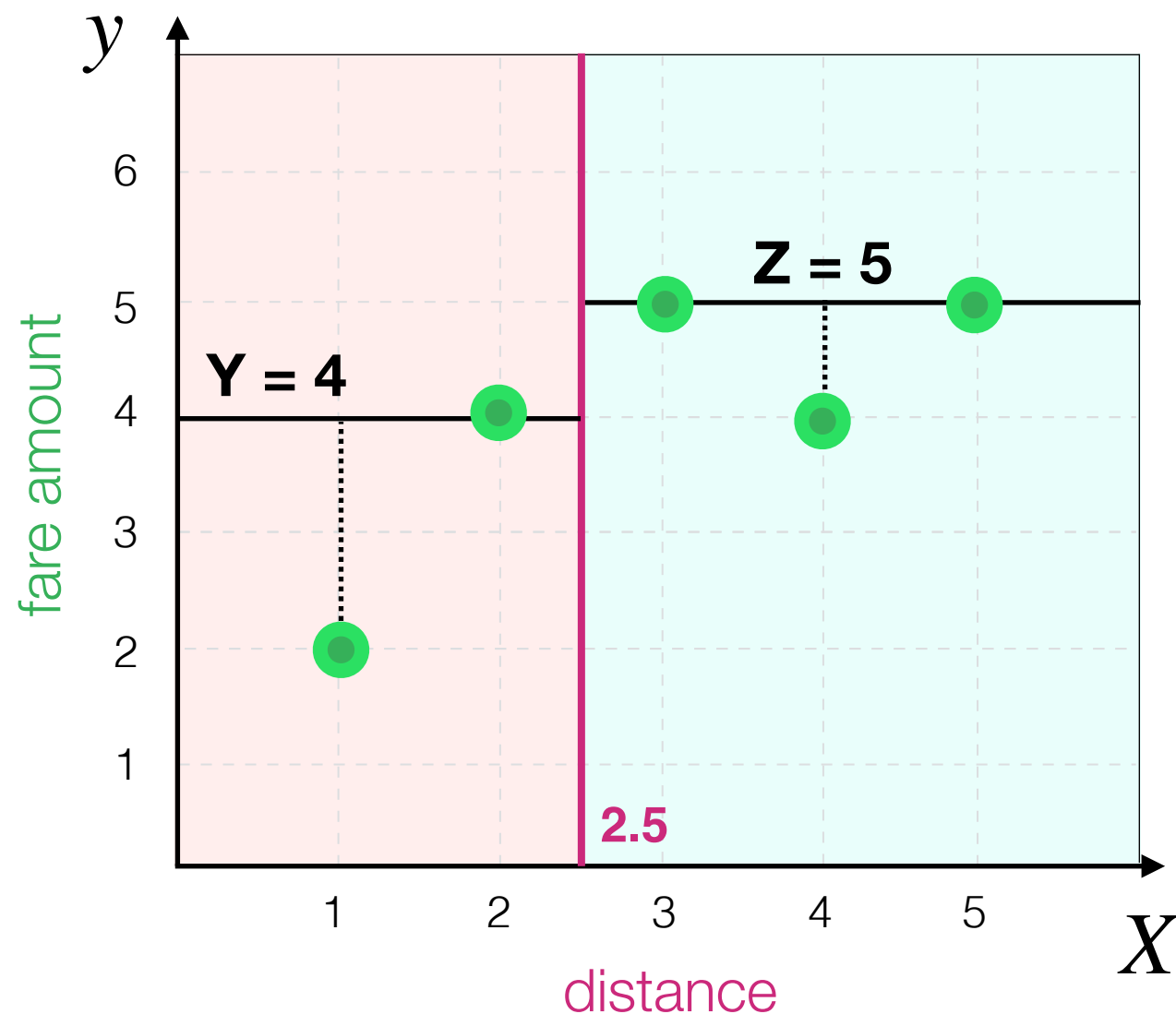
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 4)^2 + (4 - 4)^2 + (5 - 5)^2 + (4 - 5)^2 + (5 - 5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

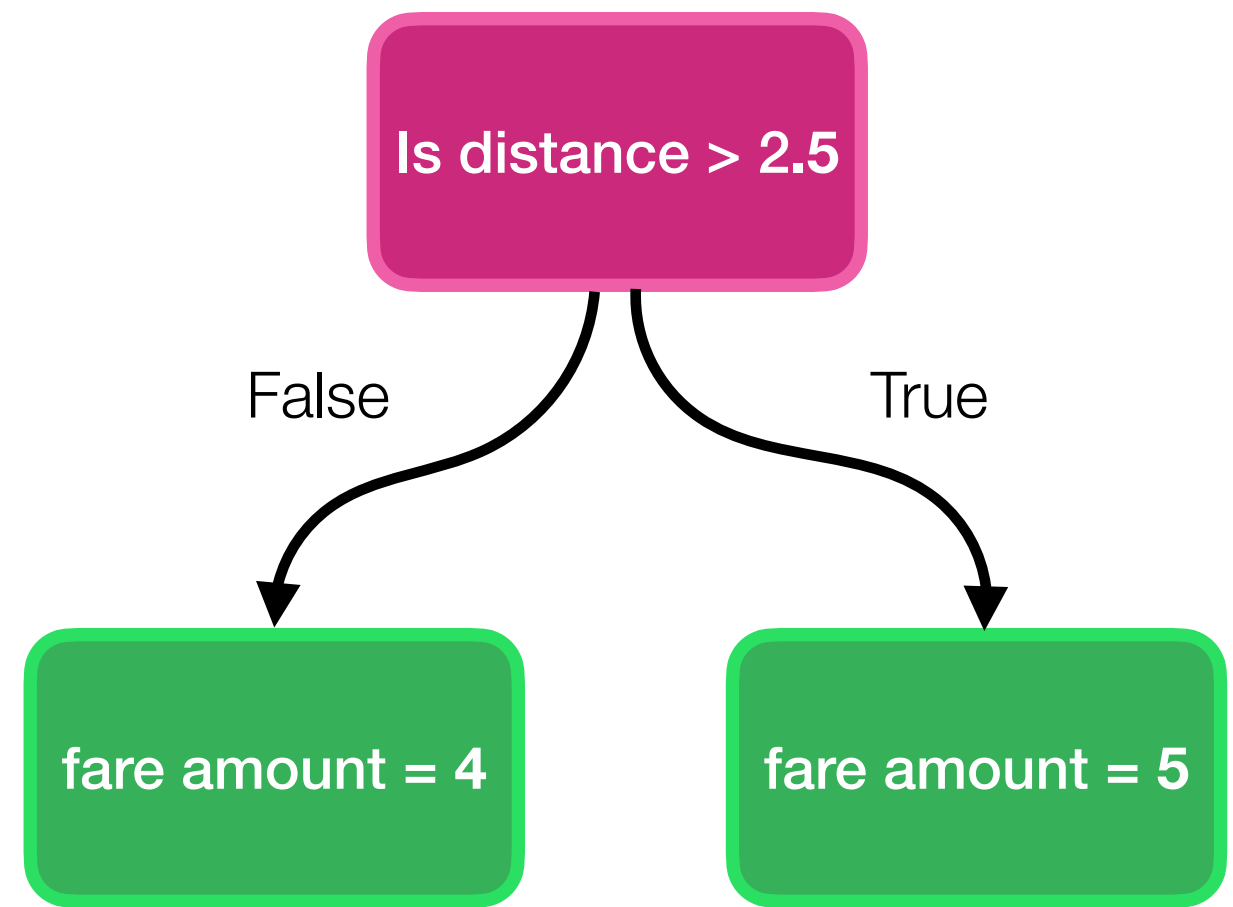
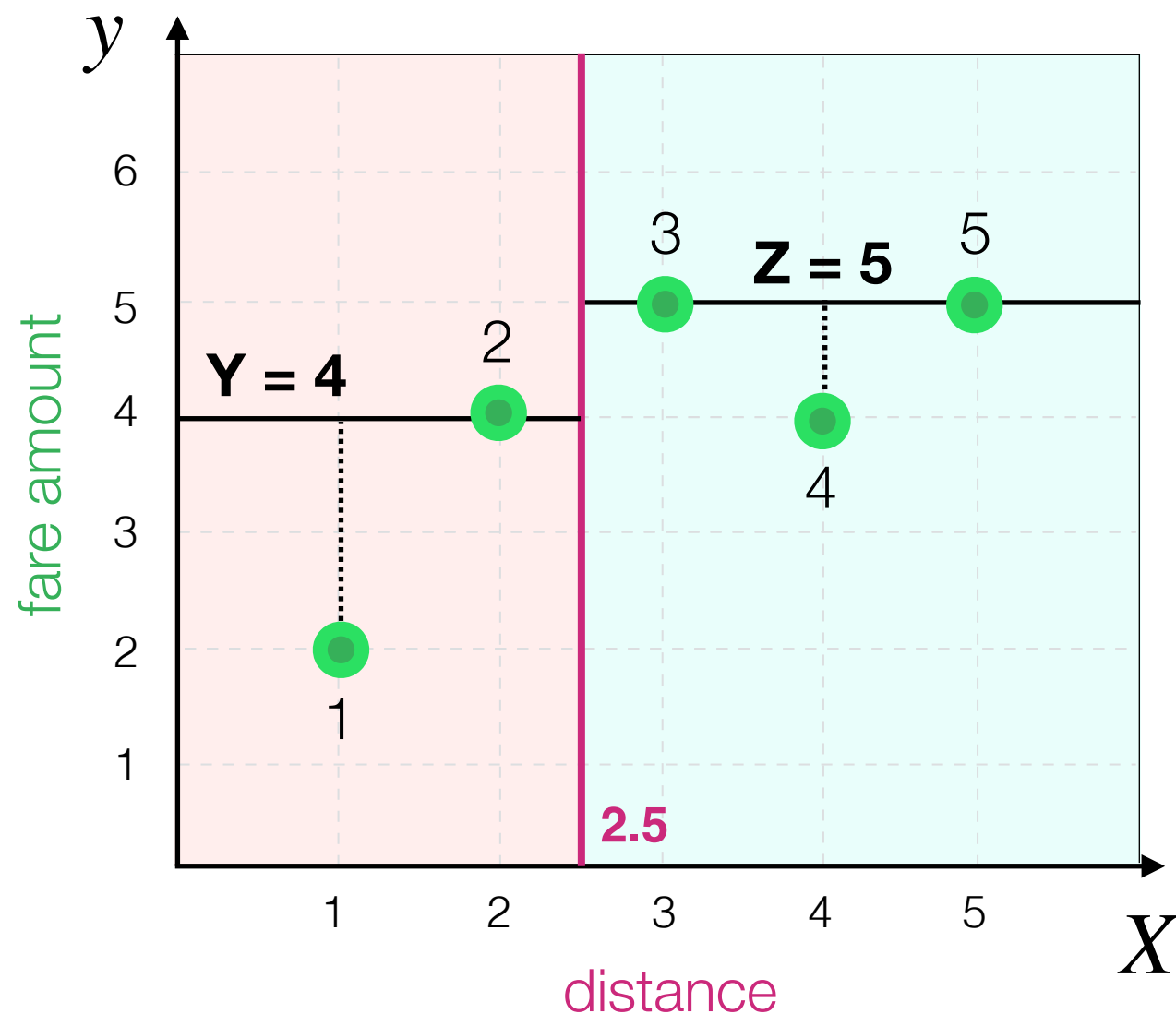
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 4)^2 + (4 - 4)^2 + (5 - 5)^2 + (4 - 5)^2 + (5 - 5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

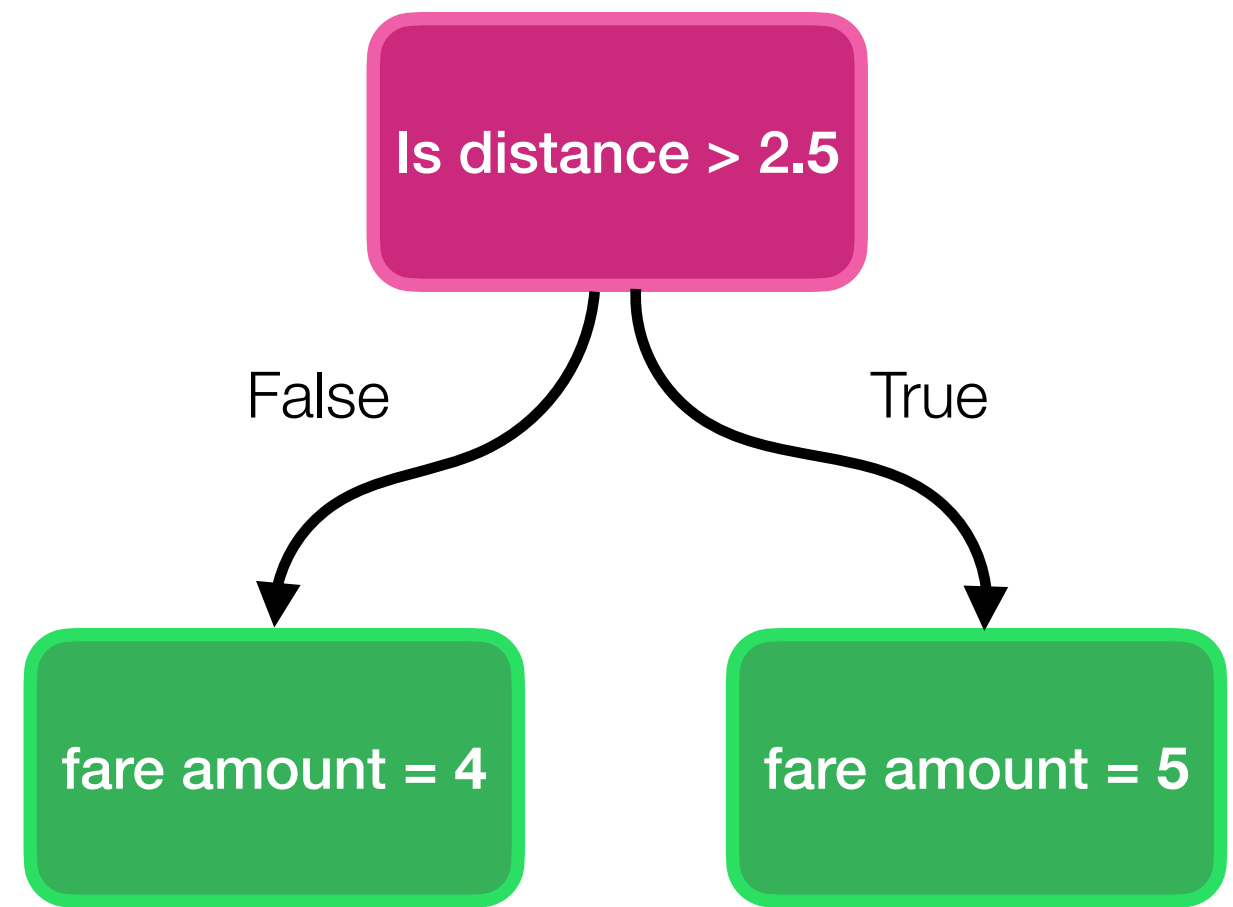
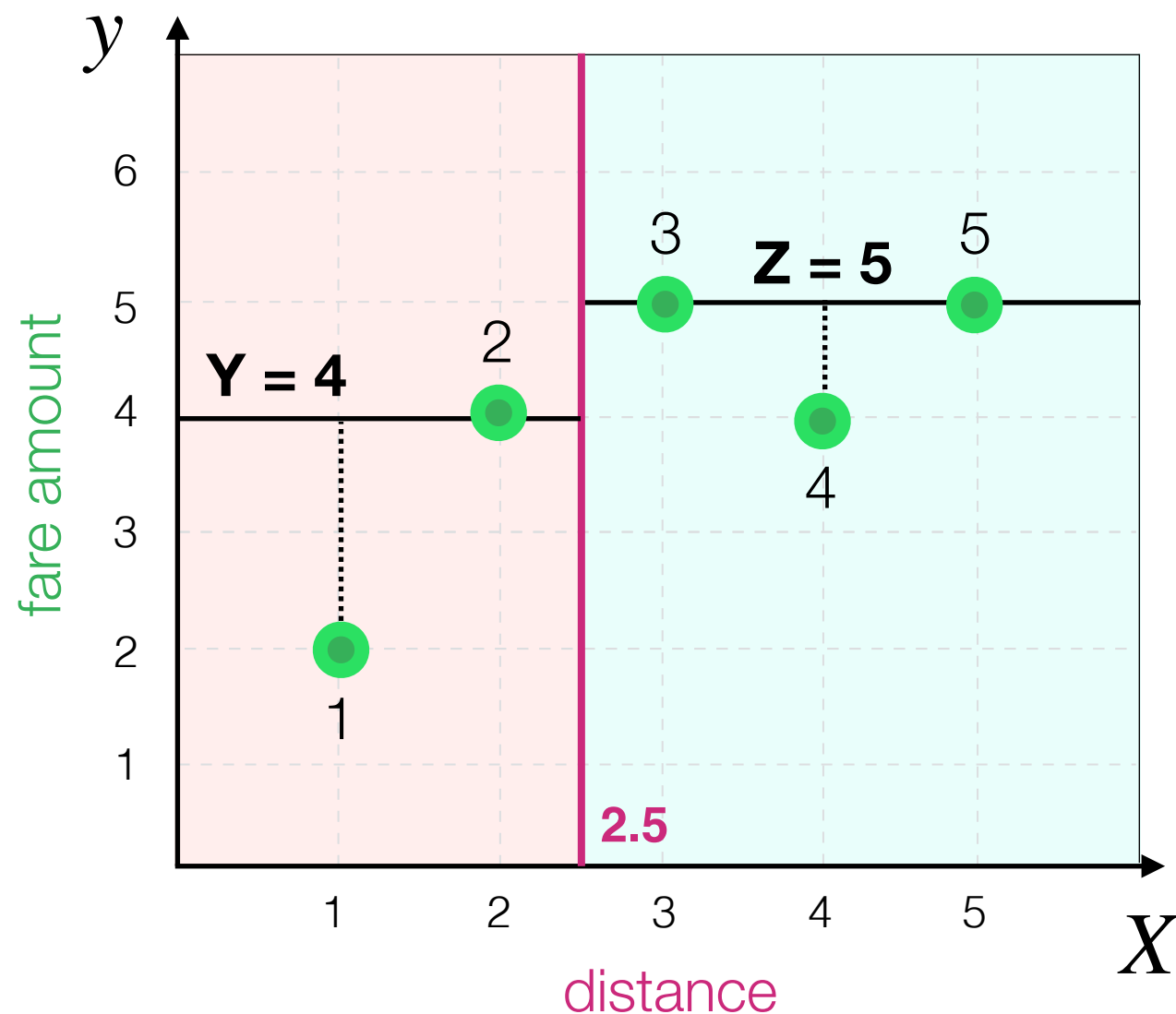
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(-2)^2 + (0)^2 + (0)^2 + (-1)^2 + (0)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

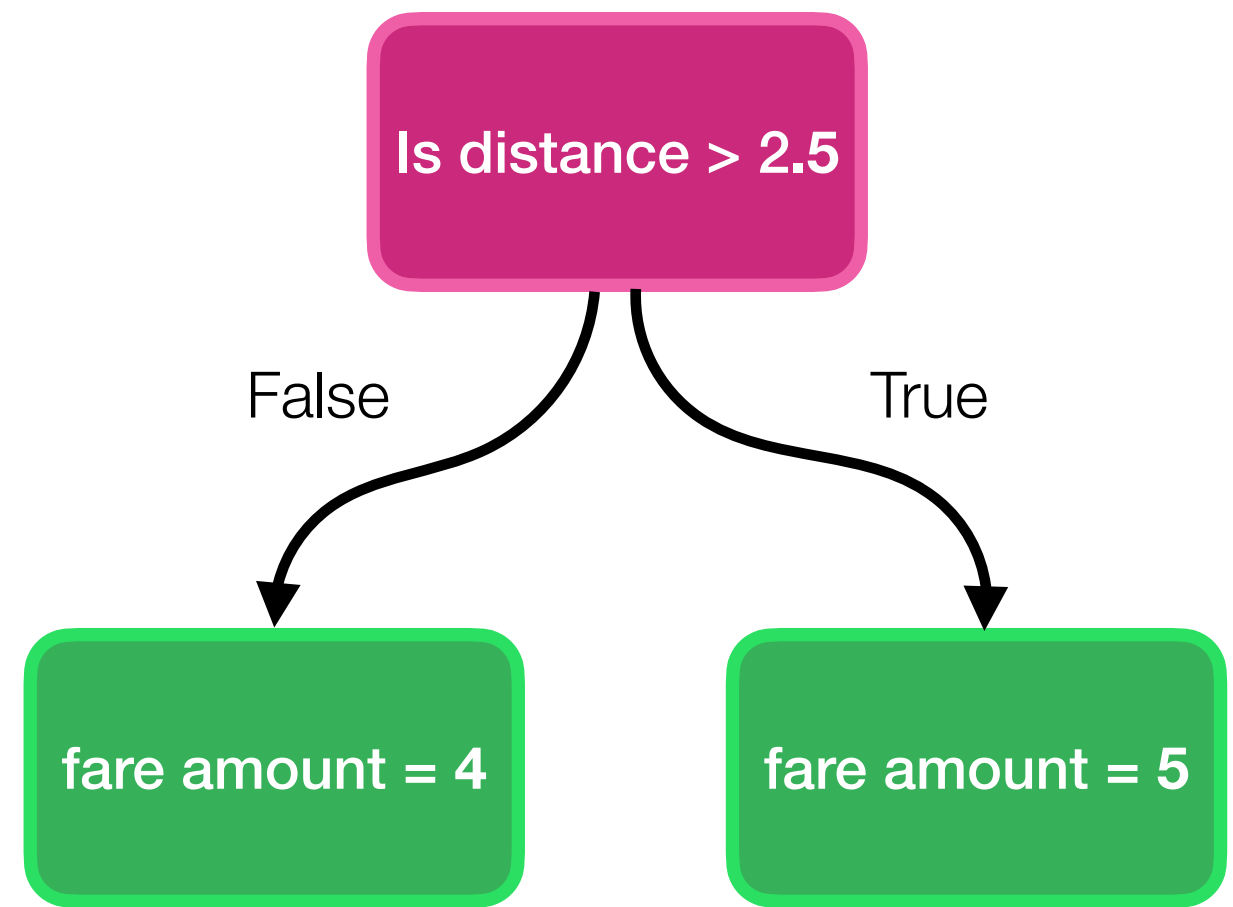
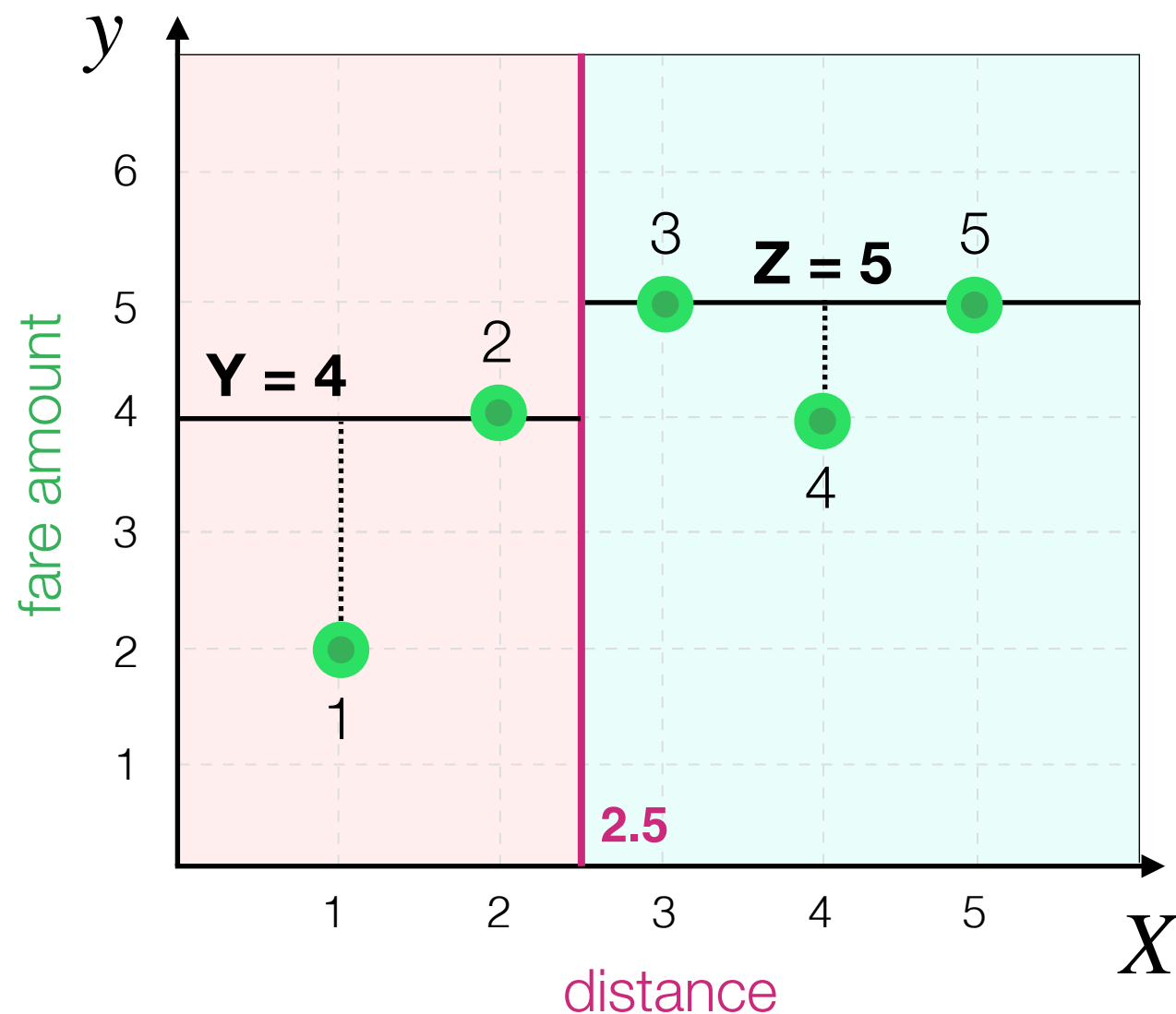
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{4 + 0 + 0 + 1 + 0}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

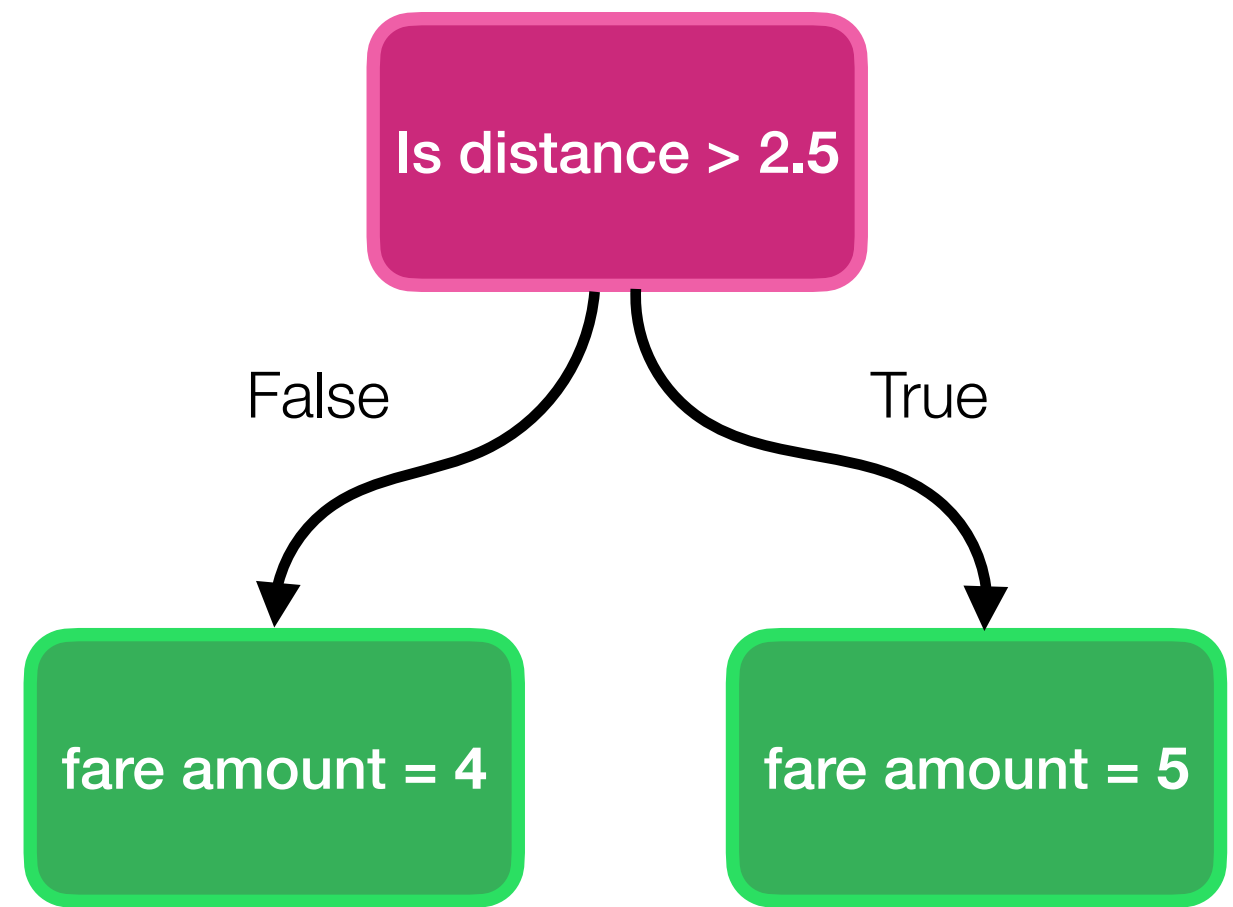
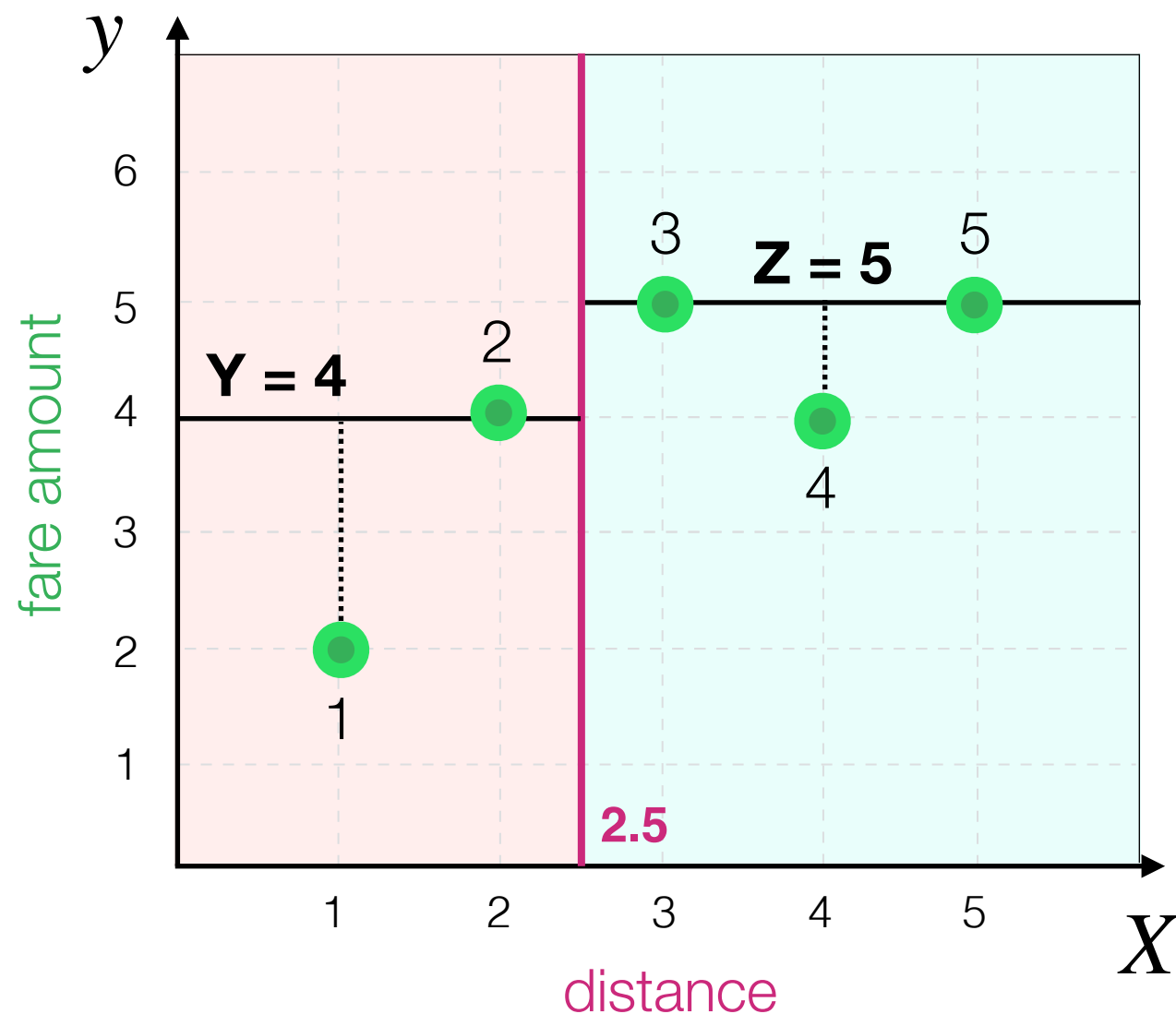
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{5}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

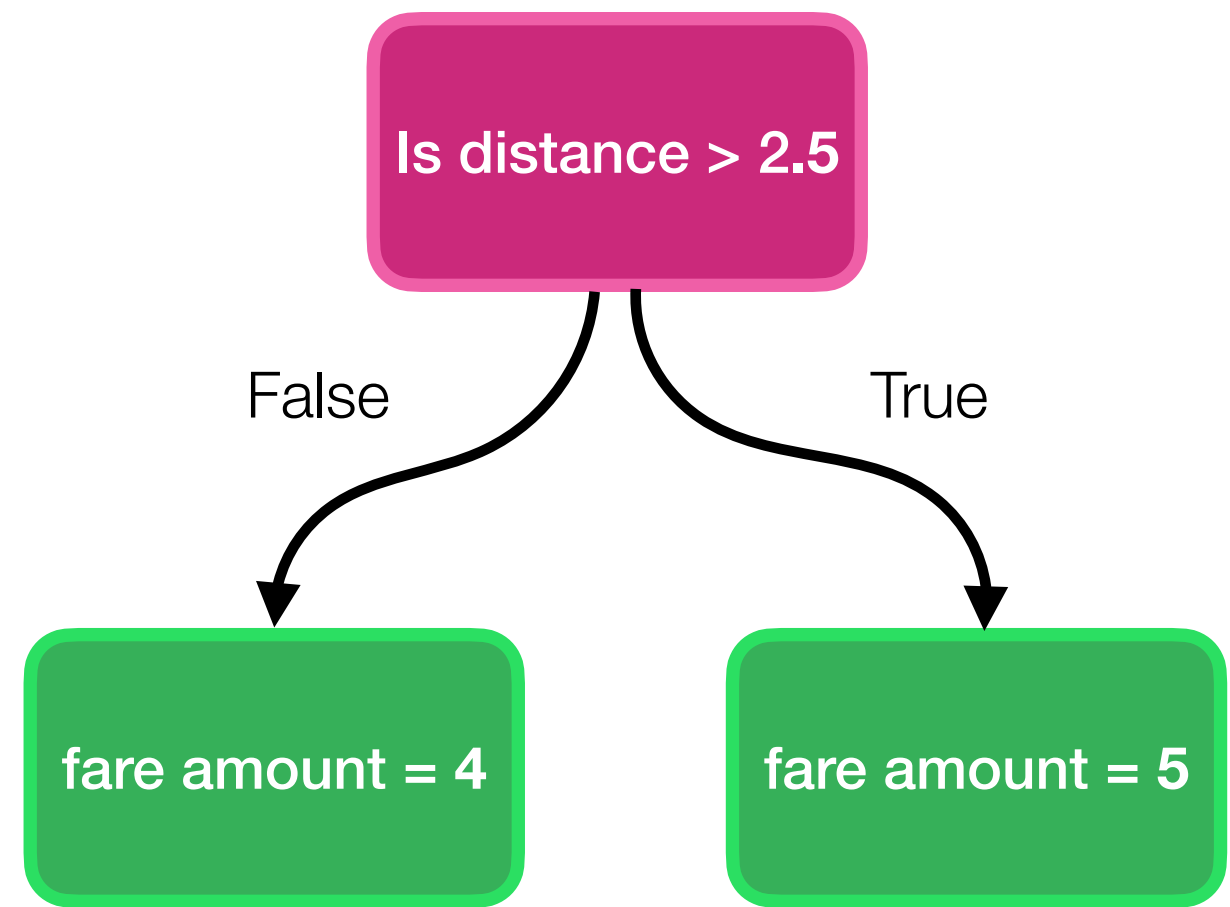
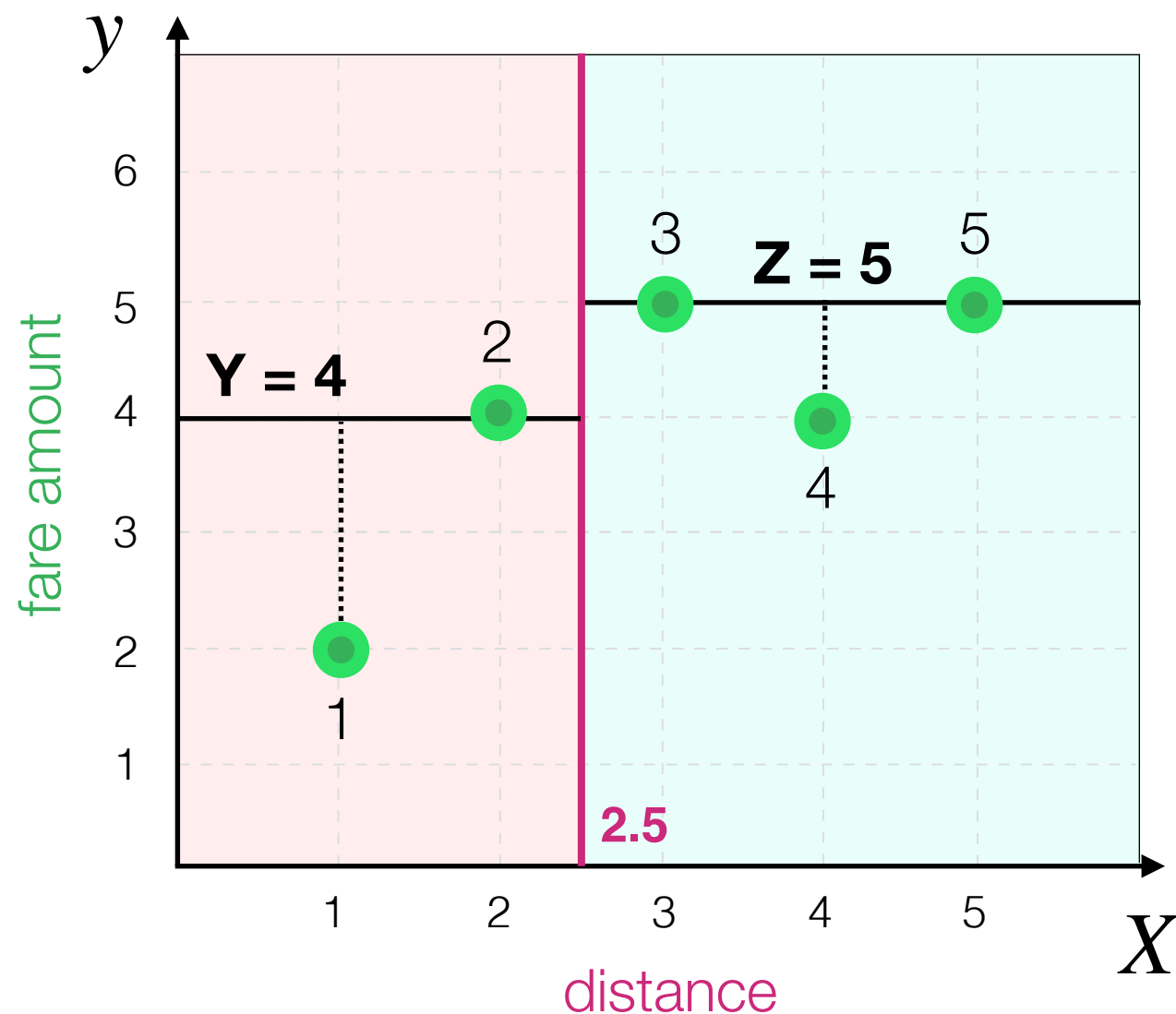
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = 1$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = 1 \quad \text{so, if } \mathbf{X} = \mathbf{2.5}, \mathbf{Y} = \mathbf{4} \text{ and } \mathbf{Z} = \mathbf{5}, \text{ MSE is } \mathbf{1}$$

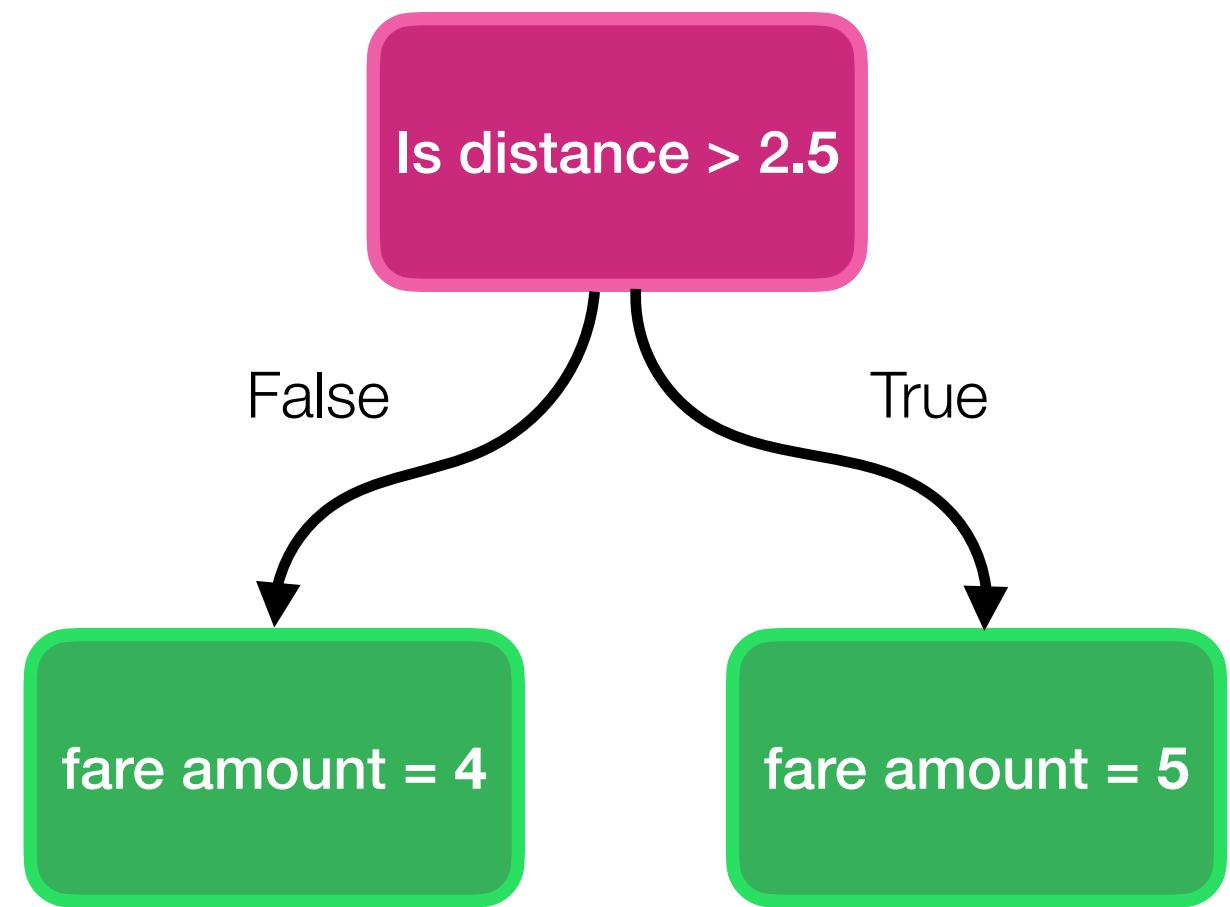
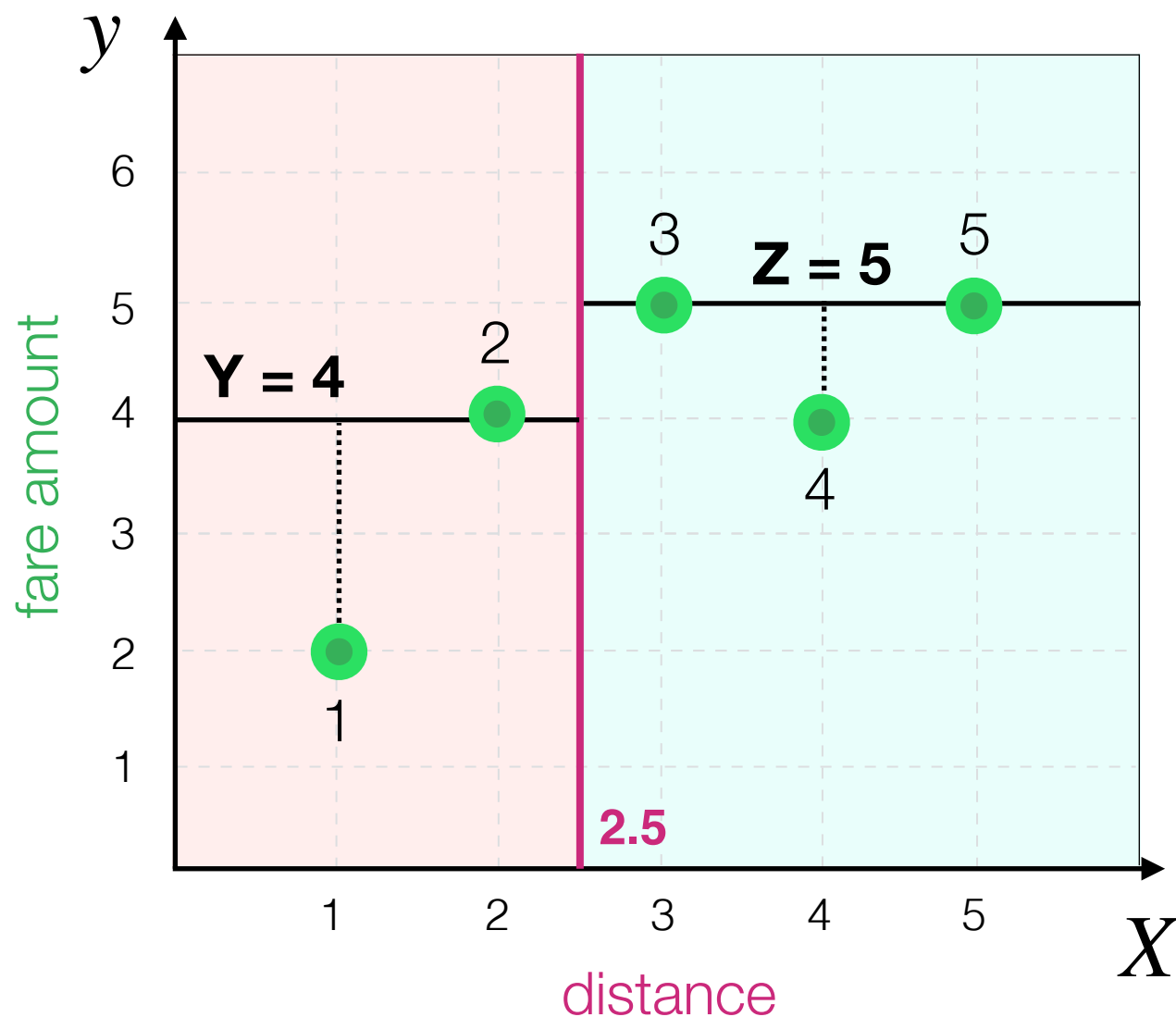


What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = 1 \quad \text{so, if } \mathbf{X} = \mathbf{2.5}, \mathbf{Y} = \mathbf{4} \text{ and } \mathbf{Z} = \mathbf{5}, \text{ MSE is } \mathbf{1}$$

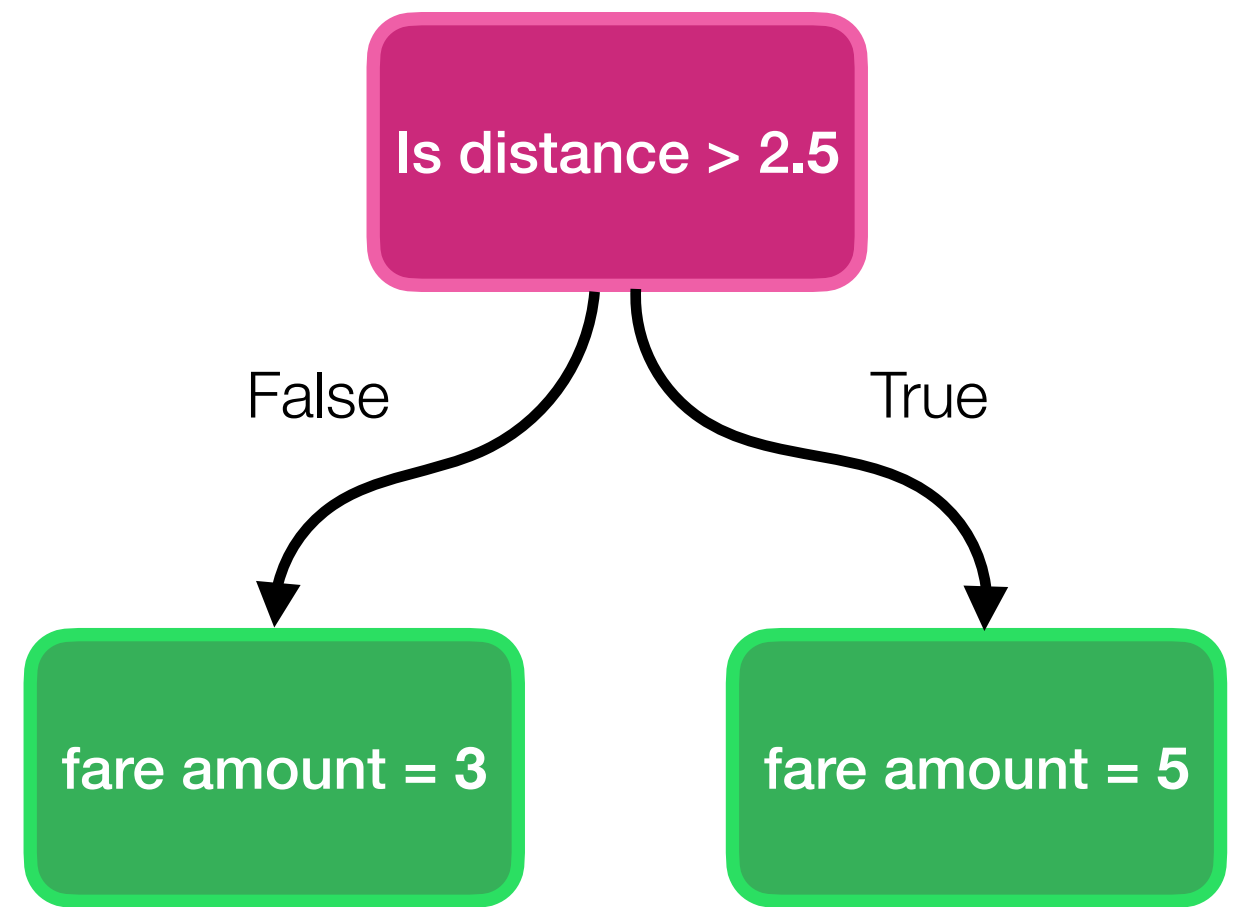
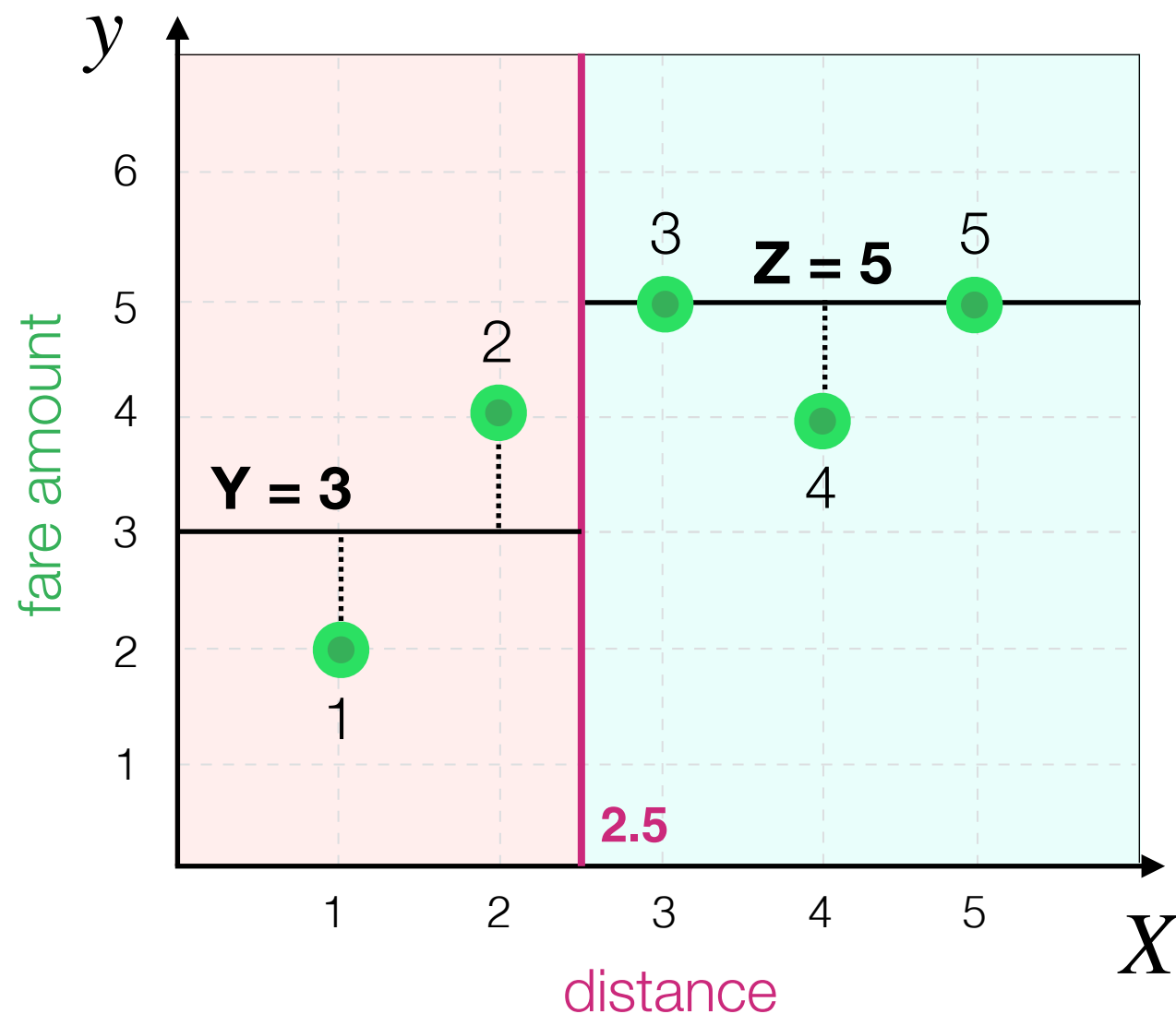
Can we find better **Y** and **Z**?



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

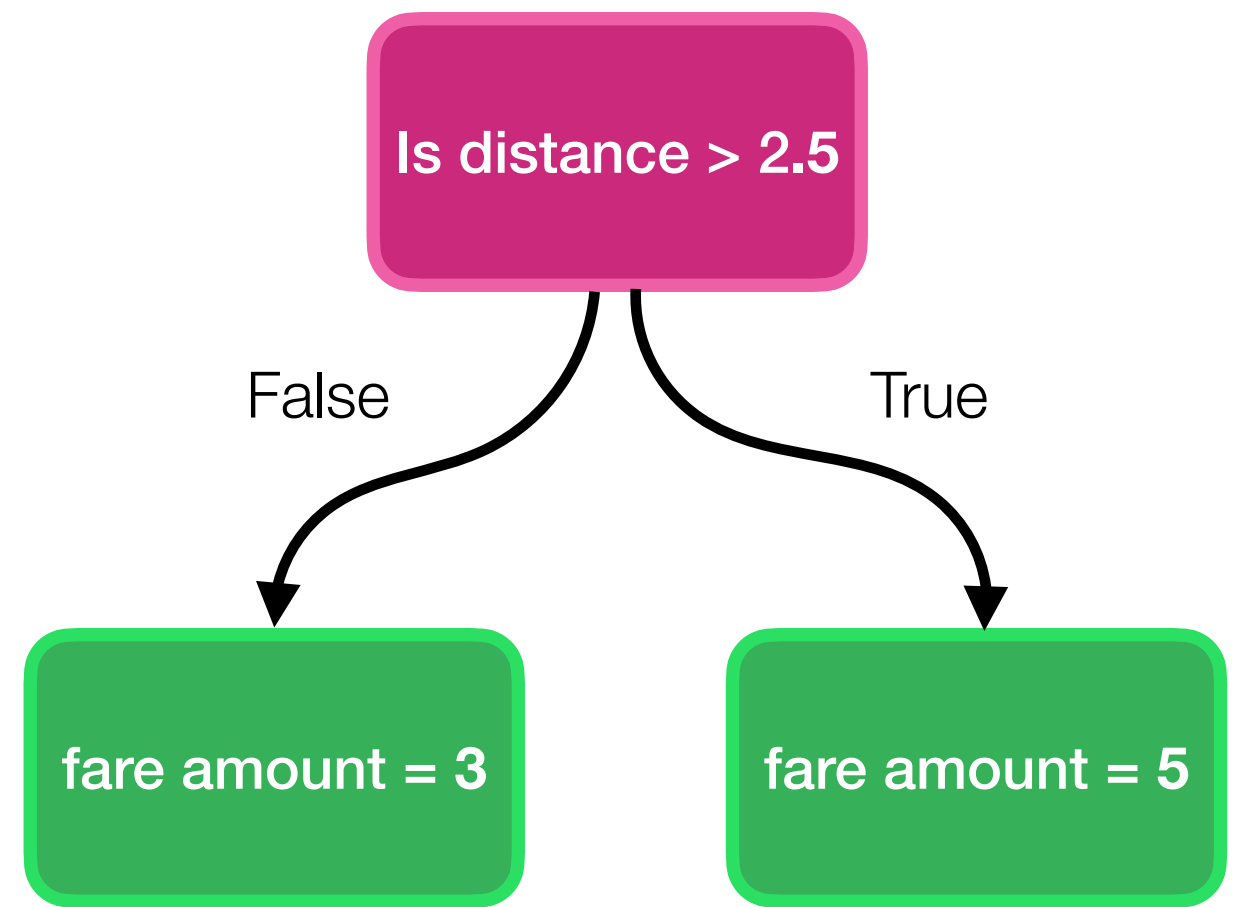
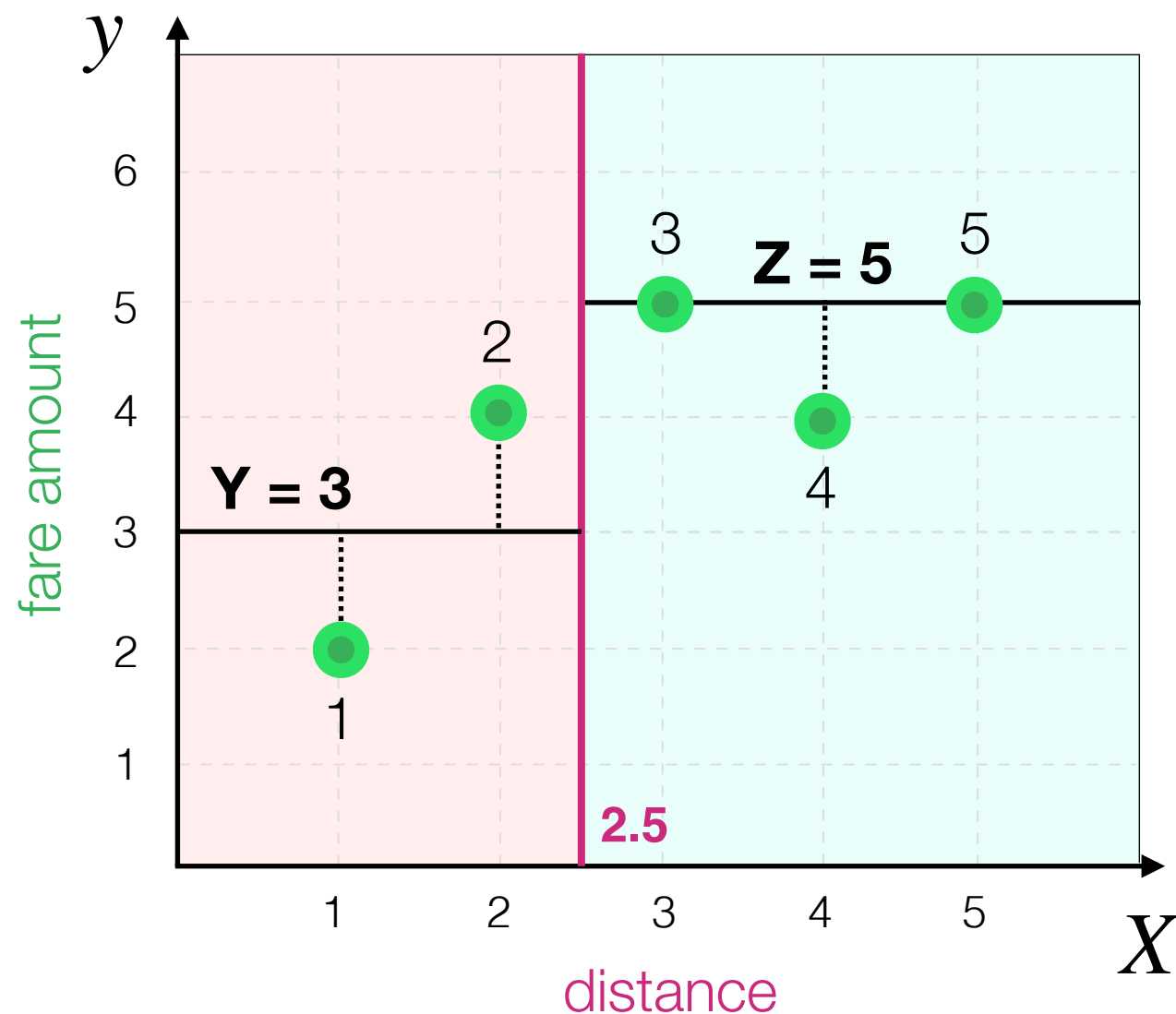
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + (y_4 - \hat{y}_4)^2 + (y_5 - \hat{y}_5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

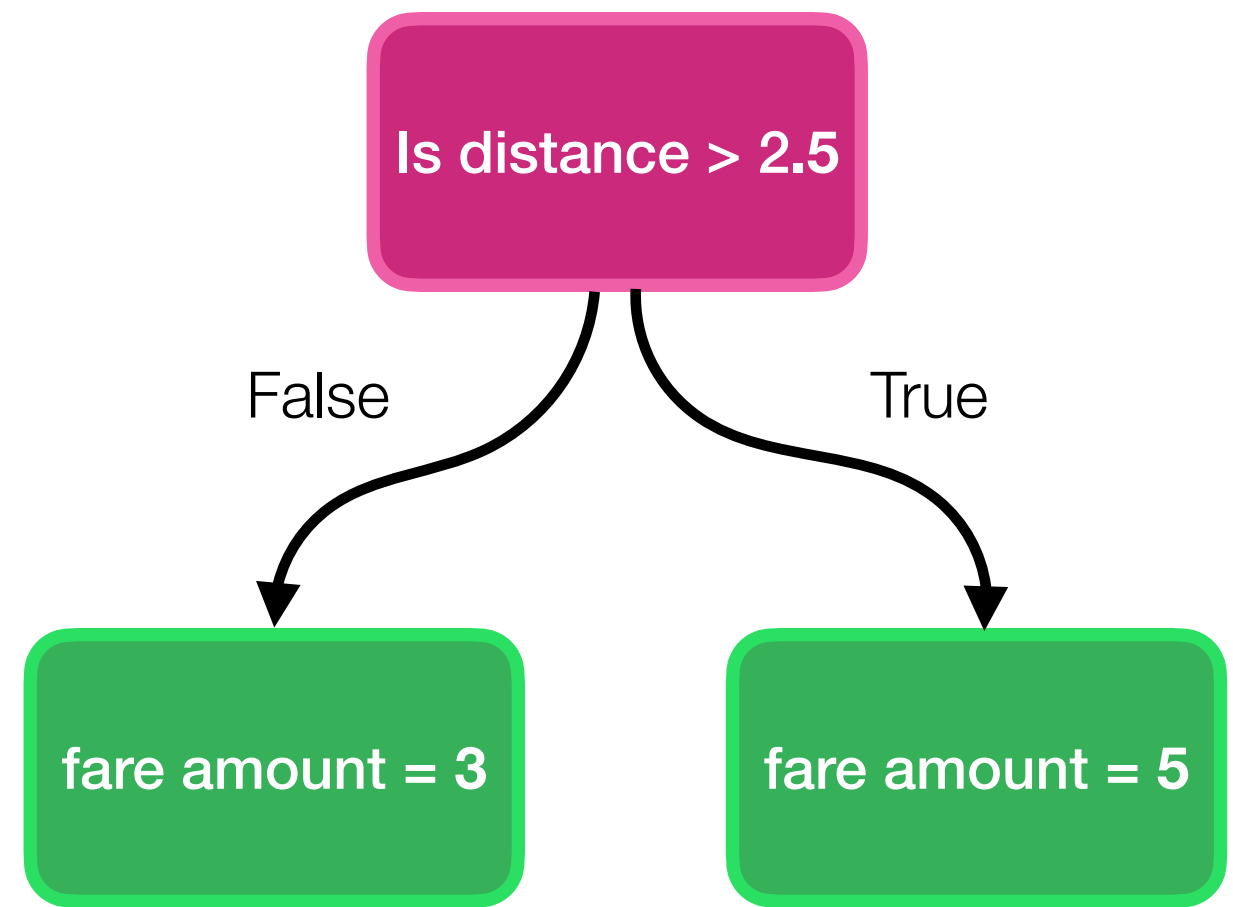
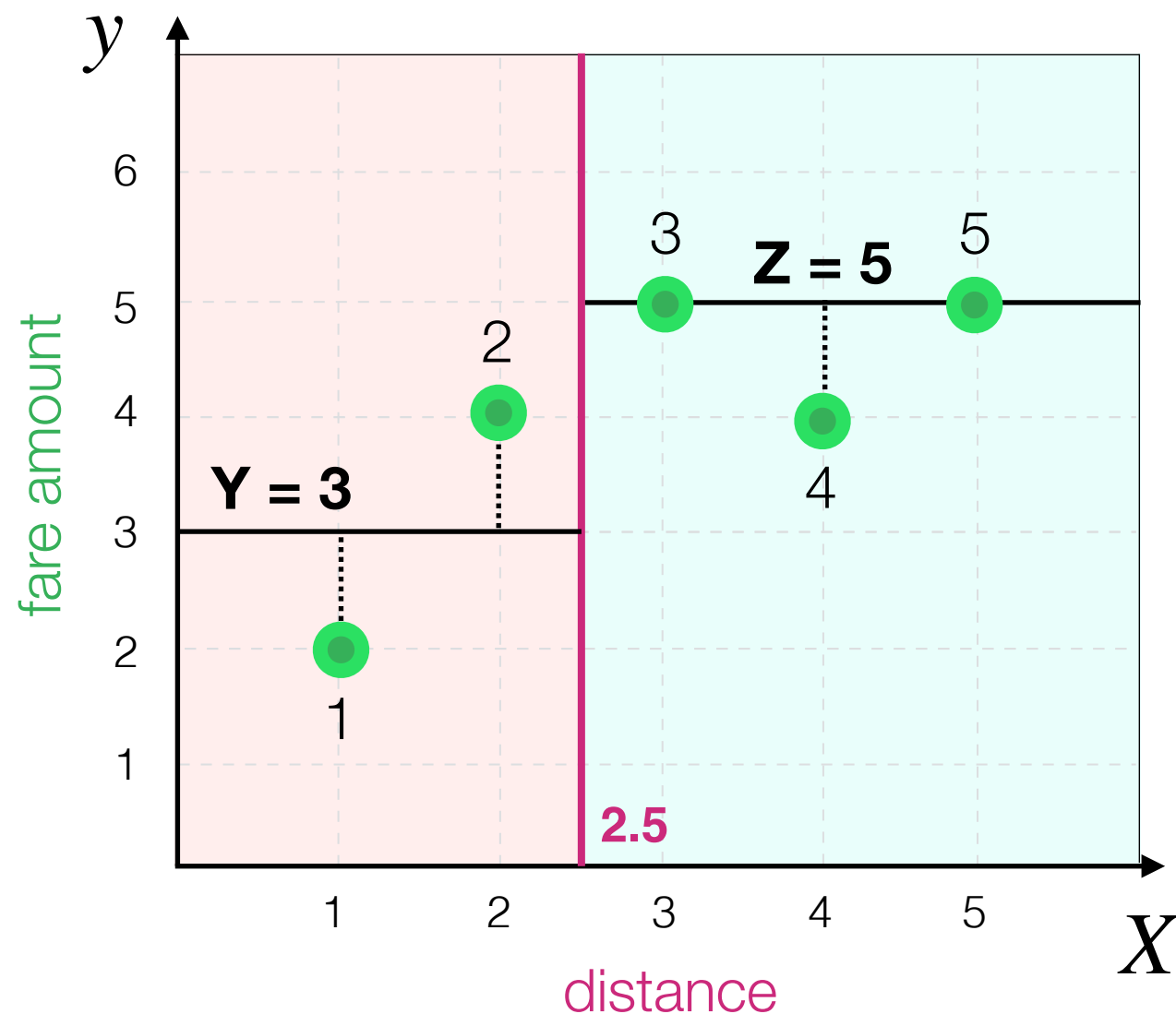
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 3)^2 + (4 - 3)^2 + (5 - 5)^2 + (4 - 5)^2 + (5 - 5)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

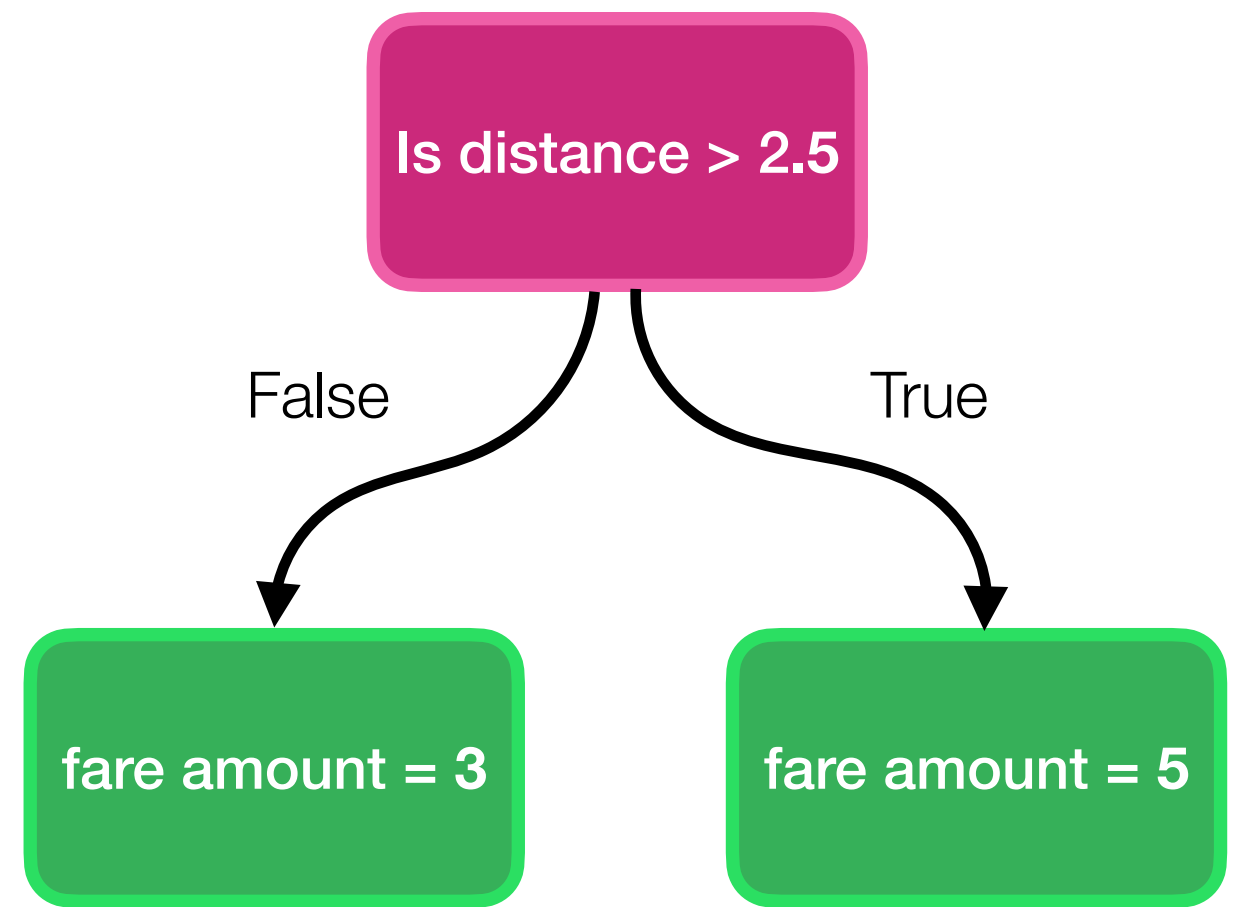
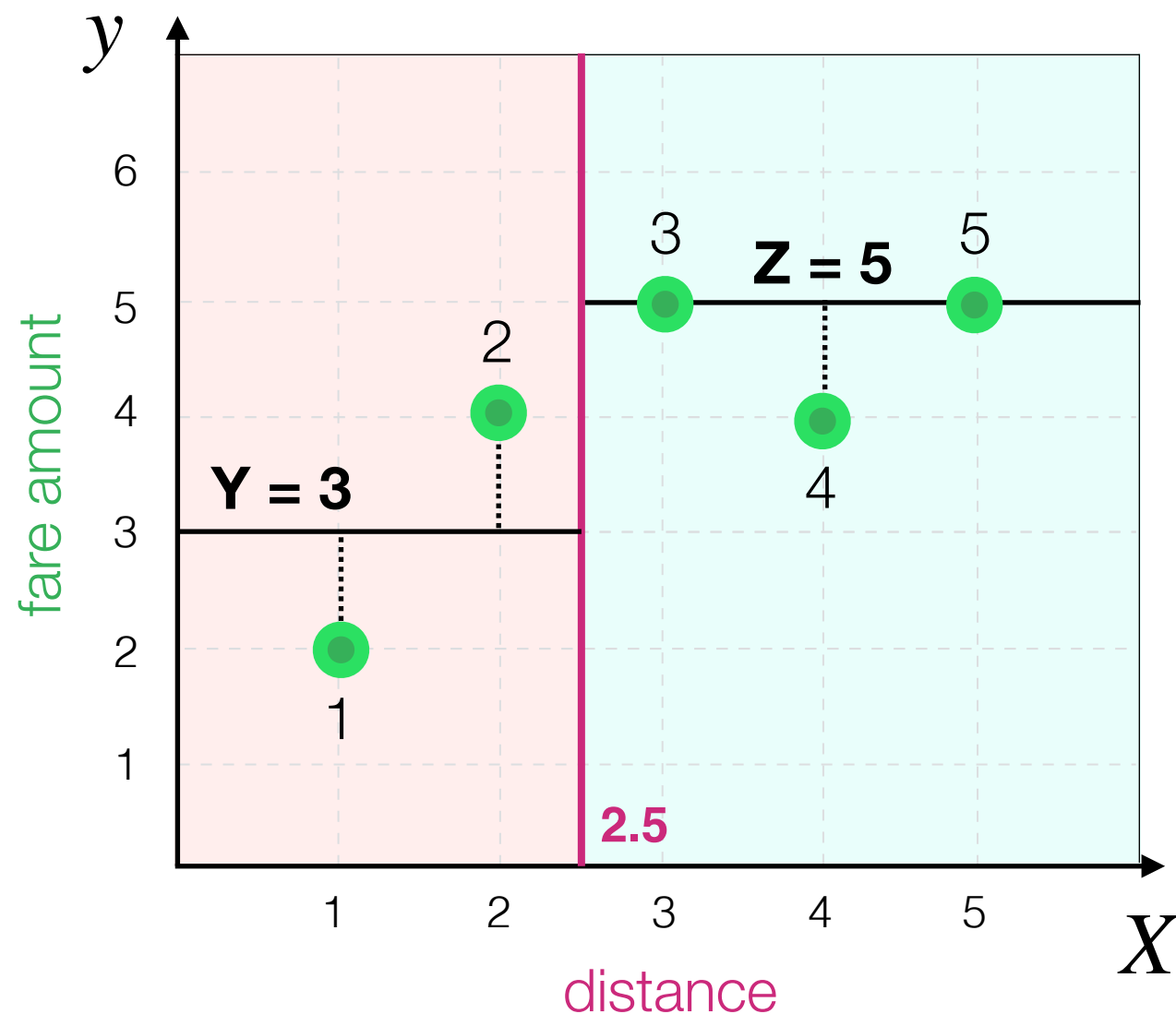
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1 + 1 + 0 + 1 + 0}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{3}{5} = 0.6$$

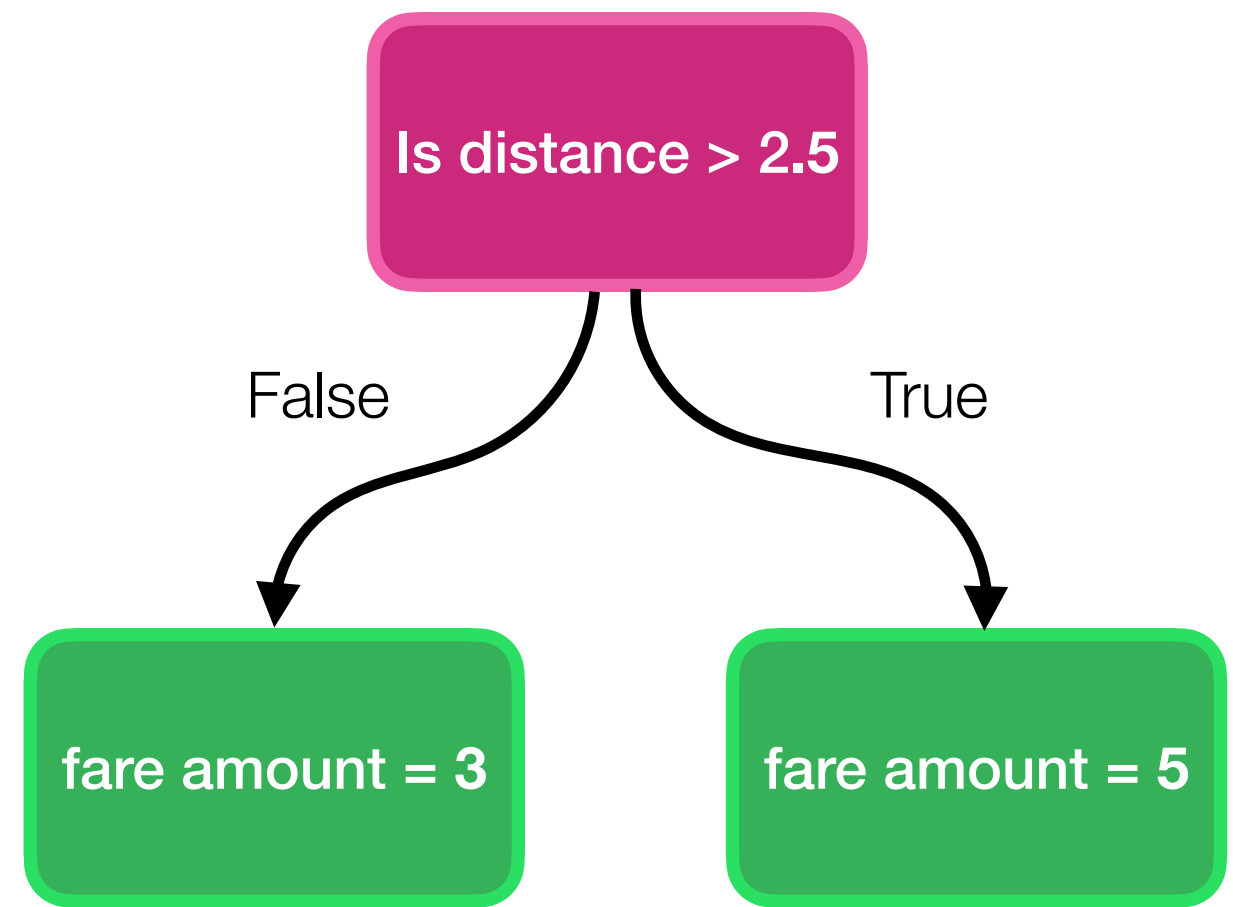
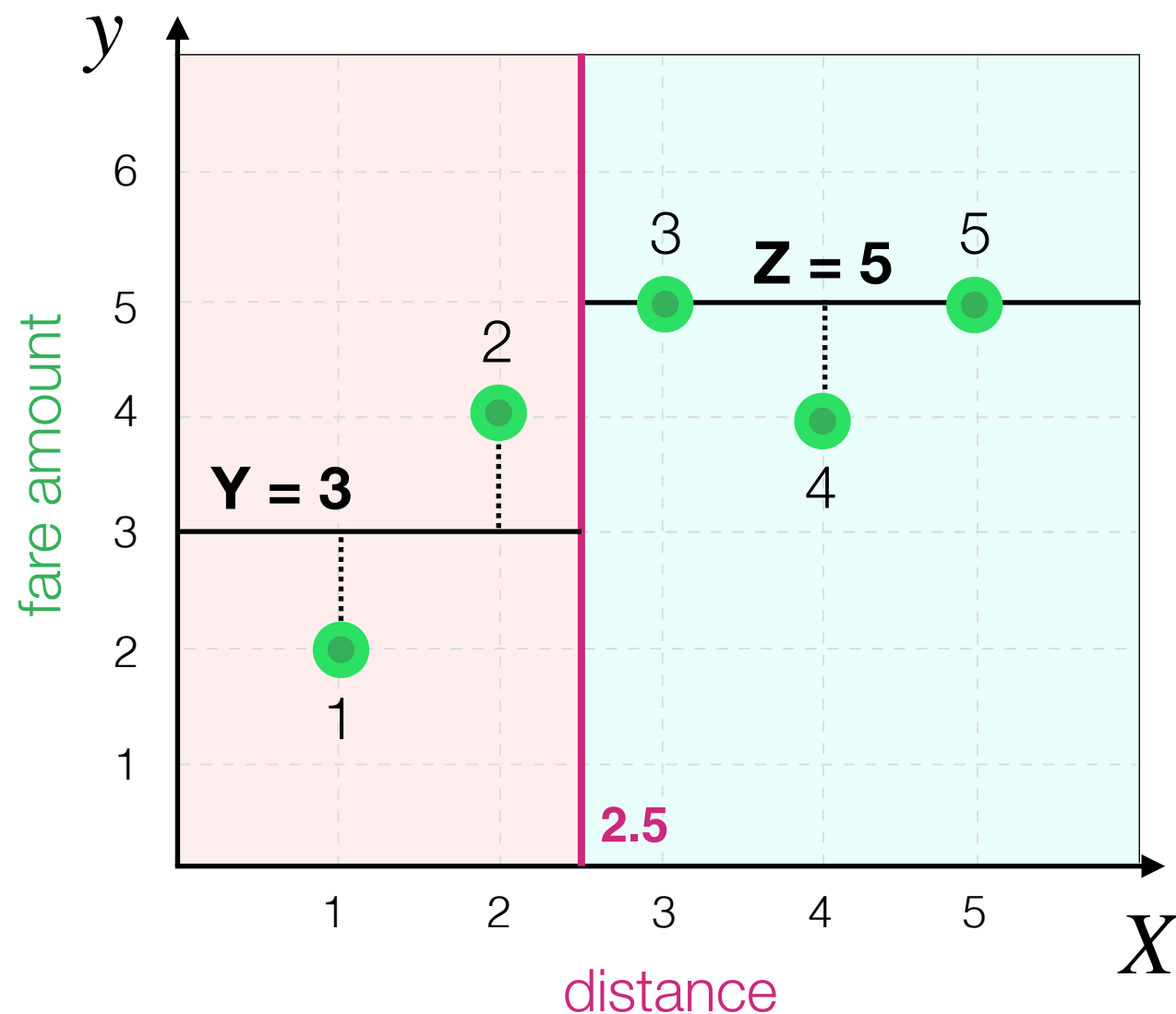


What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{3}{5} = 0.6 \quad \text{so, if } \mathbf{X} = \mathbf{2.5}, \mathbf{Y} = \mathbf{3} \text{ and } \mathbf{Z} = \mathbf{5},$$

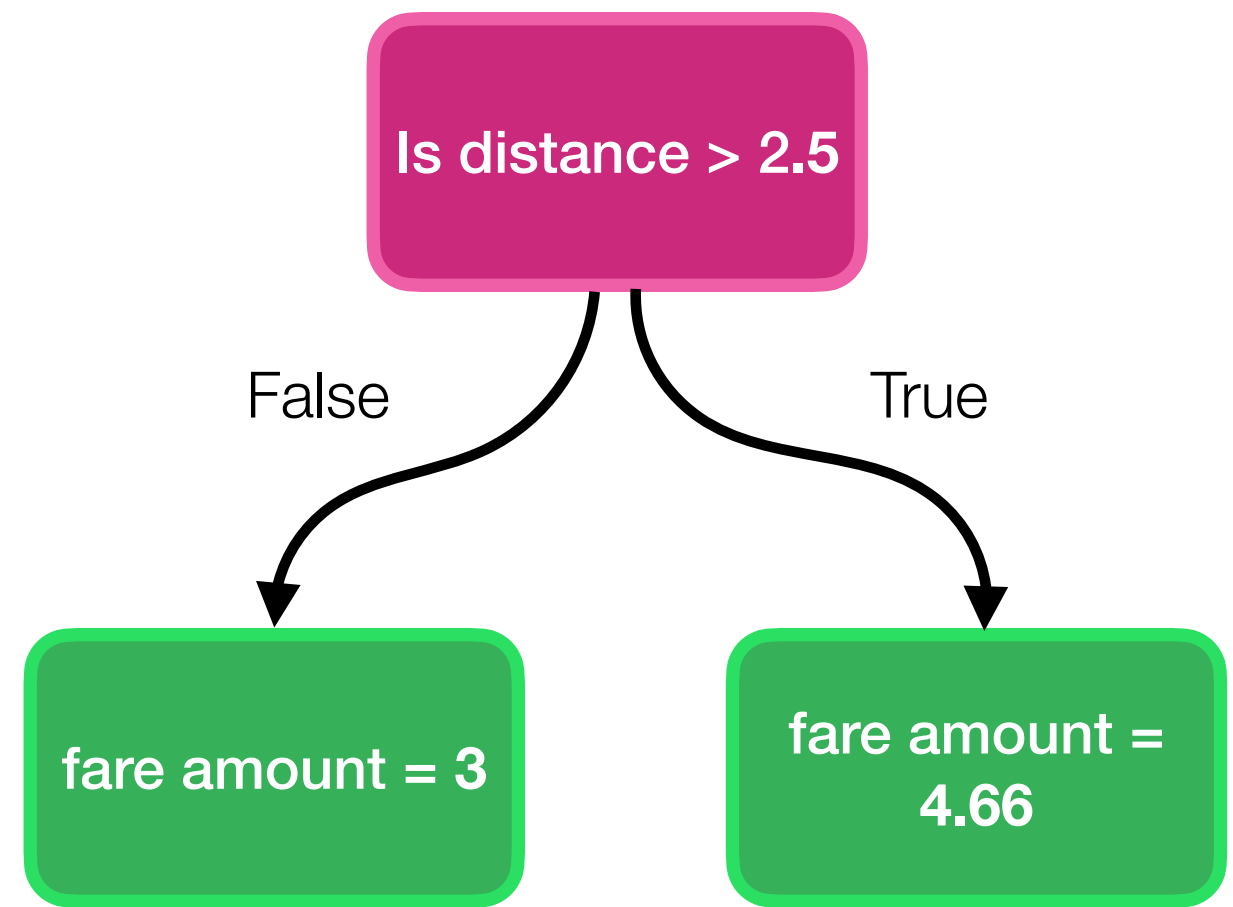
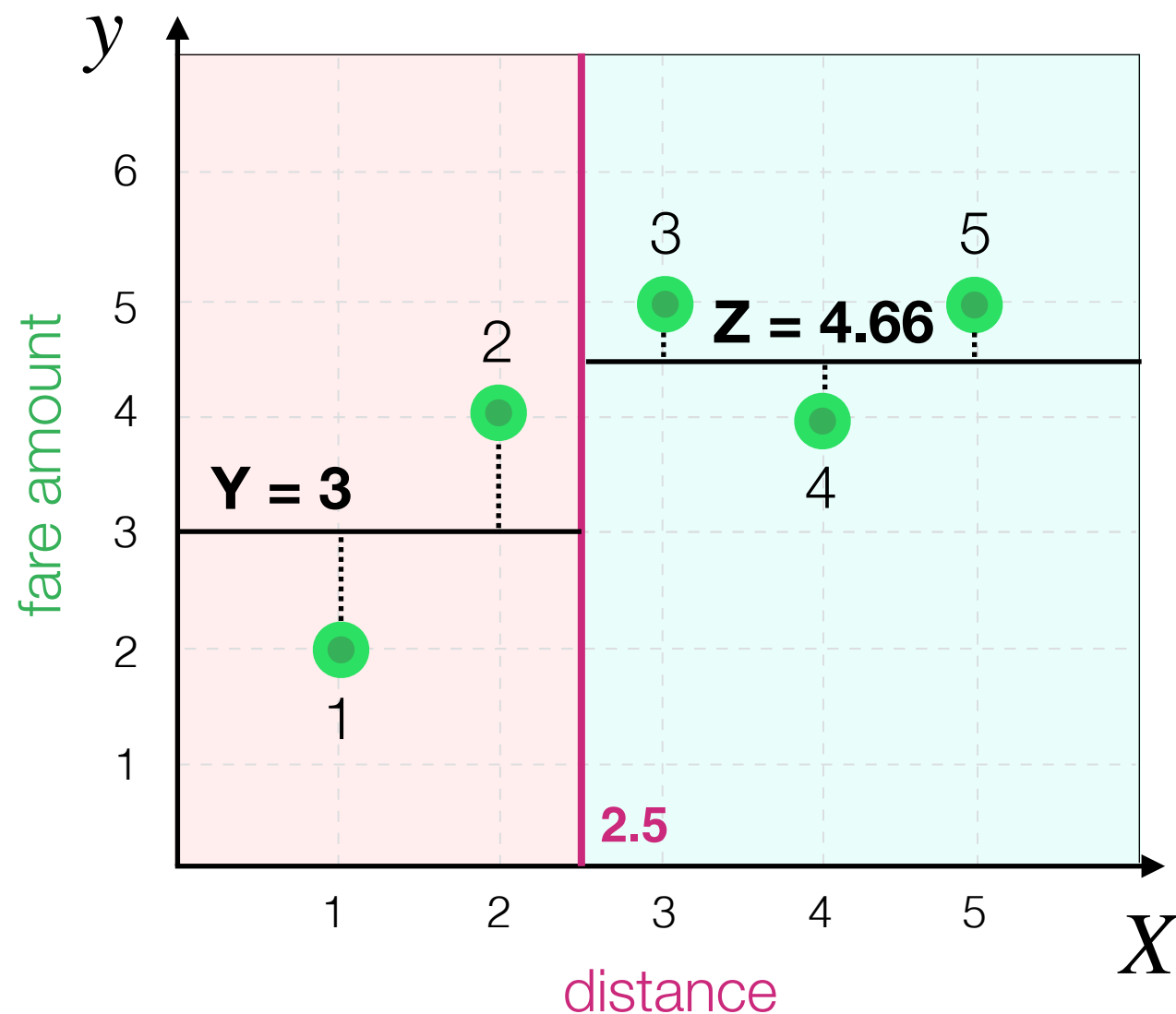
MSE is **0.6**



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

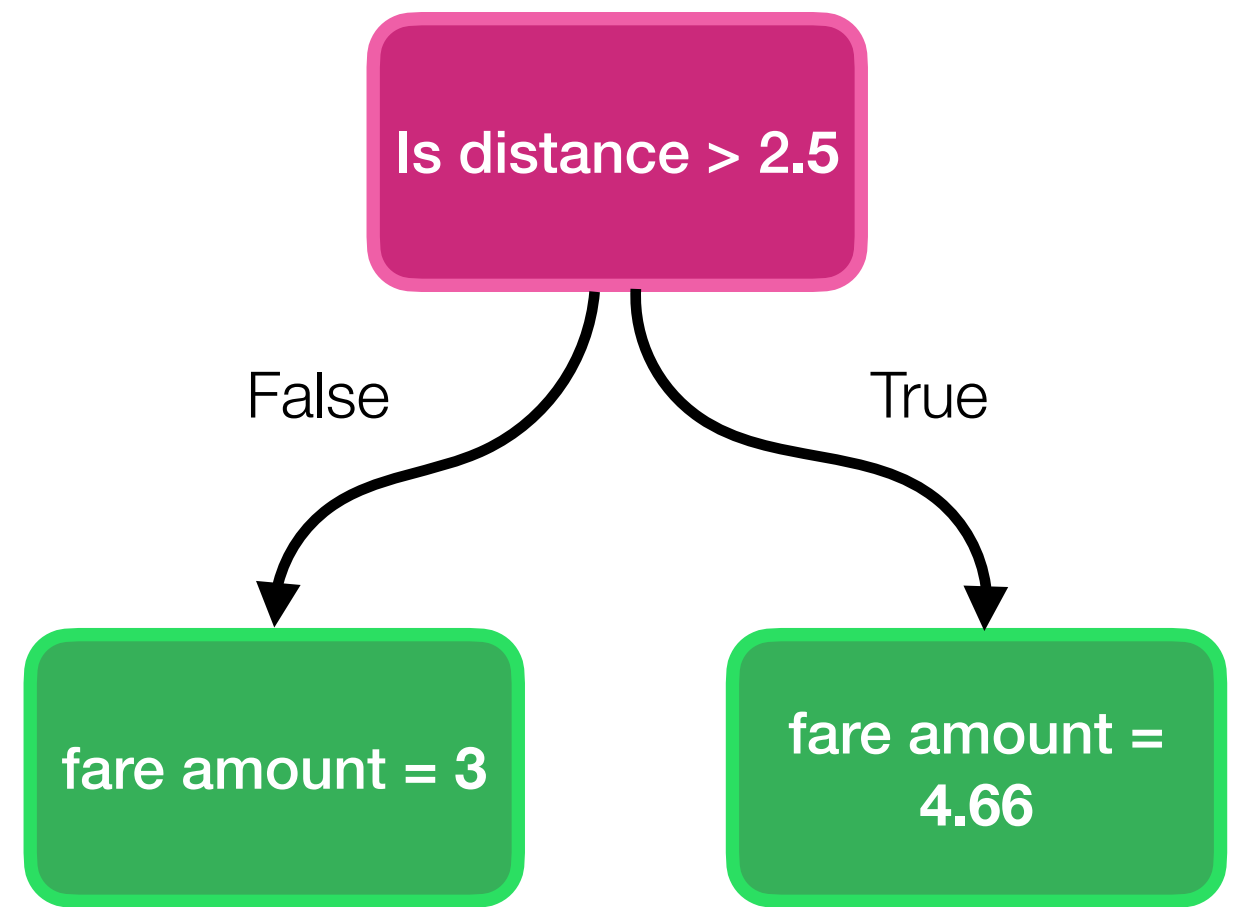
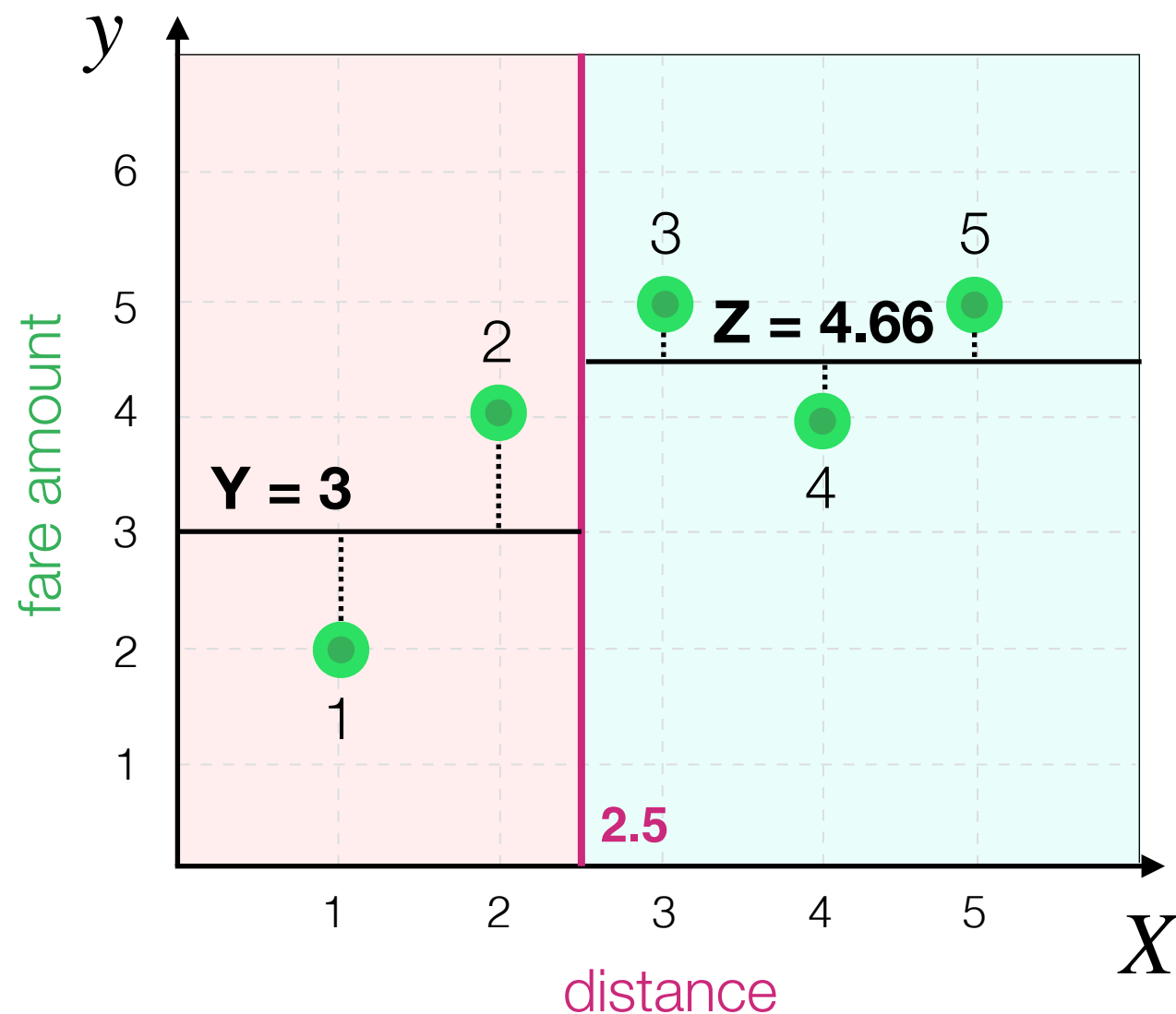
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{(2 - 3)^2 + (4 - 3)^2 + (5 - 4.66)^2 + (4 - 4.66)^2 + (5 - 4.66)^2}{5}$$



What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1 + 1 + 0.12 + 0.43 + 0.12}{5}$$

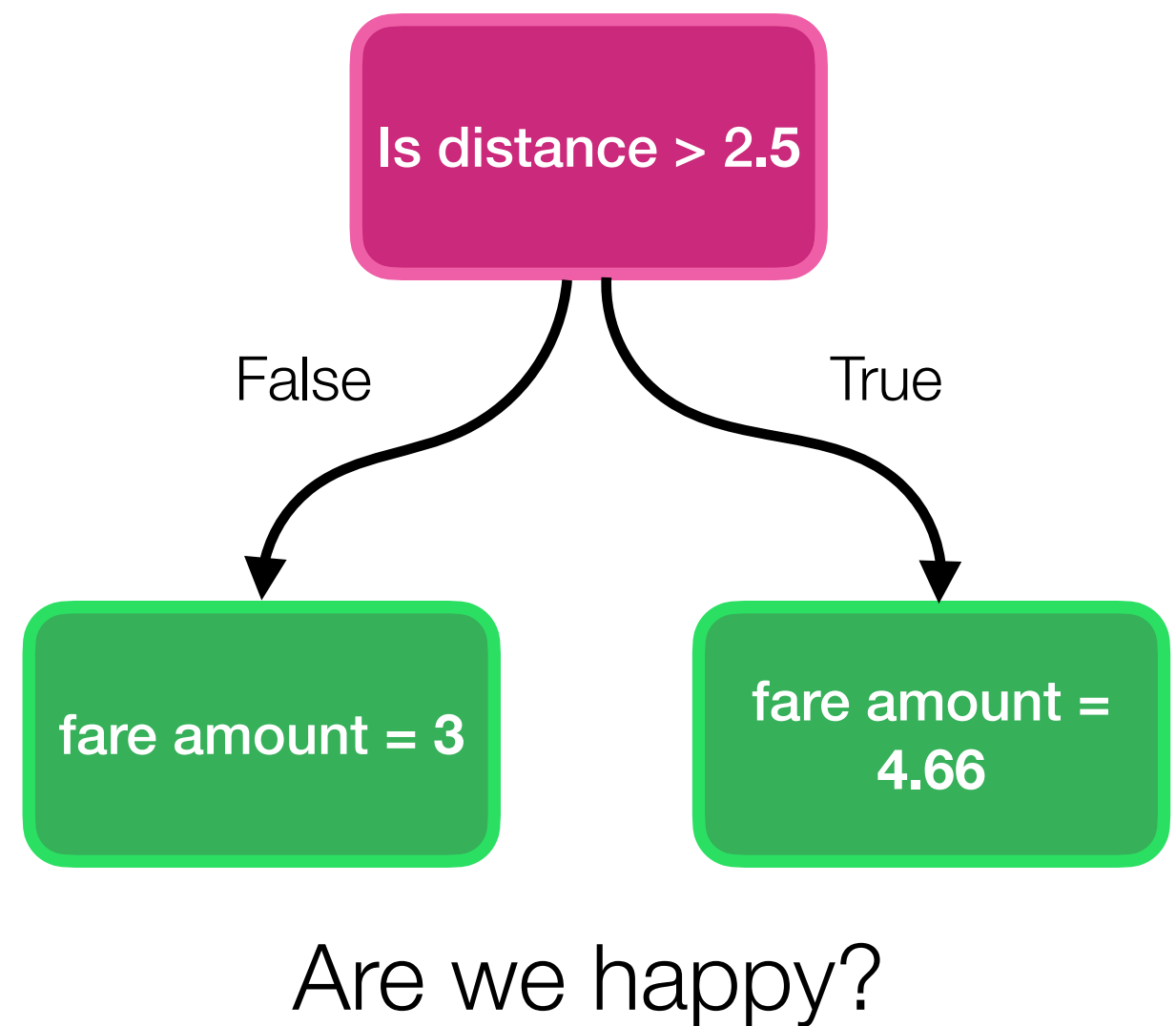
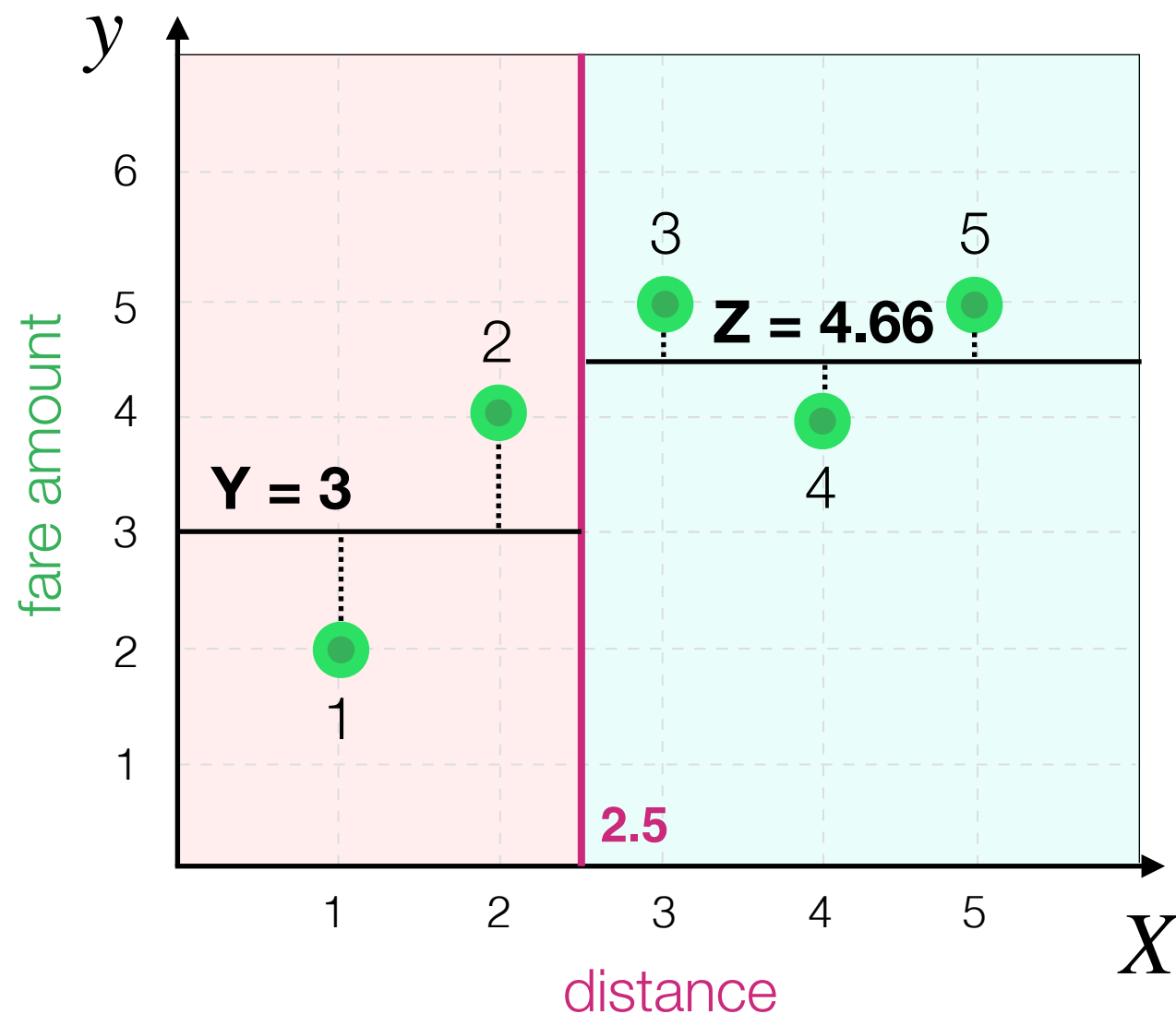


What are most reasonable values for **Y** and **Z** (that minimise total MSE)?

Decision Tree Algorithm

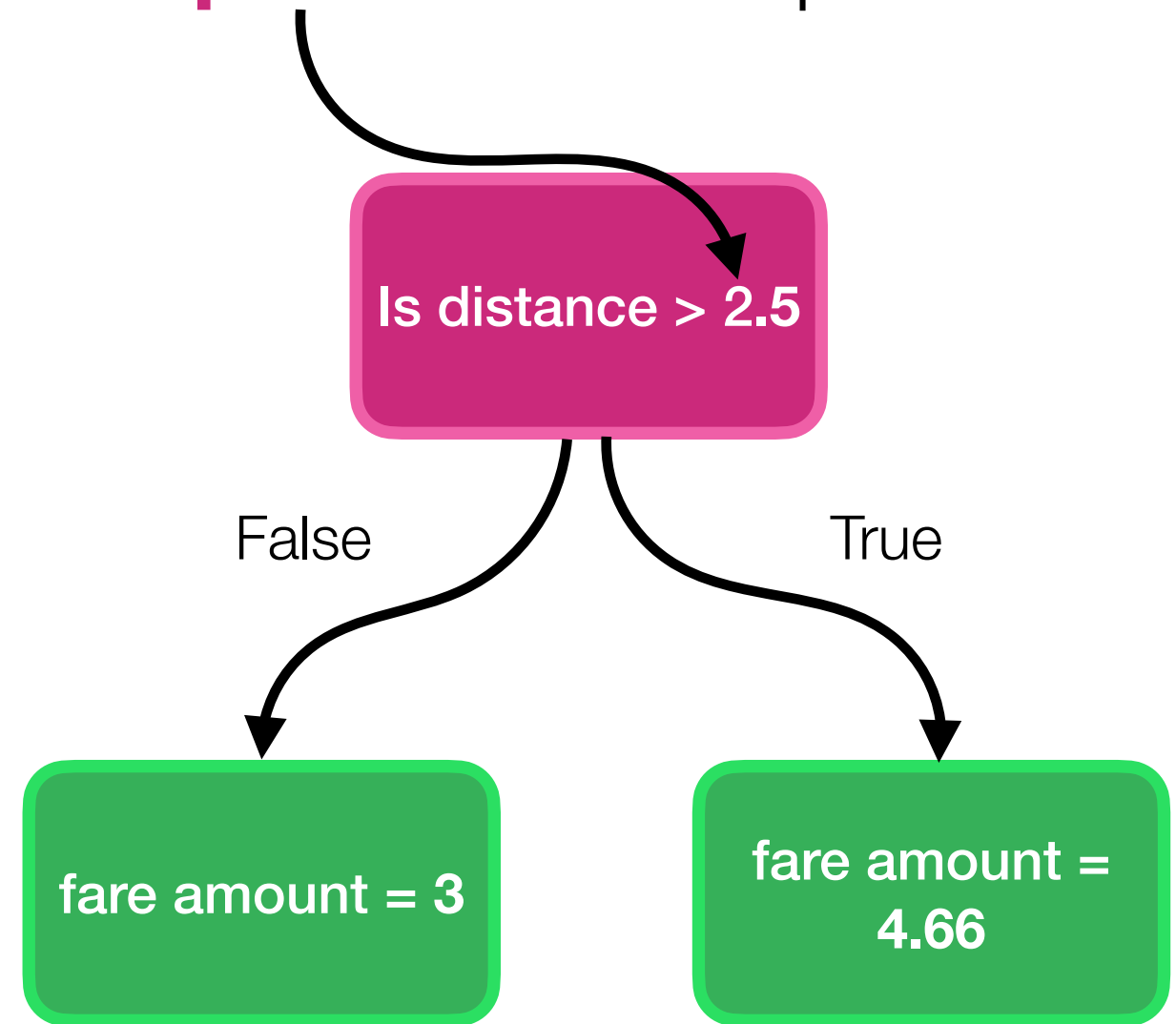
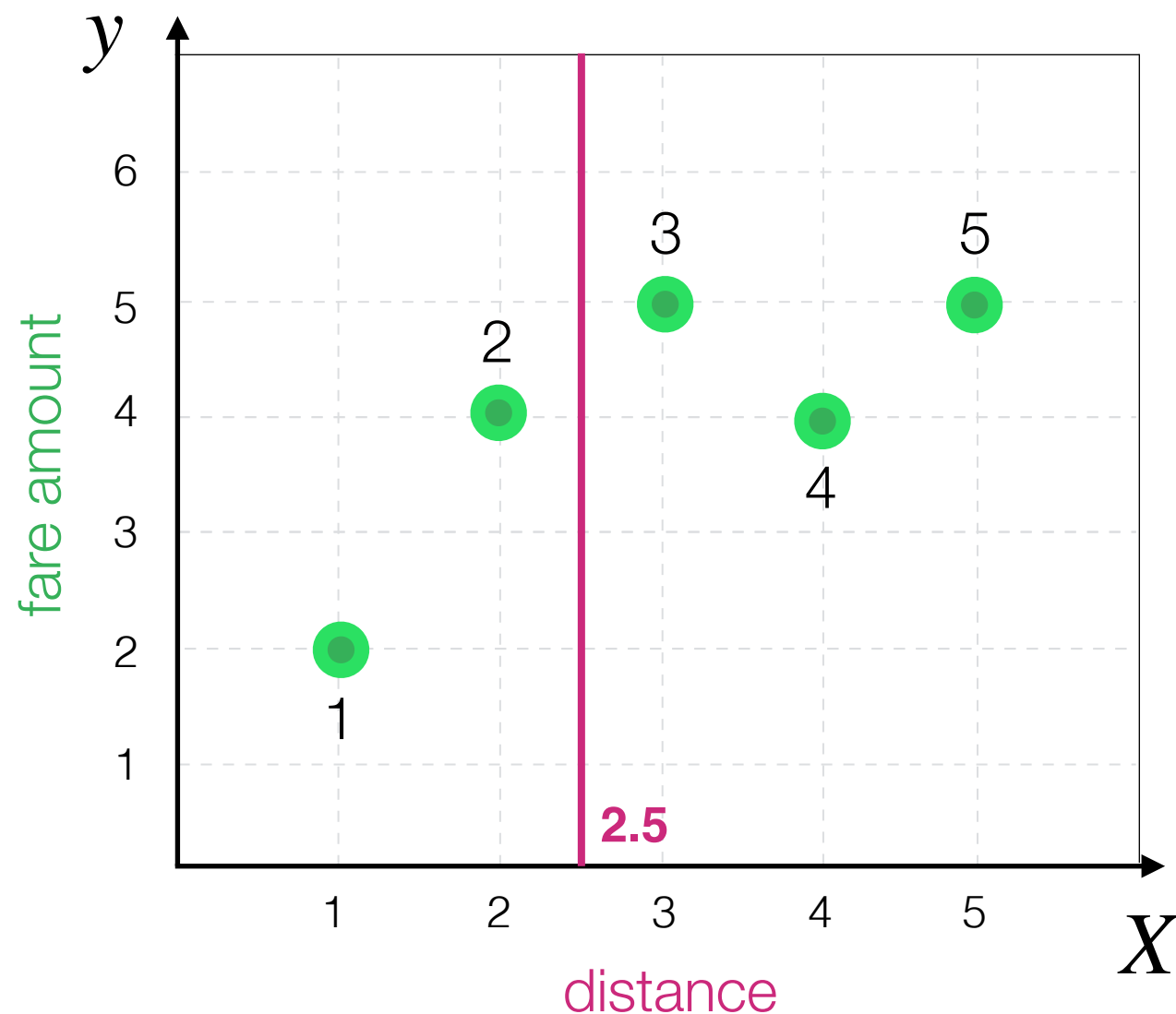
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{2.67}{5} = 0.53$$

so, if **Y = 3** and **Z = 4.66**,
MSE is **smallest**



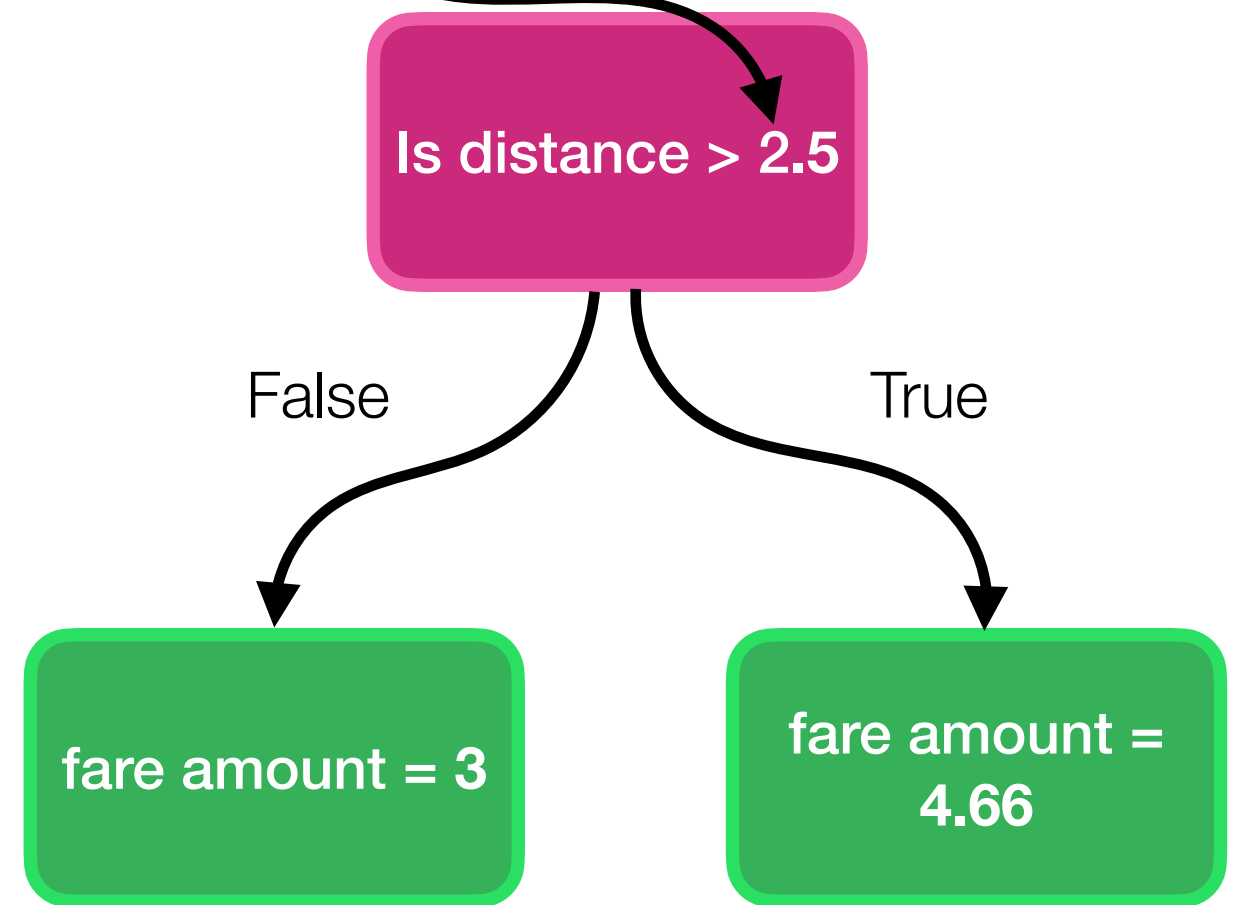
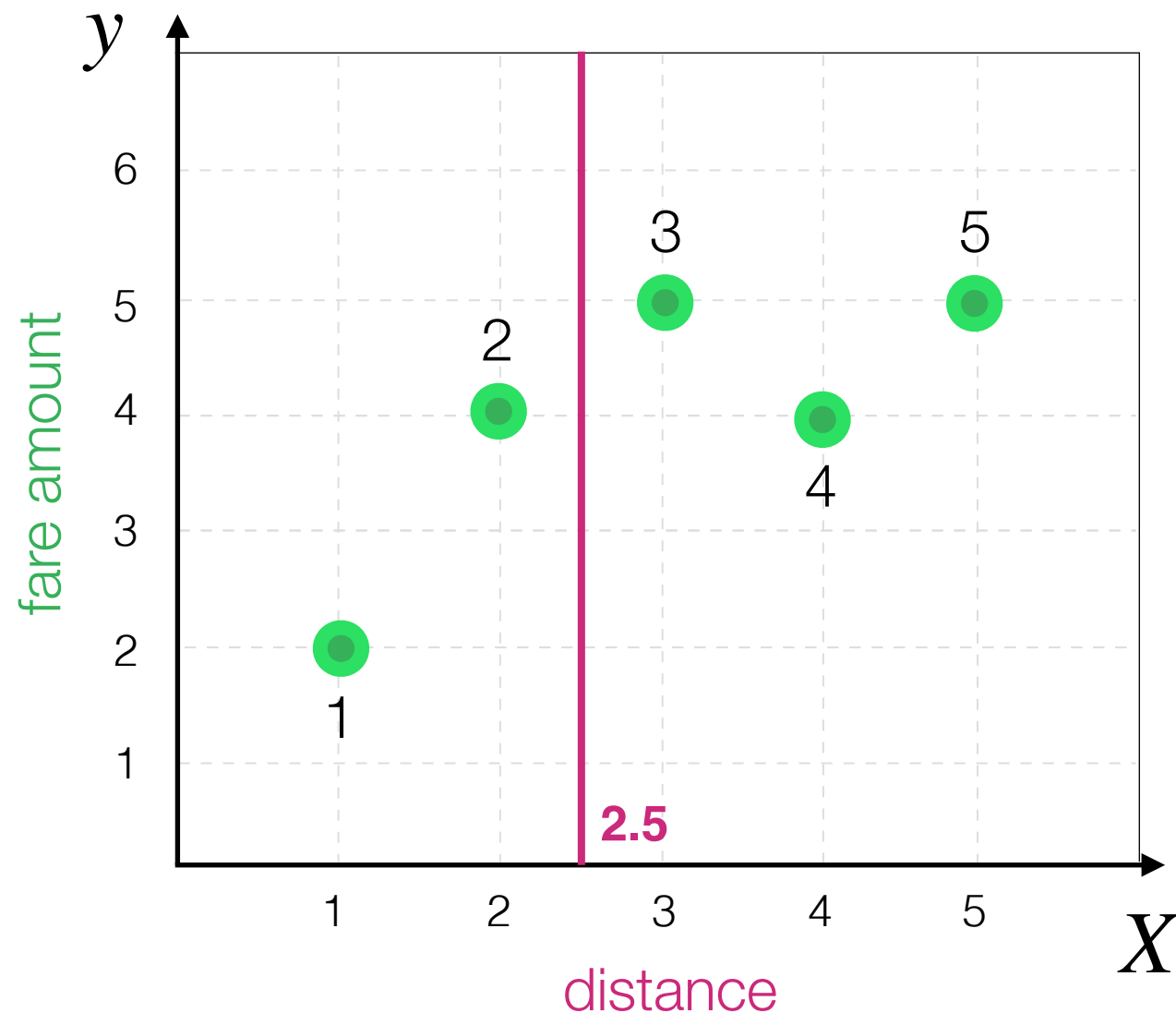
Decision Tree Algorithm

Hold on, how did we choose this **split** on the first place?



Decision Tree Algorithm

Hold on, how did we choose this **split** on the first place?
Maybe there are better options?





If I have seen further than others, it is by standing
upon the shoulders of giants.

(Isaac Newton)

References

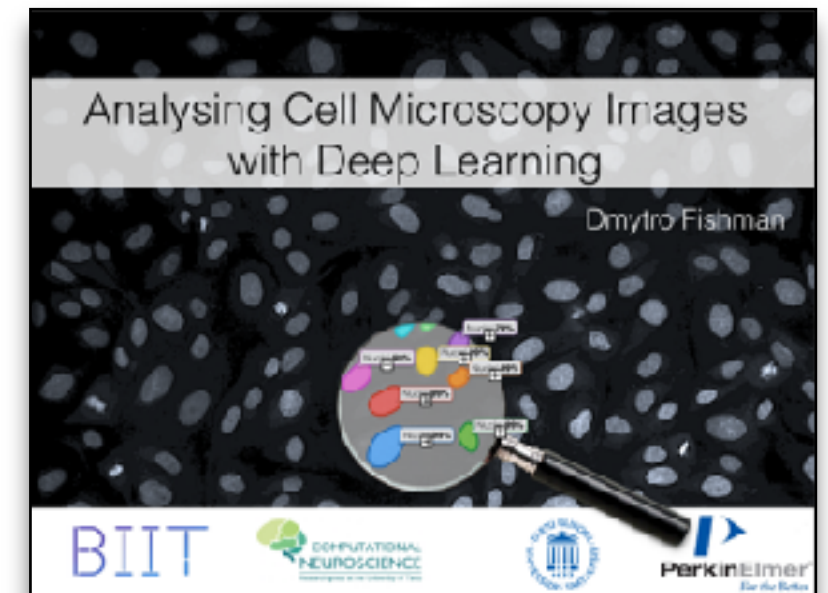
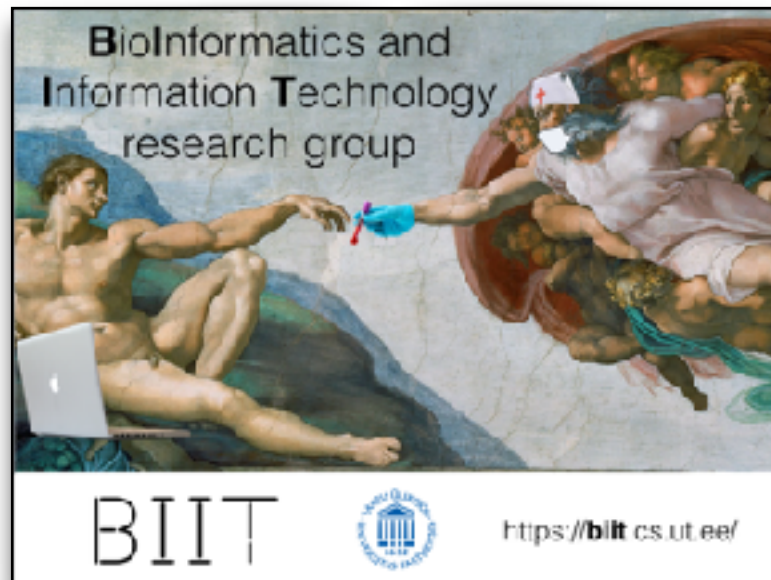
- Machine Learning by **Andrew Ng** (<https://www.coursera.org/learn/machine-learning>)
- Introduction to Machine Learning by **Pascal Vincent** given at Deep Learning Summer School, Montreal 2015 (http://videolectures.net/deeplearning2015_vincent_machine_learning/)
- Welcome to Machine Learning by **Konstantin Tretyakov** delivered at AACIMP Summer School 2015 (<http://kt.era.ee/lectures/aacimp2015/1-intro.pdf>)
- Stanford CS class: Convolutional Neural Networks for Visual Recognition by **Andrej Karpathy** (<http://cs231n.github.io/>)
- Data Mining Course by **Jaak Vilo** at University of Tartu (https://courses.cs.ut.ee/MTAT.03.183/2017_spring/uploads/Main/DM_05_Clustering.pdf)
- Machine Learning Essential Concepts by **Ilya Kuzovkin** (<https://www.slideshare.net/iljakuzovkin>)
- From the brain to deep learning and back by **Raul Vicente Zafra** and **Ilya Kuzovkin** (<http://www.uttv.ee/naita?id=23585&keel=eng>)

That's all Folks!

BIIT



www.slideshare.net/DimaFishman



www.slideshare.net/DimaFishman

